MACHINE LEARNING FOUNDATIONS (MLF)

Comprehensive Categorized Past Year Question Repository (2022 – 2026)
100% Exact Question Paper Cards with Mathematical Precision & Answer Keys

Total Unique Questions: 677	**Exam Sessions Covered:** 44 Papers	**Format:** MCQ, MSQ, NAT / SA
Week 1: ML Foundations (444 Qs)	**Week 2:** Math Foundations (19 Qs)	**Week 3:** Linear Algebra (91 Qs)
Week 4: Regression & Eigenvalues (123 Qs)	**Fidelity:** Official TCS iON Cards	**Interactive Mode:** NTA JEE CBT Online

■ WEEK 1 - ML FOUNDATIONS

■ Category: ML Concepts (415 Questions)

MLF-Q001 | [MCQ] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG1*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 53 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL: MACHINE LEARNING FOUNDATIONS"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

A. ✔ YES

B. ✖ NO

✔ Official Answer Key: Option (A)

MLF-Q002 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG1 (+1 more papers)*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 54 Question Type : SA

Correct Marks : 4

Question Label : Short Answer Question

For the data sets $(x^i, y^i) = [(1,1),(2,3),(3,4),(4,3),(5,6)], i = 1$ to 5, let the regression function be $x+1$. Consider the regression model $f(x) = x+1$. What is the mean squared loss of $f(x)$

NOTE: Enter your answer to the nearest integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1

✔ **Official Answer Key (NAT): 0.55 to 0.65**

MLF-Q003 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG1 (+1 more papers)*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 55 Question Type : SA

Correct Marks : 4

Question Label : Short Answer Question

The coach of indian cricket team hired a ML team to predict whether Kohli will be successful or not in hitting a century in 2022 by using some classification models. Therefore the team collected presence/absence of various factors $x = [x_1, x_2, x_3]$ from both successful and unsuccessful innings by Kohli in past. The data and the corresponding label is shown in the table below.

x	y
$[0,1,1]$	0
$[1,0,1]$	1
$[1,0,0]$	1
$[1,1,0]$	0

Compute the zero-one error of the model if they have used

$$u(z) = \begin{cases} 1, & \text{for } z \geq 0 \\ 0, & \text{otherwise} \end{cases}$$

and $z = 0.5x_1 - 0.8x_2 + 0.4x_3$

NOTE: Enter your answer in two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.22 to 0.28

✔ **Official Answer Key (NAT): 0**

MLF-Q004 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG1*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 58 Question Type : SA

Correct Marks : 5

Question Label : Short Answer Question

The two eigenvalues of the matrix $\begin{bmatrix} 3 & 1 \\ 1 & p \end{bmatrix}$ have a ratio 2:1 for p=3. What is another value of p for which eigenvalues have same ratio 2:1?

NOTE: Enter your answer in one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

4.4 to 4.6

✔ **Official Answer Key (NAT): 4.56 to 4.76**

MLF-Q005 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG1 (+1 more papers)*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 59 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Let $f(x, y, z) = -3x + 4ye^{6z}$. Find directional derivative of f at (1,-1,0) in the direction of unit vector along (1,0,-1)

NOTE: Enter your answer in one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

14.5 to 15.0

✔ **Official Answer Key (NAT): 0.904 to 0.910**

MLF-Q008 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG1 (+1 more papers)*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Find the projection of vector $b = \begin{bmatrix} 2 \\ 3 \\ -1 \\ 2 \end{bmatrix}$ onto the line through vector $a = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix}$

NOTE: If the answer is $[x, y, z, w]^T$, then enter the values of x, y, z and w.

Based on the above data, answer the subquestions.

Sub questions

Question Number : 64 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Enter the value of **x**:

NOTE: Enter your answer in two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.84 to 0.88

✔ **Official Answer Key (NAT): 1**

MLF-Q009 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG1 (+1 more papers)*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Find the projection of vector $b = \begin{bmatrix} 2 \\ 3 \\ -1 \\ 2 \end{bmatrix}$ onto the line through vector $a = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix}$

NOTE: If the answer is $[x, y, z, w]^T$, then enter the values of x, y, z and w.

Based on the above data, answer the subquestions.

Sub questions

Question Number : 65 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Enter the value of **y**:

NOTE: Enter your answer in two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1.71 to 1.75

✔ **Official Answer Key (NAT): 2**

MLF-Q010 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG1 (+1 more papers)*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Find the projection of vector $b = \begin{bmatrix} 2 \\ 3 \\ -1 \\ 2 \end{bmatrix}$ onto the line through vector $a = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix}$

NOTE: If the answer is $[x, y, z, w]^T$, then enter the values of x, y, z and w.

Based on the above data, answer the subquestions.

Sub questions

Question Number : 66 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Enter the value of **z**:

NOTE: Enter your answer in two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.84 to 0.88

✔ **Official Answer Key (NAT): -1**

MLF-Q011 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG1 (+1 more papers)*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

The second degree polynomial of the form $a + bx + cx^2$ which best fits the following data is

x	y
1	3
2	4
3	8

Question Number : 67 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Enter the value of **w**:

NOTE: Enter your answer in one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

2.5 to 2.7

Question Type : COMPREHENSION

Question Numbers : (68 to 70)

Question Label : Comprehension

The second degree polynomial of the form $a + bx + cx^2$ which best fits the following data is

x	y
1	3
2	4
3	8

Based on the above data, answer the subquestions.

Sub questions

✔ Official Answer Key (NAT): 3

MLF-Q012 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG1*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

The second degree polynomial of the form $a + bx + cx^2$ which best fits the following data is

x	y
1	3
2	4
3	8

Question Number : 68 Question Type : SA

Correct Marks : 1.4

Question Label : Short Answer Question

Enter the value of a:

NOTE: Enter your answer to the nearest integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

✔ **Official Answer Key (NAT): 4**

MLF-Q013 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG1*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

The second degree polynomial of the form $a + bx + cx^2$ which best fits the following data is

x	y
1	3
2	4
3	8

Question Number : 69 Question Type : SA

Correct Marks : 1.3

Question Label : Short Answer Question

Enter the value of b:

NOTE: Enter your answer in one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

-3.6 to -3.4

✔ **Official Answer Key (NAT): -3.6 to -3.4**

MLF-Q014 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG1*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

The best fit line of the form mx+c using least squares method for the data set is

x	y
1	3
2	4
3	8
4	11
5	14

Note: Enter m and c of the best fit line respectively.

Based on the above data, answer the subquestions.

Sub questions

Question Number : 70 Question Type : SA

Correct Marks : 1.3

Question Label : Short Answer Question

Enter the value of c:

NOTE: Enter your answer in two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1.45 to 1.55

Question Type : COMPREHENSION

Question Numbers : (71 to 72)

Question Label : Comprehension

The best fit line of the form mx+c using least squares method for the data set is

x	y
1	3
2	4
3	8
4	11
5	14

Note: Enter m and c of the best fit line respectively.

Based on the above data, answer the subquestions.

Sub questions

✔ Official Answer Key (NAT): 1.45 to 1.55

MLF-Q015 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG1*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

The best fit line of the form mx+c using least squares method for the data set is

x	y
1	3
2	4
3	8
4	11
5	14

Note: Enter m and c of the best fit line respectively.

Based on the above data, answer the subquestions.

Sub questions

Question Number : 71 Question Type : SA

Correct Marks : 2.5

Question Label : Short Answer Question

Enter the value of m:

NOTE: Enter your answer in one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

2.8 to 3.0

✔ **Official Answer Key (NAT): 2.8 to 3.0**

MLF-Q016 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG1*

Category: Week 1 - ML Foundations › ML Concepts

Question Type : COMPREHENSION

Question Numbers : (73 to 76)

Question Number : 72 Question Type : SA

Correct Marks : 2.5

Question Label : Short Answer Question

Enter the value of c:

NOTE: Enter your answer in one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

-0.8 to -0.6

Question Type : COMPREHENSION

Question Numbers : (73 to 76)

✔ **Official Answer Key (NAT): -1.8 to -1.6**

MLF-Q017 | [MCQ] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG2*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 53 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL: MACHINE LEARNING FOUNDATIONS"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

A. ✔ YES

B. ✖ NO

✔ Official Answer Key: Option (A)

MLF-Q018 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG2*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 58 Question Type : SA

Correct Marks : 5

Question Label : Short Answer Question

The two eigenvalues of the matrix $\begin{bmatrix} 3 & 1 \\ 1 & p \end{bmatrix}$ have a ratio 2:1 for p=3. What is another value of p for which eigenvalues have same ratio 2:1?

NOTE: Enter your answer in one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

4.4 to 4.6

✔ **Official Answer Key (NAT): 4.4 to 4.6**

MLF-Q019 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG2*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

What is the linear approximation of $f(x_1, x_2) = 2x_1^2 + 2x_2^2$ around (1,1)?
If the approximation is $ax_1 + bx_2 + c$.

Based on the above data, answer the subquestions.

Sub questions

Question Number : 61 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Enter the value of a:

NOTE: Enter your answer to the nearest integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

4

✔ **Official Answer Key (NAT): 4**

MLF-Q020 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG2*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

What is the linear approximation of $f(x_1, x_2) = 2x_1^2 + 2x_2^2$ around (1,1)?
If the approximation is $ax_1 + bx_2 + c$.

Based on the above data, answer the subquestions.

Sub questions

Question Number : 62 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Enter the value of b:

NOTE: Enter your answer to the nearest integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

4

✔ **Official Answer Key (NAT): 4**

MLF-Q021 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG2*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

What is the linear approximation of $f(x_1, x_2) = 2x_1^2 + 2x_2^2$ around (1,1)?
If the approximation is $ax_1 + bx_2 + c$.

Based on the above data, answer the subquestions.

Sub questions

Question Number : 63 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Enter the value of c:

NOTE: Enter your answer to the nearest integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

-4

Question Type : COMPREHENSION

Question Numbers : (64 to 67)

✔ **Official Answer Key (NAT): -4**

MLF-Q022 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG2*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Find the projection of vector $b = \begin{bmatrix} 2 \\ 3 \\ -1 \\ 2 \end{bmatrix}$ onto the line through vector $a = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix}$

NOTE: If the answer is $[x, y, z, w]^T$, then enter the values of x, y, z and w.

Based on the above data, answer the subquestions.

Sub questions

Question Number : 64 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Enter the value of **x**:

NOTE: Enter your answer in two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.84 to 0.88

✔ **Official Answer Key (NAT): 0.84 to 0.88**

MLF-Q023 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG2*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Find the projection of vector $b = \begin{bmatrix} 2 \\ 3 \\ -1 \\ 2 \end{bmatrix}$ onto the line through vector $a = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix}$

NOTE: If the answer is $[x, y, z, w]^T$, then enter the values of x, y, z and w.

Based on the above data, answer the subquestions.

Sub questions

Question Number : 65 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Enter the value of **y**:

NOTE: Enter your answer in two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1.71 to 1.75

✔ **Official Answer Key (NAT): 1.71 to 1.75**

MLF-Q024 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG2*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Find the projection of vector $b = \begin{bmatrix} 2 \\ 3 \\ -1 \\ 2 \end{bmatrix}$ onto the line through vector $a = \begin{bmatrix} 1 \\ 2 \\ 1 \\ 3 \end{bmatrix}$

NOTE: If the answer is $[x, y, z, w]^T$, then enter the values of x, y, z and w.

Based on the above data, answer the subquestions.

Sub questions

Question Number : 66 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Enter the value of **z**:

NOTE: Enter your answer in two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.84 to 0.88

✔ **Official Answer Key (NAT): 0.84 to 0.88**

MLF-Q025 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG2*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

The second degree polynomial of the form $a + bx + cx^2$ which best fits the following data is

x	y
1	3
2	4
3	8

Question Number : 67 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Enter the value of **w**:

NOTE: Enter your answer in one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

2.5 to 2.7

Question Type : COMPREHENSION

Question Numbers : (68 to 70)

Question Label : Comprehension

The second degree polynomial of the form $a + bx + cx^2$ which best fits the following data is

x	y
1	3
2	4
3	8

Based on the above data, answer the subquestions.

Sub questions

✔ Official Answer Key (NAT): 2.5 to 2.7

MLF-Q026 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG2*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

The second degree polynomial of the form $a + bx + cx^2$ which best fits the following data is

x	y
1	3
2	4
3	8

Question Number : 69 Question Type : SA

Correct Marks : 1.3

Question Label : Short Answer Question

Enter the value of b:

NOTE: Enter your answer in one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

-3.6 to -3.4

✔ **Official Answer Key (NAT): -3.6 to -3.4**

MLF-Q027 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG2*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

The best fit line of the form mx+c using least squares method for the data set is

x	y
1	3
2	4
3	8
4	11
5	14

Note: Enter m and c of the best fit line respectively.

Based on the above data, answer the subquestions.

Sub questions

Question Number : 70 Question Type : SA

Correct Marks : 1.3

Question Label : Short Answer Question

Enter the value of c:

NOTE: Enter your answer in two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

1.45 to 1.55

Question Type : COMPREHENSION

Question Numbers : (71 to 72)

Question Label : Comprehension

The best fit line of the form mx+c using least squares method for the data set is

x	y
1	3
2	4
3	8
4	11
5	14

Note: Enter m and c of the best fit line respectively.

Based on the above data, answer the subquestions.

Sub questions

✔ Official Answer Key (NAT): 1.45 to 1.55

MLF-Q028 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG2*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

The best fit line of the form mx+c using least squares method for the data set is

x	y
1	3
2	4
3	8
4	11
5	14

Note: Enter m and c of the best fit line respectively.

Based on the above data, answer the subquestions.

Sub questions

Question Number : 71 Question Type : SA

Correct Marks : 2.5

Question Label : Short Answer Question

Enter the value of m:

NOTE: Enter your answer in one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

2.8 to 3.0

✔ **Official Answer Key (NAT): 2.8 to 3.0**

MLF-Q029 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG2*

Category: Week 1 - ML Foundations › ML Concepts

Question Type : COMPREHENSION

Question Numbers : (73 to 76)

Question Number : 72 Question Type : SA

Correct Marks : 2.5

Question Label : Short Answer Question

Enter the value of c:

NOTE: Enter your answer in one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

-0.8 to -0.6

Question Type : COMPREHENSION

Question Numbers : (73 to 76)

✔ **Official Answer Key (NAT): -0.8 to -0.6**

MLF-Q030 | [MCQ] | Marks: +2 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 122 Question Id : 640653345257 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "MACHINE LEARNING FOUNDATIONS"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406531148395. ✔ YES

6406531148396. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	64065348777
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLF-Q031 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 123 Question Id : 640653345258 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

For the data sets $(x^i, y^i) = [(1,1),(2,3),(3,4),(4,4),(5,7)], i = 1$ to 5, Consider the regression model $f(x) = x + 2$. What is the mean squared loss of $f(x)$

NOTE: Enter your answer to the nearest integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2

✔ **Official Answer Key (NAT): 2**

MLF-Q033 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 125 Question Id : 640653345262 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

What will be the directional derivative of $f(x, y, z) = x^2y^3 - 4xz$ at point $(1, 1, 1)$ in the direction of $[-1, 2, 0]$?

NOTE: Enter your answer in one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

3.3 to 3.7

✔ **Official Answer Key (NAT): 3.3 to 3.7**

MLF-Q034 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 126 Question Id : 640653345263 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

For what value of k, will the following function be continuous?

$$f(x) = \begin{cases} 3x^2 - kx + 5, & \text{if } 0 \leq x < 2 \\ 1 - 3x, & \text{if } 2 \leq x \leq 3 \end{cases}$$

NOTE: Enter your answer to the nearest integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

11

✔ **Official Answer Key (NAT): 11**

MLF-Q035 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 127 Question Id : 640653345264 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

Find the linear approximation (L) of $f(x_1, x_2) = 2x_1^2 + 4x_2^4$ around (1, 1), and use it to compute L(1.1, 1.1). What will be the value of $f(1.1, 1.1) - L(1.1, 1.1)$?

NOTE: Enter your answer in two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.20 to 0.35

✔ **Official Answer Key (NAT): 0.20 to 0.35**

MLF-Q036 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 128 Question Id : 640653345269 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

The eigenvalues of the matrix $\begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & 0 \\ -2 & 2 & 1 \end{bmatrix}$ are a, b, c in increasing order, then the value of $a + b$ is

NOTE: Enter your answer to the nearest integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0

✔ **Official Answer Key (NAT): 0**

MLF-Q037 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 129 Question Id : 640653345270 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

If $A = PDP^{-1}$ where $P = \begin{bmatrix} 1 & 2 \\ 1 & -5 \end{bmatrix}$ and $D = \begin{bmatrix} -1 & 0 \\ 0 & 6 \end{bmatrix}$. Calculate A^4 and enter the summation of all the elements of A^4

NOTE: Enter your answer to the nearest integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1297

✔ **Official Answer Key (NAT): 1297**

MLF-Q038 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 130 Question Id : 640653345271 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

Find the values of α and β for which the matrix $A = \begin{bmatrix} 2 & \alpha \\ 2 & \beta \end{bmatrix}$ has eigenvalues equal to -1 and -3. Enter your answer as $2 \times \alpha + \beta$

NOTE: Enter your answer to the nearest integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

-21

Sub-Section Number :	3
Sub-Section Id :	64065348778
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): -21**

MLF-Q040 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 132 Question Id : 640653345261 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

Consider a function

$$f(x) = \begin{cases} \frac{x-|x|}{x}, & \text{if } x \neq 0 \\ 2, & \text{if } x = 0 \end{cases}$$

Is f continuous?

(Provide 1 as answer for 'Yes' and 0 for 'No'.)

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0

Sub-Section Number :	5
Sub-Section Id :	64065348780
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 0**

MLF-Q041 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 133 Question Id : 640653345265 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

For the matrix $S = \begin{bmatrix} 2 & -1 & 3 \\ 1 & 0 & 1 \\ 0 & 2 & -1 \\ 1 & 1 & 4 \end{bmatrix}$ what is the dimension of its row space?

NOTE: Enter your answer to the nearest integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3

Sub-Section Number :	6
Sub-Section Id :	64065348781
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 3**

MLF-Q042 | [MCQ] | Marks: +2 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 134 Question Id : 640653345266 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction

Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Consider the system $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 7 \\ 2 & 4 & 6 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$.

Under what condition on b_1, b_2, b_3 is this system solvable?

Options :

6406531148408. ✖ $b_2 = 2b_1$

6406531148409. ✖ $b_2 = 3b_1$

6406531148410. ✔ $b_3 = 2b_1$

6406531148411. ✖ $b_3 = 3b_1$

Sub-Section Number :	7
Sub-Section Id :	64065348782
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (C)

MLF-Q043 | [MCQ] | Marks: +2 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 135 Question Id : 640653345267 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Calculate the projection of vector $(2, -1, 3)^T$ onto another vector $(-1, 3, 1)^T$.

Which of the following is the squared length of the projection vector?

Options :

6406531148412. ✖ 0.8686

6406531148413. ✔ 0.3636

6406531148414. ✖ 1.0363

6406531148415. ✖ 1.3636

✔ Official Answer Key: Option (B)

MLF-Q045 | [MCQ] | Marks: +2 | *Quiz 1 Jan23*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 120 Question Id : 640653470046 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406531562038. ✔ YES

6406531562039. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	64065367686
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key: Option (A)**

MLF-Q049 | [MSQ] | Marks: +2 | *Quiz 1 Jan23*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 124 Question Id : 640653470059 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Selectable Option : 0

Question Label : Multiple Select Question

Which of the following statements is/are true?

Options :

6406531562077. ✔ There exists a $n \times n$ matrix A, n is odd, for which the rank of A is not equal to the nullity of A.

6406531562078. ✖ There exists a $n \times n$ matrix A, n is odd, for which the rank of A is equal to the nullity of A.

6406531562079. ✔ There exists a $n \times n$ matrix A, n is even, for which the rank of A is not equal to the nullity of A.

6406531562080. ✔ There exists a $n \times n$ matrix A, n is even, for which the rank of A is equal to the nullity of A.

✔ **Official Answer Key: Option (A, C, D)**

MLF-Q050 | [MSQ] | Marks: +2 | *Quiz 1 Jan23*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 125 Question Id : 640653470061 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Selectable Option : 0

Question Label : Multiple Select Question

Let P be the matrix that projects vectors in $\mathbb{R}^3$ onto the subspace spanned by the vector $a = [1, 2, 3]^T$. Which of the following are eigenvectors of P corresponding to the eigenvalue 0?

Options :

6406531562085. ✖ [1,−2, 2]

6406531562086. ✖ [2, 2,−1]

6406531562087. ✔ [−2, 1, 0]

6406531562088. ✔ [−3, 0, 1]

Sub-Section Number :	4
Sub-Section Id :	64065367688
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (C, D)

MLF-Q051 | [MCQ] | Marks: +2 | *Quiz 1 Jan23*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 126 Question Id : 640653470052 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

A quadratic approximation of the function $f(x) = \sin^2 x$ at $x = \frac{\pi}{4}$ radian is:

Options :

6406531562054. ✖ $Q_{\frac{\pi}{4}}[f(x)] = \frac{1}{2} + (x - \frac{\pi}{4}) + (x - \frac{\pi}{4})$

6406531562055. ✔ $Q_{\frac{\pi}{4}}[f(x)] = \frac{1}{2} + (x - \frac{\pi}{4})$

6406531562056. ✖ $Q_{\frac{\pi}{4}}[f(x)] = \frac{1}{2} + 2(x - \frac{\pi}{4}) + \frac{1}{2}(x - \frac{\pi}{4})$

6406531562057. ✖ $Q_{\frac{\pi}{4}}[f(x)] = \frac{\pi}{2} + 2(x - \frac{\pi}{4}) + \frac{1}{2}(x - \frac{\pi}{4})$

✔ **Official Answer Key: Option (B)**

MLF-Q052 | [MCQ] | Marks: +2 | *Quiz 1 Jan23*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 127 Question Id : 640653470053 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

The direction of steepest descent for the function $f(x, y, z) = \cos(x)\sin(y)\cos(z)$ at the point $(x, y, z) = \left(\frac{\pi}{4}, \frac{\pi}{6}, \frac{\pi}{4}\right)$ is?

Options :

6406531562058. ✖ $\left[-\frac{1}{4}, \frac{\sqrt{3}}{4}, -\frac{1}{4}\right]$

6406531562059. ✔ $\left[\frac{1}{4}, -\frac{\sqrt{3}}{4}, \frac{1}{4}\right]$

6406531562060. ✖ $\left[\frac{1}{4}, \frac{1}{4}, \frac{1}{4}\right]$

6406531562061. ✖ $\left[-\frac{1}{4}, -\frac{1}{4}, -\frac{1}{4}\right]$

✔ Official Answer Key: Option (B)

MLF-Q053 | [MCQ] | Marks: +2 | *Quiz 1 Jan23*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 128 Question Id : 640653470054 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Which of the following is the equation of the line that passes through point $[1,2,3,4]$ and is tangent to the function $f(p,q,r,s) = p^2+q^2+r^2+s^2+pq+rs-2p-2q$ at the point $[p,q,r,s] = [1,3,5,1]$?

Options :

6406531562062. ✔ $[1,2,3,4] + \alpha[3,5,11,7]$

6406531562063. ✖ $[3,5,11,7] + \alpha[1,2,3,4]$

6406531562064. ✖ $[1,2,3,4] + \alpha[0,4,10,2]$

6406531562065. ✖ $[1,2,3,4] + \alpha[5,3,11,7]$

6406531562066. ✖ $[5,3,11,7] + \alpha[1,2,3,4]$

✔ **Official Answer Key: Option (A)**

MLF-Q055 | [MCQ] | Marks: +2 | *Quiz 1 Jan23*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 130 Question Id : 640653470056 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

If null space of a matrix A is span$\left\{\begin{bmatrix}7/17\\22/17\\-1\end{bmatrix}\right\}$, then which of the following can be row space of matrix A?

Options :

6406531562068. ✔ $\text{Span}\left\{\begin{bmatrix}1\\0\\\frac{7}{17}\end{bmatrix}, \begin{bmatrix}0\\1\\\frac{22}{17}\end{bmatrix}\right\}$

6406531562069. ✖ $\text{Span}\left\{\begin{bmatrix}-1\\0\\\frac{7}{17}\end{bmatrix}, \begin{bmatrix}0\\1\\\frac{22}{17}\end{bmatrix}\right\}$

6406531562070. ✖ $\text{Span}\left\{\begin{bmatrix}1\\0\\\frac{7}{17}\end{bmatrix}, \begin{bmatrix}0\\-1\\\frac{22}{17}\end{bmatrix}\right\}$

6406531562071. ✖ $\text{Span}\left\{\begin{bmatrix}-1\\0\\\frac{7}{17}\end{bmatrix}, \begin{bmatrix}0\\-1\\\frac{22}{17}\end{bmatrix}\right\}$

✔ Official Answer Key: Option (A)

MLF-Q056 | [MCQ] | Marks: +2 | *Quiz 1 Jan23*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 131 Question Id : 640653470058 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Let $A = \begin{bmatrix} 1 & 1 \\ 2 & 0 \\ 3 & 1 \\ 5 & 1 \end{bmatrix}$, then which among the following is orthogonal to $C(A)$?

Options :

6406531562073. ✔ $\text{Span}\left\{\begin{bmatrix} -3 \\ -5 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix}\right\}$

6406531562074. ✖ $\text{Span}\left\{\begin{bmatrix} 3 \\ 5 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} -1 \\ -1 \\ 1 \\ 0 \end{bmatrix}\right\}$

6406531562075. ✖ $\text{Span}\left\{\begin{bmatrix} 3 \\ 5 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}\right\}$

6406531562076. ✖ $\text{Span}\left\{\begin{bmatrix} -3 \\ -5 \\ 1 \\ 2 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ 1 \\ 0 \end{bmatrix}\right\}$

Sub-Section Number :	6
Sub-Section Id :	64065367690
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A)

MLF-Q057 | [SA] | Marks: +2 | *Quiz 1 Jan23*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 132 Question Id : 640653470049 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

For the data set $\{(x^i, y^i)\} = \{(1,1),(2,6),(3,8),(4,15),(5,26)\}, i = 1$ to 5, Consider the regression model $f(x) = x^2 + 1$. What is the mean squared loss of $f(x)$?(Round your answer off to the nearest integer).

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

2

✔ **Official Answer Key (NAT): 2**

MLF-Q058 | [SA] | Marks: +2 | *Quiz 1 Jan23*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 133 Question Id : 640653470050 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

What will be the average misclassification error when the functions $g(X) = sign((x_1 - x_2)^2 - 8)$ is used to classify the data points into classes +1 or −1. Enter the answer closest to one decimal accuracy.

X	y
(5,2)	1
(5,6)	-1
(6,3)	1
(6,4)	-1
(4,8)	-1

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.2

✔ **Official Answer Key (NAT): 0.2**

MLF-Q059 | [SA] | Marks: +2 | *Quiz 1 Jan23*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 134 Question Id : 640653470055 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

For a 3×3 matrix A whose nullity is equal to 1, what is the value of $R(A) + \dim(C(A)) + \dim(C(A^T))$?
Here $R(A)$ stands for Rank of A, $\dim(C(A))$ stands for dimension of column space of A,
and $\dim(C(A^T))$ stands for dimension of column space of A^T. Enter the answer as an integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

6

Sub-Section Number :	7
Sub-Section Id :	64065367691
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key (NAT): 6**

MLF-Q060 | [SA] | Marks: +2 | *Quiz 1 Jan23*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 135 Question Id : 640653470057 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

If projection of vector A of length 10 units, onto vector B of length 8 units, has a magnitude of 5 units, then length of projection of vector B onto A is?

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

4

✔ **Official Answer Key (NAT): 4**

MLF-Q061 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 134 Question Id : 640653737411 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406532467966. ✔ YES

6406532467967. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653107631
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653737412 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (135 to 136)

✔ Official Answer Key: Option (A)

MLF-Q064 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 137 Question Id : 640653737417 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$. Find a matrix P such that $P^{-1}AP$ is diagonal.

Options :

6406532467984. ✖ $\begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$

6406532467985. ✔ $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

6406532467986. ✖ $\begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$

6406532467987. ✖ $\begin{bmatrix} -2 & -2 \\ 0 & 0 \end{bmatrix}$

✔ **Official Answer Key: Option (B)**

MLF-Q065 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 138 Question Id : 640653737423 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following is the best approximation of $\sqrt[4]{16.1}$? (Use linear approximation around 16).

Options :

6406532468005. ✖ 2.1

6406532468006. ✔ 2.003125

6406532468007. ✖ 2.031

6406532468008. ✖ 1.9968

Sub-Section Number :	4
Sub-Section Id :	640653107633
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key: Option (B)**

MLF-Q067 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 140 Question Id : 640653737419 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a multivariable scalar-valued continuous function, such that

$\frac{\partial f}{\partial x} = 2x + 1$ and $\frac{\partial f}{\partial y} = 2y + 1$.

Suppose $\lim_{(x,y)\to(0,0)} f(x,y) = 5$.

Find out the directional derivative of f at the point $(0,0)$ in the direction of the vector $(1,1)$.

Options :

6406532467992. ✔ $\sqrt{2}$

6406532467993. ✖ $2\sqrt{2}$

6406532467994. ✖ $-2\sqrt{2}$

6406532467995. ✖ $-\sqrt{2}$

Sub-Section Number :	5
Sub-Section Id :	640653107634
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A)

MLF-Q071 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 144 Question Id : 640653737422 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a function $f : \mathbb{R} \longrightarrow \mathbb{R}$ defined as $f(x) = |x(x-2)|$

Which of the following options is/are true?

Options :

6406532468001. ✔ $\lim_{x \to 2^+} f(x) = \lim_{x \to 2^-} f(x)$.

6406532468002. ✖ f is differentiable at $x = 0$.

6406532468003. ✔ $f'(3) = 4$.

6406532468004. ✔ f is continuous at $x = 0$.

Sub-Section Number :	7
Sub-Section Id :	640653107636
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A, C, D)

MLF-Q072 | [SA] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 145 Question Id : 640653737420 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

Consider an encoder-decoder pair used for dimensionality reduction, where the encoder function is denoted as $f(x_1, x_2, x_3) = (x_1 + x_2 - x_3, x_1 + x_3)$ and the decoder function is denoted as $g(u) = (u_1, u_2, u_1 + u_2)$

Compute the reconstruction error $\left(R(f,g) = \frac{1}{n}\sum_{i=1}^{n} \|X_i - g(f(X_i))\|^2\right)$ for the following dataset:

$$X_1 = [1,2,3], \quad X_2 = [0,2,2], \quad X_3 = [1,0,-1]$$

Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

3.12 to 3.20

✔ **Official Answer Key (NAT): 3.12 to 3.20**

MLF-Q073 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 134 Question Id : 640653737697 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406532468790. ✔ YES

6406532468791. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653107744
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653737698 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (135 to 136)

✔ Official Answer Key: Option (A)

MLF-Q076 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 137 Question Id : 640653737703 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$. Find a matrix P such that $P^{-1}AP$ is diagonal.

Options :

6406532468808. ✖ $\begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$

6406532468809. ✔ $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

6406532468810. ✖ $\begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$

6406532468811. ✖ $\begin{bmatrix} -2 & -2 \\ 0 & 0 \end{bmatrix}$

✔ **Official Answer Key: Option (B)**

MLF-Q077 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 138 Question Id : 640653737709 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following is the best approximation of $\sqrt[4]{16.1}$? (Use linear approximation around 16).

Options :

6406532468829. ✖ 2.1

6406532468830. ✔ 2.003125

6406532468831. ✖ 2.031

6406532468832. ✖ 1.9968

Sub-Section Number :	4
Sub-Section Id :	640653107746
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key: Option (B)**

MLF-Q079 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 140 Question Id : 640653737705 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a multivariable scalar-valued continuous function, such that

$\frac{\partial f}{\partial x} = 2x + 1$ and $\frac{\partial f}{\partial y} = 2y + 1$.

Suppose $\lim_{(x,y)\to(0,0)} f(x,y) = 5$.

Find out the directional derivative of f at the point $(0,0)$ in the direction of the vector $(1,1)$.

Options :

6406532468816. ✔ $\sqrt{2}$

6406532468817. ✖ $2\sqrt{2}$

6406532468818. ✖ $-2\sqrt{2}$

6406532468819. ✖ $-\sqrt{2}$

Sub-Section Number :	5
Sub-Section Id :	640653107747
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A)

MLF-Q083 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 144 Question Id : 640653737708 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a function $f : \mathbb{R} \longrightarrow \mathbb{R}$ defined as $f(x) = |x(x-2)|$

Which of the following options is/are true?

Options :

6406532468825. ✔ $\lim_{x \to 2^+} f(x) = \lim_{x \to 2^-} f(x)$.

6406532468826. ✖ f is differentiable at $x = 0$.

6406532468827. ✔ $f'(3) = 4$.

6406532468828. ✔ f is continuous at $x = 0$.

Sub-Section Number :	7
Sub-Section Id :	640653107749
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A, C, D)

MLF-Q084 | [SA] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 145 Question Id : 640653737706 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

Consider an encoder-decoder pair used for dimensionality reduction, where the encoder function is denoted as $f(x_1, x_2, x_3) = (x_1 + x_2 - x_3, x_1 + x_3)$ and the decoder function is denoted as $g(u) = (u_1, u_2, u_1 + u_2)$

Compute the reconstruction error $\left(R(f,g) = \frac{1}{n}\sum_{i=1}^{n} \|X_i - g(f(X_i))\|^2\right)$ for the following dataset:

$$X_1 = [1,2,3], \quad X_2 = [0,2,2], \quad X_3 = [1,0,-1]$$

Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

3.12 to 3.20

✔ **Official Answer Key (NAT): 3.12 to 3.20**

MLF-Q085 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QDD4 25 Feb 2024*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 134 Question Id : 640653737983 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

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CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406532469614. ✔ YES

6406532469615. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653107857
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653737984 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (135 to 136)

✔ Official Answer Key: Option (A)

MLF-Q088 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QDD4 25 Feb 2024*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 137 Question Id : 640653737989 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Let $A = \begin{bmatrix} 1 & 2 \\ 0 & 3 \end{bmatrix}$. Find a matrix P such that $P^{-1}AP$ is diagonal.

Options :

6406532469632. ✖ $\begin{bmatrix} 1 & 0 \\ 2 & 3 \end{bmatrix}$

6406532469633. ✔ $\begin{bmatrix} 1 & 1 \\ 1 & 0 \end{bmatrix}$

6406532469634. ✖ $\begin{bmatrix} 1 & 1 \\ -1 & 0 \end{bmatrix}$

6406532469635. ✖ $\begin{bmatrix} -2 & -2 \\ 0 & 0 \end{bmatrix}$

✔ **Official Answer Key: Option (B)**

MLF-Q089 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QDD4 25 Feb 2024*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 138 Question Id : 640653737995 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following is the best approximation of $\sqrt[4]{16.1}$? (Use linear approximation around 16).

Options :

6406532469653. ✖ 2.1

6406532469654. ✔ 2.003125

6406532469655. ✖ 2.031

6406532469656. ✖ 1.9968

Sub-Section Number :	4
Sub-Section Id :	640653107859
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key: Option (B)**

MLF-Q091 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QDD4 25 Feb 2024*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 140 Question Id : 640653737991 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Let $f : \mathbb{R}^2 \to \mathbb{R}$ be a multivariable scalar-valued continuous function, such that

$\frac{\partial f}{\partial x} = 2x + 1$ and $\frac{\partial f}{\partial y} = 2y + 1$.

Suppose $\lim_{(x,y)\to(0,0)} f(x,y) = 5$.

Find out the directional derivative of f at the point $(0,0)$ in the direction of the vector $(1,1)$.

Options :

6406532469640. ✔ $\sqrt{2}$

6406532469641. ✖ $2\sqrt{2}$

6406532469642. ✖ $-2\sqrt{2}$

6406532469643. ✖ $-\sqrt{2}$

Sub-Section Number :	5
Sub-Section Id :	640653107860
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A)

MLF-Q095 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QDD4 25 Feb 2024*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 144 Question Id : 640653737994 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a function $f : \mathbb{R} \longrightarrow \mathbb{R}$ defined as $f(x) = |x(x-2)|$

Which of the following options is/are true?

Options :

6406532469649. ✔ $\lim_{x \to 2^+} f(x) = \lim_{x \to 2^-} f(x)$.

6406532469650. ✖ f is differentiable at $x = 0$.

6406532469651. ✔ $f'(3) = 4$.

6406532469652. ✔ f is continuous at $x = 0$.

Sub-Section Number :	7
Sub-Section Id :	640653107862
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A, C, D)

MLF-Q096 | [SA] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QDD4 25 Feb 2024*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 145 Question Id : 640653737992 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

Consider an encoder-decoder pair used for dimensionality reduction, where the encoder function is denoted as $f(x_1, x_2, x_3) = (x_1 + x_2 - x_3, x_1 + x_3)$ and the decoder function is denoted as $g(u) = (u_1, u_2, u_1 + u_2)$

Compute the reconstruction error $\left(R(f,g) = \frac{1}{n}\sum_{i=1}^{n} \|X_i - g(f(X_i))\|^2\right)$ for the following dataset:

$$X_1 = [1,2,3], \quad X_2 = [0,2,2], \quad X_3 = [1,0,-1]$$

Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

3.12 to 3.20

✔ **Official Answer Key (NAT): 3.12 to 3.20**

MLF-Q097 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 83 Question Id : 6406531113776 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)
Options :
6406533774204. ✖ YES

6406533774205. ✔ NO

Sub-Section Number :	2
Sub-Section Id :	640653168515
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B)**

MLF-Q098 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 84 Question Id : 6406531113777 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Consider the function $f(x, y) = \sin(xy)$. Let $L(x, y)$ be the linear approximation of $f(x, y)$ around the point $\left(\frac{\pi}{2}, 0\right)$. What is the value of $L\left(\frac{\pi}{2} + 0.1, 0.1\right)$?
hint: Use the approximation of $\pi = 3.14$
Options :
6406533774206. ✖ 0.162
6406533774207. ✖ 0.257
6406533774208. ✔ 0.157
6406533774209. ✖ 1.57

✔ **Official Answer Key: Option (C)**

MLF-Q100 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 86 Question Id : 6406531113779 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
The directional derivative of a scalar function $f(x, y)$ at a point (x_0, y_0) in the direction of a unit vector u represents:

Options :
6406533774214. ✔
The rate of change of f in the direction of u at (x_0, y_0).

6406533774215. ✖ The magnitude of the gradient vector ∇f at (x_0, y_0).

6406533774216. ✖ The maximum value of the gradient at (x_0, y_0).

6406533774217. ✖ The angle between the vector u and ∇f.

✔ Official Answer Key: Option (A)

MLF-Q101 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 87 Question Id : 6406531113780 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Which of the following is the best second-order polynomial that fits the dataset by the least sum of squares method?

$$\min_{\theta_0,\theta_1,\theta_2} \sum_{i=1}^{3} (y_i - (\theta_0 + \theta_1 x_i + \theta_2 x_i^2))^2$$

x	1	0	-1
y	-1	1	1

Options :

6406533774218. ✖ $2 - \frac{1}{2}x + \frac{5}{2}x^2$

6406533774219. ✔ $2 - \frac{1}{2}x - \frac{5}{2}x^2$

6406533774220. ✖ $2 + \frac{1}{2}x - \frac{5}{2}x^2$

6406533774221. ✖ $2 + \frac{1}{2}x + \frac{5}{2}x^2$

Sub-Section Number :	3
Sub-Section Id :	640653168516
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B)**

MLF-Q102 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 88 Question Id : 6406531113781 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question
Which of the following options is/are true?
Options :

6406533774222. ✔ $\begin{bmatrix}0\\0\\1\end{bmatrix}$ is orthogonal to the vectors $\begin{bmatrix}1\\0\\0\end{bmatrix}$ and $\begin{bmatrix}1\\1\\0\end{bmatrix}$.

6406533774223. ✖ For a matrix A, vectors of column space of A are perpendicular to the vectors of null space of A.

6406533774224. ✖ Length of the vector, $a = \begin{bmatrix}1\\1\\-1\end{bmatrix}$ is 3.

6406533774225. ✔ $\begin{bmatrix}1\\2\\3\end{bmatrix}$ and $\begin{bmatrix}1\\1\\-1\end{bmatrix}$ are orthogonal vectors.

✔ Official Answer Key: Option (A, D)

MLF-Q103 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 89 Question Id : 6406531113782 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Which of the following options is/are true when the vector $\begin{bmatrix}1\\2\\1\end{bmatrix}$ projected onto the vector $\begin{bmatrix}1\\-1\\0\end{bmatrix}$?

Options :

6406533774226. ✖ The projection matrix is $\frac{1}{2}\begin{bmatrix}-1 & -1 & 0\\1 & 1 & 0\\0 & 0 & 0\end{bmatrix}$

6406533774227. ✔ The projection matrix is $\frac{1}{2}\begin{bmatrix}1 & -1 & 0\\-1 & 1 & 0\\0 & 0 & 0\end{bmatrix}$

6406533774228. ✔ Projected vector is $\frac{1}{2}\begin{bmatrix}-1\\1\\0\end{bmatrix}$

6406533774229. ✖ Projected vector is $\frac{1}{2}\begin{bmatrix}1\\-1\\0\end{bmatrix}$

✔ **Official Answer Key: Option (B, C)**

MLF-Q104 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 90 Question Id : 6406531113784 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following statements is/are correct?
Options :

6406533774234. ✖ $f(x) = \ln(x)$ is continuous and differentiable in the domain $[0, \infty)$.

6406533774235. ✔ $f(x) = e^x$ is continuous and differentiable for all $x \in \mathbb{R}$.

6406533774236. ✔ $f(x) = \sqrt{x}$ is continuous but not differentiable at $x = 0$.

6406533774237. ✔ $f(x) = \frac{1}{x}$ is continuous and differentiable for all $x \neq 0$.

Sub-Section Number :	4
Sub-Section Id :	640653168517
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B, C, D)

MLF-Q105 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 91 Question Id : 6406531113783 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider a matrix

$$A = \begin{bmatrix} 2 & 5 \\ 6 & 15 \end{bmatrix}.$$

Which of the following options is/are true?

Options :

6406533774230. ✔ Null space of A is span $\left\{\begin{bmatrix} -5 \\ 2 \end{bmatrix}\right\}$

6406533774231. ✔ Column space of A is span $\left\{\begin{bmatrix} 1 \\ 3 \end{bmatrix}\right\}$

6406533774232. ✔ Left null space of A is span $\left\{\begin{bmatrix} -3 \\ 1 \end{bmatrix}\right\}$

6406533774233. ✖ Row space of A is span $\left\{\begin{bmatrix} 2 \\ -5 \end{bmatrix}\right\}$

Sub-Section Number :	5
Sub-Section Id :	640653168518
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A, B, C)**

MLF-Q106 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 92 Question Id : 6406531113785 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question
Consider a matrix $A_{20\times25}$. If the nullity of the matrix is 10, then find the rank of the matrix.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
15

Sub-Section Number :	6
Sub-Section Id :	640653168519
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 15**

MLF-Q107 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 93 Question Id : 6406531113786 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question
If $f(x, y, z) = \sin(xy) + e^{yz} + xz^2$, what is the length of the gradient at the point, $(1, 0, 2)$? Enter the answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
6.37 to 6.43

Sub-Section Number :	7
Sub-Section Id :	640653168520
Question Shuffling Allowed :	No

Question Id : 6406531113787 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (94 to 95)
✔ **Official Answer Key (NAT): 6.37 to 6.43**

MLF-Q108 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

A bank wants to predict whether a customer will default on a loan using a classification model. The bank collects data from past customers, which includes the following features:

- x_1: Credit score (scaled between 0 and 1)
- x_2: Monthly income (in \$1000s)
- x_3: Number of late payments in the past year

The target variable y represents whether the customer defaulted (1 for default, 0 for no default). The data is provided in the table below:

x	y
[0.9, 5, 1]	0
[0.4, 3, 5]	1
[0.8, 4, 2]	0
[0.2, 2, 7]	1
[0.7, 6, 0]	0

The bank uses the following linear combination to compute z:

$$z = 0.5x_1 + 0.3x_2 - 0.2x_3$$

The step function $u(z)$ is used to classify the outcome:

$$u(z) = \begin{cases} 1 & \text{if } z \geq 1.5 \\ 0 & \text{otherwise} \end{cases}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 94 Question Id : 6406531113788 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
For the input *x* = [0.9, 5, 1], what will be the predicted output *u(z)*?
Options :
6406533774240. ✖ 0
6406533774241. ✔ 1
6406533774242. ✖ 1.75
6406533774243. ✖ Cannot be determined

✔ **Official Answer Key: Option (B)**

MLF-Q112 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 83 Question Id : 6406531114403 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)
Options :
6406533776017. ✖ YES

6406533776018. ✔ NO

Sub-Section Number :	2
Sub-Section Id :	640653168719
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B)**

MLF-Q113 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 84 Question Id : 6406531114404 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Consider the function $f(x, y) = \sin(xy)$. Let $L(x, y)$ be the linear approximation of $f(x, y)$ around the point $\left(\frac{\pi}{2}, 0\right)$. What is the value of $L\left(\frac{\pi}{2} + 0.1, 0.1\right)$?
hint: Use the approximation of $\pi = 3.14$
Options :
6406533776019. ✖ 0.162
6406533776020. ✖ 0.257
6406533776021. ✔ 0.157
6406533776022. ✖ 1.57

✔ **Official Answer Key: Option (C)**

MLF-Q115 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 86 Question Id : 6406531114406 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
The directional derivative of a scalar function $f(x, y)$ at a point (x_0, y_0) in the direction of a unit vector u represents:

Options :
6406533776027. ✔
The rate of change of f in the direction of u at (x_0, y_0).

6406533776028. ✖ The magnitude of the gradient vector ∇f at (x_0, y_0).

6406533776029. ✖ The maximum value of the gradient at (x_0, y_0).

6406533776030. ✖ The angle between the vector u and ∇f.

✔ Official Answer Key: Option (A)

MLF-Q116 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 87 Question Id : 6406531114407 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Which of the following is the best second-order polynomial that fits the dataset by the least sum of squares method?

$$\min_{\theta_0,\theta_1,\theta_2} \sum_{i=1}^{3} (y_i - (\theta_0 + \theta_1 x_i + \theta_2 x_i^2))^2$$

x	1	0	-1
y	-1	1	1

Options :

6406533776031. ✖ $2 - \frac{1}{2}x + \frac{5}{2}x^2$

6406533776032. ✔ $2 - \frac{1}{2}x - \frac{5}{2}x^2$

6406533776033. ✖ $2 + \frac{1}{2}x - \frac{5}{2}x^2$

6406533776034. ✖ $2 + \frac{1}{2}x + \frac{5}{2}x^2$

Sub-Section Number :	3
Sub-Section Id :	640653168720
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B)**

MLF-Q117 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 88 Question Id : 6406531114408 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question
Which of the following options is/are true?
Options :

6406533776035. ✔ $\begin{bmatrix}0\\0\\1\end{bmatrix}$ is orthogonal to the vectors $\begin{bmatrix}1\\0\\0\end{bmatrix}$ and $\begin{bmatrix}1\\1\\0\end{bmatrix}$.

6406533776036. ✖ For a matrix A, vectors of column space of A are perpendicular to the vectors of null space of A.

6406533776037. ✖ Length of the vector, $a = \begin{bmatrix}1\\1\\-1\end{bmatrix}$ is 3.

6406533776038. ✔ $\begin{bmatrix}1\\2\\3\end{bmatrix}$ and $\begin{bmatrix}1\\1\\-1\end{bmatrix}$ are orthogonal vectors.

✔ Official Answer Key: Option (A, D)

MLF-Q118 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 89 Question Id : 6406531114409 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Which of the following options is/are true when the vector $\begin{bmatrix}1\\2\\1\end{bmatrix}$ projected onto the vector $\begin{bmatrix}1\\-1\\0\end{bmatrix}$?

Options :

6406533776039. ✖ The projection matrix is $\frac{1}{2}\begin{bmatrix}-1 & -1 & 0\\1 & 1 & 0\\0 & 0 & 0\end{bmatrix}$

6406533776040. ✔ The projection matrix is $\frac{1}{2}\begin{bmatrix}1 & -1 & 0\\-1 & 1 & 0\\0 & 0 & 0\end{bmatrix}$

6406533776041. ✔ Projected vector is $\frac{1}{2}\begin{bmatrix}-1\\1\\0\end{bmatrix}$

6406533776042. ✖ Projected vector is $\frac{1}{2}\begin{bmatrix}1\\-1\\0\end{bmatrix}$

✔ **Official Answer Key: Option (B, C)**

MLF-Q119 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 90 Question Id : 6406531114411 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following statements is/are correct?
Options :

6406533776047. ✖ $f(x) = \ln(x)$ is continuous and differentiable in the domain $[0, \infty)$.

6406533776048. ✔ $f(x) = e^x$ is continuous and differentiable for all $x \in \mathbb{R}$.

6406533776049. ✔ $f(x) = \sqrt{x}$ is continuous but not differentiable at $x = 0$.

6406533776050. ✔ $f(x) = \frac{1}{x}$ is continuous and differentiable for all $x \neq 0$.

Sub-Section Number :	4
Sub-Section Id :	640653168721
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B, C, D)

MLF-Q120 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 91 Question Id : 6406531114410 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider a matrix

$$A = \begin{bmatrix} 2 & 5 \\ 6 & 15 \end{bmatrix}.$$

Which of the following options is/are true?

Options :

6406533776043. ✔ Null space of A is span $\left\{\begin{bmatrix} -5 \\ 2 \end{bmatrix}\right\}$

6406533776044. ✔ Column space of A is span $\left\{\begin{bmatrix} 1 \\ 3 \end{bmatrix}\right\}$

6406533776045. ✔ Left null space of A is span $\left\{\begin{bmatrix} -3 \\ 1 \end{bmatrix}\right\}$

6406533776046. ✖ Row space of A is span $\left\{\begin{bmatrix} 2 \\ -5 \end{bmatrix}\right\}$

Sub-Section Number :	5
Sub-Section Id :	640653168722
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A, B, C)**

MLF-Q121 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 92 Question Id : 6406531114412 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question
Consider a matrix $A_{20\times25}$. If the nullity of the matrix is 10, then find the rank of the matrix.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
15

Sub-Section Number :	6
Sub-Section Id :	640653168723
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 15**

MLF-Q122 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 93 Question Id : 6406531114413 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question
If $f(x, y, z) = \sin(xy) + e^{yz} + xz^2$, what is the length of the gradient at the point, $(1, 0, 2)$? Enter the answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
6.37 to 6.43

Sub-Section Number :	7
Sub-Section Id :	640653168724
Question Shuffling Allowed :	No

Question Id : 6406531114414 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (94 to 95)
✔ **Official Answer Key (NAT): 6.37 to 6.43**

MLF-Q123 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

A bank wants to predict whether a customer will default on a loan using a classification model. The bank collects data from past customers, which includes the following features:

- x_1: Credit score (scaled between 0 and 1)
- x_2: Monthly income (in \$1000s)
- x_3: Number of late payments in the past year

The target variable y represents whether the customer defaulted (1 for default, 0 for no default). The data is provided in the table below:

x	y
[0.9, 5, 1]	0
[0.4, 3, 5]	1
[0.8, 4, 2]	0
[0.2, 2, 7]	1
[0.7, 6, 0]	0

The bank uses the following linear combination to compute z:

$$z = 0.5x_1 + 0.3x_2 - 0.2x_3$$

The step function $u(z)$ is used to classify the outcome:

$$u(z) = \begin{cases} 1 & \text{if } z \geq 1.5 \\ 0 & \text{otherwise} \end{cases}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 94 Question Id : 6406531114415 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
For the input x = [0.9, 5, 1], what will be the predicted output $u(z)$?
Options :
6406533776053. ✖ 0
6406533776054. ✔ 1
6406533776055. ✖ 1.75
6406533776056. ✖ Cannot be determined

✔ **Official Answer Key: Option (B)**

MLF-Q127 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 83 Question Id : 6406531114116 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)
Options :
6406533775188. ✖ YES

6406533775189. ✔ NO

Sub-Section Number :	2
Sub-Section Id :	640653168620
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B)**

MLF-Q128 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 84 Question Id : 6406531114117 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Consider the function $f(x, y) = \sin(xy)$. Let $L(x, y)$ be the linear approximation of $f(x, y)$ around the point $\left(\frac{\pi}{2}, 0\right)$. What is the value of $L\left(\frac{\pi}{2} + 0.1, 0.1\right)$?
hint: Use the approximation of $\pi = 3.14$
Options :
6406533775190. ✖ 0.162
6406533775191. ✖ 0.257
6406533775192. ✔ 0.157
6406533775193. ✖ 1.57

✔ **Official Answer Key: Option (C)**

MLF-Q130 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 86 Question Id : 6406531114119 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

The directional derivative of a scalar function $f(x, y)$ at a point (x_0, y_0) in the direction of a unit vector u represents:

Options :

6406533775198. ✔

The rate of change of f in the direction of u at (x_0, y_0).

6406533775199. ✖ The magnitude of the gradient vector ∇f at (x_0, y_0).

6406533775200. ✖ The maximum value of the gradient at (x_0, y_0).

6406533775201. ✖ The angle between the vector u and ∇f.

✔ Official Answer Key: Option (A)

MLF-Q131 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 87 Question Id : 6406531114120 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Which of the following is the best second-order polynomial that fits the dataset by the least sum of squares method?

$$\min_{\theta_0,\theta_1,\theta_2} \sum_{i=1}^{3} (y_i - (\theta_0 + \theta_1 x_i + \theta_2 x_i^2))^2$$

x	1	0	-1
y	-1	1	1

Options :

6406533775202. ✖ $2 - \frac{1}{2}x + \frac{5}{2}x^2$

6406533775203. ✔ $2 - \frac{1}{2}x - \frac{5}{2}x^2$

6406533775204. ✖ $2 + \frac{1}{2}x - \frac{5}{2}x^2$

6406533775205. ✖ $2 + \frac{1}{2}x + \frac{5}{2}x^2$

Sub-Section Number :	3
Sub-Section Id :	640653168621
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B)**

MLF-Q132 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 88 Question Id : 6406531114121 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question
Which of the following options is/are true?
Options :

6406533775206. ✔ $\begin{bmatrix}0\\0\\1\end{bmatrix}$ is orthogonal to the vectors $\begin{bmatrix}1\\0\\0\end{bmatrix}$ and $\begin{bmatrix}1\\1\\0\end{bmatrix}$.

6406533775207. ✖ For a matrix A, vectors of column space of A are perpendicular to the vectors of null space of A.

6406533775208. ✖ Length of the vector, $a = \begin{bmatrix}1\\1\\-1\end{bmatrix}$ is 3.

6406533775209. ✔ $\begin{bmatrix}1\\2\\3\end{bmatrix}$ and $\begin{bmatrix}1\\1\\-1\end{bmatrix}$ are orthogonal vectors.

✔ Official Answer Key: Option (A, D)

MLF-Q133 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 89 Question Id : 6406531114122 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Which of the following options is/are true when the vector $\begin{bmatrix}1\\2\\1\end{bmatrix}$ projected onto the vector $\begin{bmatrix}1\\-1\\0\end{bmatrix}$?

Options :

6406533775210. ✖ The projection matrix is $\frac{1}{2}\begin{bmatrix}-1 & -1 & 0\\1 & 1 & 0\\0 & 0 & 0\end{bmatrix}$

6406533775211. ✔ The projection matrix is $\frac{1}{2}\begin{bmatrix}1 & -1 & 0\\-1 & 1 & 0\\0 & 0 & 0\end{bmatrix}$

6406533775212. ✔ Projected vector is $\frac{1}{2}\begin{bmatrix}-1\\1\\0\end{bmatrix}$

6406533775213. ✖ Projected vector is $\frac{1}{2}\begin{bmatrix}1\\-1\\0\end{bmatrix}$

✔ **Official Answer Key: Option (B, C)**

MLF-Q134 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 90 Question Id : 6406531114124 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following statements is/are correct?
Options :

6406533775218. ✖ $f(x) = \ln(x)$ is continuous and differentiable in the domain $[0, \infty)$.

6406533775219. ✔ $f(x) = e^x$ is continuous and differentiable for all $x \in \mathbb{R}$.

6406533775220. ✔ $f(x) = \sqrt{x}$ is continuous but not differentiable at $x = 0$.

6406533775221. ✔ $f(x) = \frac{1}{x}$ is continuous and differentiable for all $x \neq 0$.

Sub-Section Number :	4
Sub-Section Id :	640653168622
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B, C, D)

MLF-Q135 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 91 Question Id : 6406531114123 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider a matrix

$$A = \begin{bmatrix} 2 & 5 \\ 6 & 15 \end{bmatrix}.$$

Which of the following options is/are true?

Options :

6406533775214. ✔ Null space of A is span $\left\{\begin{bmatrix} -5 \\ 2 \end{bmatrix}\right\}$

6406533775215. ✔ Column space of A is span $\left\{\begin{bmatrix} 1 \\ 3 \end{bmatrix}\right\}$

6406533775216. ✔ Left null space of A is span $\left\{\begin{bmatrix} -3 \\ 1 \end{bmatrix}\right\}$

6406533775217. ✖ Row space of A is span $\left\{\begin{bmatrix} 2 \\ -5 \end{bmatrix}\right\}$

Sub-Section Number :	5
Sub-Section Id :	640653168623
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A, B, C)**

MLF-Q136 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 92 Question Id : 6406531114125 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question
Consider a matrix $A_{20\times25}$. If the nullity of the matrix is 10, then find the rank of the matrix.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
15

Sub-Section Number :	6
Sub-Section Id :	640653168624
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 15**

MLF-Q137 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 93 Question Id : 6406531114126 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question
If $f(x, y, z) = \sin(xy) + e^{yz} + xz^2$, what is the length of the gradient at the point, $(1, 0, 2)$? Enter the answer correct to two decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
6.37 to 6.43

Sub-Section Number :	7
Sub-Section Id :	640653168625
Question Shuffling Allowed :	No

Question Id : 6406531114127 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (94 to 95)

✔ **Official Answer Key (NAT): 6.37 to 6.43**

MLF-Q138 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

A bank wants to predict whether a customer will default on a loan using a classification model. The bank collects data from past customers, which includes the following features:

- x_1: Credit score (scaled between 0 and 1)
- x_2: Monthly income (in \$1000s)
- x_3: Number of late payments in the past year

The target variable y represents whether the customer defaulted (1 for default, 0 for no default). The data is provided in the table below:

x	y
[0.9, 5, 1]	0
[0.4, 3, 5]	1
[0.8, 4, 2]	0
[0.2, 2, 7]	1
[0.7, 6, 0]	0

The bank uses the following linear combination to compute z:

$$z = 0.5x_1 + 0.3x_2 - 0.2x_3$$

The step function $u(z)$ is used to classify the outcome:

$$u(z) = \begin{cases} 1 & \text{if } z \geq 1.5 \\ 0 & \text{otherwise} \end{cases}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 94 Question Id : 6406531114128 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
For the input *x* = [0.9, 5, 1], what will be the predicted output *u(z)*?
Options :
6406533775224. ✖ 0
6406533775225. ✔ 1
6406533775226. ✖ 1.75
6406533775227. ✖ Cannot be determined

✔ **Official Answer Key: Option (B)**

MLF-Q142 | [MCQ] | Marks: +2 | *MLF*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 1 Question Id : 6406531951380

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406536344496. ✔ Yes

6406536344497. ✖ No

Sub-Section Number :	2
Sub-Section Id :	640653360281
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLF-Q148 | [MCQ] | Marks: +2 | *MLF*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 7 Question Id : 6406531951386
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
The matrix $P = \begin{bmatrix} m & 2 \\ 1 & n \end{bmatrix}$ has eigenvalues 0 and 3. Which of the following options is/are correct?
Options :

6406536344517. ✖ $mn = 0$

6406536344518. ✔ $mn = 2$

6406536344519. ✔ $m + n = 3$

6406536344520. ✖ $m + n = 2$

✔ Official Answer Key: Option (B, C)

MLF-Q150 | [MCQ] | Marks: +2 | *MLF*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 9 Question Id : 6406531951388
Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Let L_1 and L_2 be two lines in $\mathbb{R}^3$ defined in parametric form:

$$L_1: \quad x = 1 + 4t,\ y = 5 - 4t,\ z = -1 + 5t,\ \ t \in \mathbb{R}$$

$$L_2: \quad x = 2 + 8s,\ y = 4 - 3s,\ z = 5 + s,\ \ s \in \mathbb{R}$$

For example, every point on line L_1 is of the form $(1 + 4t, 5 - 4t, -1 + 5t)$. Which of the following is/are true?

Options :

6406536344525. ✖ L_1 and L_2 are parallel

6406536344526. ✔ L_1 and L_2 are not parallel

6406536344527. ✖ L_1 and L_2 intersect at a point

6406536344528. ✔ L_1 and L_2 do not intersect at any point

✔ **Official Answer Key: Option (B, D)**

MLF-Q152 | [MCQ] | Marks: +2 | *MLF*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 11 Question Id : 6406531951389

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Suppose you are building a credit card fraud detection system and have two approaches:

(i) Manually writing rules such as "If transaction amount is above Rs. 50,000 and the location is overseas, mark it as fraud."
(ii) Training a model using millions of past transactions labeled as fraud or not fraud.

Which of the following options is/are correct?

Options :

6406536344529. ✔ Approach (i) is not machine learning.

6406536344530. ✔ Approach (ii) is machine learning.

6406536344531. ✖ Both approaches are machine learning.

6406536344532. ✖ Neither approach is machine learning.

Sub-Section Number :	6
Sub-Section Id :	640653360285
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A, B)**

MLF-Q155 | [MCQ] | Marks: +2 | *MLF*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $P = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be the projection matrix that projects vectors in $\mathbb{R}^2$ onto the line $y = 3x$.

Let $A = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$ be the matrix that projects vectors in $\mathbb{R}^2$ onto the nullspace of P.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 14 Question Id : 6406531951394

Correct Marks : 3

Question Label : Short Answer Question

Find $10(a+b+c+d)$. Your answer should be an integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

16

MLF-Q156 | [MCQ] | Marks: +2 | *MLF*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $P = \begin{bmatrix} a & b \\ c & d \end{bmatrix}$ be the projection matrix that projects vectors in $\mathbb{R}^2$ onto the line $y = 3x$.

Let $A = \begin{bmatrix} e & f \\ g & h \end{bmatrix}$ be the matrix that projects vectors in $\mathbb{R}^2$ onto the nullspace of P.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 15 Question Id : 6406531951395

Correct Marks : 1

Question Label : Short Answer Question

Find $10(e + f + g + h)$. Your answer should be an integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

4

Question Id : 6406531951396 Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (16 to 17)

MLF-Q159 | [MCQ] | Marks: +2 | *Quiz1 May23*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 119 Question Id : 640653577871 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406531929719. ✔ YES

6406531929720. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	64065382595
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key: Option (A)**

MLF-Q160 | [MSQ] | Marks: +2 | *Quiz1 May23*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 120 Question Id : 640653577872 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following statements are correct?

Options :

6406531929721. ✔ $\mathbf{I}(444\%3 = 0 \text{ and } \pi \in \mathbb{R}) = 1.$

6406531929722. ✖ Mean Squared Error (MSE) is a suitable loss function for classification problems.

6406531929723. ✖ The loss obtained after applying encoder and decoder functions to a given data can be negative.

6406531929724. ✔ Gaussian mixture model is a family of distributions for modeling multimodal data.

✔ Official Answer Key: Option (A, D)

MLF-Q162 | [MSQ] | Marks: +2 | *Quiz1 May23*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 122 Question Id : 640653577886 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following matrices have eigenvectors as $\begin{bmatrix}2\\1\\2\end{bmatrix}$, $\begin{bmatrix}1\\-2\\0\end{bmatrix}$ and $\begin{bmatrix}1\\0\\-1\end{bmatrix}$?

Options :

6406531929753. ✔ $\begin{bmatrix}3&2&4\\2&0&2\\4&2&3\end{bmatrix}$

6406531929754. ✔ $\begin{bmatrix}6&4&8\\4&0&4\\8&4&6\end{bmatrix}$

6406531929755. ✖ $\begin{bmatrix}2&1&4\\4&0&4\\8&4&6\end{bmatrix}$

6406531929756. ✖ $\begin{bmatrix}3&1&0\\2&0&2\\4&2&3\end{bmatrix}$

✔ Official Answer Key: Option (A, B)

MLF-Q163 | [MSQ] | Marks: +2 | *Quiz1 May23*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 123 Question Id : 640653577887 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Let the matrix $A \in \mathbb{R}^{d \times d}$ be diagonalizable, then which of the following statements is/are true for A?

Options :

6406531929757. ✖ A must be symmetric.

6406531929758. ✔ We can find a basis of eigenvectors for $\mathbb{R}^d$.

6406531929759. ✖ We can find an orthonormal basis of eigenvectors for $\mathbb{R}^d$.

6406531929760. ✔ A can have some repeated eigenvalues but must have distinct independent eigenvectors.

Sub-Section Number :	3
Sub-Section Id :	64065382596
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (B, D)

MLF-Q164 | [MSQ] | Marks: +2 | *Quiz1 May23*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 124 Question Id : 640653577879 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Let $A, B \in \mathbb{R}^{n \times n}$ be two matrices, then which among the following statements are correct?

Options :

6406531929737. ✔ $x \in \text{Nullspace}(A) \implies x \in \text{Nullspace}(BA)$.

6406531929738. ✔ $x \in \text{Columnspace}(A) \implies x \in \text{Columnspace}(B)$, where $A = BC, C \in \mathbb{R}^{n \times n}$.

6406531929739. ✔ $x \in \text{Nullspace}(AB) \implies Bx \in \text{Nullspace}(A)$.

6406531929740. ✖ $x \in \text{Nullspace}(AB) \implies x \in \text{Nullspace}(B)$.

✔ Official Answer Key: Option (A, B, C)

MLF-Q165 | [MSQ] | Marks: +2 | *Quiz1 May23*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 125 Question Id : 640653577885 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider the solution of the linear regression problem $(Ax = b)$ using least squares method, where $A = \begin{bmatrix} 1 & 5 & 5 \\ 2 & 3 & 5 \\ 0 & 7 & 1 \end{bmatrix}$ and $b = \begin{bmatrix} 7 \\ 8 \\ 4 \end{bmatrix}$.

Based on this data, which of the following statements is/are true?

Options :

6406531929749. ✔ The error value is zero as we can get an x such that Ax is exactly equal to b.

6406531929750. ✖ The error value is non-zero as we cannot get an x such that Ax is exactly equal to b.

6406531929751. ✔ Least squares solution for x is $\begin{bmatrix} 2 \\ 0.5 \\ 0.5 \end{bmatrix}$.

6406531929752. ✖ Least squares solution for x is $\begin{bmatrix} 1 \\ 0.6 \\ 0.4 \end{bmatrix}$.

Sub-Section Number :	4
Sub-Section Id :	64065382597
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A, C)

MLF-Q166 | [MCQ] | Marks: +2 | *Quiz1 May23*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 126 Question Id : 640653577877 Question Type : MCQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Find the linear approximation of the function $f(x, y) = ye^{xy}$ in the neighbourhood of (0,1).

Options :

6406531929729. ✔ $x + y$

6406531929730. ✖ $x + 2y$

6406531929731. ✖ $2x + y$

6406531929732. ✖ $2x + 3y$

✔ Official Answer Key: Option (A)

MLF-Q167 | [MCQ] | Marks: +2 | *Quiz1 May23*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 127 Question Id : 640653577878 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Let $f : \mathbb{R}^2 \to \mathbb{R}$, given by $f(x, y) = x^2 + y^2$. Consider the following algorithm:

$$X_{n+1} = X_n - h\nabla f(X_n), \quad X_i \in \mathbb{R}^2, i : 1, 2, \ldots \text{ and } h > 0,$$

where h is the step size and $\nabla f(X_n)$ is the gradient of f evaluated at X_n. Fix $h = 0.25$ and $X_0 = (1, 3)$, which among the following points does the algorithm converge to?

Options :

6406531929733. ✖ (1, 3)

6406531929734. ✖ (1/2, 1/6)

6406531929735. ✔ (0, 0)

6406531929736. ✖ The algorithm does not converge

Sub-Section Number :	5
Sub-Section Id :	64065382598
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key: Option (C)**

MLF-Q168 | [SA] | Marks: +2 | *Quiz1 May23*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 128 Question Id : 640653577873 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

Consider the following dataset with 5 data points:

X	y
$(2,3)$	5
$(-1,1)$	2
$(4,2)$	7
$(0,-2)$	1
$(-3,5)$	4

We want to fit a linear regression model of the form $f(X) = w^T x$ to this dataset, where $w = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$. Compute the value of the loss function L for this dataset which is defined as $L = \max_i |f(x^i) - y^i|$.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

5

✔ **Official Answer Key (NAT): 5**

MLF-Q169 | [SA] | Marks: +2 | *Quiz1 May23*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 129 Question Id : 640653577874 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

Let $f(x_1, x_2, x_3) = \frac{x_1 + x_2 + x_3}{2}$ be used as an encoder function and $g(u) = [u, u, u]$ be used as a decoder function for dimensionality reduction of the dataset $X_1 = [1, 2, 3]$, $X_2 = [2, 4, 6], X_3 = [3, 6, 9], X_4 = [4, 8, 12]$. Calculate the reconstruction error $\left(R(f, g) = \frac{1}{n}\sum_{i=1}^{n} \|x_i - g(f(x_i))\|^2\right)$ for this encoder decoder pair. Enter the answer correct to one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

37 to 38

✔ **Official Answer Key (NAT): 37 to 38**

MLF-Q171 | [SA] | Marks: +2 | *Quiz1 May23*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 131 Question Id : 640653577875 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

Find the directional derivative of $f(x, y, z) = x(x^2 - y^2) - z$ at a point D(1, -1, 0) in the direction of vector $\hat{h} = \frac{2}{7}i - \frac{3}{7}j + \frac{6}{7}k$. Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

-1.16 to -1.12

Sub-Section Number :	7
Sub-Section Id :	64065382600
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key (NAT): -1.16 to -1.12**

MLF-Q172 | [SA] | Marks: +2 | *Quiz1 May23*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 132 Question Id : 640653577876 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Short Answer Question

Let the function $f(x) = \begin{cases} ax - 3, & x < 2 \\ 4, & x = 2 \\ 2x, & x > 2 \end{cases}$

be continuous for all $x \in \mathbb{R}$. Then, find the value of a.

Enter the answer correct to one decimal place.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3.5

Sub-Section Number :	8
Sub-Section Id :	64065382601
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653577880 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (133 to 134)

✔ **Official Answer Key (NAT): 3.5**

MLF-Q173 | [MCQ] | Marks: +2 | *Quiz1 May23*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}$ and $b = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 133 Question Id : 640653577881 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Is the system $Ax = b$ consistent?

Options :

6406531929741. ✖ Yes

6406531929742. ✔ No

✔ **Official Answer Key: Option (B)**

MLF-Q175 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 13 July 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 80 Question Id : 6406531297911 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)
Options :
6406534364696. ✔ YES
6406534364697. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653206543
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLF-Q176 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 81 Question Id : 6406531297912 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider a matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Which of the following options is/are true?

Options :

6406534364698. ✔ Null space of A is span $\left\{ \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \right\}$

6406534364699. ✖ Dimension of the column space of A is 1

6406534364700. ✔ Left null space of A is span $\left\{ \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} \right\}$

6406534364701. ✔ Row space of A is span $\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0.5 \end{bmatrix} \right\}$

✔ Official Answer Key: Option (A, C, D)

MLF-Q178 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 83 Question Id : 6406531297919 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider a matrix $A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$. Which of the following options is/are true?

Options :

6406534364722. ✔ 1 is an eigenvalue of A.

6406534364723. ✔ 2 is an eigenvalue of A.

6406534364724. ✔ $\begin{bmatrix} -3 \\ -1 \\ 2 \end{bmatrix}$ is an eigenvector of A.

6406534364725. ✖ $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ is an eigenvector of A.

✔ Official Answer Key: Option (A, B, C)

MLF-Q179 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 84 Question Id : 6406531297920 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider a subspace $S = \{(x, y, z) \mid x + 2y + 2z = 0\}$ of the vector space $\mathbb{R}^3$.

Which of the following options is/are true?

Options :

6406534364726. ✖ $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ is a vector of S.

6406534364727. ✔ $\left\{ \begin{bmatrix} 1 \\ 2 \\ -2.5 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right\}$ is an orthogonal basis of S.

6406534364728. ✖ The dimension of S is 1.

6406534364729. ✔ $\left\{ \begin{bmatrix} 1 \\ 2 \\ -2.5 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right\}$ is a basis of S.

✔ Official Answer Key: Option (B, D)

MLF-Q180 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 85 Question Id : 6406531297924 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

The function

$$f(x) = \begin{cases} px^2 + qx + 2, & \text{if } x < 1 \\ rx + s, & \text{if } x \geq 1 \end{cases}$$

is differentiable at $x = 1$. If p, q, r, and s are constants, which of the following is/are must be true?

Options :

6406534364739. ✔ $r = 2p + q$ and $s = 2 - p$

6406534364740. ✖ $p + q = r + s$

6406534364741. ✖ $q = r$ and $s = 2$

6406534364742. ✔ $p + q + 2 = r + s$ and $2p + q = r$

✔ Official Answer Key: Option (A, D)

MLF-Q182 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 87 Question Id : 6406531297926 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider a matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $A = PDP^{-1}$.

Which of the following options is/are true?

Options :

6406534364747. ✔ A is an orthogonally diagonalizable matrix.

6406534364748. ✔ $P = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

6406534364749. ✖ $P = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$

6406534364750. ✖ $P^{-1} = \frac{1}{2}\begin{bmatrix} 1 & 0 & -1 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$

Sub-Section Number :	3
Sub-Section Id :	640653206544
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A, B)

MLF-Q183 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 88 Question Id : 6406531297913 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question

Let $\hat{y}$ be the regression line (the best fit line) for the dataset given below using the least squares method. Which of the following options is/are true?

x	0	2	3
y	-2	5	4

Options :

6406534364702. ✖ $\hat{y} = \frac{1}{14}(13x - 19)$

6406534364703. ✖ $\hat{y} = \frac{1}{14}(31x - 91)$

6406534364704. ✔ If $x = 11$, then predicted value of y is 23.

6406534364705. ✔ If $x = 7$, then predicted value of y is $\frac{99}{7}$.

Sub-Section Number : 4
Sub-Section Id : 640653206545
Question Shuffling Allowed : No

Question Id : 6406531297914 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (89 to 91)

✔ Official Answer Key: Option (C, D)

MLF-Q185 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 13 July 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

Sub questions

Question Number : 90 Question Id : 6406531297916 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following options is/are true ?
Options :

6406534364710. ✔ 1 is an eigenvalue of A.

6406534364711. ✔ $1 + i\sqrt{2}$ is an eigenvalue of A.

6406534364712. ✖ The rank of the matrix A is 2.

6406534364713. ✖ Nullity of the matrix A is 1.

✔ **Official Answer Key: Option (A, B)**

MLF-Q186 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

Sub questions

Question Number : 91 Question Id : 6406531297917 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following options is/are true ?
Options :

6406534364714. ✔ Matrix A is diagonalizable.

6406534364715. ✔ Eigenvectors of A are orthogonal.

6406534364716. ✔ If A is diagonalizable and diagonal form of $A = P^{-1}DP$, then P is an orthogonal matrix.

6406534364717. ✖ Entries of the eigenvector v corresponding to the eigenvalue $1 - i\sqrt{2}$ of A are complex numbers.

Sub-Section Number :	5
Sub-Section Id :	640653206546
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A, B, C)**

MLF-Q187 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

Sub questions

Question Number : 92 Question Id : 6406531297921 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following is/are true?
Options :

6406534364730. ✔ If $a_n = -1 + \frac{n}{n+1}$ then $a_n \to 0$ as $n \to \infty$.

6406534364731. ✖ If $a_n = \sin(n)$ then $a_n \to 0$ as $n \to \infty$.

6406534364732. ✔ If $a_n = n!$ then $a_n \to \infty$ as $n \to \infty$.

6406534364733. ✔ $a_n = \sin(n)$ is not a convergent sequence.

Sub-Section Number :	6
Sub-Section Id :	640653206547
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A, C, D)

MLF-Q188 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

Sub questions

Question Number : 93 Question Id : 6406531297922 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Let $f(x, y) = \ln(x^2 + y^2)$. Use the linear approximation at the point $(1, 2)$ to estimate $f(1.02, 2.01)$. Write the answer correctly to 4 decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
1.6251 to 1.6258

Sub-Section Number :	7
Sub-Section Id :	640653206548
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 1.6251 to 1.6258**

MLF-Q190 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 80 Question Id : 6406531298181 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)
Options :
6406534365467. ✔ YES
6406534365468. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653206644
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLF-Q191 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 81 Question Id : 6406531298182 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider a matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Which of the following options is/are true?

Options :

6406534365469. ✔ Null space of A is span $\left\{ \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \right\}$

6406534365470. ✖ Dimension of the column space of A is 1

6406534365471. ✔ Left null space of A is span $\left\{ \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} \right\}$

6406534365472. ✔ Row space of A is span $\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0.5 \end{bmatrix} \right\}$

✔ Official Answer Key: Option (A, C, D)

MLF-Q193 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 83 Question Id : 6406531298189 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider a matrix $A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$. Which of the following options is/are true?

Options :

6406534365493. ✔ 1 is an eigenvalue of A.

6406534365494. ✔ 2 is an eigenvalue of A.

6406534365495. ✔ $\begin{bmatrix} -3 \\ -1 \\ 2 \end{bmatrix}$ is an eigenvector of A.

6406534365496. ✖ $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ is an eigenvector of A.

✔ Official Answer Key: Option (A, B, C)

MLF-Q194 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 84 Question Id : 6406531298190 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider a subspace $S = \{(x, y, z) \mid x + 2y + 2z = 0\}$ of the vector space $\mathbb{R}^3$.

Which of the following options is/are true?

Options :

6406534365497. ✖ $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ is a vector of S.

6406534365498. ✔ $\left\{ \begin{bmatrix} 1 \\ 2 \\ -2.5 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right\}$ is an orthogonal basis of S.

6406534365499. ✖ The dimension of S is 1.

6406534365500. ✔ $\left\{ \begin{bmatrix} 1 \\ 2 \\ -2.5 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right\}$ is a basis of S.

✔ Official Answer Key: Option (B, D)

MLF-Q195 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 85 Question Id : 6406531298194 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

The function

$$f(x) = \begin{cases} px^2 + qx + 2, & \text{if } x < 1 \\ rx + s, & \text{if } x \geq 1 \end{cases}$$

is differentiable at $x = 1$. If p, q, r, and s are constants, which of the following is/are must be true?

Options :

6406534365510. ✔ $r = 2p + q$ and $s = 2 - p$

6406534365511. ✖ $p + q = r + s$

6406534365512. ✖ $q = r$ and $s = 2$

6406534365513. ✔ $p + q + 2 = r + s$ and $2p + q = r$

✔ Official Answer Key: Option (A, D)

MLF-Q197 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 87 Question Id : 6406531298196 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider a matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $A = PDP^{-1}$.

Which of the following options is/are true?

Options :

6406534365518. ✔ A is an orthogonally diagonalizable matrix.

6406534365519. ✔ $P = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

6406534365520. ✖ $P = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$

6406534365521. ✖ $P^{-1} = \frac{1}{2}\begin{bmatrix} 1 & 0 & -1 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$

Sub-Section Number :	3
Sub-Section Id :	640653206645
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A, B)

MLF-Q198 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 88 Question Id : 6406531298183 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question

Let $\hat{y}$ be the regression line (the best fit line) for the dataset given below using the least squares method. Which of the following options is/are true?

x	0	2	3
y	-2	5	4

Options :

6406534365473. ✖ $\hat{y} = \frac{1}{14}(13x - 19)$

6406534365474. ✖ $\hat{y} = \frac{1}{14}(31x - 91)$

6406534365475. ✔ If $x = 11$, then predicted value of y is 23.

6406534365476. ✔ If $x = 7$, then predicted value of y is $\frac{99}{7}$.

Sub-Section Number : 4
Sub-Section Id : 640653206646
Question Shuffling Allowed : No

Question Id : 6406531298184 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (89 to 91)

✔ **Official Answer Key: Option (C, D)**

MLF-Q200 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

Sub questions

Question Number : 90 Question Id : 6406531298186 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following options is/are true ?
Options :
6406534365481. ✔ 1 is an eigenvalue of A.

6406534365482. ✔ $1 + i\sqrt{2}$ is an eigenvalue of A.

6406534365483. ✖ The rank of the matrix A is 2.

6406534365484. ✖ Nullity of the matrix A is 1.

✔ **Official Answer Key: Option (A, B)**

MLF-Q201 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

Sub questions

Question Number : 91 Question Id : 6406531298187 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following options is/are true ?
Options :

6406534365485. ✔ Matrix A is diagonalizable.

6406534365486. ✔ Eigenvectors of A are orthogonal.

6406534365487. ✔ If A is diagonalizable and diagonal form of $A = P^{-1}DP$, then P is an orthogonal matrix.

6406534365488. ✖ Entries of the eigenvector v corresponding to the eigenvalue $1 - i\sqrt{2}$ of A are complex numbers.

Sub-Section Number :	5
Sub-Section Id :	640653206647
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A, B, C)

MLF-Q202 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

Sub questions

Question Number : 92 Question Id : 6406531298191 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following is/are true?
Options :

6406534365501. ✔ If $a_n = -1 + \frac{n}{n+1}$ then $a_n \to 0$ as $n \to \infty$.

6406534365502. ✖ If $a_n = \sin(n)$ then $a_n \to 0$ as $n \to \infty$.

6406534365503. ✔ If $a_n = n!$ then $a_n \to \infty$ as $n \to \infty$.

6406534365504. ✔ $a_n = \sin(n)$ is not a convergent sequence.

Sub-Section Number :	6
Sub-Section Id :	640653206648
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A, C, D)

MLF-Q203 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

Sub questions

Question Number : 93 Question Id : 6406531298192 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Let $f(x, y) = \ln(x^2 + y^2)$. Use the linear approximation at the point $(1, 2)$ to estimate $f(1.02, 2.01)$. Write the answer correctly to 4 decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
1.6251 to 1.6258

Sub-Section Number :	7
Sub-Section Id :	640653206649
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 1.6251 to 1.6258**

MLF-Q205 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 80 Question Id : 6406531298451 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)
Options :
6406534366238. ✔ YES
6406534366239. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653206745
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLF-Q206 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 81 Question Id : 6406531298452 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider a matrix $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 1 & 3 & 2 \end{bmatrix}$. Which of the following options is/are true?

Options :

6406534366240. ✔ Null space of A is span $\left\{ \begin{bmatrix} 1 \\ 1 \\ -2 \end{bmatrix} \right\}$

6406534366241. ✖ Dimension of the column space of A is 1

6406534366242. ✔ Left null space of A is span $\left\{ \begin{bmatrix} -1 \\ -1 \\ 2 \end{bmatrix} \right\}$

6406534366243. ✔ Row space of A is span $\left\{ \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ 0.5 \end{bmatrix} \right\}$

✔ **Official Answer Key: Option (A, C, D)**

MLF-Q208 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 83 Question Id : 6406531298459 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider a matrix $A = \begin{bmatrix} 1 & 2 & 1 \\ 1 & 0 & 1 \\ 0 & 0 & 1 \end{bmatrix}$. Which of the following options is/are true?

Options :

6406534366264. ✔ 1 is an eigenvalue of A.

6406534366265. ✔ 2 is an eigenvalue of A.

6406534366266. ✔ $\begin{bmatrix} -3 \\ -1 \\ 2 \end{bmatrix}$ is an eigenvector of A.

6406534366267. ✖ $\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}$ is an eigenvector of A.

✔ Official Answer Key: Option (A, B, C)

MLF-Q209 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 84 Question Id : 6406531298460 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider a subspace $S = \{(x, y, z) \mid x + 2y + 2z = 0\}$ of the vector space $\mathbb{R}^3$.

Which of the following options is/are true?

Options :

6406534366268. ✖ $\begin{bmatrix} 0 \\ 1 \\ 1 \end{bmatrix}$ is a vector of S.

6406534366269. ✔ $\left\{ \begin{bmatrix} 1 \\ 2 \\ -2.5 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right\}$ is an orthogonal basis of S.

6406534366270. ✖ The dimension of S is 1.

6406534366271. ✔ $\left\{ \begin{bmatrix} 1 \\ 2 \\ -2.5 \end{bmatrix}, \begin{bmatrix} 0 \\ 1 \\ -1 \end{bmatrix} \right\}$ is a basis of S.

✔ Official Answer Key: Option (B, D)

MLF-Q210 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 85 Question Id : 6406531298464 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

The function

$$f(x) = \begin{cases} px^2 + qx + 2, & \text{if } x < 1 \\ rx + s, & \text{if } x \geq 1 \end{cases}$$

is differentiable at $x = 1$. If p, q, r, and s are constants, which of the following is/are must be true?

Options :

6406534366281. ✔ $r = 2p + q$ and $s = 2 - p$

6406534366282. ✖ $p + q = r + s$

6406534366283. ✖ $q = r$ and $s = 2$

6406534366284. ✔ $p + q + 2 = r + s$ and $2p + q = r$

✔ Official Answer Key: Option (A, D)

MLF-Q212 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 87 Question Id : 6406531298466 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider a matrix $A = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$ and $A = PDP^{-1}$.

Which of the following options is/are true?

Options :

6406534366289. ✔ A is an orthogonally diagonalizable matrix.

6406534366290. ✔ $P = \begin{bmatrix} 1 & 0 & -1 \\ 1 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$

6406534366291. ✖ $P = \begin{bmatrix} 1 & 0 & -1 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$

6406534366292. ✖ $P^{-1} = \frac{1}{2}\begin{bmatrix} 1 & 0 & -1 \\ -1 & 0 & 1 \\ 0 & -1 & 0 \end{bmatrix}$

Sub-Section Number :	3
Sub-Section Id :	640653206746
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A, B)

MLF-Q213 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 88 Question Id : 6406531298453 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question

Let $\hat{y}$ be the regression line (the best fit line) for the dataset given below using the least squares method. Which of the following options is/are true?

x	0	2	3
y	-2	5	4

Options :

6406534366244. ✖ $\hat{y} = \frac{1}{14}(13x - 19)$

6406534366245. ✖ $\hat{y} = \frac{1}{14}(31x - 91)$

6406534366246. ✔ If $x = 11$, then predicted value of y is 23.

6406534366247. ✔ If $x = 7$, then predicted value of y is $\frac{99}{7}$.

Sub-Section Number :	4
Sub-Section Id :	640653206747
Question Shuffling Allowed :	No

Question Id : 6406531298454 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (89 to 91)

✔ Official Answer Key: Option (C, D)

MLF-Q215 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

Sub questions

Question Number : 90 Question Id : 6406531298456 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following options is/are true ?
Options :
6406534366252. ✔ 1 is an eigenvalue of A.

6406534366253. ✔ $1 + i\sqrt{2}$ is an eigenvalue of A.

6406534366254. ✖ The rank of the matrix A is 2.

6406534366255. ✖ Nullity of the matrix A is 1.

✔ **Official Answer Key: Option (A, B)**

MLF-Q216 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

Sub questions

Question Number : 91 Question Id : 6406531298457 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following options is/are true ?
Options :

6406534366256. ✔ Matrix A is diagonalizable.

6406534366257. ✔ Eigenvectors of A are orthogonal.

6406534366258. ✔ If A is diagonalizable and diagonal form of $A = P^{-1}DP$, then P is an orthogonal matrix.

6406534366259. ✖ Entries of the eigenvector v corresponding to the eigenvalue $1 - i\sqrt{2}$ of A are complex numbers.

Sub-Section Number :	5
Sub-Section Id :	640653206748
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A, B, C)**

MLF-Q217 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

Sub questions

Question Number : 92 Question Id : 6406531298461 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following is/are true?
Options :

6406534366272. ✔ If $a_n = -1 + \frac{n}{n+1}$ then $a_n \to 0$ as $n \to \infty$.

6406534366273. ✖ If $a_n = \sin(n)$ then $a_n \to 0$ as $n \to \infty$.

6406534366274. ✔ If $a_n = n!$ then $a_n \to \infty$ as $n \to \infty$.

6406534366275. ✔ $a_n = \sin(n)$ is not a convergent sequence.

Sub-Section Number :	6
Sub-Section Id :	640653206749
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A, C, D)

MLF-Q218 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

Sub questions

Question Number : 93 Question Id : 6406531298462 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Let $f(x, y) = \ln(x^2 + y^2)$. Use the linear approximation at the point $(1, 2)$ to estimate $f(1.02, 2.01)$. Write the answer correctly to 4 decimal places.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
1.6251 to 1.6258

Sub-Section Number :	7
Sub-Section Id :	640653206750
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 1.6251 to 1.6258**

MLF-Q220 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024*
Category: Week 1 - ML Foundations › ML Concepts

Question Id : 640653825037 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Number : 50 Question Id : 640653825036 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)
Options :
6406532774792. ✔ YES
6406532774793. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653122726
Question Shuffling Allowed :	No

Question Id : 640653825037 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (51 to 52)

✔ **Official Answer Key: Option (A)**

MLF-Q221 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 2 & 1 & 1 \\ a & 3 & 2 \\ 3 & b & c \end{pmatrix}$ be a 3×3 matrix. Let α be the eigenvalue corresponding to the eigenvector $v_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ and β is the eigenvalue corresponding to the eigenvector $v_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ of matrix A. Based on this information, answer the given sub-questions.

Sub questions

Question Number : 51 Question Id : 640653825038 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question
Find the value of $\alpha + 2\beta$.
Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
3

✔ **Official Answer Key (NAT): 3**

MLF-Q222 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 2 & 1 & 1 \\ a & 3 & 2 \\ 3 & b & c \end{pmatrix}$ be a 3×3 matrix. Let α be the eigenvalue corresponding to the eigenvector $v_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ and β is the eigenvalue corresponding to the eigenvector $v_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ of matrix A. Based on this information, answer the given sub-questions.

Sub questions

Question Number : 52 Question Id : 640653825039 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following options is/are true ?
Options :

6406532774795. ✔ Eigenvectors, v_1 and v_2 are linearly independent.

6406532774796. ✖ Eigenvectors, v_1 and v_2 are orthogonal.

6406532774797. ✔ A third eigenvector of A is $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

6406532774798. ✔ $a + b + c = 9$.

Question Id : 640653825040 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (53 to 54)

✔ **Official Answer Key: Option (A, C, D)**

MLF-Q224 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 54 Question Id : 640653825042 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Let v_1 and v_2 are the eigenvectors of A corresponding to distinct eigen-values.
We construct a new matrix B using these eigenvectors as its columns, such that $B = [v_1, v_2]$. Based on this information, which of the following options are correct?

Options :

6406532774803. ✔ Null space of the matrix B is zero space.

6406532774804. ✖ Column space of the matrix B is $\mathbb{R}^3$.

6406532774805. ✔ Column space of the matrix B represents a plane in $\mathbb{R}^3$.

6406532774806. ✖ Null space of the matrix B represents a line in $\mathbb{R}^3$.

Sub-Section Number :	3
Sub-Section Id :	640653122727
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A, C)

MLF-Q225 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 55 Question Id : 640653825043 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Let P be the projection matrix for a vector $a = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. Which of the following options is/are true?

Options :

6406532774807. ✔ P is orthogonally diagonalizable.

6406532774808. ✔ $P^4 = P$.

6406532774809. ✔ Projection of $b = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ on the vector a is $-\frac{1}{5}a$.

6406532774810. ✖ $Rank(P) = 2$

Sub-Section Number :	4
Sub-Section Id :	640653122728
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A, B, C)

MLF-Q226 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 56 Question Id : 640653825044 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Find the best-fit line for the dataset given below using the least squares method.

x	0	2	3	5
y	1	2	4	3

Options :

6406532774811. ✔ $\hat{y} = 1.34 + 0.46x$

6406532774812. ✖ $\hat{y} = 0.46 + 1.34x$

6406532774813. ✖ $\hat{y} = 1.34 - 0.46x$

6406532774814. ✖ $\hat{y} = 2.16 + 0.74x$

✔ **Official Answer Key: Option (A)**

MLF-Q227 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 57 Question Id : 640653825045 Question Type : MCQ
Correct Marks : 4

Question Label : Multiple Choice Question

The function

$$f(x) = \begin{cases} ax^2 + bx + 1, & \text{if } x < 0 \\ cx + d, & \text{if } x \geq 0 \end{cases}$$

is differentiable at $(x = 0)$. If a, b, c, and d are constants, which of the following must be true?

Options :

6406532774815. ✖ $a = c$ and $b = 1$

6406532774816. ✔ $b = c$ and $d = 1$

6406532774817. ✖ $a = c$ and $d = 1$

6406532774818. ✖ $b = d = 1$

Sub-Section Number :	5
Sub-Section Id :	640653122729
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B)**

MLF-Q230 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 60 Question Id : 640653825048 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

If $f(x, y) = x^2y + 3xy^2$, what is the length of the gradient at the point (1,2)? Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

20.58 to 20.64

✔ **Official Answer Key (NAT): 20.58 to 20.64**

MLF-Q231 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 61 Question Id : 640653825049 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

In a sports analytics company, the team is interested in predicting whether a basketball player will score above 20 points in a game using classification models. They collected data on various factors $x = [x_1, x_2, x_3]$ from past games of different players. The factors include minutes played, shots attempted, and free throw percentage. The data and corresponding labels are shown in the table below: Compute the misclassification

x	y
[1, 5, 0.8]	1
[0, 3, 0.9]	0
[1, 2, 0.7]	0
[1, 8, 0.6]	1
[0, 7, 0.5]	0

rate if they use the following step function:

$$u(z) = \begin{cases} 1, & \text{if } z \geq 3 \\ 0, & \text{otherwise} \end{cases}$$

and the linear combination z is given by: $0.6x_1 + 0.3x_2 + 0.1x_3$. Enter the answer correct to one decimal place.

Note: Here, x_1 represents whether the player is a starter (1) or not (0), x_2 is the number of shots attempted, and x_3 is the free throw percentage. The label y indicates whether the player scored more than 20 points (1) or not (0). The misclassification rate is the number of labels misclassified.e.g. if 3 out of 5 players are mislabeled then the misclassification rate is 3/5.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0.2

Sub-Section Number :	7
Sub-Section Id :	640653122731
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 0.2**

MLF-Q233 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 7 July 2024*
Category: Week 1 - ML Foundations › ML Concepts

Question Id : 640653827506 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Number : 50 Question Id : 640653827505 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)
Options :
6406532781316. ✔ YES
6406532781317. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653123477
Question Shuffling Allowed :	No

Question Id : 640653827506 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (51 to 52)

✔ **Official Answer Key: Option (A)**

MLF-Q234 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 7 July 2024*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 2 & 1 & 1 \\ a & 3 & 2 \\ 3 & b & c \end{pmatrix}$ be a 3×3 matrix. Let α be the eigenvalue corresponding to the eigenvector $v_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ and β is the eigenvalue corresponding to the eigenvector $v_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ of matrix A. Based on this information, answer the given sub-questions.

Sub questions

Question Number : 51 Question Id : 640653827507 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question
Find the value of $\alpha + 2\beta$.
Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
3

✔ **Official Answer Key (NAT): 3**

MLF-Q235 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 7 July 2024*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 2 & 1 & 1 \\ a & 3 & 2 \\ 3 & b & c \end{pmatrix}$ be a 3×3 matrix. Let α be the eigenvalue corresponding to the eigenvector $v_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ and β is the eigenvalue corresponding to the eigenvector $v_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ of matrix A. Based on this information, answer the given sub-questions.

Sub questions

Question Number : 52 Question Id : 640653827508 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following options is/are true ?
Options :

6406532781319. ✔ Eigenvectors, v_1 and v_2 are linearly independent.

6406532781320. ✖ Eigenvectors, v_1 and v_2 are orthogonal.

6406532781321. ✔ A third eigenvector of A is $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

6406532781322. ✔ $a + b + c = 9$.

Question Id : 640653827509 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (53 to 54)

✔ **Official Answer Key: Option (A, C, D)**

MLF-Q237 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 7 July 2024*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 54 Question Id : 640653827511 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Let v_1 and v_2 are the eigenvectors of A corresponding to distinct eigen-values.
We construct a new matrix B using these eigenvectors as its columns, such that $B = [v_1, v_2]$. Based on this information, which of the following options are correct?

Options :

6406532781327. ✔ Null space of the matrix B is zero space.

6406532781328. ✖ Column space of the matrix B is $\mathbb{R}^3$.

6406532781329. ✔ Column space of the matrix B represents a plane in $\mathbb{R}^3$.

6406532781330. ✖ Null space of the matrix B represents a line in $\mathbb{R}^3$.

Sub-Section Number :	3
Sub-Section Id :	640653123478
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A, C)

MLF-Q238 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 7 July 2024*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 55 Question Id : 640653827512 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Let P be the projection matrix for a vector $a = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. Which of the following options is/are true?

Options :

6406532781331. ✔ P is orthogonally diagonalizable.

6406532781332. ✔ $P^4 = P$.

6406532781333. ✔ Projection of $b = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ on the vector a is $-\frac{1}{5}a$.

6406532781334. ✖ $Rank(P) = 2$

Sub-Section Number :	4
Sub-Section Id :	640653123479
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A, B, C)**

MLF-Q239 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 7 July 2024*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 56 Question Id : 640653827513 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Find the best-fit line for the dataset given below using the least squares method.

x	0	2	3	5
y	1	2	4	3

Options :

6406532781335. ✔ $\hat{y} = 1.34 + 0.46x$

6406532781336. ✖ $\hat{y} = 0.46 + 1.34x$

6406532781337. ✖ $\hat{y} = 1.34 - 0.46x$

6406532781338. ✖ $\hat{y} = 2.16 + 0.74x$

✔ **Official Answer Key: Option (A)**

MLF-Q240 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 7 July 2024*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 57 Question Id : 640653827514 Question Type : MCQ
Correct Marks : 4

Question Label : Multiple Choice Question

The function

$$f(x) = \begin{cases} ax^2 + bx + 1, & \text{if } x < 0 \\ cx + d, & \text{if } x \geq 0 \end{cases}$$

is differentiable at $(x = 0)$. If a, b, c, and d are constants, which of the following must be true?

Options :

6406532781339. ✖ $a = c$ and $b = 1$

6406532781340. ✔ $b = c$ and $d = 1$

6406532781341. ✖ $a = c$ and $d = 1$

6406532781342. ✖ $b = d = 1$

Sub-Section Number :	5
Sub-Section Id :	640653123480
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B)

MLF-Q243 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 7 July 2024*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 60 Question Id : 640653827517 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

If $f(x, y) = x^2y + 3xy^2$, what is the length of the gradient at the point (1,2)? Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

20.58 to 20.64

✔ **Official Answer Key (NAT): 20.58 to 20.64**

MLF-Q244 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 7 July 2024*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 61 Question Id : 640653827518 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

In a sports analytics company, the team is interested in predicting whether a basketball player will score above 20 points in a game using classification models. They collected data on various factors $x = [x_1, x_2, x_3]$ from past games of different players. The factors include minutes played, shots attempted, and free throw percentage. The data and corresponding labels are shown in the table below: Compute the misclassification

x	y
[1, 5, 0.8]	1
[0, 3, 0.9]	0
[1, 2, 0.7]	0
[1, 8, 0.6]	1
[0, 7, 0.5]	0

rate if they use the following step function:

$$u(z) = \begin{cases} 1, & \text{if } z \geq 3 \\ 0, & \text{otherwise} \end{cases}$$

and the linear combination z is given by: $0.6x_1 + 0.3x_2 + 0.1x_3$. Enter the answer correct to one decimal place.

Note: Here, x_1 represents whether the player is a starter (1) or not (0), x_2 is the number of shots attempted, and x_3 is the free throw percentage. The label y indicates whether the player scored more than 20 points (1) or not (0). The misclassification rate is the number of labels misclassified.e.g. if 3 out of 5 players are mislabeled then the misclassification rate is 3/5.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0.2

Sub-Section Number :	7
Sub-Section Id :	640653123482
Question Shuffling Allowed :	Yes

✔ Official Answer Key (NAT): 0.2

MLF-Q246 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 7 July 2024(1) (+1 more papers)*
Category: Week 1 - ML Foundations › ML Concepts

Question Id : 640653827794 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Number : 50 Question Id : 640653827793 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)
Options :
6406532782112. ✔ YES
6406532782113. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653123584
Question Shuffling Allowed :	No

Question Id : 640653827794 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (51 to 52)

✔ **Official Answer Key: Option (A)**

MLF-Q247 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 7 July 2024(1) (+1 more papers)*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 2 & 1 & 1 \\ a & 3 & 2 \\ 3 & b & c \end{pmatrix}$ be a 3×3 matrix. Let α be the eigenvalue corresponding to the eigenvector $v_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ and β is the eigenvalue corresponding to the eigenvector $v_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ of matrix A. Based on this information, answer the given sub-questions.

Sub questions

Question Number : 51 Question Id : 640653827795 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question
Find the value of $\alpha + 2\beta$.
Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
3

✔ **Official Answer Key (NAT): 3**

MLF-Q248 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 7 July 2024(1) (+1 more papers)*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 2 & 1 & 1 \\ a & 3 & 2 \\ 3 & b & c \end{pmatrix}$ be a 3×3 matrix. Let α be the eigenvalue corresponding to the eigenvector $v_1 = \begin{pmatrix} 1 \\ 0 \\ -1 \end{pmatrix}$ and β is the eigenvalue corresponding to the eigenvector $v_2 = \begin{pmatrix} 0 \\ 1 \\ -1 \end{pmatrix}$ of matrix A. Based on this information, answer the given sub-questions.

Sub questions

Question Number : 52 Question Id : 640653827796 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following options is/are true ?
Options :

6406532782115. ✔ Eigenvectors, v_1 and v_2 are linearly independent.

6406532782116. ✖ Eigenvectors, v_1 and v_2 are orthogonal.

6406532782117. ✔ A third eigenvector of A is $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$

6406532782118. ✔ $a + b + c = 9$.

Question Id : 640653827797 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (53 to 54)

✔ Official Answer Key: Option (A, C, D)

MLF-Q250 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 7 July 2024(1) (+1 more papers)*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 54 Question Id : 640653827799 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

Let v_1 and v_2 are the eigenvectors of A corresponding to distinct eigen-values.
We construct a new matrix B using these eigenvectors as its columns, such that $B = [v_1, v_2]$. Based on this information, which of the following options are correct?

Options :

6406532782123. ✔ Null space of the matrix B is zero space.

6406532782124. ✖ Column space of the matrix B is $\mathbb{R}^3$.

6406532782125. ✔ Column space of the matrix B represents a plane in $\mathbb{R}^3$.

6406532782126. ✖ Null space of the matrix B represents a line in $\mathbb{R}^3$.

Sub-Section Number :	3
Sub-Section Id :	640653123585
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A, C)

MLF-Q251 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 7 July 2024(1) (+1 more papers)*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 55 Question Id : 640653827800 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Let P be the projection matrix for a vector $a = \begin{pmatrix} 1 \\ 2 \end{pmatrix}$. Which of the following options is/are true?

Options :

6406532782127. ✔ P is orthogonally diagonalizable.

6406532782128. ✔ $P^4 = P$.

6406532782129. ✔ Projection of $b = \begin{pmatrix} 1 \\ -1 \end{pmatrix}$ on the vector a is $-\frac{1}{5}a$.

6406532782130. ✖ $Rank(P) = 2$

Sub-Section Number :	4
Sub-Section Id :	640653123586
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A, B, C)

MLF-Q252 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 7 July 2024(1) (+1 more papers)*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 56 Question Id : 640653827801 Question Type : MCQ

Correct Marks : 4

Question Label : Multiple Choice Question

Find the best-fit line for the dataset given below using the least squares method.

x	0	2	3	5
y	1	2	4	3

Options :

6406532782131. ✔ $\hat{y} = 1.34 + 0.46x$

6406532782132. ✖ $\hat{y} = 0.46 + 1.34x$

6406532782133. ✖ $\hat{y} = 1.34 - 0.46x$

6406532782134. ✖ $\hat{y} = 2.16 + 0.74x$

✔ **Official Answer Key: Option (A)**

MLF-Q253 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 7 July 2024(1) (+1 more papers)*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 57 Question Id : 640653827802 Question Type : MCQ
Correct Marks : 4

Question Label : Multiple Choice Question

The function

$$f(x) = \begin{cases} ax^2 + bx + 1, & \text{if } x < 0 \\ cx + d, & \text{if } x \geq 0 \end{cases}$$

is differentiable at $(x = 0)$. If a, b, c, and d are constants, which of the following must be true?

Options :

6406532782135. ✖ $a = c$ and $b = 1$

6406532782136. ✔ $b = c$ and $d = 1$

6406532782137. ✖ $a = c$ and $d = 1$

6406532782138. ✖ $b = d = 1$

Sub-Section Number :	5
Sub-Section Id :	640653123587
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B)**

MLF-Q256 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 7 July 2024(1) (+1 more papers)*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 60 Question Id : 640653827805 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

If $f(x, y) = x^2y + 3xy^2$, what is the length of the gradient at the point (1,2)? Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

20.58 to 20.64

✔ **Official Answer Key (NAT): 20.58 to 20.64**

MLF-Q257 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 7 July 2024(1) (+1 more papers)*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 61 Question Id : 640653827806 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

In a sports analytics company, the team is interested in predicting whether a basketball player will score above 20 points in a game using classification models. They collected data on various factors $x = [x_1, x_2, x_3]$ from past games of different players. The factors include minutes played, shots attempted, and free throw percentage. The data and corresponding labels are shown in the table below: Compute the misclassification

x	y
[1, 5, 0.8]	1
[0, 3, 0.9]	0
[1, 2, 0.7]	0
[1, 8, 0.6]	1
[0, 7, 0.5]	0

rate if they use the following step function:

$$u(z) = \begin{cases} 1, & \text{if } z \geq 3 \\ 0, & \text{otherwise} \end{cases}$$

and the linear combination z is given by: $0.6x_1 + 0.3x_2 + 0.1x_3$. Enter the answer correct to one decimal place.

Note: Here, x_1 represents whether the player is a starter (1) or not (0), x_2 is the number of shots attempted, and x_3 is the free throw percentage. The label y indicates whether the player scored more than 20 points (1) or not (0). The misclassification rate is the number of labels misclassified.e.g. if 3 out of 5 players are mislabeled then the misclassification rate is 3/5.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0.2

Sub-Section Number :	7
Sub-Section Id :	640653123589
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 0.2**

MLF-Q259 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 110 Question Id : 6406531527839 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406535115674. ✔ YES

6406535115675. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653244922
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLF-Q261 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 112 Question Id : 6406531527841 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the dataset $(x, y) : (-1, 2), (0, 1), (1, 2)$. Perform polynomial regression of degree 2 by fitting a quadratic model of the form $y = \theta_0 + \theta_1 x + \theta_2 x^2$. Using the method of least squares, find the estimates of θ_0, θ_1 and θ_2.

Options :

6406535115680. ✔ $y = x^2 + 1$

6406535115681. ✖ $y = x^2$

6406535115682. ✖ $y = x^2 - 1$

6406535115683. ✖ $y = 2x$

Sub-Section Number :	4
Sub-Section Id :	640653244924
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A)**

MLF-Q262 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 113 Question Id : 6406531527842 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Two vectors v_1 and v_2 are such that $v_1 = (1, 2, 2)^T$ and $v_2 = (2, -1, 0)^T$.
Which of the following options is/are correct?

Options :

6406535115684. ✖ Vectors v_1 and v_2 are parallel to each other.

6406535115685. ✔ Vectors v_1 and v_2 are orthogonal to each other.

6406535115686. ✔ Projection of v_1 onto v_2 is the zero vector.

6406535115687. ✔ Projection of v_2 onto v_1 is the zero vector.

Sub-Section Number :	5
Sub-Section Id :	640653244925
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B, C, D)

MLF-Q263 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 114 Question Id : 6406531527843 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question
Consider the following dataset where each data point consists of three features x_1, x_2 and x_3 as

$$(2,4,3),(5,5,6),(2,2,1),(4,4,6)$$

Suppose the encoder and decoder functions, f and g respectively, are defined as

$$f(x_1,x_2,x_3) = 0.5x_1 + 0.5x_2 - x_3 \quad \text{and} \quad g(u) = [u,u,u].$$

Find the reconstruction loss.
Hint : "The reconstruction loss of the encoder decoder pair is the mean of the squared distance between the actual input and reconstructed input."

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal

Text Areas : PlainText
Possible Answers :
72

✔ **Official Answer Key (NAT): 72**

MLF-Q264 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Id : 6406531527846 Question Type : COMPREHENSION Sub Question Shuffling

Question Number : 115 Question Id : 6406531527844 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Does there exist a matrix A of shape 10×10 such that every non-zero vector in $\mathbb{R}^{10}$ is an eigenvector of A with eigenvalue 2? If yes, compute the value of $\det(A)$. If not, enter -1 as the answer.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1024

Sub-Section Number :	6
Sub-Section Id :	640653244926
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 1024**

MLF-Q265 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Id : 6406531527846 Question Type : COMPREHENSION Sub Question Shuffling

Question Number : 116 Question Id : 6406531527845 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question

If the sum of the columns of a square matrix A is equal to the zero vector, then λ is certainly an eigenvalue of A. What is the value of λ?

Note that the term "column" here refers to a column vector, that is, each column of A is treated as a vector.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0

Sub-Section Number :	7
Sub-Section Id :	640653244927
Question Shuffling Allowed :	No

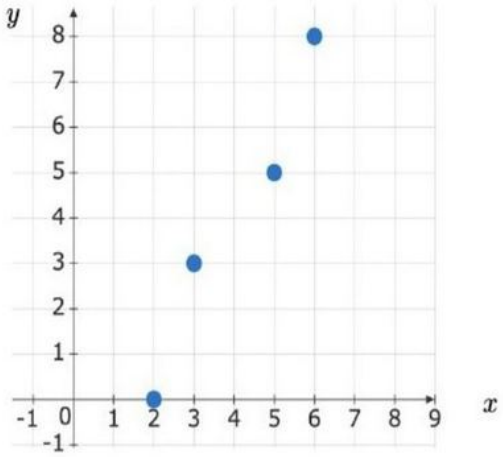
Question Id : 6406531527846 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (117 to 118)

✔ **Official Answer Key (NAT): 0**

MLF-Q267 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 118 Question Id : 6406531527848 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Is there a model of the form $f(x) = wx + b$, with $w, b \in \mathbb{R}$ such that the squared loss is exactly zero on this training dataset?

Options :

6406535115695. ✔ No

6406535115696. ✖ Yes

6406535115697. ✖ Insufficient information to answer the question

Question Id : 6406531527853 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (119 to 120)

✔ Official Answer Key: Option (A)

MLF-Q269 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 120 Question Id : 6406531527855 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following linear relations is valid among the rows (R_1, R_2, R_3) of matrix A?

Options :

6406535115709. ✖ $R_2 - R_1 - R_3 = 0$

6406535115710. ✖ $R_1 + R_2 - 2R_3 = 0$

6406535115711. ✔ $0.R_1 + 0.R_2 + 0.R_3 = 0$

6406535115712. ✖ $R_2 = 2R_1 + R_3$

Question Id : 6406531527856 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (121 to 122)

✔ **Official Answer Key: Option (C)**

MLF-Q275 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 126 Question Id : 6406531527850 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following is
$L_{(0,0)}[f](x)$?

Options :

6406535115698. ✔ $1 + x_1 + x_2$

6406535115699. ✖ $x_1 + 2x_2$

6406535115700. ✖ $1 - x_1 - x_2$

6406535115701. ✖ $x_1 - x_2$

✔ **Official Answer Key: Option (A)**

MLF-Q276 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 127 Question Id : 6406531527851 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question

Recall that $L_{(0,0)}[f](x)$ corresponds to a plane in $\mathbb{R}^3$ and can be represented as the set of all points that satisfy the equation $w^T x + b = 0$, where $w \in \mathbb{R}^3, b \in \mathbb{R}$, with w being normal to the plane. If $w^T w = 3$, what is b?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

1

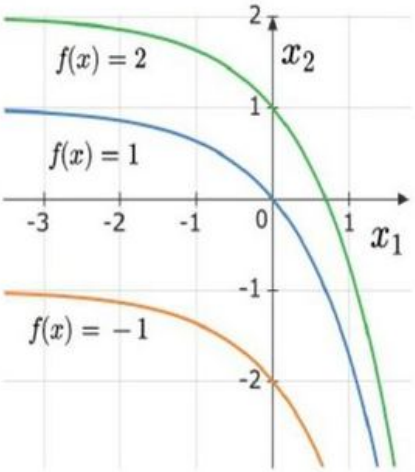
✔ **Official Answer Key (NAT): 1**

MLF-Q277 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

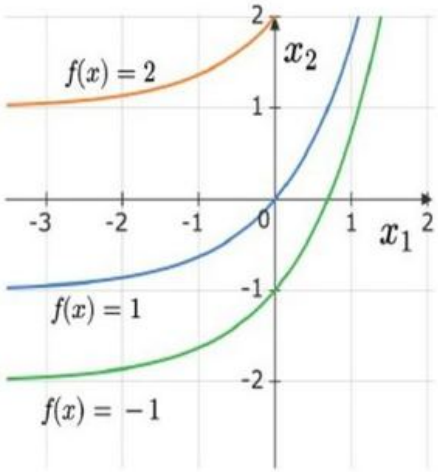
Question Label : Comprehension

Question Number : 128 Question Id : 6406531527852 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following represents the contours of f?



(A)



(B)

Options :

6406535115703. ✔ (A)

6406535115704. ✖ (B)

System commands

✔ **Official Answer Key: Option (A)**

MLF-Q278 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 110 Question Id : 6406531528142 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406535116517. ✔ YES

6406535116518. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653245018
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLF-Q280 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 112 Question Id : 6406531528144 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the dataset $(x, y) : (-1, 2), (0, 1), (1, 2)$. Perform polynomial regression of degree 2 by fitting a quadratic model of the form $y = \theta_0 + \theta_1 x + \theta_2 x^2$. Using the method of least squares, find the estimates of θ_0, θ_1 and θ_2.

Options :

6406535116523. ✔ $y = x^2 + 1$

6406535116524. ✖ $y = x^2$

6406535116525. ✖ $y = x^2 - 1$

6406535116526. ✖ $y = 2x$

Sub-Section Number :	4
Sub-Section Id :	640653245020
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A)**

MLF-Q281 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 113 Question Id : 6406531528145 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Two vectors v_1 and v_2 are such that $v_1 = (1, 2, 2)^T$ and $v_2 = (2, -1, 0)^T$.
Which of the following options is/are correct?

Options :

6406535116527. ✖ Vectors v_1 and v_2 are parallel to each other.

6406535116528. ✔ Vectors v_1 and v_2 are orthogonal to each other.

6406535116529. ✔ Projection of v_1 onto v_2 is the zero vector.

6406535116530. ✔ Projection of v_2 onto v_1 is the zero vector.

Sub-Section Number :	5
Sub-Section Id :	640653245021
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B, C, D)**

MLF-Q282 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 114 Question Id : 6406531528146 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question
Consider the following dataset where each data point consists of three features x_1, x_2 and x_3 as

$$(2,4,3),(5,5,6),(2,2,1),(4,4,6)$$

Suppose the encoder and decoder functions, f and g respectively, are defined as

$$f(x_1,x_2,x_3) = 0.5x_1 + 0.5x_2 - x_3 \quad \text{and} \quad g(u) = [u,u,u].$$

Find the reconstruction loss.
Hint : "The reconstruction loss of the encoder decoder pair is the mean of the squared distance between the actual input and reconstructed input."

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal

Text Areas : PlainText
Possible Answers :
72

✔ **Official Answer Key (NAT): 72**

MLF-Q283 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Id : 6406531528149 Question Type : COMPREHENSION Sub Question Shuffling

Question Number : 115 Question Id : 6406531528147 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Does there exist a matrix A of shape 10×10 such that every non-zero vector in $\mathbb{R}^{10}$ is an eigenvector of A with eigenvalue 2? If yes, compute the value of $\det(A)$. If not, enter -1 as the answer.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1024

Sub-Section Number :	6
Sub-Section Id :	640653245022
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 1024**

MLF-Q284 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Id : 6406531528149 Question Type : COMPREHENSION Sub Question Shuffling

Question Number : 116 Question Id : 6406531528148 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question

If the sum of the columns of a square matrix A is equal to the zero vector, then λ is certainly an eigenvalue of A. What is the value of λ?

Note that the term "column" here refers to a column vector, that is, each column of A is treated as a vector.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0

Sub-Section Number :	7
Sub-Section Id :	640653245023
Question Shuffling Allowed :	No

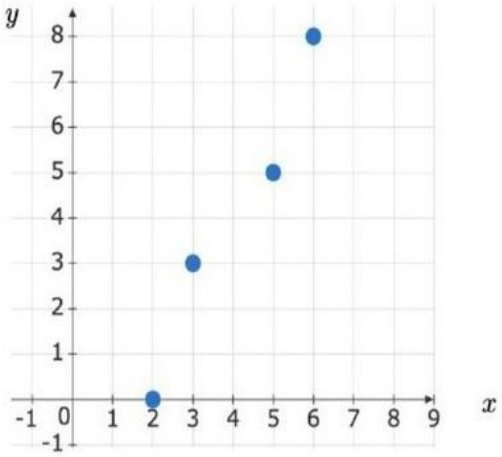
Question Id : 6406531528149 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (117 to 118)

✔ **Official Answer Key (NAT): 0**

MLF-Q286 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 118 Question Id : 6406531528151 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Is there a model of the form $f(x) = wx + b$, with $w, b \in \mathbb{R}$ such that the squared loss is exactly zero on this training dataset?

Options :

6406535116538. ✔ No

6406535116539. ✖ Yes

6406535116540. ✖ Insufficient information to answer the question

Question Id : 6406531528156 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (119 to 120)

✔ Official Answer Key: Option (A)

MLF-Q288 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 120 Question Id : 6406531528158 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following linear relations is valid among the rows (R_1, R_2, R_3) of matrix A?

Options :

6406535116552. ✖ $R_2 - R_1 - R_3 = 0$

6406535116553. ✖ $R_1 + R_2 - 2R_3 = 0$

6406535116554. ✔ $0.R_1 + 0.R_2 + 0.R_3 = 0$

6406535116555. ✖ $R_2 = 2R_1 + R_3$

Question Id : 6406531528159 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (121 to 122)

✔ **Official Answer Key: Option (C)**

MLF-Q294 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 126 Question Id : 6406531528153 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following is
$L_{(0,0)}[f](x)$?

Options :

6406535116541. ✔ $1 + x_1 + x_2$

6406535116542. ✖ $x_1 + 2x_2$

6406535116543. ✖ $1 - x_1 - x_2$

6406535116544. ✖ $x_1 - x_2$

✔ **Official Answer Key: Option (A)**

MLF-Q295 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 127 Question Id : 6406531528154 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question
Recall that $L_{(0,0)}[f](x)$ corresponds to a plane in $\mathbb{R}^3$ and can be represented as the set of all points that satisfy the equation $w^T x + b = 0$, where $w \in \mathbb{R}^3, b \in \mathbb{R}$, with w being normal to the plane. If $w^T w = 3$, what is b?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

1

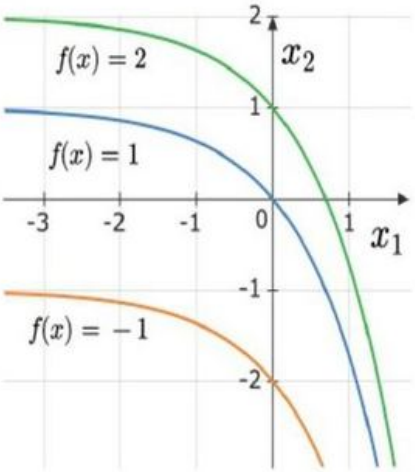
✔ **Official Answer Key (NAT): 1**

MLF-Q296 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

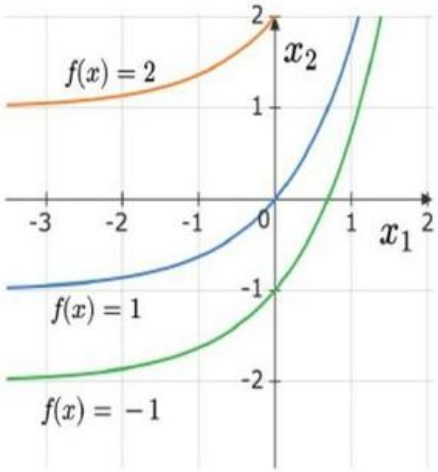
Question Label : Comprehension

Question Number : 128 Question Id : 6406531528155 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following represents the contours of f?



(A)



(B)

Options :

6406535116546. ✔ (A)

6406535116547. ✖ (B)

System commands

✔ **Official Answer Key: Option (A)**

MLF-Q297 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 110 Question Id : 6406531528804 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406535118448. ✔ YES

6406535118449. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653245203
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLF-Q299 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 112 Question Id : 6406531528806 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the dataset $(x, y) : (-1, 2), (0, 1), (1, 2)$. Perform polynomial regression of degree 2 by fitting a quadratic model of the form $y = \theta_0 + \theta_1 x + \theta_2 x^2$. Using the method of least squares, find the estimates of θ_0, θ_1 and θ_2.

Options :

6406535118454. ✔ $y = x^2 + 1$

6406535118455. ✖ $y = x^2$

6406535118456. ✖ $y = x^2 - 1$

6406535118457. ✖ $y = 2x$

Sub-Section Number :	4
Sub-Section Id :	640653245205
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A)**

MLF-Q300 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 113 Question Id : 6406531528807 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Two vectors v_1 and v_2 are such that $v_1 = (1, 2, 2)^T$ and $v_2 = (2, -1, 0)^T$.
Which of the following options is/are correct?

Options :

6406535118458. ✖ Vectors v_1 and v_2 are parallel to each other.

6406535118459. ✔ Vectors v_1 and v_2 are orthogonal to each other.

6406535118460. ✔ Projection of v_1 onto v_2 is the zero vector.

6406535118461. ✔ Projection of v_2 onto v_1 is the zero vector.

Sub-Section Number :	5
Sub-Section Id :	640653245206
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B, C, D)**

MLF-Q301 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 114 Question Id : 6406531528808 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question
Consider the following dataset where each data point consists of three features x_1, x_2 and x_3 as

$$(2,4,3),(5,5,6),(2,2,1),(4,4,6)$$

Suppose the encoder and decoder functions, f and g respectively, are defined as

$$f(x_1,x_2,x_3) = 0.5x_1 + 0.5x_2 - x_3 \quad \text{and} \quad g(u) = [u,u,u].$$

Find the reconstruction loss.
Hint : "The reconstruction loss of the encoder decoder pair is the mean of the squared distance between the actual input and reconstructed input."

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal

Text Areas : PlainText
Possible Answers :
72

✔ **Official Answer Key (NAT): 72**

MLF-Q302 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Id : 6406531528811 Question Type : COMPREHENSION Sub Question Shuffling

Question Number : 115 Question Id : 6406531528809 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Does there exist a matrix A of shape 10×10 such that every non-zero vector in $\mathbb{R}^{10}$ is an eigenvector of A with eigenvalue 2? If yes, compute the value of $\det(A)$. If not, enter -1 as the answer.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1024

Sub-Section Number :	6
Sub-Section Id :	640653245207
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 1024**

MLF-Q303 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Id : 6406531528811 Question Type : COMPREHENSION Sub Question Shuffling

Question Number : 116 Question Id : 6406531528810 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question

If the sum of the columns of a square matrix A is equal to the zero vector, then λ is certainly an eigenvalue of A. What is the value of λ?

Note that the term "column" here refers to a column vector, that is, each column of A is treated as a vector.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0

Sub-Section Number :	7
Sub-Section Id :	640653245208
Question Shuffling Allowed :	No

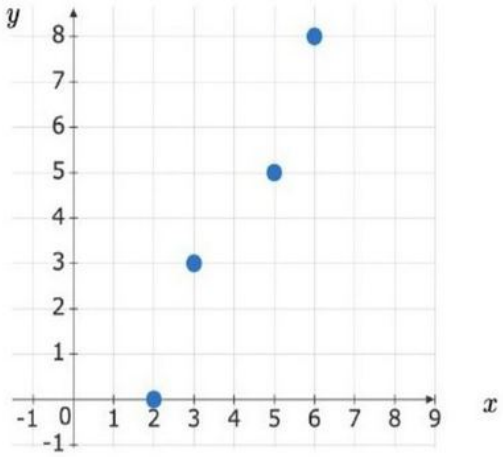
Question Id : 6406531528811 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (117 to 118)

✔ **Official Answer Key (NAT): 0**

MLF-Q305 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 118 Question Id : 6406531528813 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Is there a model of the form $f(x) = wx + b$, with $w, b \in \mathbb{R}$ such that the squared loss is exactly zero on this training dataset?

Options :

6406535118469. ✔ No

6406535118470. ✖ Yes

6406535118471. ✖ Insufficient information to answer the question

Question Id : 6406531528818 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (119 to 120)

✔ Official Answer Key: Option (A)

MLF-Q307 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 120 Question Id : 6406531528820 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following linear relations is valid among the rows (R_1, R_2, R_3) of matrix A?

Options :

6406535118483. ✖ $R_2 - R_1 - R_3 = 0$

6406535118484. ✖ $R_1 + R_2 - 2R_3 = 0$

6406535118485. ✔ $0.R_1 + 0.R_2 + 0.R_3 = 0$

6406535118486. ✖ $R_2 = 2R_1 + R_3$

Question Id : 6406531528821 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (121 to 122)

✔ **Official Answer Key: Option (C)**

MLF-Q313 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 126 Question Id : 6406531528815 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following is
$L_{(0,0)}[f](x)$?

Options :

6406535118472. ✔ $1 + x_1 + x_2$

6406535118473. ✖ $x_1 + 2x_2$

6406535118474. ✖ $1 - x_1 - x_2$

6406535118475. ✖ $x_1 - x_2$

✔ **Official Answer Key: Option (A)**

MLF-Q314 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 127 Question Id : 6406531528816 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question
Recall that $L_{(0,0)}[f](x)$ corresponds to a plane in $\mathbb{R}^3$ and can be represented as the set of all points that satisfy the equation $w^T x + b = 0$, where $w \in \mathbb{R}^3, b \in \mathbb{R}$, with w being normal to the plane. If $w^T w = 3$, what is b?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

1

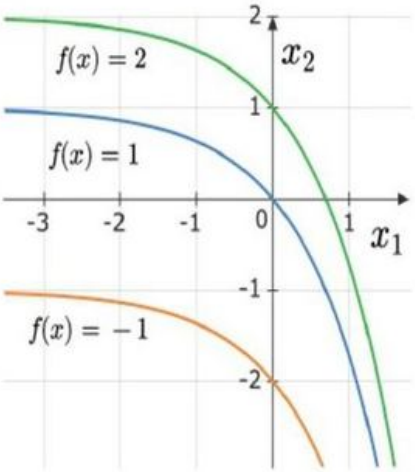
✔ **Official Answer Key (NAT): 1**

MLF-Q315 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

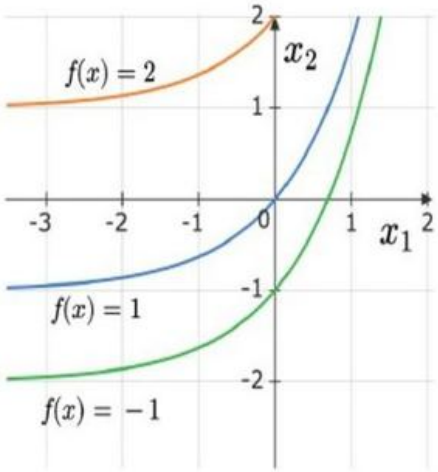
Question Label : Comprehension

Question Number : 128 Question Id : 6406531528817 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following represents the contours of f?



(A)



(B)

Options :

6406535118477. ✔ (A)

6406535118478. ✖ (B)

System commands

✔ **Official Answer Key: Option (A)**

MLF-Q316 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 136 Question Id : 6406531525150 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)
Options :
6406535108149. ✔ YES
6406535108150. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653244042
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLF-Q318 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 138 Question Id : 6406531525152 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the dataset $(x, y) : (-1, 2), (0, 1), (1, 2)$. Perform polynomial regression of degree 2 by fitting a quadratic model of the form $y = \theta_0 + \theta_1 x + \theta_2 x^2$. Using the method of least squares, find the estimates of θ_0, θ_1 and θ_2.

Options :

6406535108155. ✔ $y = x^2 + 1$

6406535108156. ✖ $y = x^2$

6406535108157. ✖ $y = x^2 - 1$

6406535108158. ✖ $y = 2x$

Sub-Section Number :	4
Sub-Section Id :	640653244044
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A)**

MLF-Q319 | [MSQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 139 Question Id : 6406531525153 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Two vectors v_1 and v_2 are such that $v_1 = (1, 2, 2)^T$ and $v_2 = (2, -1, 0)^T$.
Which of the following options is/are correct?

Options :

6406535108159. ✖ Vectors v_1 and v_2 are parallel to each other.

6406535108160. ✔ Vectors v_1 and v_2 are orthogonal to each other.

6406535108161. ✔ Projection of v_1 onto v_2 is the zero vector.

6406535108162. ✔ Projection of v_2 onto v_1 is the zero vector.

Sub-Section Number :	5
Sub-Section Id :	640653244045
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B, C, D)

MLF-Q320 | [SA] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 140 Question Id : 6406531525154 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Consider the following dataset where each data point consists of three features x_1, x_2 and x_3 as

$$(2,4,3),(5,5,6),(2,2,1),(4,4,6)$$

Suppose the encoder and decoder functions, f and g respectively, are defined as

$$f(x_1,x_2,x_3) = 0.5x_1 + 0.5x_2 - x_3 \quad \text{and} \quad g(u) = [u,u,u].$$

Find the reconstruction loss.
Hint : "The reconstruction loss of the encoder decoder pair is the mean of the squared distance between the actual input and reconstructed input."

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
72

✔ **Official Answer Key (NAT): 72**

MLF-Q321 | [SA] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 141 Question Id : 6406531525155 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Does there exist a matrix A of shape 10×10 such that every non-zero vector in $\mathbb{R}^{10}$ is an eigenvector of A with eigenvalue 2? If yes, compute the value of $\det(A)$. If not, enter -1 as the answer.

Response Type : Numeric

Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1024

Sub-Section Number :	6
Sub-Section Id :	640653244046
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 1024**

MLF-Q322 | [SA] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 142 Question Id : 6406531525156 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question

If the sum of the columns of a square matrix A is equal to the zero vector, then λ is certainly an eigenvalue of A. What is the value of λ?

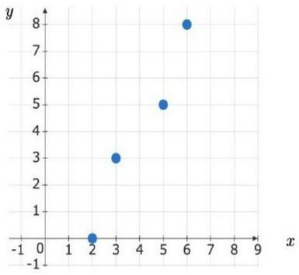
Note that the term "column" here refers to a column vector, that is, each column of A is treated as a vector.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0

Sub-Section Number :	7
Sub-Section Id :	640653244047
Question Shuffling Allowed :	No

Question Id : 6406531525157 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (143 to 144)
Question Label : Comprehension

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Based on the above data, answer the given subquestions.
Sub questions

✔ Official Answer Key (NAT): 0

MLF-Q324 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 144 Question Id : 6406531525159 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Is there a model of the form $f(x) = wx + b$,
with $w, b \in \mathbb{R}$ such that the squared loss
is exactly zero on this training dataset?

Options :

6406535108170. ✔ No

6406535108171. ✖ Yes

6406535108172. ✖ Insufficient information to answer the question

Question Id : 6406531525164 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (145 to 146)

✔ **Official Answer Key: Option (A)**

MLF-Q326 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 146 Question Id : 6406531525166 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following linear relations is valid among the rows (R_1, R_2, R_3) of matrix A?

Options :

6406535108184. ✖ $R_2 - R_1 - R_3 = 0$

6406535108185. ✖ $R_1 + R_2 - 2R_3 = 0$

6406535108186. ✔ $0.R_1 + 0.R_2 + 0.R_3 = 0$

6406535108187. ✖ $R_2 = 2R_1 + R_3$

Question Id : 6406531525167 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (147 to 148)

✔ **Official Answer Key: Option (C)**

MLF-Q332 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 152 Question Id : 6406531525161 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following is $L_{(0,0)}[f](x)$?

Options :

6406535108173. ✔ $1 + x_1 + x_2$

6406535108174. ✖ $x_1 + 2x_2$

6406535108175. ✖ $1 - x_1 - x_2$

6406535108176. ✖ $x_1 - x_2$

✔ **Official Answer Key: Option (A)**

MLF-Q333 | [SA] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 153 Question Id : 6406531525162 Question Type : SA
Correct Marks : 2

Question Label : Short Answer Question

Recall that $L_{(0,0)}[f](x)$ corresponds to a plane in $\mathbb{R}^3$ and can be represented as the set of all points that satisfy the equation $w^T x + b = 0$, where $w \in \mathbb{R}^3, b \in \mathbb{R}$, with w being normal to the plane. If $w^T w = 3$, what is b?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

✔ **Official Answer Key (NAT): 1**

MLF-Q334 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

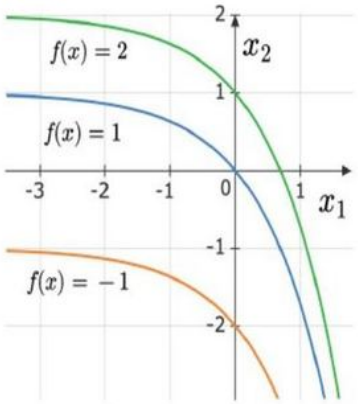
Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

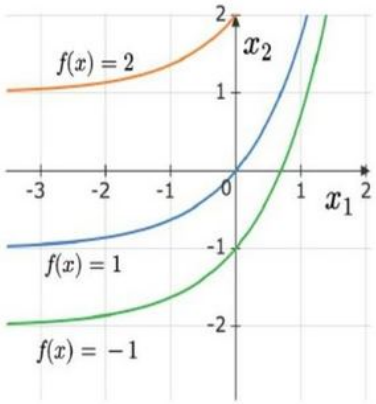
Sub questions

Question Number : 154 Question Id : 6406531525163 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following represents the contours of f?



(A)



(B)

Options :

6406535108178. ✔ (A)

6406535108179. ✖ (B)

✔ **Official Answer Key: Option (A)**

MLF-Q335 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 136 Question Id : 6406531525777 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)
Options :
6406535109926. ✔ YES
6406535109927. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653244235
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLF-Q337 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 138 Question Id : 6406531525779 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the dataset $(x, y) : (-1, 2), (0, 1), (1, 2)$. Perform polynomial regression of degree 2 by fitting a quadratic model of the form $y = \theta_0 + \theta_1 x + \theta_2 x^2$. Using the method of least squares, find the estimates of θ_0, θ_1 and θ_2.

Options :

6406535109932. ✔ $y = x^2 + 1$

6406535109933. ✖ $y = x^2$

6406535109934. ✖ $y = x^2 - 1$

6406535109935. ✖ $y = 2x$

Sub-Section Number :	4
Sub-Section Id :	640653244237
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A)**

MLF-Q338 | [MSQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 139 Question Id : 6406531525780 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Two vectors v_1 and v_2 are such that $v_1 = (1, 2, 2)^T$ and $v_2 = (2, -1, 0)^T$.
Which of the following options is/are correct?

Options :

6406535109936. ✖ Vectors v_1 and v_2 are parallel to each other.

6406535109937. ✔ Vectors v_1 and v_2 are orthogonal to each other.

6406535109938. ✔ Projection of v_1 onto v_2 is the zero vector.

6406535109939. ✔ Projection of v_2 onto v_1 is the zero vector.

Sub-Section Number :	5
Sub-Section Id :	640653244238
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B, C, D)

MLF-Q339 | [SA] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 140 Question Id : 6406531525781 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Consider the following dataset where each data point consists of three features x_1, x_2 and x_3 as

$$(2,4,3),(5,5,6),(2,2,1),(4,4,6)$$

Suppose the encoder and decoder functions, f and g respectively, are defined as

$$f(x_1,x_2,x_3) = 0.5x_1 + 0.5x_2 - x_3 \quad \text{and} \quad g(u) = [u,u,u].$$

Find the reconstruction loss.
Hint : "The reconstruction loss of the encoder decoder pair is the mean of the squared distance between the actual input and reconstructed input."

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
72

✔ **Official Answer Key (NAT): 72**

MLF-Q340 | [SA] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 141 Question Id : 6406531525782 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Does there exist a matrix A of shape 10×10 such that every non-zero vector in $\mathbb{R}^{10}$ is an eigenvector of A with eigenvalue 2? If yes, compute the value of $\det(A)$. If not, enter -1 as the answer.

Response Type : Numeric

Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1024

Sub-Section Number :	6
Sub-Section Id :	640653244239
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 1024**

MLF-Q341 | [SA] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 142 Question Id : 6406531525783 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question

If the sum of the columns of a square matrix A is equal to the zero vector, then λ is certainly an eigenvalue of A. What is the value of λ?

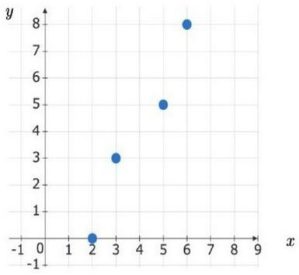
Note that the term "column" here refers to a column vector, that is, each column of A is treated as a vector.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0

Sub-Section Number :	7
Sub-Section Id :	640653244240
Question Shuffling Allowed :	No

Question Id : 6406531525784 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (143 to 144)
Question Label : Comprehension

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Based on the above data, answer the given subquestions.
Sub questions

✔ Official Answer Key (NAT): 0

MLF-Q343 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 144 Question Id : 6406531525786 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Is there a model of the form $f(x) = wx + b$,
with $w, b \in \mathbb{R}$ such that the squared loss
is exactly zero on this training dataset?

Options :

6406535109947. ✔ No

6406535109948. ✖ Yes

6406535109949. ✖ Insufficient information to answer the question

Question Id : 6406531525791 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (145 to 146)

✔ **Official Answer Key: Option (A)**

MLF-Q345 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 146 Question Id : 6406531525793 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following linear relations is valid among the rows (R_1, R_2, R_3) of matrix A?

Options :

6406535109961. ✖ $R_2 - R_1 - R_3 = 0$

6406535109962. ✖ $R_1 + R_2 - 2R_3 = 0$

6406535109963. ✔ $0.R_1 + 0.R_2 + 0.R_3 = 0$

6406535109964. ✖ $R_2 = 2R_1 + R_3$

Question Id : 6406531525794 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (147 to 148)

✔ **Official Answer Key: Option (C)**

MLF-Q351 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 152 Question Id : 6406531525788 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following is $L_{(0,0)}[f](x)$?

Options :

6406535109950. ✔ $1 + x_1 + x_2$

6406535109951. ✖ $x_1 + 2x_2$

6406535109952. ✖ $1 - x_1 - x_2$

6406535109953. ✖ $x_1 - x_2$

✔ **Official Answer Key: Option (A)**

MLF-Q352 | [SA] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 153 Question Id : 6406531525789 Question Type : SA
Correct Marks : 2

Question Label : Short Answer Question

Recall that $L_{(0,0)}[f](x)$ corresponds to a plane in $\mathbb{R}^3$ and can be represented as the set of all points that satisfy the equation $w^T x + b = 0$, where $w \in \mathbb{R}^3, b \in \mathbb{R}$, with w being normal to the plane. If $w^T w = 3$, what is b?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

✔ **Official Answer Key (NAT): 1**

MLF-Q353 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

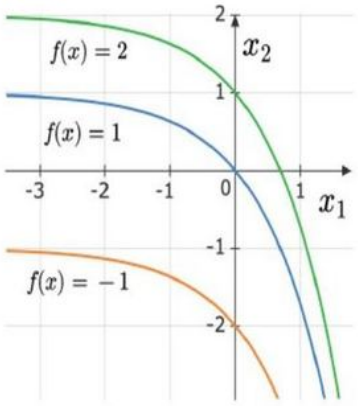
Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

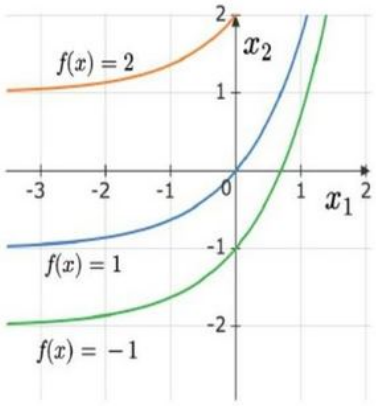
Sub questions

Question Number : 154 Question Id : 6406531525790 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following represents the contours of f?



(A)



(B)

Options :

6406535109955. ✔ (A)

6406535109956. ✖ (B)

✔ Official Answer Key: Option (A)

MLF-Q354 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 136 Question Id : 6406531526706 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)
Options :
6406535112571. ✔ YES
6406535112572. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653244526
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLF-Q356 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 138 Question Id : 6406531526708 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the dataset $(x, y) : (-1, 2), (0, 1), (1, 2)$. Perform polynomial regression of degree 2 by fitting a quadratic model of the form $y = \theta_0 + \theta_1 x + \theta_2 x^2$. Using the method of least squares, find the estimates of θ_0, θ_1 and θ_2.

Options :

6406535112577. ✔ $y = x^2 + 1$

6406535112578. ✖ $y = x^2$

6406535112579. ✖ $y = x^2 - 1$

6406535112580. ✖ $y = 2x$

Sub-Section Number :	4
Sub-Section Id :	640653244528
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A)**

MLF-Q357 | [MSQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 139 Question Id : 6406531526709 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Two vectors v_1 and v_2 are such that $v_1 = (1, 2, 2)^T$ and $v_2 = (2, -1, 0)^T$.
Which of the following options is/are correct?

Options :

6406535112581. ✖ Vectors v_1 and v_2 are parallel to each other.

6406535112582. ✔ Vectors v_1 and v_2 are orthogonal to each other.

6406535112583. ✔ Projection of v_1 onto v_2 is the zero vector.

6406535112584. ✔ Projection of v_2 onto v_1 is the zero vector.

Sub-Section Number :	5
Sub-Section Id :	640653244529
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B, C, D)

MLF-Q358 | [SA] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 140 Question Id : 6406531526710 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Consider the following dataset where each data point consists of three features x_1, x_2 and x_3 as

$$(2,4,3),(5,5,6),(2,2,1),(4,4,6)$$

Suppose the encoder and decoder functions, f and g respectively, are defined as

$$f(x_1,x_2,x_3) = 0.5x_1 + 0.5x_2 - x_3 \quad \text{and} \quad g(u) = [u,u,u].$$

Find the reconstruction loss.
Hint : "The reconstruction loss of the encoder decoder pair is the mean of the squared distance between the actual input and reconstructed input."

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
72

✔ **Official Answer Key (NAT): 72**

MLF-Q359 | [SA] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 141 Question Id : 6406531526711 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Does there exist a matrix A of shape 10×10 such that every non-zero vector in $\mathbb{R}^{10}$ is an eigenvector of A with eigenvalue 2? If yes, compute the value of $\det(A)$. If not, enter -1 as the answer.

Response Type : Numeric

Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1024

Sub-Section Number :	6
Sub-Section Id :	640653244530
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 1024**

MLF-Q360 | [SA] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 142 Question Id : 6406531526712 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question

If the sum of the columns of a square matrix A is equal to the zero vector, then λ is certainly an eigenvalue of A. What is the value of λ?

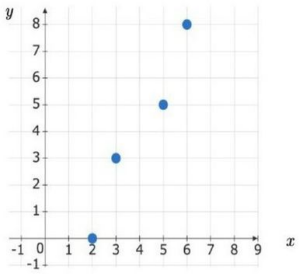
Note that the term "column" here refers to a column vector, that is, each column of A is treated as a vector.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0

Sub-Section Number :	7
Sub-Section Id :	640653244531
Question Shuffling Allowed :	No

Question Id : 6406531526713 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (143 to 144)
Question Label : Comprehension

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Based on the above data, answer the given subquestions.
Sub questions

✔ Official Answer Key (NAT): 0

MLF-Q362 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 144 Question Id : 6406531526715 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Is there a model of the form $f(x) = wx + b$,
with $w, b \in \mathbb{R}$ such that the squared loss
is exactly zero on this training dataset?

Options :

6406535112592. ✔ No

6406535112593. ✖ Yes

6406535112594. ✖ Insufficient information to answer the question

Question Id : 6406531526720 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (145 to 146)

✔ **Official Answer Key: Option (A)**

MLF-Q364 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 146 Question Id : 6406531526722 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following linear relations is valid among the rows (R_1, R_2, R_3) of matrix A?

Options :

6406535112606. ✖ $R_2 - R_1 - R_3 = 0$

6406535112607. ✖ $R_1 + R_2 - 2R_3 = 0$

6406535112608. ✔ $0.R_1 + 0.R_2 + 0.R_3 = 0$

6406535112609. ✖ $R_2 = 2R_1 + R_3$

Question Id : 6406531526723 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (147 to 148)

✔ **Official Answer Key: Option (C)**

MLF-Q370 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 152 Question Id : 6406531526717 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following is $L_{(0,0)}[f](x)$?

Options :

6406535112595. ✔ $1 + x_1 + x_2$

6406535112596. ✖ $x_1 + 2x_2$

6406535112597. ✖ $1 - x_1 - x_2$

6406535112598. ✖ $x_1 - x_2$

✔ **Official Answer Key: Option (A)**

MLF-Q371 | [SA] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 153 Question Id : 6406531526718 Question Type : SA
Correct Marks : 2

Question Label : Short Answer Question

Recall that $L_{(0,0)}[f](x)$ corresponds to a plane in $\mathbb{R}^3$ and can be represented as the set of all points that satisfy the equation $w^T x + b = 0$, where $w \in \mathbb{R}^3, b \in \mathbb{R}$, with w being normal to the plane. If $w^T w = 3$, what is b?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

✔ **Official Answer Key (NAT): 1**

MLF-Q372 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

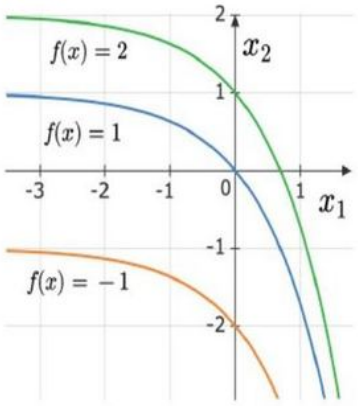
Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

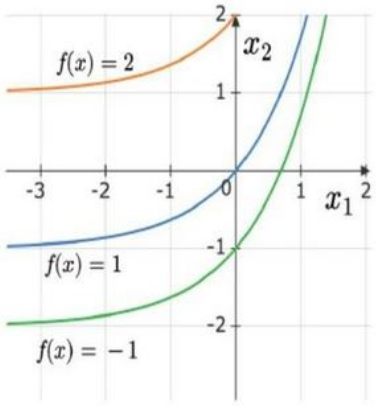
Sub questions

Question Number : 154 Question Id : 6406531526719 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following represents the contours of f?



(A)



(B)

Options :

6406535112600. ✔ (A)

6406535112601. ✖ (B)

✔ **Official Answer Key: Option (A)**

MLF-Q373 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 135 Question Id : 6406531534275 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)
Options :
6406535133759. ✔ YES
6406535133760. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653246971
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLF-Q375 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 137 Question Id : 6406531534277 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the dataset $(x, y) : (-1, 2), (0, 1), (1, 2)$. Perform polynomial regression of degree 2 by fitting a quadratic model of the form $y = \theta_0 + \theta_1 x + \theta_2 x^2$. Using the method of least squares, find the estimates of θ_0, θ_1 and θ_2.

Options :

6406535133765. ✔ $y = x^2 + 1$

6406535133766. ✖ $y = x^2$

6406535133767. ✖ $y = x^2 - 1$

6406535133768. ✖ $y = 2x$

Sub-Section Number :	4
Sub-Section Id :	640653246973
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A)**

MLF-Q376 | [MSQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 138 Question Id : 6406531534278 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Two vectors v_1 and v_2 are such that $v_1 = (1, 2, 2)^T$ and $v_2 = (2, -1, 0)^T$.
Which of the following options is/are correct?

Options :

6406535133769. ✖ Vectors v_1 and v_2 are parallel to each other.

6406535133770. ✔ Vectors v_1 and v_2 are orthogonal to each other.

6406535133771. ✔ Projection of v_1 onto v_2 is the zero vector.

6406535133772. ✔ Projection of v_2 onto v_1 is the zero vector.

Sub-Section Number :	5
Sub-Section Id :	640653246974
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B, C, D)**

MLF-Q377 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 139 Question Id : 6406531534279 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Consider the following dataset where each data point consists of three features x_1, x_2 and x_3 as

$$(2,4,3),(5,5,6),(2,2,1),(4,4,6)$$

Suppose the encoder and decoder functions, f and g respectively, are defined as

$$f(x_1,x_2,x_3) = 0.5x_1 + 0.5x_2 - x_3 \quad \text{and} \quad g(u) = [u,u,u].$$

Find the reconstruction loss.
Hint : "The reconstruction loss of the encoder decoder pair is the mean of the squared distance between the actual input and reconstructed input."

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
72

✔ **Official Answer Key (NAT): 72**

MLF-Q378 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 140 Question Id : 6406531534280 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Does there exist a matrix A of shape 10×10 such that every non-zero vector in $\mathbb{R}^{10}$ is an eigenvector of A with eigenvalue 2? If yes, compute the value of $\det(A)$. If not, enter -1 as the answer.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1024

Sub-Section Number :	6
Sub-Section Id :	640653246975
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 1024**

MLF-Q379 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 141 Question Id : 6406531534281 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question

If the sum of the columns of a square matrix A is equal to the zero vector, then λ is certainly an eigenvalue of A. What is the value of λ?

Note that the term "column" here refers to a column vector, that is, each column of A is treated as a vector.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0

Sub-Section Number :	7
Sub-Section Id :	640653246976
Question Shuffling Allowed :	No

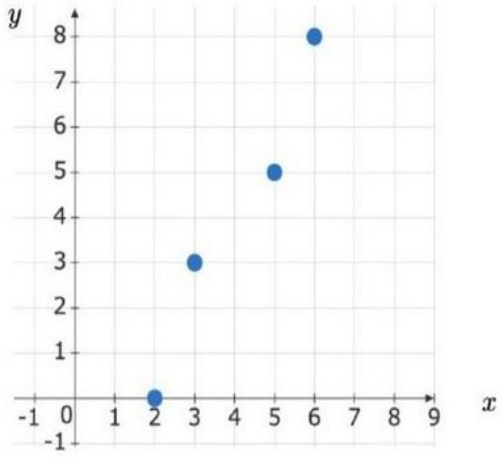
Question Id : 6406531534282 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (142 to 143)

✔ **Official Answer Key (NAT): 0**

MLF-Q381 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 143 Question Id : 6406531534284 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Is there a model of the form $f(x) = wx + b$, with $w, b \in \mathbb{R}$ such that the squared loss is exactly zero on this training dataset?

Options :

6406535133780. ✔ No

6406535133781. ✖ Yes

6406535133782. ✖ Insufficient information to answer the question

Question Id : 6406531534289 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (144 to 145)

✔ Official Answer Key: Option (A)

MLF-Q383 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA2 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 145 Question Id : 6406531534291 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following linear relations is valid among the rows (R_1, R_2, R_3) of matrix A?

Options :

6406535133794. ✖ $R_2 - R_1 - R_3 = 0$

6406535133795. ✖ $R_1 + R_2 - 2R_3 = 0$

6406535133796. ✔ $0.R_1 + 0.R_2 + 0.R_3 = 0$

6406535133797. ✖ $R_2 = 2R_1 + R_3$

Question Id : 6406531534292 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (146 to 147)

✔ **Official Answer Key: Option (C)**

MLF-Q389 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 151 Question Id : 6406531534286 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following is
$L_{(0,0)}[f](x)$?

Options :

6406535133783. ✔ $1 + x_1 + x_2$

6406535133784. ✖ $x_1 + 2x_2$

6406535133785. ✖ $1 - x_1 - x_2$

6406535133786. ✖ $x_1 - x_2$

✔ **Official Answer Key: Option (A)**

MLF-Q390 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 152 Question Id : 6406531534287 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question
Recall that $L_{(0,0)}[f](x)$ corresponds to a plane in $\mathbb{R}^3$ and can be represented as the set of all points that satisfy the equation $w^T x + b = 0$, where $w \in \mathbb{R}^3, b \in \mathbb{R}$, with w being normal to the plane. If $w^T w = 3$, what is b?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

1

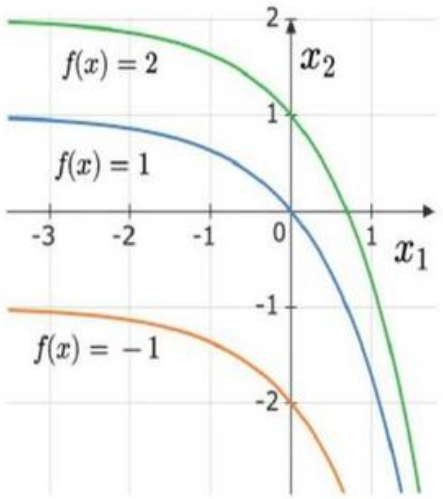
✔ **Official Answer Key (NAT): 1**

MLF-Q391 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

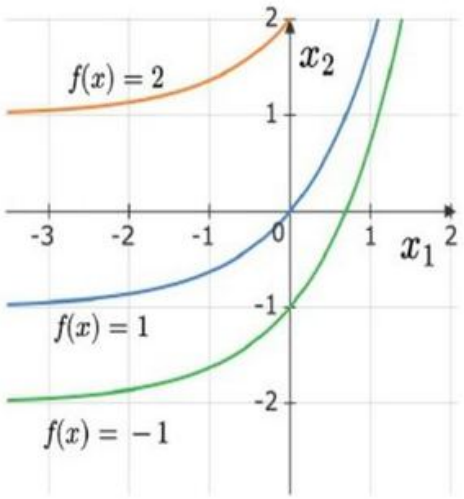
Question Label : Comprehension

Question Number : 153 Question Id : 6406531534288 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following represents the contours of f?



(A)



(B)

Options :

6406535133788. ✔ (A)

6406535133789. ✖ (B)

PSOSM

✔ **Official Answer Key: Option (A)**

MLF-Q392 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 135 Question Id : 6406531534614 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)
Options :
6406535134729. ✔ YES
6406535134730. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653247073
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A)**

MLF-Q394 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 137 Question Id : 6406531534616 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the dataset $(x, y) : (-1, 2), (0, 1), (1, 2)$. Perform polynomial regression of degree 2 by fitting a quadratic model of the form $y = \theta_0 + \theta_1 x + \theta_2 x^2$. Using the method of least squares, find the estimates of θ_0, θ_1 and θ_2.

Options :

6406535134735. ✔ $y = x^2 + 1$

6406535134736. ✖ $y = x^2$

6406535134737. ✖ $y = x^2 - 1$

6406535134738. ✖ $y = 2x$

Sub-Section Number :	4
Sub-Section Id :	640653247075
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A)**

MLF-Q395 | [MSQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 138 Question Id : 6406531534617 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Two vectors v_1 and v_2 are such that $v_1 = (1, 2, 2)^T$ and $v_2 = (2, -1, 0)^T$.
Which of the following options is/are correct?

Options :

6406535134739. ✖ Vectors v_1 and v_2 are parallel to each other.

6406535134740. ✔ Vectors v_1 and v_2 are orthogonal to each other.

6406535134741. ✔ Projection of v_1 onto v_2 is the zero vector.

6406535134742. ✔ Projection of v_2 onto v_1 is the zero vector.

Sub-Section Number :	5
Sub-Section Id :	640653247076
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B, C, D)**

MLF-Q396 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 139 Question Id : 6406531534618 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Consider the following dataset where each data point consists of three features x_1, x_2 and x_3 as

$$(2,4,3),(5,5,6),(2,2,1),(4,4,6)$$

Suppose the encoder and decoder functions, f and g respectively, are defined as

$$f(x_1,x_2,x_3) = 0.5x_1 + 0.5x_2 - x_3 \quad \text{and} \quad g(u) = [u,u,u].$$

Find the reconstruction loss.
Hint : "The reconstruction loss of the encoder decoder pair is the mean of the squared distance between the actual input and reconstructed input."

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
72

✔ **Official Answer Key (NAT): 72**

MLF-Q397 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 140 Question Id : 6406531534619 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Does there exist a matrix A of shape 10×10 such that every non-zero vector in $\mathbb{R}^{10}$ is an eigenvector of A with eigenvalue 2? If yes, compute the value of $\det(A)$. If not, enter -1 as the answer.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1024

Sub-Section Number :	6
Sub-Section Id :	640653247077
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 1024**

MLF-Q398 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 141 Question Id : 6406531534620 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question

If the sum of the columns of a square matrix A is equal to the zero vector, then λ is certainly an eigenvalue of A. What is the value of λ?

Note that the term "column" here refers to a column vector, that is, each column of A is treated as a vector.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0

Sub-Section Number :	7
Sub-Section Id :	640653247078
Question Shuffling Allowed :	No

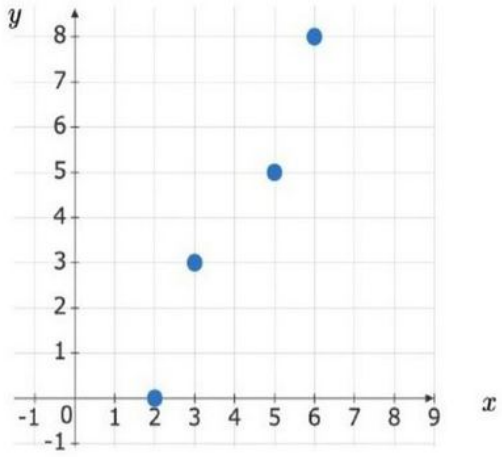
Question Id : 6406531534621 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (142 to 143)

✔ **Official Answer Key (NAT): 0**

MLF-Q400 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 143 Question Id : 6406531534623 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Is there a model of the form $f(x) = wx + b$, with $w, b \in \mathbb{R}$ such that the squared loss is exactly zero on this training dataset?

Options :

6406535134750. ✔ No

6406535134751. ✖ Yes

6406535134752. ✖ Insufficient information to answer the question

Question Id : 6406531534628 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (144 to 145)

✔ Official Answer Key: Option (A)

MLF-Q402 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA3 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 145 Question Id : 6406531534630 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following linear relations is valid among the rows (R_1, R_2, R_3) of matrix A?

Options :

6406535134764. ✖ $R_2 - R_1 - R_3 = 0$

6406535134765. ✖ $R_1 + R_2 - 2R_3 = 0$

6406535134766. ✔ $0.R_1 + 0.R_2 + 0.R_3 = 0$

6406535134767. ✖ $R_2 = 2R_1 + R_3$

Question Id : 6406531534631 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (146 to 147)

✔ **Official Answer Key: Option (C)**

MLF-Q408 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA3 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 151 Question Id : 6406531534625 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following is
$L_{(0,0)}[f](x)$?

Options :

6406535134753. ✔ $1 + x_1 + x_2$

6406535134754. ✖ $x_1 + 2x_2$

6406535134755. ✖ $1 - x_1 - x_2$

6406535134756. ✖ $x_1 - x_2$

✔ **Official Answer Key: Option (A)**

MLF-Q409 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 152 Question Id : 6406531534626 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question
Recall that $L_{(0,0)}[f](x)$ corresponds to a plane in $\mathbb{R}^3$ and can be represented as the set of all points that satisfy the equation $w^T x + b = 0$, where $w \in \mathbb{R}^3, b \in \mathbb{R}$, with w being normal to the plane. If $w^T w = 3$, what is b?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

1

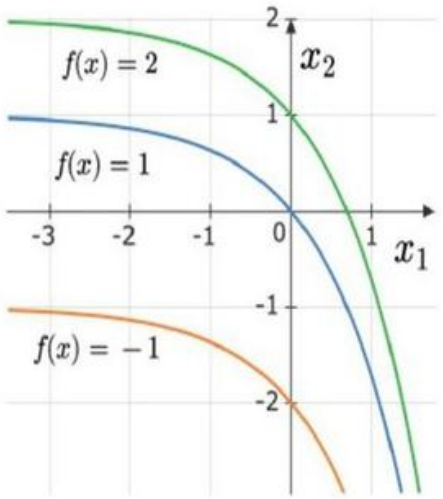
✔ **Official Answer Key (NAT): 1**

MLF-Q410 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

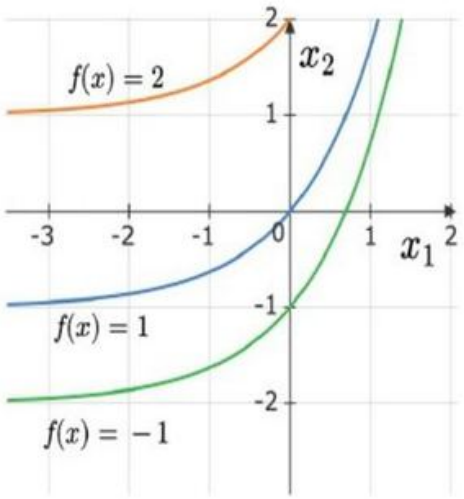
Question Label : Comprehension

Question Number : 153 Question Id : 6406531534627 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following represents the contours of f?



(A)



(B)

Options :

6406535134758. ✔ (A)

6406535134759. ✖ (B)

PSOSM

✔ **Official Answer Key: Option (A)**

MLF-Q411 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 135 Question Id : 6406531535051 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING

FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)
Options :
6406535135875. ✔ YES
6406535135876. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653247203
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLF-Q413 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 137 Question Id : 6406531535053 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the dataset $(x, y) : (-1, 2), (0, 1), (1, 2)$. Perform polynomial regression of degree 2 by fitting a quadratic model of the form $y = \theta_0 + \theta_1 x + \theta_2 x^2$. Using the method of least squares, find the estimates of θ_0, θ_1 and θ_2.

Options :

6406535135881. ✔ $y = x^2 + 1$

6406535135882. ✖ $y = x^2$

6406535135883. ✖ $y = x^2 - 1$

6406535135884. ✖ $y = 2x$

Sub-Section Number :	4
Sub-Section Id :	640653247205
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A)**

MLF-Q414 | [MSQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 138 Question Id : 6406531535054 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Two vectors v_1 and v_2 are such that $v_1 = (1, 2, 2)^T$ and $v_2 = (2, -1, 0)^T$.
Which of the following options is/are correct?

Options :

6406535135885. ✖ Vectors v_1 and v_2 are parallel to each other.

6406535135886. ✔ Vectors v_1 and v_2 are orthogonal to each other.

6406535135887. ✔ Projection of v_1 onto v_2 is the zero vector.

6406535135888. ✔ Projection of v_2 onto v_1 is the zero vector.

Sub-Section Number :	5
Sub-Section Id :	640653247206
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B, C, D)**

MLF-Q415 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 139 Question Id : 6406531535055 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Consider the following dataset where each data point consists of three features x_1, x_2 and x_3 as

$$(2,4,3),(5,5,6),(2,2,1),(4,4,6)$$

Suppose the encoder and decoder functions, f and g respectively, are defined as

$$f(x_1,x_2,x_3) = 0.5x_1 + 0.5x_2 - x_3 \quad \text{and} \quad g(u) = [u,u,u].$$

Find the reconstruction loss.
Hint : "The reconstruction loss of the encoder decoder pair is the mean of the squared distance between the actual input and reconstructed input."

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
72

✔ **Official Answer Key (NAT): 72**

MLF-Q416 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 140 Question Id : 6406531535056 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Does there exist a matrix A of shape 10×10 such that every non-zero vector in $\mathbb{R}^{10}$ is an eigenvector of A with eigenvalue 2? If yes, compute the value of $\det(A)$. If not, enter -1 as the answer.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1024

Sub-Section Number :	6
Sub-Section Id :	640653247207
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 1024**

MLF-Q417 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 141 Question Id : 6406531535057 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question

If the sum of the columns of a square matrix A is equal to the zero vector, then λ is certainly an eigenvalue of A. What is the value of λ?

Note that the term "column" here refers to a column vector, that is, each column of A is treated as a vector.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0

Sub-Section Number :	7
Sub-Section Id :	640653247208
Question Shuffling Allowed :	No

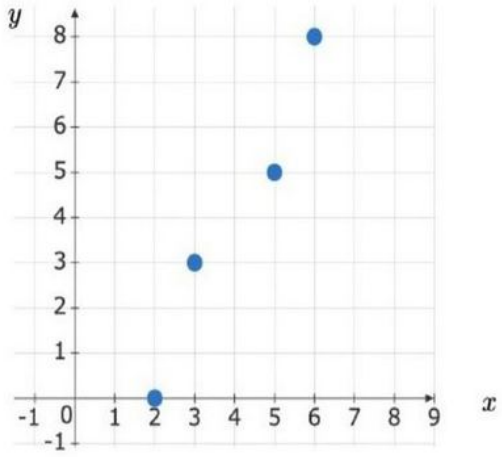
Question Id : 6406531535058 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (142 to 143)

✔ **Official Answer Key (NAT): 0**

MLF-Q419 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 143 Question Id : 6406531535060 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Is there a model of the form $f(x) = wx + b$, with $w, b \in \mathbb{R}$ such that the squared loss is exactly zero on this training dataset?

Options :

6406535135896. ✔ No

6406535135897. ✖ Yes

6406535135898. ✖ Insufficient information to answer the question

Question Id : 6406531535065 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (144 to 145)

✔ Official Answer Key: Option (A)

MLF-Q421 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA4 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 145 Question Id : 6406531535067 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following linear relations is valid among the rows (R_1, R_2, R_3) of matrix A?

Options :

6406535135910. ✖ $R_2 - R_1 - R_3 = 0$

6406535135911. ✖ $R_1 + R_2 - 2R_3 = 0$

6406535135912. ✔ $0.R_1 + 0.R_2 + 0.R_3 = 0$

6406535135913. ✖ $R_2 = 2R_1 + R_3$

Question Id : 6406531535068 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (146 to 147)

✔ **Official Answer Key: Option (C)**

MLF-Q427 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 151 Question Id : 6406531535062 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following is
$L_{(0,0)}[f](x)$?

Options :

6406535135899. ✔ $1 + x_1 + x_2$

6406535135900. ✖ $x_1 + 2x_2$

6406535135901. ✖ $1 - x_1 - x_2$

6406535135902. ✖ $x_1 - x_2$

✔ **Official Answer Key: Option (A)**

MLF-Q428 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 152 Question Id : 6406531535063 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question
Recall that $L_{(0,0)}[f](x)$ corresponds to a plane in $\mathbb{R}^3$ and can be represented as the set of all points that satisfy the equation $w^T x + b = 0$, where $w \in \mathbb{R}^3, b \in \mathbb{R}$, with w being normal to the plane. If $w^T w = 3$, what is b?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

1

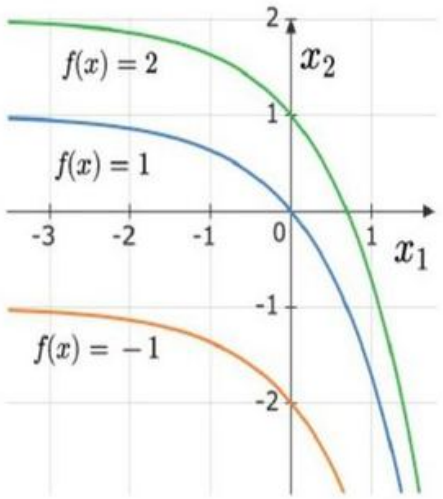
✔ **Official Answer Key (NAT): 1**

MLF-Q429 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

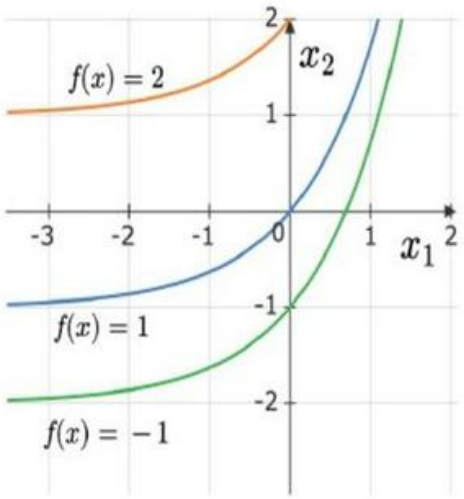
Question Label : Comprehension

Question Number : 153 Question Id : 6406531535064 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following represents the contours of f?



(A)



(B)

Options :

6406535135904. ✔ (A)

6406535135905. ✖ (B)

PSOSM

✔ **Official Answer Key: Option (A)**

MLF-Q430 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 129 Question Id : 6406531535787 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)
Options :
6406535138055. ✔ YES

6406535138056. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653247429
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLF-Q432 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 131 Question Id : 6406531535789 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the dataset $(x, y) : (-1, 2), (0, 1), (1, 2)$. Perform polynomial regression of degree 2 by fitting a quadratic model of the form $y = \theta_0 + \theta_1 x + \theta_2 x^2$. Using the method of least squares, find the estimates of θ_0, θ_1 and θ_2.

Options :

6406535138061. ✔ $y = x^2 + 1$

6406535138062. ✖ $y = x^2$

6406535138063. ✖ $y = x^2 - 1$

6406535138064. ✖ $y = 2x$

Sub-Section Number :	4
Sub-Section Id :	640653247431
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A)**

MLF-Q433 | [MSQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 132 Question Id : 6406531535790 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Two vectors v_1 and v_2 are such that $v_1 = (1, 2, 2)^T$ and $v_2 = (2, -1, 0)^T$.
Which of the following options is/are correct?

Options :

6406535138065. ✖ Vectors v_1 and v_2 are parallel to each other.

6406535138066. ✔ Vectors v_1 and v_2 are orthogonal to each other.

6406535138067. ✔ Projection of v_1 onto v_2 is the zero vector.

6406535138068. ✔ Projection of v_2 onto v_1 is the zero vector.

Sub-Section Number :	5
Sub-Section Id :	640653247432
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B, C, D)

MLF-Q434 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 133 Question Id : 6406531535791 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Consider the following dataset where each data point consists of three features x_1, x_2 and x_3 as

$$(2,4,3),(5,5,6),(2,2,1),(4,4,6)$$

Suppose the encoder and decoder functions, f and g respectively, are defined as

$$f(x_1,x_2,x_3) = 0.5x_1 + 0.5x_2 - x_3 \quad \text{and} \quad g(u) = [u,u,u].$$

Find the reconstruction loss.
Hint : "The reconstruction loss of the encoder decoder pair is the mean of the squared distance between the actual input and reconstructed input."

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
72

✔ **Official Answer Key (NAT): 72**

MLF-Q435 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 134 Question Id : 6406531535792 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Does there exist a matrix A of shape 10×10 such that every non-zero vector in $\mathbb{R}^{10}$ is an eigenvector of A with eigenvalue 2? If yes, compute the value of $\det(A)$. If not, enter -1 as the answer.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1024

Sub-Section Number :	6
Sub-Section Id :	640653247433
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 1024**

MLF-Q436 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 135 Question Id : 6406531535793 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question

If the sum of the columns of a square matrix A is equal to the zero vector, then λ is certainly an eigenvalue of A. What is the value of λ?

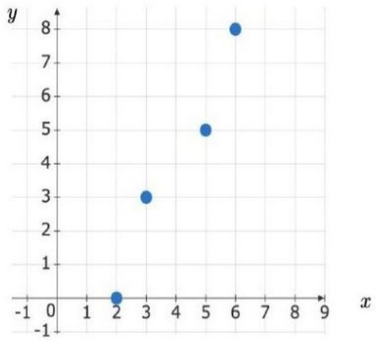
Note that the term "column" here refers to a column vector, that is, each column of A is treated as a vector.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0

Sub-Section Number :	7
Sub-Section Id :	640653247434
Question Shuffling Allowed :	No

Question Id : 6406531535794 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (136 to 137)
Question Label : Comprehension

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Based on the above data, answer the given subquestions.
Sub questions

✔ Official Answer Key (NAT): 0

MLF-Q438 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 137 Question Id : 6406531535796 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Is there a model of the form $f(x) = wx + b$, with $w, b \in \mathbb{R}$ such that the squared loss is exactly zero on this training dataset?

Options :

6406535138076. ✔ No

6406535138077. ✖ Yes

6406535138078. ✖ Insufficient information to answer the question

Question Id : 6406531535801 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (138 to 139)

✔ **Official Answer Key: Option (A)**

MLF-Q440 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 139 Question Id : 6406531535803 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following linear relations is valid among the rows (R_1, R_2, R_3) of matrix A?

Options :

6406535138090. ✖ $R_2 - R_1 - R_3 = 0$

6406535138091. ✖ $R_1 + R_2 - 2R_3 = 0$

6406535138092. ✔ $0.R_1 + 0.R_2 + 0.R_3 = 0$

6406535138093. ✖ $R_2 = 2R_1 + R_3$

Question Id : 6406531535804 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (140 to 141)

✔ **Official Answer Key: Option (C)**

MLF-Q446 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 145 Question Id : 6406531535798 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following is $L_{(0,0)}[f](x)$?

Options :

6406535138079. ✔ $1 + x_1 + x_2$

6406535138080. ✖ $x_1 + 2x_2$

6406535138081. ✖ $1 - x_1 - x_2$

6406535138082. ✖ $x_1 - x_2$

✔ **Official Answer Key: Option (A)**

MLF-Q447 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 146 Question Id : 6406531535799 Question Type : SA
Correct Marks : 2

Question Label : Short Answer Question

Recall that $L_{(0,0)}[f](x)$ corresponds to a plane in $\mathbb{R}^3$ and can be represented as the set of all points that satisfy the equation $w^T x + b = 0$, where $w \in \mathbb{R}^3, b \in \mathbb{R}$, with w being normal to the plane. If $w^T w = 3$, what is b?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

✔ **Official Answer Key (NAT): 1**

MLF-Q448 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

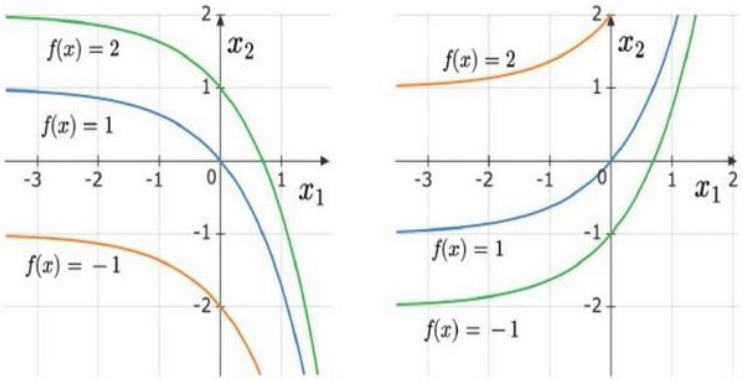
Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 147 Question Id : 6406531535800 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following represents the contours of f?



(A) (B)

Options :

6406535138084. ✔ (A)

6406535138085. ✖ (B)

OS

✔ **Official Answer Key: Option (A)**

MLF-Q449 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 129 Question Id : 6406531536127 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)
Options :
6406535139155. ✔ YES

6406535139156. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653247531
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLF-Q451 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 131 Question Id : 6406531536129 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the dataset $(x, y) : (-1, 2), (0, 1), (1, 2)$. Perform polynomial regression of degree 2 by fitting a quadratic model of the form $y = \theta_0 + \theta_1 x + \theta_2 x^2$. Using the method of least squares, find the estimates of θ_0, θ_1 and θ_2.

Options :

6406535139161. ✔ $y = x^2 + 1$

6406535139162. ✖ $y = x^2$

6406535139163. ✖ $y = x^2 - 1$

6406535139164. ✖ $y = 2x$

Sub-Section Number :	4
Sub-Section Id :	640653247533
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A)**

MLF-Q452 | [MSQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 132 Question Id : 6406531536130 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Two vectors v_1 and v_2 are such that $v_1 = (1, 2, 2)^T$ and $v_2 = (2, -1, 0)^T$.
Which of the following options is/are correct?

Options :

6406535139165. ✖ Vectors v_1 and v_2 are parallel to each other.

6406535139166. ✔ Vectors v_1 and v_2 are orthogonal to each other.

6406535139167. ✔ Projection of v_1 onto v_2 is the zero vector.

6406535139168. ✔ Projection of v_2 onto v_1 is the zero vector.

Sub-Section Number :	5
Sub-Section Id :	640653247534
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B, C, D)

MLF-Q453 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 133 Question Id : 6406531536131 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Consider the following dataset where each data point consists of three features x_1, x_2 and x_3 as

$$(2,4,3),(5,5,6),(2,2,1),(4,4,6)$$

Suppose the encoder and decoder functions, f and g respectively, are defined as

$$f(x_1,x_2,x_3) = 0.5x_1 + 0.5x_2 - x_3 \quad \text{and} \quad g(u) = [u,u,u].$$

Find the reconstruction loss.
Hint : "The reconstruction loss of the encoder decoder pair is the mean of the squared distance between the actual input and reconstructed input."

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
72

✔ **Official Answer Key (NAT): 72**

MLF-Q454 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 134 Question Id : 6406531536132 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Does there exist a matrix A of shape 10×10 such that every non-zero vector in $\mathbb{R}^{10}$ is an eigenvector of A with eigenvalue 2? If yes, compute the value of $\det(A)$. If not, enter -1 as the answer.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1024

Sub-Section Number :	6
Sub-Section Id :	640653247535
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 1024**

MLF-Q455 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 135 Question Id : 6406531536133 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question

If the sum of the columns of a square matrix A is equal to the zero vector, then λ is certainly an eigenvalue of A. What is the value of λ?

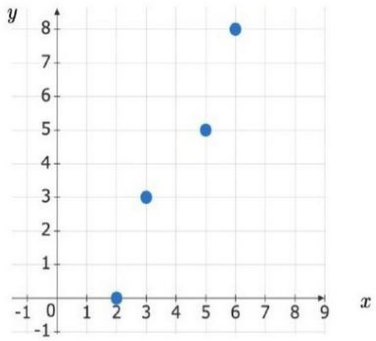
Note that the term "column" here refers to a column vector, that is, each column of A is treated as a vector.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0

Sub-Section Number :	7
Sub-Section Id :	640653247536
Question Shuffling Allowed :	No

Question Id : 6406531536134 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (136 to 137)
Question Label : Comprehension

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Based on the above data, answer the given subquestions.
Sub questions

✔ Official Answer Key (NAT): 0

MLF-Q457 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 137 Question Id : 6406531536136 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Is there a model of the form $f(x) = wx + b$, with $w, b \in \mathbb{R}$ such that the squared loss is exactly zero on this training dataset?

Options :

6406535139176. ✔ No

6406535139177. ✖ Yes

6406535139178. ✖ Insufficient information to answer the question

Question Id : 6406531536141 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (138 to 139)

✔ **Official Answer Key: Option (A)**

MLF-Q459 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 139 Question Id : 6406531536143 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following linear relations is valid among the rows (R_1, R_2, R_3) of matrix A?

Options :

6406535139190. ✖ $R_2 - R_1 - R_3 = 0$

6406535139191. ✖ $R_1 + R_2 - 2R_3 = 0$

6406535139192. ✔ $0.R_1 + 0.R_2 + 0.R_3 = 0$

6406535139193. ✖ $R_2 = 2R_1 + R_3$

Question Id : 6406531536144 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (140 to 141)

✔ **Official Answer Key: Option (C)**

MLF-Q465 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 145 Question Id : 6406531536138 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following is $L_{(0,0)}[f](x)$?

Options :

6406535139179. ✔ $1 + x_1 + x_2$

6406535139180. ✖ $x_1 + 2x_2$

6406535139181. ✖ $1 - x_1 - x_2$

6406535139182. ✖ $x_1 - x_2$

✔ **Official Answer Key: Option (A)**

MLF-Q466 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 146 Question Id : 6406531536139 Question Type : SA
Correct Marks : 2

Question Label : Short Answer Question

Recall that $L_{(0,0)}[f](x)$ corresponds to a plane in $\mathbb{R}^3$ and can be represented as the set of all points that satisfy the equation $w^T x + b = 0$, where $w \in \mathbb{R}^3, b \in \mathbb{R}$, with w being normal to the plane. If $w^T w = 3$, what is b?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

✔ **Official Answer Key (NAT): 1**

MLF-Q467 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

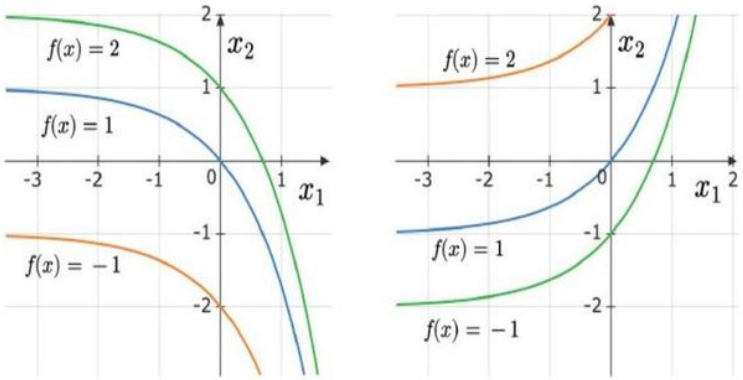
Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 147 Question Id : 6406531536140 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following represents the contours of f?



(A) (B)

Options :

6406535139184. ✔ (A)

6406535139185. ✖ (B)

OS

✔ **Official Answer Key: Option (A)**

MLF-Q468 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 129 Question Id : 6406531536417 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)
Options :
6406535139986. ✔ YES

6406535139987. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653247633
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLF-Q470 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 131 Question Id : 6406531536419 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the dataset $(x, y) : (-1, 2), (0, 1), (1, 2)$. Perform polynomial regression of degree 2 by fitting a quadratic model of the form $y = \theta_0 + \theta_1 x + \theta_2 x^2$. Using the method of least squares, find the estimates of θ_0, θ_1 and θ_2.

Options :

6406535139992. ✔ $y = x^2 + 1$

6406535139993. ✖ $y = x^2$

6406535139994. ✖ $y = x^2 - 1$

6406535139995. ✖ $y = 2x$

Sub-Section Number :	4
Sub-Section Id :	640653247635
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A)**

MLF-Q471 | [MSQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 132 Question Id : 6406531536420 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Two vectors v_1 and v_2 are such that $v_1 = (1, 2, 2)^T$ and $v_2 = (2, -1, 0)^T$.
Which of the following options is/are correct?

Options :

6406535139996. ✖ Vectors v_1 and v_2 are parallel to each other.

6406535139997. ✔ Vectors v_1 and v_2 are orthogonal to each other.

6406535139998. ✔ Projection of v_1 onto v_2 is the zero vector.

6406535139999. ✔ Projection of v_2 onto v_1 is the zero vector.

Sub-Section Number :	5
Sub-Section Id :	640653247636
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B, C, D)

MLF-Q472 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 133 Question Id : 6406531536421 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Consider the following dataset where each data point consists of three features x_1, x_2 and x_3 as

$$(2,4,3),(5,5,6),(2,2,1),(4,4,6)$$

Suppose the encoder and decoder functions, f and g respectively, are defined as

$$f(x_1,x_2,x_3) = 0.5x_1 + 0.5x_2 - x_3 \quad \text{and} \quad g(u) = [u,u,u].$$

Find the reconstruction loss.
Hint : "The reconstruction loss of the encoder decoder pair is the mean of the squared distance between the actual input and reconstructed input."

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
72

✔ **Official Answer Key (NAT): 72**

MLF-Q473 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 134 Question Id : 6406531536422 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Does there exist a matrix A of shape 10×10 such that every non-zero vector in $\mathbb{R}^{10}$ is an eigenvector of A with eigenvalue 2? If yes, compute the value of $\det(A)$. If not, enter -1 as the answer.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1024

Sub-Section Number :	6
Sub-Section Id :	640653247637
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 1024**

MLF-Q474 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 135 Question Id : 6406531536423 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question

If the sum of the columns of a square matrix A is equal to the zero vector, then λ is certainly an eigenvalue of A. What is the value of λ?

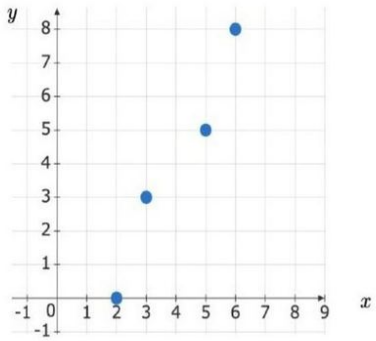
Note that the term "column" here refers to a column vector, that is, each column of A is treated as a vector.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0

Sub-Section Number :	7
Sub-Section Id :	640653247638
Question Shuffling Allowed :	No

Question Id : 6406531536424 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (136 to 137)
Question Label : Comprehension

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Based on the above data, answer the given subquestions.
Sub questions

✔ Official Answer Key (NAT): 0

MLF-Q476 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 137 Question Id : 6406531536426 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Is there a model of the form $f(x) = wx + b$, with $w, b \in \mathbb{R}$ such that the squared loss is exactly zero on this training dataset?

Options :

6406535140007. ✔ No

6406535140008. ✖ Yes

6406535140009. ✖ Insufficient information to answer the question

Question Id : 6406531536431 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (138 to 139)

✔ **Official Answer Key: Option (A)**

MLF-Q478 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 139 Question Id : 6406531536433 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following linear relations is valid among the rows (R_1, R_2, R_3) of matrix A?

Options :

6406535140021. ✖ $R_2 - R_1 - R_3 = 0$

6406535140022. ✖ $R_1 + R_2 - 2R_3 = 0$

6406535140023. ✔ $0.R_1 + 0.R_2 + 0.R_3 = 0$

6406535140024. ✖ $R_2 = 2R_1 + R_3$

Question Id : 6406531536434 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (140 to 141)

✔ **Official Answer Key: Option (C)**

MLF-Q484 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 145 Question Id : 6406531536428 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following is $L_{(0,0)}[f](x)$?

Options :

6406535140010. ✔ $1 + x_1 + x_2$

6406535140011. ✖ $x_1 + 2x_2$

6406535140012. ✖ $1 - x_1 - x_2$

6406535140013. ✖ $x_1 - x_2$

✔ **Official Answer Key: Option (A)**

MLF-Q485 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 146 Question Id : 6406531536429 Question Type : SA
Correct Marks : 2

Question Label : Short Answer Question

Recall that $L_{(0,0)}[f](x)$ corresponds to a plane in $\mathbb{R}^3$ and can be represented as the set of all points that satisfy the equation $w^T x + b = 0$, where $w \in \mathbb{R}^3, b \in \mathbb{R}$, with w being normal to the plane. If $w^T w = 3$, what is b?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1

✔ **Official Answer Key (NAT): 1**

MLF-Q486 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

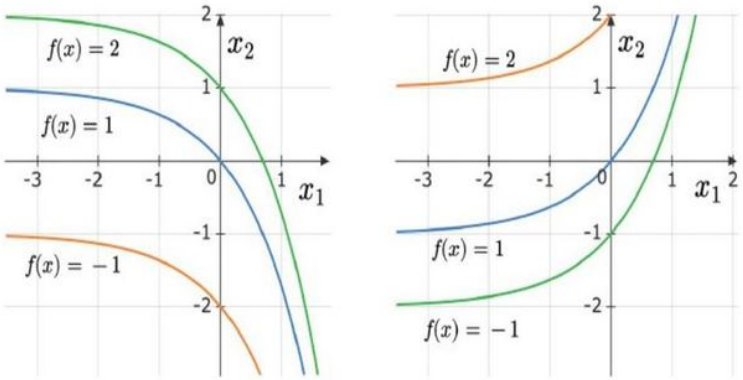
Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 147 Question Id : 6406531536430 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following represents the contours of f?



(A) (B)

Options :

6406535140015. ✔ (A)

6406535140016. ✖ (B)

OS

✔ **Official Answer Key: Option (A)**

MLF-Q487 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 133 Question Id : 6406531527030 Question Type : MCQ
Correct Marks : 0
Question Label : Multiple Choice Question
THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)
Options :
6406535113448. ✔ YES
6406535113449. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653244630
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A)**

MLF-Q489 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 135 Question Id : 6406531527032 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the dataset $(x, y) : (-1, 2), (0, 1), (1, 2)$. Perform polynomial regression of degree 2 by fitting a quadratic model of the form $y = \theta_0 + \theta_1 x + \theta_2 x^2$. Using the method of least squares, find the estimates of θ_0, θ_1 and θ_2.

Options :

6406535113454. ✔ $y = x^2 + 1$

6406535113455. ✖ $y = x^2$

6406535113456. ✖ $y = x^2 - 1$

6406535113457. ✖ $y = 2x$

Sub-Section Number :	4
Sub-Section Id :	640653244632
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A)**

MLF-Q490 | [MSQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 136 Question Id : 6406531527033 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Two vectors v_1 and v_2 are such that $v_1 = (1, 2, 2)^T$ and $v_2 = (2, -1, 0)^T$.
Which of the following options is/are correct?

Options :

6406535113458. ✖ Vectors v_1 and v_2 are parallel to each other.

6406535113459. ✔ Vectors v_1 and v_2 are orthogonal to each other.

6406535113460. ✔ Projection of v_1 onto v_2 is the zero vector.

6406535113461. ✔ Projection of v_2 onto v_1 is the zero vector.

Sub-Section Number :	5
Sub-Section Id :	640653244633
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B, C, D)**

MLF-Q491 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 137 Question Id : 6406531527034 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question
Consider the following dataset where each data point consists of three features x_1, x_2 and x_3 as

$$(2,4,3),(5,5,6),(2,2,1),(4,4,6)$$

Suppose the encoder and decoder functions, f and g respectively, are defined as

$$f(x_1,x_2,x_3) = 0.5x_1 + 0.5x_2 - x_3 \quad \text{and} \quad g(u) = [u,u,u].$$

Find the reconstruction loss.
Hint : "The reconstruction loss of the encoder decoder pair is the mean of the squared distance between the actual input and reconstructed input."

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal

Text Areas : PlainText
Possible Answers :
72

✔ **Official Answer Key (NAT): 72**

MLF-Q492 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Id : 6406531527037 Question Type : COMPREHENSION Sub Question Shuffling

Question Number : 138 Question Id : 6406531527035 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Does there exist a matrix A of shape 10×10 such that every non-zero vector in $\mathbb{R}^{10}$ is an eigenvector of A with eigenvalue 2? If yes, compute the value of $\det(A)$. If not, enter -1 as the answer.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1024

Sub-Section Number :	6
Sub-Section Id :	640653244634
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 1024**

MLF-Q493 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Id : 6406531527037 Question Type : COMPREHENSION Sub Question Shuffling

Question Number : 139 Question Id : 6406531527036 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question

If the sum of the columns of a square matrix A is equal to the zero vector, then λ is certainly an eigenvalue of A. What is the value of λ?

Note that the term "column" here refers to a column vector, that is, each column of A is treated as a vector.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0

Sub-Section Number :	7
Sub-Section Id :	640653244635
Question Shuffling Allowed :	No

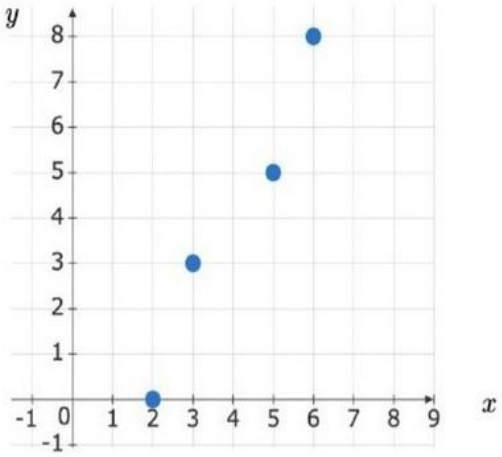
Question Id : 6406531527037 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (140 to 141)

✔ **Official Answer Key (NAT): 0**

MLF-Q495 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 141 Question Id : 6406531527039 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Is there a model of the form $f(x) = wx + b$, with $w, b \in \mathbb{R}$ such that the squared loss is exactly zero on this training dataset?

Options :

6406535113469. ✔ No

6406535113470. ✖ Yes

6406535113471. ✖ Insufficient information to answer the question

Question Id : 6406531527044 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (142 to 143)

✔ Official Answer Key: Option (A)

MLF-Q497 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 143 Question Id : 6406531527046 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following linear relations is valid among the rows (R_1, R_2, R_3) of matrix A?

Options :

6406535113483. ✖ $R_2 - R_1 - R_3 = 0$

6406535113484. ✖ $R_1 + R_2 - 2R_3 = 0$

6406535113485. ✔ $0.R_1 + 0.R_2 + 0.R_3 = 0$

6406535113486. ✖ $R_2 = 2R_1 + R_3$

Question Id : 6406531527047 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (144 to 145)

✔ **Official Answer Key: Option (C)**

MLF-Q503 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 149 Question Id : 6406531527041 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following is
$L_{(0,0)}[f](x)$?

Options :

6406535113472. ✔ $1 + x_1 + x_2$

6406535113473. ✖ $x_1 + 2x_2$

6406535113474. ✖ $1 - x_1 - x_2$

6406535113475. ✖ $x_1 - x_2$

✔ **Official Answer Key: Option (A)**

MLF-Q504 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 150 Question Id : 6406531527042 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question
Recall that $L_{(0,0)}[f](x)$ corresponds to a plane in $\mathbb{R}^3$ and can be represented as the set of all points that satisfy the equation $w^T x + b = 0$, where $w \in \mathbb{R}^3, b \in \mathbb{R}$, with w being normal to the plane. If $w^T w = 3$, what is b?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

1

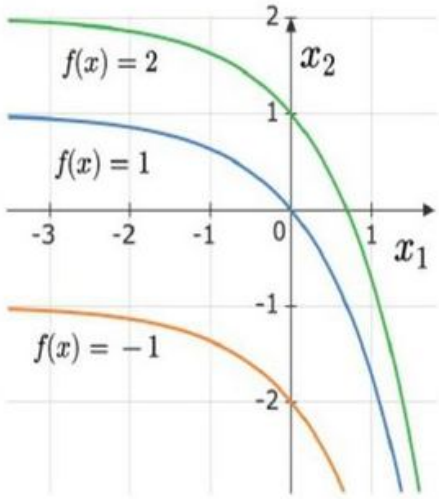
✔ **Official Answer Key (NAT): 1**

MLF-Q505 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

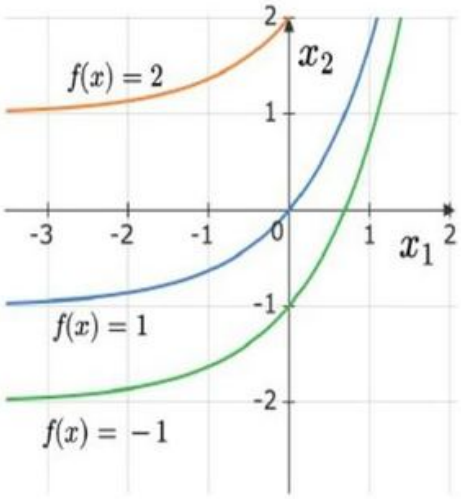
Question Label : Comprehension

Question Number : 151 Question Id : 6406531527043 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following represents the contours of f?



(A)



(B)

Options :

6406535113477. ✔ (A)

6406535113478. ✖ (B)

✔ **Official Answer Key: Option (A)**

MLF-Q506 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 133 Question Id : 6406531527309 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406535114206. ✔ YES

6406535114207. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653244734
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLF-Q508 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 135 Question Id : 6406531527311 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the dataset $(x, y) : (-1, 2), (0, 1), (1, 2)$. Perform polynomial regression of degree 2 by fitting a quadratic model of the form $y = \theta_0 + \theta_1 x + \theta_2 x^2$. Using the method of least squares, find the estimates of θ_0, θ_1 and θ_2.

Options :

6406535114212. ✔ $y = x^2 + 1$

6406535114213. ✖ $y = x^2$

6406535114214. ✖ $y = x^2 - 1$

6406535114215. ✖ $y = 2x$

Sub-Section Number :	4
Sub-Section Id :	640653244736
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A)**

MLF-Q509 | [MSQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 136 Question Id : 6406531527312 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Two vectors v_1 and v_2 are such that $v_1 = (1, 2, 2)^T$ and $v_2 = (2, -1, 0)^T$.
Which of the following options is/are correct?

Options :

6406535114216. ✖ Vectors v_1 and v_2 are parallel to each other.

6406535114217. ✔ Vectors v_1 and v_2 are orthogonal to each other.

6406535114218. ✔ Projection of v_1 onto v_2 is the zero vector.

6406535114219. ✔ Projection of v_2 onto v_1 is the zero vector.

Sub-Section Number :	5
Sub-Section Id :	640653244737
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B, C, D)

MLF-Q510 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 137 Question Id : 6406531527313 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question
Consider the following dataset where each data point consists of three features x_1, x_2 and x_3 as

$$(2,4,3),(5,5,6),(2,2,1),(4,4,6)$$

Suppose the encoder and decoder functions, f and g respectively, are defined as

$$f(x_1,x_2,x_3) = 0.5x_1 + 0.5x_2 - x_3 \quad \text{and} \quad g(u) = [u,u,u].$$

Find the reconstruction loss.
Hint : "The reconstruction loss of the encoder decoder pair is the mean of the squared distance between the actual input and reconstructed input."

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal

Text Areas : PlainText
Possible Answers :
72

✔ **Official Answer Key (NAT): 72**

MLF-Q511 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Id : 6406531527316 Question Type : COMPREHENSION Sub Question Shuffling

Question Number : 138 Question Id : 6406531527314 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Does there exist a matrix A of shape 10×10 such that every non-zero vector in $\mathbb{R}^{10}$ is an eigenvector of A with eigenvalue 2? If yes, compute the value of $\det(A)$. If not, enter -1 as the answer.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1024

Sub-Section Number :	6
Sub-Section Id :	640653244738
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 1024**

MLF-Q512 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Id : 6406531527316 Question Type : COMPREHENSION Sub Question Shuffling

Question Number : 139 Question Id : 6406531527315 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question

If the sum of the columns of a square matrix A is equal to the zero vector, then λ is certainly an eigenvalue of A. What is the value of λ?

Note that the term "column" here refers to a column vector, that is, each column of A is treated as a vector.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0

Sub-Section Number :	7
Sub-Section Id :	640653244739
Question Shuffling Allowed :	No

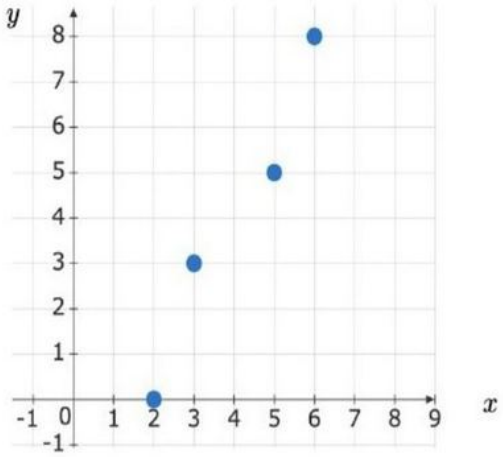
Question Id : 6406531527316 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (140 to 141)

✔ **Official Answer Key (NAT): 0**

MLF-Q514 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 141 Question Id : 6406531527318 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Is there a model of the form $f(x) = wx + b$, with $w, b \in \mathbb{R}$ such that the squared loss is exactly zero on this training dataset?

Options :

6406535114227. ✔ No

6406535114228. ✖ Yes

6406535114229. ✖ Insufficient information to answer the question

Question Id : 6406531527323 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (142 to 143)

✔ Official Answer Key: Option (A)

MLF-Q516 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 143 Question Id : 6406531527325 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following linear relations is valid among the rows (R_1, R_2, R_3) of matrix A?

Options :

6406535114241. ✖ $R_2 - R_1 - R_3 = 0$

6406535114242. ✖ $R_1 + R_2 - 2R_3 = 0$

6406535114243. ✔ $0.R_1 + 0.R_2 + 0.R_3 = 0$

6406535114244. ✖ $R_2 = 2R_1 + R_3$

Question Id : 6406531527326 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (144 to 145)

✔ **Official Answer Key: Option (C)**

MLF-Q522 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 149 Question Id : 6406531527320 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following is
$L_{(0,0)}[f](x)$?

Options :

6406535114230. ✔ $1 + x_1 + x_2$

6406535114231. ✖ $x_1 + 2x_2$

6406535114232. ✖ $1 - x_1 - x_2$

6406535114233. ✖ $x_1 - x_2$

✔ **Official Answer Key: Option (A)**

MLF-Q523 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 150 Question Id : 6406531527321 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question
Recall that $L_{(0,0)}[f](x)$ corresponds to a plane in $\mathbb{R}^3$ and can be represented as the set of all points that satisfy the equation $w^T x + b = 0$, where $w \in \mathbb{R}^3, b \in \mathbb{R}$, with w being normal to the plane. If $w^T w = 3$, what is b?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

1

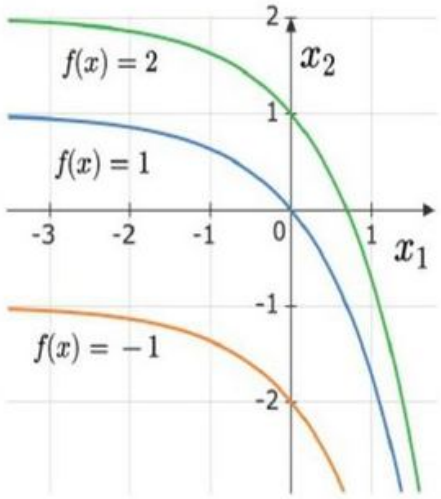
✔ **Official Answer Key (NAT): 1**

MLF-Q524 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

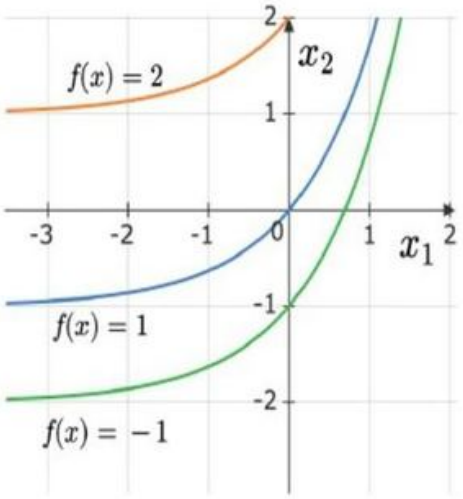
Question Label : Comprehension

Question Number : 151 Question Id : 6406531527322 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following represents the contours of f?



(A)



(B)

Options :

6406535114235. ✔ (A)

6406535114236. ✖ (B)

✔ **Official Answer Key: Option (A)**

MLF-Q525 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 133 Question Id : 6406531527588 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?

CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406535114964. ✔ YES

6406535114965. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653244838
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLF-Q527 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 135 Question Id : 6406531527590 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the dataset $(x, y) : (-1, 2), (0, 1), (1, 2)$. Perform polynomial regression of degree 2 by fitting a quadratic model of the form $y = \theta_0 + \theta_1 x + \theta_2 x^2$. Using the method of least squares, find the estimates of θ_0, θ_1 and θ_2.

Options :

6406535114970. ✔ $y = x^2 + 1$

6406535114971. ✖ $y = x^2$

6406535114972. ✖ $y = x^2 - 1$

6406535114973. ✖ $y = 2x$

Sub-Section Number :	4
Sub-Section Id :	640653244840
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A)**

MLF-Q528 | [MSQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 136 Question Id : 6406531527591 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question

Two vectors v_1 and v_2 are such that $v_1 = (1, 2, 2)^T$ and $v_2 = (2, -1, 0)^T$.
Which of the following options is/are correct?

Options :

6406535114974. ✖ Vectors v_1 and v_2 are parallel to each other.

6406535114975. ✔ Vectors v_1 and v_2 are orthogonal to each other.

6406535114976. ✔ Projection of v_1 onto v_2 is the zero vector.

6406535114977. ✔ Projection of v_2 onto v_1 is the zero vector.

Sub-Section Number :	5
Sub-Section Id :	640653244841
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B, C, D)

MLF-Q529 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 137 Question Id : 6406531527592 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question
Consider the following dataset where each data point consists of three features x_1, x_2 and x_3 as

$$(2,4,3),(5,5,6),(2,2,1),(4,4,6)$$

Suppose the encoder and decoder functions, f and g respectively, are defined as

$$f(x_1,x_2,x_3) = 0.5x_1 + 0.5x_2 - x_3 \quad \text{and} \quad g(u) = [u,u,u].$$

Find the reconstruction loss.
Hint : "The reconstruction loss of the encoder decoder pair is the mean of the squared distance between the actual input and reconstructed input."

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal

Text Areas : PlainText
Possible Answers :
72

✔ **Official Answer Key (NAT): 72**

MLF-Q530 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Id : 6406531527595 Question Type : COMPREHENSION Sub Question Shuffling

Question Number : 138 Question Id : 6406531527593 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Does there exist a matrix A of shape 10×10 such that every non-zero vector in $\mathbb{R}^{10}$ is an eigenvector of A with eigenvalue 2? If yes, compute the value of $\det(A)$. If not, enter -1 as the answer.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1024

Sub-Section Number :	6
Sub-Section Id :	640653244842
Question Shuffling Allowed :	Yes

✔ **Official Answer Key (NAT): 1024**

MLF-Q531 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Id : 6406531527595 Question Type : COMPREHENSION Sub Question Shuffling

Question Number : 139 Question Id : 6406531527594 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question

If the sum of the columns of a square matrix A is equal to the zero vector, then λ is certainly an eigenvalue of A. What is the value of λ?

Note that the term "column" here refers to a column vector, that is, each column of A is treated as a vector.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0

Sub-Section Number :	7
Sub-Section Id :	640653244843
Question Shuffling Allowed :	No

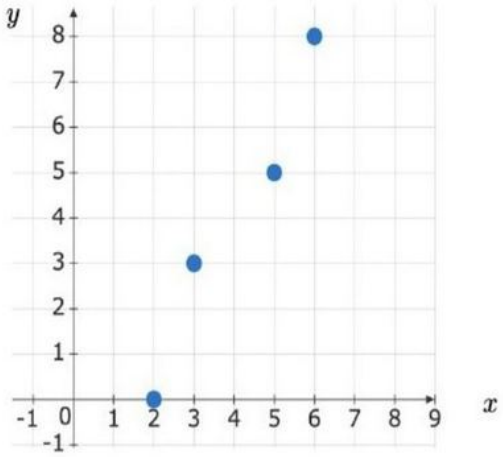
Question Id : 6406531527595 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (140 to 141)

✔ **Official Answer Key (NAT): 0**

MLF-Q533 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 141 Question Id : 6406531527597 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Is there a model of the form $f(x) = wx + b$, with $w, b \in \mathbb{R}$ such that the squared loss is exactly zero on this training dataset?

Options :

6406535114985. ✔ No

6406535114986. ✖ Yes

6406535114987. ✖ Insufficient information to answer the question

Question Id : 6406531527602 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (142 to 143)

✔ Official Answer Key: Option (A)

MLF-Q535 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 143 Question Id : 6406531527604 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

Which of the following linear relations is valid among the rows (R_1, R_2, R_3) of matrix A?

Options :

6406535114999. ✖ $R_2 - R_1 - R_3 = 0$

6406535115000. ✖ $R_1 + R_2 - 2R_3 = 0$

6406535115001. ✔ $0.R_1 + 0.R_2 + 0.R_3 = 0$

6406535115002. ✖ $R_2 = 2R_1 + R_3$

Question Id : 6406531527605 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (144 to 145)

✔ **Official Answer Key: Option (C)**

MLF-Q541 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 149 Question Id : 6406531527599 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following is
$L_{(0,0)}[f](x)$?

Options :

6406535114988. ✔ $1 + x_1 + x_2$

6406535114989. ✖ $x_1 + 2x_2$

6406535114990. ✖ $1 - x_1 - x_2$

6406535114991. ✖ $x_1 - x_2$

✔ **Official Answer Key: Option (A)**

MLF-Q542 | [SA] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

Question Label : Comprehension

Question Number : 150 Question Id : 6406531527600 Question Type : SA
Correct Marks : 2
Question Label : Short Answer Question
Recall that $L_{(0,0)}[f](x)$ corresponds to a plane in $\mathbb{R}^3$ and can be represented as the set of all points that satisfy the equation $w^T x + b = 0$, where $w \in \mathbb{R}^3, b \in \mathbb{R}$, with w being normal to the plane. If $w^T w = 3$, what is b?

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :

1

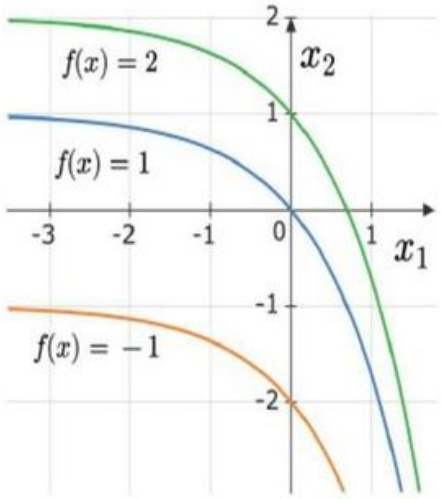
✔ **Official Answer Key (NAT): 1**

MLF-Q543 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Concepts

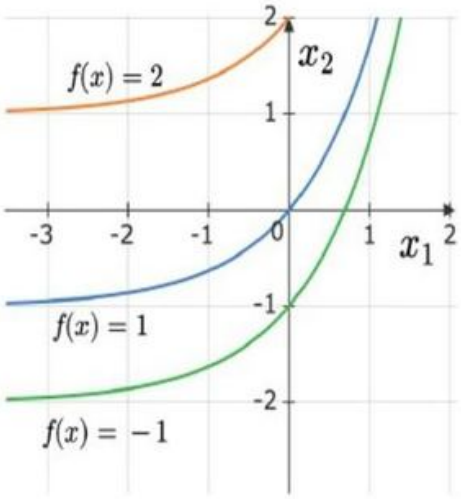
Question Label : Comprehension

Question Number : 151 Question Id : 6406531527601 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question

Which of the following represents the contours of f?



(A)



(B)

Options :

6406535114993. ✔ (A)

6406535114994. ✖ (B)

✔ Official Answer Key: Option (A)

MLF-Q544 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 118 Question Id : 640653668558 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406532239871. ✔ YES

6406532239872. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	64065395171
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key: Option (A)**

MLF-Q547 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 121 Question Id : 640653668566 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Consider a vector $v = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$. Does v belong to the columnspace of $A = \begin{bmatrix} -8 & -2 & -9 \\ 6 & 4 & 8 \\ 4 & 0 & 4 \end{bmatrix}$?

Options :

6406532239892. ✔ Yes

6406532239893. ✖ No

✔ Official Answer Key: Option (A)

MLF-Q548 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 122 Question Id : 640653668572 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Let $A = \begin{bmatrix} 8 & 3 \\ 2 & 7 \end{bmatrix}$. Find a matrix P such that $P^{-1}AP$ is diagonal.

Options :

6406532239907. ✖ $\begin{bmatrix} 1 & 1 \\ 2 & -3 \end{bmatrix}$

6406532239908. ✖ $\begin{bmatrix} 10 & 0 \\ 0 & 5 \end{bmatrix}$

6406532239909. ✖ $\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$

6406532239910. ✔ $\begin{bmatrix} 3 & 1 \\ 2 & -1 \end{bmatrix}$

Sub-Section Number :	4
Sub-Section Id :	64065395173
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (D)

MLF-Q549 | [SA] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 123 Question Id : 640653668561 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

Consider a regression model $h(x_1, x_2) = \frac{x_1 + x_2}{2}$. For the below input data points, find the mean squared loss of h. Enter the answer correct to two decimal places.

X	Y
$[1, 2]$	2
$[2, 3]$	3
$[3, 4]$	4
$[4, 5]$	5

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.25

✔ **Official Answer Key (NAT): 0.25**

MLF-Q550 | [SA] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 124 Question Id : 640653668563 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

The curve $y = ax^2 + bx + c$ passes through the points $(2, -1)$ and $(0, 0)$, and is tangent to the line $y = -x$ at the origin. Find the absolute value of the product ab. Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.25

✔ **Official Answer Key (NAT): 0.25**

MLF-Q551 | [SA] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 125 Question Id : 640653668564 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

At the point $P_0 = (1, 2)$, the function $f(x, y)$ has a derivative of $2\sqrt{2}$ in the direction $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ and a derivative of 5 in the direction $(0, 1)$. Find the derivative of $f(x, y)$ at point P_0 in the direction $\left(\frac{-1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$. Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

4.88 to 4.95

✔ **Official Answer Key (NAT): 4.88 to 4.95**

MLF-Q552 | [SA] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 126 Question Id : 640653668573 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

Consider a 2×2 matrix A whose eigenvectors are $v_1 = [1, 1]^T$ and $v_2 = [1, 2]^T$ with eigenvalues 2 and 3, respectively. Find the sum of square of all the entries of A.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

22

Sub-Section Number :	5
Sub-Section Id :	64065395174
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key (NAT): 22**

MLF-Q554 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 128 Question Id : 640653668565 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Find the second order approximation for $f(x,y) = e^{2y-x}$ in the neighbourhood of $(0,0)$.

Options :

6406532239888. ✖ $1 - x + 2y - 2xy + y^2 + \frac{x^2}{2}$

6406532239889. ✖ $1 - 2xy + 2y^2 + \frac{x^2}{2}$

6406532239890. ✖ $2y^2 + \frac{x^2}{2}$

6406532239891. ✔ $1 - x + 2y - 2xy + 2y^2 + \frac{x^2}{2}$

✔ **Official Answer Key: Option (D)**

MLF-Q555 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 129 Question Id : 640653668571 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Find the best fit line for the dataset given below using least squares method.

x	1	3	4	5	7
y	-3	7	1	0	4

Options :

6406532239903. ✔ $\hat{y} = 0.7x - 1$

6406532239904. ✖ $\hat{y} = x + 0.5$

6406532239905. ✖ $\hat{y} = x + 0.2$

6406532239906. ✖ $\hat{y} = 0.2x + 1$

✔ **Official Answer Key: Option (A)**

MLF-Q556 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 130 Question Id : 640653668574 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Let A be a $n \times n$ real matrix. Then, which among the following statements are true?

Options :

6406532239912. ✖ If the eigenvalues of A are $\lambda_1, \ldots, \lambda_n$, then A is similar to a diagonal matrix with $\lambda_1, \ldots, \lambda_n$ along the diagonal.

6406532239913. ✖ If $\text{rank}(A) = r$, then A has r non-zero eigenvalues.

6406532239914. ✔ A may have complex eigenvalues.

6406532239915. ✖ If A has a repeated eigenvalue, then A is not diagonalizable.

Sub-Section Number :	7
Sub-Section Id :	64065395176
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (C)

MLF-Q557 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 131 Question Id : 640653668567 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select the correct options from the following.

Options :

6406532239894. ✖ There exists a matrix whose row space contains $\begin{bmatrix}1\\0\\-1\end{bmatrix}$ and whose null space contains $\begin{bmatrix}0\\1\\1\end{bmatrix}$.

6406532239895. ✖ There exists a matrix with row space $= \mathbb{R}^3$ and column space $= \mathbb{R}^4$.

6406532239896. ✔ There exists a matrix whose column space contains $\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}$ and row space contains $\begin{bmatrix}1\\1\end{bmatrix}, \begin{bmatrix}1\\2\end{bmatrix}$.

6406532239897. ✔ There exists a matrix whose nullspace consists of all vectors in $\mathbb{R}^3$ such that $x_1 + 2x_2 + 4x_3 = 0$.

Sub-Section Number :	8
Sub-Section Id :	64065395177
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653668568 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (132 to 133)

✔ Official Answer Key: Option (C, D)

MLF-Q560 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QPD3 29 Oct*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 118 Question Id : 640653668919 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406532240967. ✔ YES

6406532240968. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	64065395276
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key: Option (A)**

MLF-Q563 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QPD3 29 Oct*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 121 Question Id : 640653668927 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Consider a vector $v = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$. Does v belong to the columnspace of $A = \begin{bmatrix} -8 & -2 & -9 \\ 6 & 4 & 8 \\ 4 & 0 & 4 \end{bmatrix}$?

Options :

6406532240988. ✔ Yes

6406532240989. ✖ No

✔ Official Answer Key: Option (A)

MLF-Q564 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QPD3 29 Oct*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 122 Question Id : 640653668933 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Let $A = \begin{bmatrix} 8 & 3 \\ 2 & 7 \end{bmatrix}$. Find a matrix P such that $P^{-1}AP$ is diagonal.

Options :

6406532241003. ✖ $\begin{bmatrix} 1 & 1 \\ 2 & -3 \end{bmatrix}$

6406532241004. ✖ $\begin{bmatrix} 10 & 0 \\ 0 & 5 \end{bmatrix}$

6406532241005. ✖ $\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$

6406532241006. ✔ $\begin{bmatrix} 3 & 1 \\ 2 & -1 \end{bmatrix}$

Sub-Section Number :	4
Sub-Section Id :	64065395278
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (D)

MLF-Q565 | [SA] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QPD3 29 Oct*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 123 Question Id : 640653668922 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

Consider a regression model $h(x_1, x_2) = \frac{x_1 + x_2}{2}$. For the below input data points, find the mean squared loss of h. Enter the answer correct to two decimal places.

X	Y
$[1, 2]$	2
$[2, 3]$	3
$[3, 4]$	4
$[4, 5]$	5

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.25

✔ **Official Answer Key (NAT): 0.25**

MLF-Q566 | [SA] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QPD3 29 Oct*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 124 Question Id : 640653668924 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

The curve $y = ax^2 + bx + c$ passes through the points $(2, -1)$ and $(0, 0)$, and is tangent to the line $y = -x$ at the origin. Find the absolute value of the product ab. Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.25

✔ **Official Answer Key (NAT): 0.25**

MLF-Q567 | [SA] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QPD3 29 Oct*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 125 Question Id : 640653668925 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

At the point $P_0 = (1, 2)$, the function $f(x, y)$ has a derivative of $2\sqrt{2}$ in the direction $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ and a derivative of 5 in the direction $(0, 1)$. Find the derivative of $f(x, y)$ at point P_0 in the direction $\left(\frac{-1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$. Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

4.88 to 4.95

✔ **Official Answer Key (NAT): 4.88 to 4.95**

MLF-Q568 | [SA] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QPD3 29 Oct*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 126 Question Id : 640653668934 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

Consider a 2×2 matrix A whose eigenvectors are $v_1 = [1, 1]^T$ and $v_2 = [1, 2]^T$ with eigenvalues 2 and 3, respectively. Find the sum of square of all the entries of A.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

22

Sub-Section Number :	5
Sub-Section Id :	64065395279
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key (NAT): 22**

MLF-Q570 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QPD3 29 Oct*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 128 Question Id : 640653668926 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Find the second order approximation for $f(x,y) = e^{2y-x}$ in the neighbourhood of $(0,0)$.

Options :

6406532240984. ✖ $1 - x + 2y - 2xy + y^2 + \frac{x^2}{2}$

6406532240985. ✖ $1 - 2xy + 2y^2 + \frac{x^2}{2}$

6406532240986. ✖ $2y^2 + \frac{x^2}{2}$

6406532240987. ✔ $1 - x + 2y - 2xy + 2y^2 + \frac{x^2}{2}$

✔ **Official Answer Key: Option (D)**

MLF-Q571 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QPD3 29 Oct*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 129 Question Id : 640653668932 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Find the best fit line for the dataset given below using least squares method.

x	1	3	4	5	7
y	-3	7	1	0	4

Options :

6406532240999. ✔ $\hat{y} = 0.7x - 1$

6406532241000. ✖ $\hat{y} = x + 0.5$

6406532241001. ✖ $\hat{y} = x + 0.2$

6406532241002. ✖ $\hat{y} = 0.2x + 1$

✔ Official Answer Key: Option (A)

MLF-Q572 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QPD3 29 Oct*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 130 Question Id : 640653668935 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Let A be a $n \times n$ real matrix. Then, which among the following statements are true?

Options :

6406532241008. ✖ If the eigenvalues of A are $\lambda_1, \ldots, \lambda_n$, then A is similar to a diagonal matrix with $\lambda_1, \ldots, \lambda_n$ along the diagonal.

6406532241009. ✖ If $\text{rank}(A) = r$, then A has r non-zero eigenvalues.

6406532241010. ✔ A may have complex eigenvalues.

6406532241011. ✖ If A has a repeated eigenvalue, then A is not diagonalizable.

Sub-Section Number :	7
Sub-Section Id :	64065395281
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (C)

MLF-Q573 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QPD3 29 Oct*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 131 Question Id : 640653668928 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select the correct options from the following.

Options :

6406532240990. ✖ There exists a matrix whose row space contains $\begin{bmatrix}1\\0\\-1\end{bmatrix}$ and whose null space contains $\begin{bmatrix}0\\1\\1\end{bmatrix}$.

6406532240991. ✖ There exists a matrix with row space $= \mathbb{R}^3$ and column space $= \mathbb{R}^4$.

6406532240992. ✔ There exists a matrix whose column space contains $\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}$ and row space contains $\begin{bmatrix}1\\1\end{bmatrix}, \begin{bmatrix}1\\2\end{bmatrix}$

6406532240993. ✔ There exists a matrix whose nullspace consists of all vectors in $\mathbb{R}^3$ such that $x_1 + 2x_2 + 4x_3 = 0$.

Sub-Section Number :	8
Sub-Section Id :	64065395282
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653668929 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (132 to 133)

✔ Official Answer Key: Option (C, D)

MLF-Q576 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QPD4 29 Oct*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 118 Question Id : 640653669180 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406532241719. ✔ YES

6406532241720. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	64065395375
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key: Option (A)**

MLF-Q579 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QPD4 29 Oct*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 121 Question Id : 640653669188 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Consider a vector $v = \begin{bmatrix} 2 \\ 1 \\ -2 \end{bmatrix}$. Does v belong to the columnspace of $A = \begin{bmatrix} -8 & -2 & -9 \\ 6 & 4 & 8 \\ 4 & 0 & 4 \end{bmatrix}$?

Options :

6406532241740. ✔ Yes

6406532241741. ✖ No

✔ Official Answer Key: Option (A)

MLF-Q580 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QPD4 29 Oct*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 122 Question Id : 640653669194 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Let $A = \begin{bmatrix} 8 & 3 \\ 2 & 7 \end{bmatrix}$. Find a matrix P such that $P^{-1}AP$ is diagonal.

Options :

6406532241755. ✖ $\begin{bmatrix} 1 & 1 \\ 2 & -3 \end{bmatrix}$

6406532241756. ✖ $\begin{bmatrix} 10 & 0 \\ 0 & 5 \end{bmatrix}$

6406532241757. ✖ $\begin{bmatrix} 1 & 1 \\ 2 & 3 \end{bmatrix}$

6406532241758. ✔ $\begin{bmatrix} 3 & 1 \\ 2 & -1 \end{bmatrix}$

Sub-Section Number :	4
Sub-Section Id :	64065395377
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (D)

MLF-Q581 | [SA] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QPD4 29 Oct*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 123 Question Id : 640653669183 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

Consider a regression model $h(x_1, x_2) = \frac{x_1 + x_2}{2}$. For the below input data points, find the mean squared loss of h. Enter the answer correct to two decimal places.

X	Y
$[1, 2]$	2
$[2, 3]$	3
$[3, 4]$	4
$[4, 5]$	5

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.25

✔ **Official Answer Key (NAT): 0.25**

MLF-Q582 | [SA] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QPD4 29 Oct*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 124 Question Id : 640653669185 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

The curve $y = ax^2 + bx + c$ passes through the points $(2, -1)$ and $(0, 0)$, and is tangent to the line $y = -x$ at the origin. Find the absolute value of the product ab. Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

0.25

✔ **Official Answer Key (NAT): 0.25**

MLF-Q583 | [SA] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QPD4 29 Oct*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 125 Question Id : 640653669186 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

At the point $P_0 = (1, 2)$, the function $f(x, y)$ has a derivative of $2\sqrt{2}$ in the direction $\left(\frac{1}{\sqrt{2}}, \frac{1}{\sqrt{2}}\right)$ and a derivative of 5 in the direction $(0, 1)$. Find the derivative of $f(x, y)$ at point P_0 in the direction $\left(\frac{-1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$. Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

4.88 to 4.95

✔ **Official Answer Key (NAT): 4.88 to 4.95**

MLF-Q584 | [SA] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QPD4 29 Oct*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 126 Question Id : 640653669195 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

Consider a 2×2 matrix A whose eigenvectors are $v_1 = [1, 1]^T$ and $v_2 = [1, 2]^T$ with eigenvalues 2 and 3, respectively. Find the sum of square of all the entries of A.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

22

Sub-Section Number :	5
Sub-Section Id :	64065395378
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key (NAT): 22**

MLF-Q586 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QPD4 29 Oct*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 128 Question Id : 640653669187 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Find the second order approximation for $f(x,y) = e^{2y-x}$ in the neighbourhood of $(0,0)$.

Options :

6406532241736. ✖ $1 - x + 2y - 2xy + y^2 + \frac{x^2}{2}$

6406532241737. ✖ $1 - 2xy + 2y^2 + \frac{x^2}{2}$

6406532241738. ✖ $2y^2 + \frac{x^2}{2}$

6406532241739. ✔ $1 - x + 2y - 2xy + 2y^2 + \frac{x^2}{2}$

✔ **Official Answer Key: Option (D)**

MLF-Q587 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QPD4 29 Oct*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 129 Question Id : 640653669193 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Find the best fit line for the dataset given below using least squares method.

x	1	3	4	5	7
y	-3	7	1	0	4

Options :

6406532241751. ✔ $\hat{y} = 0.7x - 1$

6406532241752. ✖ $\hat{y} = x + 0.5$

6406532241753. ✖ $\hat{y} = x + 0.2$

6406532241754. ✖ $\hat{y} = 0.2x + 1$

✔ **Official Answer Key: Option (A)**

MLF-Q588 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QPD4 29 Oct*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 130 Question Id : 640653669196 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Let A be a $n \times n$ real matrix. Then, which among the following statements are true?

Options :

6406532241760. ✖ If the eigenvalues of A are $\lambda_1, \ldots, \lambda_n$, then A is similar to a diagonal matrix with $\lambda_1, \ldots, \lambda_n$ along the diagonal.

6406532241761. ✖ If $\text{rank}(A) = r$, then A has r non-zero eigenvalues.

6406532241762. ✔ A may have complex eigenvalues.

6406532241763. ✖ If A has a repeated eigenvalue, then A is not diagonalizable.

Sub-Section Number :	7
Sub-Section Id :	64065395380
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (C)

MLF-Q589 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QPD4 29 Oct*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 131 Question Id : 640653669189 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Select the correct options from the following.

Options :

6406532241742. ✖ There exists a matrix whose row space contains $\begin{bmatrix}1\\0\\-1\end{bmatrix}$ and whose null space contains $\begin{bmatrix}0\\1\\1\end{bmatrix}$.

6406532241743. ✖ There exists a matrix with row space $= \mathbb{R}^3$ and column space $= \mathbb{R}^4$.

6406532241744. ✔ There exists a matrix whose column space contains $\begin{bmatrix}1\\0\\0\end{bmatrix}, \begin{bmatrix}0\\0\\1\end{bmatrix}$ and row space contains $\begin{bmatrix}1\\1\end{bmatrix}, \begin{bmatrix}1\\2\end{bmatrix}$

6406532241745. ✔ There exists a matrix whose nullspace consists of all vectors in $\mathbb{R}^3$ such that $x_1 + 2x_2 + 4x_3 = 0$.

Sub-Section Number :	8
Sub-Section Id :	64065395381
Question Shuffling Allowed :	No
Is Section Default? :	null

Question Id : 640653669190 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Question Numbers : (132 to 133)

✔ Official Answer Key: Option (C, D)

MLF-Q592 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 1 Question Id : 640653995423 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406533360569. ✔ YES

6406533360570. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653146770
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLF-Q593 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 2 Question Id : 640653995424 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following 3 × 3 matrices have rank 2 with column space equal to row space?
Options :

6406533360571. ✔ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

6406533360572. ✖ $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$

6406533360573. ✖ $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

6406533360574. ✔ $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

✔ **Official Answer Key: Option (A, D)**

MLF-Q595 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 4 Question Id : 640653995425 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider a matrix $A = \begin{bmatrix} 6 & 3 \\ -2 & 1 \end{bmatrix}$ and a set $S = \{v_1 = (1, -1), v_2 = (1, 1)\}$.
Which of the following options is/are correct?

Options :

6406533360575. ✖ S is the set of eigenvectors of A.

6406533360576. ✔ Only v_1 is an eigenvector of A.

6406533360577. ✔ $A^{50} = \begin{pmatrix} 1 & 3 \\ -1 & -2 \end{pmatrix} \begin{pmatrix} 3^{50} & 0 \\ 0 & 4^{50} \end{pmatrix} \begin{pmatrix} -2 & -3 \\ 1 & 1 \end{pmatrix}$

6406533360578. ✖ $A^{50} = \begin{pmatrix} 3 & 1 \\ -2 & -1 \end{pmatrix} \begin{pmatrix} 3^{50} & 0 \\ 0 & 4^{50} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -2 & -3 \end{pmatrix}$

✔ **Official Answer Key: Option (B, C)**

MLF-Q596 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 5 Question Id : 640653995426 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider two vectors $a = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Let P be the projection matrix that projects a vector onto the line through the vector a. Which of the following options is/are true?

Options :

6406533360579. ✔ P is symmetric matrix.

6406533360580. ✖ Projection of b onto a is $\frac{1}{4}(-a)$.

6406533360581. ✔ P is rank one matrix.

6406533360582. ✔ Plane $x = y$ is orthogonal to a.

✔ Official Answer Key: Option (A, C, D)

MLF-Q597 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 6 Question Id : 640653995427 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question

Which of the following options is/are correct for the following dataset?

x	1	1	-1
y	1	1	1

Options :

6406533360583. ✔ The best-fit line for the dataset given using the least squares method is, $\hat{y} = 1$.

6406533360584. ✖ The best-fit line for the dataset given using the least squares method is, $\hat{y} = -1$.

6406533360585. ✖ The best-fit line for the dataset given using the least squares method is, $\hat{y} = x - 1$.

6406533360586. ✖ The best-fit line for the dataset given using the least squares method is, $\hat{y} = x + 1$.

6406533360587. ✖ Error vector (the difference between predicted value and actual value) is $\begin{bmatrix} 0 \\ 0 \\ -2 \end{bmatrix}$.

6406533360588. ✔ Error vector(the difference between predicted value and actual value) is $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

✔ **Official Answer Key: Option (A, F)**

MLF-Q598 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 7 Question Id : 640653995433 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Determine which of the following limits yields a finite value.

Options :

6406533360605. ✔ $\lim_{x\to 1} \frac{x^{\frac{1}{3}}-1}{x^{\frac{1}{2}}-1}$

6406533360606. ✔ $\lim_{x\to 1} \frac{1-x^{\frac{1}{2}}}{1-x}$

6406533360607. ✖ $\lim_{x\to \infty} \frac{x^2-7x}{x+1}$

6406533360608. ✔ $\lim_{x\to a} \frac{x^2-a^2}{x^4-a^4}$, where $a>0$

6406533360609. ✖ None of these

Sub-Section Number :	4
Sub-Section Id :	640653146772
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A, B, D)**

MLF-Q599 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 8 Question Id : 640653995429 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Let $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 6 \end{bmatrix}$. Is A diagonalizable?

Options :

6406533360593. ✔ Yes

6406533360594. ✖ No

6406533360595. ✖ Insufficient information

✔ Official Answer Key: Option (A)

MLF-Q600 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 9 Question Id : 640653995430 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the following input data points:

X	Y
[0, 1, 1]	-1
[0, 1, 0]	-1
[1, 1, 3]	+1
[1, 0, 5]	+1
[0, 0, 6]	+1
[1, 0, 4]	+1

Which among the following models has the least misclassification error?
Hint: Consider $Sign(0) = 1$

Options :

6406533360596. ✔ $Sign(x_1 + x_2 + x_3 - 5)$

6406533360597. ✖ $Sign(x_1 + x_2 - 2)$

6406533360598. ✖ $Sign(x_1 + x_2 + x_3)$

6406533360599. ✖ $Sign(x_3 - x_2 - x_1)$

✔ **Official Answer Key: Option (A)**

MLF-Q601 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 10 Question Id : 640653995432 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

For what value of b is

$$f(x) = \begin{cases} \frac{x-b}{b+1}, & \text{if } x < 0 \\ x^2 + b, & \text{if } x \geq 0 \end{cases}$$

continuous at every x?

Options :

6406533360601. ✖ 3

6406533360602. ✖ 2

6406533360603. ✔ -2

6406533360604. ✖ 1

✔ Official Answer Key: Option (C)

MLF-Q602 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 11 Question Id : 640653995435 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the function, $f(x, y) = x^2 + 3y^2$, in which direction does the maximum directional derivative occur at the point $(1, 2)$?

Options :

6406533360611. ✖ $\left(\frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}}\right)$

6406533360612. ✔ $\left(\frac{1}{\sqrt{37}}, \frac{6}{\sqrt{37}}\right)$

6406533360613. ✖ $\left(\frac{2}{\sqrt{5}}, \frac{24}{\sqrt{5}}\right)$

6406533360614. ✖ $\left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$

Sub-Section Number :	5
Sub-Section Id :	640653146773
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B)**

MLF-Q603 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 12 Question Id : 640653995431 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Consider the function, $f(x, y) = e^x \cos(y)$. Let $L(x, y)$ be the linear approximation of $f(x, y)$ at the point $(0, 0)$, then find the value $L(0.5, 0)$. Enter the answer correct to one decimal place.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1.5

✔ **Official Answer Key (NAT): 1.5**

MLF-Q604 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 13 Question Id : 640653995434 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Let $f : \mathbb{R} \to \mathbb{R}$ be the function, $f(x) = 1 + e^{-x}$. What is the value of the derivative of f at x where $f(x) = 1.4$? Enter the answer correct to one decimal place.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0.4

✔ **Official Answer Key (NAT): 0.4**

MLF-Q605 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 1 Question Id : 640653995705 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406533361353. ✔ YES

6406533361354. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653146859
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLF-Q606 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 2 Question Id : 640653995706 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following 3 × 3 matrices have rank 2 with column space equal to row space?
Options :

6406533361355. ✔ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

6406533361356. ✖ $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$

6406533361357. ✖ $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

6406533361358. ✔ $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

✔ Official Answer Key: Option (A, D)

MLF-Q608 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 4 Question Id : 640653995707 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider a matrix $A = \begin{bmatrix} 6 & 3 \\ -2 & 1 \end{bmatrix}$ and a set $S = \{v_1 = (1, -1), v_2 = (1, 1)\}$.
Which of the following options is/are correct?

Options :

6406533361359. ✖ S is the set of eigenvectors of A.

6406533361360. ✔ Only v_1 is an eigenvector of A.

6406533361361. ✔ $A^{50} = \begin{pmatrix} 1 & 3 \\ -1 & -2 \end{pmatrix} \begin{pmatrix} 3^{50} & 0 \\ 0 & 4^{50} \end{pmatrix} \begin{pmatrix} -2 & -3 \\ 1 & 1 \end{pmatrix}$

6406533361362. ✖ $A^{50} = \begin{pmatrix} 3 & 1 \\ -2 & -1 \end{pmatrix} \begin{pmatrix} 3^{50} & 0 \\ 0 & 4^{50} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -2 & -3 \end{pmatrix}$

✔ **Official Answer Key: Option (B, C)**

MLF-Q609 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 5 Question Id : 640653995708 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider two vectors $a = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Let P be the projection matrix that projects a vector onto the line through the vector a. Which of the following options is/are true?

Options :

6406533361363. ✔ P is symmetric matrix.

6406533361364. ✖ Projection of b onto a is $\frac{1}{4}(-a)$.

6406533361365. ✔ P is rank one matrix.

6406533361366. ✔ Plane $x = y$ is orthogonal to a.

✔ Official Answer Key: Option (A, C, D)

MLF-Q610 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 6 Question Id : 640653995709 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question

Which of the following options is/are correct for the following dataset?

x	1	1	-1
y	1	1	1

Options :

6406533361367. ✔ The best-fit line for the dataset given using the least squares method is, $\hat{y} = 1$.

6406533361368. ✖ The best-fit line for the dataset given using the least squares method is, $\hat{y} = -1$.

6406533361369. ✖ The best-fit line for the dataset given using the least squares method is, $\hat{y} = x - 1$.

6406533361370. ✖ The best-fit line for the dataset given using the least squares method is, $\hat{y} = x + 1$.

6406533361371. ✖ Error vector (the difference between predicted value and actual value) is $\begin{bmatrix} 0 \\ 0 \\ -2 \end{bmatrix}$.

6406533361372. ✔ Error vector(the difference between predicted value and actual value) is $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

✔ **Official Answer Key: Option (A, F)**

MLF-Q611 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 7 Question Id : 640653995715 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Determine which of the following limits yields a finite value.

Options :

6406533361389. ✔ $\lim_{x\to 1} \frac{x^{\frac{1}{3}}-1}{x^{\frac{1}{2}}-1}$

6406533361390. ✔ $\lim_{x\to 1} \frac{1-x^{\frac{1}{2}}}{1-x}$

6406533361391. ✖ $\lim_{x\to \infty} \frac{x^2-7x}{x+1}$

6406533361392. ✔ $\lim_{x\to a} \frac{x^2-a^2}{x^4-a^4}$, where $a>0$

6406533361393. ✖ None of these

Sub-Section Number :	4
Sub-Section Id :	640653146861
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A, B, D)**

MLF-Q612 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 8 Question Id : 640653995711 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Let $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 6 \end{bmatrix}$. Is A diagonalizable?

Options :

6406533361377. ✔ Yes

6406533361378. ✖ No

6406533361379. ✖ Insufficient information

✔ Official Answer Key: Option (A)

MLF-Q613 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 9 Question Id : 640653995712 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the following input data points:

X	Y
[0, 1, 1]	-1
[0, 1, 0]	-1
[1, 1, 3]	+1
[1, 0, 5]	+1
[0, 0, 6]	+1
[1, 0, 4]	+1

Which among the following models has the least misclassification error?
Hint: Consider $Sign(0) = 1$

Options :

6406533361380. ✔ $Sign(x_1 + x_2 + x_3 - 5)$

6406533361381. ✖ $Sign(x_1 + x_2 - 2)$

6406533361382. ✖ $Sign(x_1 + x_2 + x_3)$

6406533361383. ✖ $Sign(x_3 - x_2 - x_1)$

✔ **Official Answer Key: Option (A)**

MLF-Q614 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 10 Question Id : 640653995714 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

For what value of b is

$$f(x) = \begin{cases} \frac{x-b}{b+1}, & \text{if } x < 0 \\ x^2 + b, & \text{if } x \geq 0 \end{cases}$$

continuous at every x?

Options :

6406533361385. ✖ 3

6406533361386. ✖ 2

6406533361387. ✔ -2

6406533361388. ✖ 1

✔ Official Answer Key: Option (C)

MLF-Q615 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 11 Question Id : 640653995717 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the function, $f(x, y) = x^2 + 3y^2$, in which direction does the maximum directional derivative occur at the point $(1, 2)$?

Options :

6406533361395. ✖ $\left(\frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}}\right)$

6406533361396. ✔ $\left(\frac{1}{\sqrt{37}}, \frac{6}{\sqrt{37}}\right)$

6406533361397. ✖ $\left(\frac{2}{\sqrt{5}}, \frac{24}{\sqrt{5}}\right)$

6406533361398. ✖ $\left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$

Sub-Section Number :	5
Sub-Section Id :	640653146862
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B)**

MLF-Q616 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 12 Question Id : 640653995713 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Consider the function, $f(x, y) = e^x \cos(y)$. Let $L(x, y)$ be the linear approximation of $f(x, y)$ at the point $(0, 0)$, then find the value $L(0.5, 0)$. Enter the answer correct to one decimal place.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1.5

✔ **Official Answer Key (NAT): 1.5**

MLF-Q617 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 13 Question Id : 640653995716 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Let $f : \mathbb{R} \to \mathbb{R}$ be the function, $f(x) = 1 + e^{-x}$. What is the value of the derivative of f at x where $f(x) = 1.4$? Enter the answer correct to one decimal place.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0.4

✔ **Official Answer Key (NAT): 0.4**

MLF-Q618 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 1 Question Id : 640653995987 Question Type : MCQ

Correct Marks : 0

Question Label : Multiple Choice Question

THIS IS QUESTION PAPER FOR THE SUBJECT "DIPLOMA LEVEL : MACHINE LEARNING FOUNDATIONS (COMPUTER BASED EXAM)"

ARE YOU SURE YOU HAVE TO WRITE EXAM FOR THIS SUBJECT?
CROSS CHECK YOUR HALL TICKET TO CONFIRM THE SUBJECTS TO BE WRITTEN.

(IF IT IS NOT THE CORRECT SUBJECT, PLS CHECK THE SECTION AT THE TOP FOR THE SUBJECTS REGISTERED BY YOU)

Options :

6406533362137. ✔ YES

6406533362138. ✖ NO

Sub-Section Number :	2
Sub-Section Id :	640653146948
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A)

MLF-Q619 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 2 Question Id : 640653995988 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following 3 × 3 matrices have rank 2 with column space equal to row space?
Options :

6406533362139. ✔ $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

6406533362140. ✖ $\begin{bmatrix} 1 & 2 & 3 \\ 2 & 4 & 5 \\ 3 & 5 & 6 \end{bmatrix}$

6406533362141. ✖ $\begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix}$

6406533362142. ✔ $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 2 \end{bmatrix}$

✔ **Official Answer Key: Option (A, D)**

MLF-Q621 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 4 Question Id : 640653995989 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider a matrix $A = \begin{bmatrix} 6 & 3 \\ -2 & 1 \end{bmatrix}$ and a set $S = \{v_1 = (1, -1), v_2 = (1, 1)\}$.
Which of the following options is/are correct?

Options :

6406533362143. ✖ S is the set of eigenvectors of A.

6406533362144. ✔ Only v_1 is an eigenvector of A.

6406533362145. ✔ $A^{50} = \begin{pmatrix} 1 & 3 \\ -1 & -2 \end{pmatrix} \begin{pmatrix} 3^{50} & 0 \\ 0 & 4^{50} \end{pmatrix} \begin{pmatrix} -2 & -3 \\ 1 & 1 \end{pmatrix}$

6406533362146. ✖ $A^{50} = \begin{pmatrix} 3 & 1 \\ -2 & -1 \end{pmatrix} \begin{pmatrix} 3^{50} & 0 \\ 0 & 4^{50} \end{pmatrix} \begin{pmatrix} 1 & 1 \\ -2 & -3 \end{pmatrix}$

✔ **Official Answer Key: Option (B, C)**

MLF-Q622 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 5 Question Id : 640653995990 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question

Consider two vectors $a = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}$ and $b = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$. Let P be the projection matrix that projects a vector onto the line through the vector a. Which of the following options is/are true?

Options :

6406533362147. ✔ P is symmetric matrix.

6406533362148. ✖ Projection of b onto a is $\frac{1}{4}(-a)$.

6406533362149. ✔ P is rank one matrix.

6406533362150. ✔ Plane $x = y$ is orthogonal to a.

✔ Official Answer Key: Option (A, C, D)

MLF-Q623 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 6 Question Id : 640653995991 Question Type : MSQ
Correct Marks : 4 Max. Selectable Options : 0
Question Label : Multiple Select Question

Which of the following options is/are correct for the following dataset?

x	1	1	-1
y	1	1	1

Options :

6406533362151. ✔ The best-fit line for the dataset given using the least squares method is, $\hat{y} = 1$.

6406533362152. ✖ The best-fit line for the dataset given using the least squares method is, $\hat{y} = -1$.

6406533362153. ✖ The best-fit line for the dataset given using the least squares method is, $\hat{y} = x - 1$.

6406533362154. ✖ The best-fit line for the dataset given using the least squares method is, $\hat{y} = x + 1$.

6406533362155. ✖ Error vector (the difference between predicted value and actual value) is $\begin{bmatrix} 0 \\ 0 \\ -2 \end{bmatrix}$.

6406533362156. ✔ Error vector(the difference between predicted value and actual value) is $\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$

✔ **Official Answer Key: Option (A, F)**

MLF-Q624 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 7 Question Id : 640653995997 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Determine which of the following limits yields a finite value.

Options :

6406533362173. ✔ $\lim_{x\to 1} \frac{x^{\frac{1}{3}}-1}{x^{\frac{1}{2}}-1}$

6406533362174. ✔ $\lim_{x\to 1} \frac{1-x^{\frac{1}{2}}}{1-x}$

6406533362175. ✖ $\lim_{x\to \infty} \frac{x^2-7x}{x+1}$

6406533362176. ✔ $\lim_{x\to a} \frac{x^2-a^2}{x^4-a^4}$, where $a > 0$

6406533362177. ✖ None of these

Sub-Section Number :	4
Sub-Section Id :	640653146950
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A, B, D)**

MLF-Q625 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 8 Question Id : 640653995993 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Let $A = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 2 & 1 \\ 0 & 0 & 6 \end{bmatrix}$. Is A diagonalizable?

Options :

6406533362161. ✔ Yes

6406533362162. ✖ No

6406533362163. ✖ Insufficient information

✔ Official Answer Key: Option (A)

MLF-Q626 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 9 Question Id : 640653995994 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the following input data points:

X	Y
[0, 1, 1]	-1
[0, 1, 0]	-1
[1, 1, 3]	+1
[1, 0, 5]	+1
[0, 0, 6]	+1
[1, 0, 4]	+1

Which among the following models has the least misclassification error?
Hint: Consider $Sign(0) = 1$

Options :

6406533362164. ✔ $Sign(x_1 + x_2 + x_3 - 5)$

6406533362165. ✖ $Sign(x_1 + x_2 - 2)$

6406533362166. ✖ $Sign(x_1 + x_2 + x_3)$

6406533362167. ✖ $Sign(x_3 - x_2 - x_1)$

✔ **Official Answer Key: Option (A)**

MLF-Q627 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 10 Question Id : 640653995996 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

For what value of b is

$$f(x) = \begin{cases} \frac{x-b}{b+1}, & \text{if } x < 0 \\ x^2 + b, & \text{if } x \geq 0 \end{cases}$$

continuous at every x?

Options :

6406533362169. ✖ 3

6406533362170. ✖ 2

6406533362171. ✔ -2

6406533362172. ✖ 1

✔ Official Answer Key: Option (C)

MLF-Q628 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4*

Category: Week 1 - ML Foundations › ML Concepts

Question Number : 11 Question Id : 640653995999 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the function, $f(x, y) = x^2 + 3y^2$, in which direction does the maximum directional derivative occur at the point $(1, 2)$?

Options :

6406533362179. ✖ $\left(\frac{1}{\sqrt{10}}, \frac{3}{\sqrt{10}}\right)$

6406533362180. ✔ $\left(\frac{1}{\sqrt{37}}, \frac{6}{\sqrt{37}}\right)$

6406533362181. ✖ $\left(\frac{2}{\sqrt{5}}, \frac{24}{\sqrt{5}}\right)$

6406533362182. ✖ $\left(\frac{1}{\sqrt{5}}, \frac{2}{\sqrt{5}}\right)$

Sub-Section Number :	5
Sub-Section Id :	640653146951
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B)**

MLF-Q629 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 12 Question Id : 640653995995 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Consider the function, $f(x, y) = e^x \cos(y)$. Let $L(x, y)$ be the linear approximation of $f(x, y)$ at the point $(0, 0)$, then find the value $L(0.5, 0)$. Enter the answer correct to one decimal place.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
1.5

✔ **Official Answer Key (NAT): 1.5**

MLF-Q630 | [SA] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4*
Category: Week 1 - ML Foundations › ML Concepts

Question Number : 13 Question Id : 640653995998 Question Type : SA
Correct Marks : 3
Question Label : Short Answer Question

Let $f : \mathbb{R} \to \mathbb{R}$ be the function, $f(x) = 1 + e^{-x}$. What is the value of the derivative of f at x where $f(x) = 1.4$? Enter the answer correct to one decimal place.

Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Equal
Text Areas : PlainText
Possible Answers :
0.4

✔ **Official Answer Key (NAT): 0.4**

■ Category: ML Tasks (23 Questions)

MLF-Q039 | [MSQ] | Marks: +2 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*
Category: Week 1 - ML Foundations › ML Tasks

Question Number : 131 Question Id : 640653345260 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Select Question

Identify which of the following requires use of classification technique.

Options :

6406531148399. ✖ Predicting the amount of rainfall in May 2022 in North India based on precipitation data of the year 2021.

6406531148400. ✖ Predicting the price of a land based on its area and distance from the market.

6406531148401. ✔ Predicting whether an email is spam or not.

6406531148402. ✔ Predicting whether the patient is having cancer or not.

Sub-Section Number :	4
Sub-Section Id :	64065348779
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (C, D)

MLF-Q047 | [MSQ] | Marks: +2 | *Quiz 1 Jan23*
Category: Week 1 - ML Foundations › ML Tasks

Question Number : 122 Question Id : 640653470048 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3 Selectable Option : 0

Question Label : Multiple Select Question

Which of the following statements is/are true?

Options :

6406531562044. ✔ Email spam detection falls under classification problem.

6406531562045. ✖ Predicting the price of a house based on previous datasets falls under classificiation problem.

6406531562046. ✔ Classifying whether handwriting belongs to a particular person or not based on previous data falls under the classification problem.

6406531562047. ✔ Finding out the probability that a particular poem is written by Shakespeare falls under unsupervised learning.

✔ **Official Answer Key: Option (A, C, D)**

MLF-Q189 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 13 July 2025*

Category: Week 1 - ML Foundations › ML Tasks

Question Label : Comprehension

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

Sub questions

Question Number : 94 Question Id : 6406531297923 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

A hospital collects patient data including symptoms, vital signs, and lab test results. They want to find patterns to improve treatment plans. What type of task is clustering most suitable for?

Options :

6406534364735. ✖ Predicting the recovery time for a new patient.

6406534364736. ✖ Diagnosing diseases based on labeled medical records.

6406534364737. ✔ Grouping patients with similar symptoms and test results without prior labels.

6406534364738. ✖ Determining the most important variable for predicting disease.

✔ **Official Answer Key: Option (C)**

MLF-Q204 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 13 July 2025*

Category: Week 1 - ML Foundations › ML Tasks

Question Label : Comprehension

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

Sub questions

Question Number : 94 Question Id : 6406531298193 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

A hospital collects patient data including symptoms, vital signs, and lab test results. They want to find patterns to improve treatment plans. What type of task is clustering most suitable for?

Options :

6406534365506. ✖ Predicting the recovery time for a new patient.

6406534365507. ✖ Diagnosing diseases based on labeled medical records.

6406534365508. ✔ Grouping patients with similar symptoms and test results without prior labels.

6406534365509. ✖ Determining the most important variable for predicting disease.

✔ **Official Answer Key: Option (C)**

MLF-Q219 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025*

Category: Week 1 - ML Foundations › ML Tasks

Question Label : Comprehension

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

Sub questions

Question Number : 94 Question Id : 6406531298463 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

A hospital collects patient data including symptoms, vital signs, and lab test results. They want to find patterns to improve treatment plans. What type of task is clustering most suitable for?

Options :

6406534366277. ✖ Predicting the recovery time for a new patient.

6406534366278. ✖ Diagnosing diseases based on labeled medical records.

6406534366279. ✔ Grouping patients with similar symptoms and test results without prior labels.

6406534366280. ✖ Determining the most important variable for predicting disease.

✔ Official Answer Key: Option (C)

MLF-Q232 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024*

Category: Week 1 - ML Foundations › ML Tasks

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 62 Question Id : 640653825050 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Consider a dataset from an e-commerce company such as Flipkart that has details of customer behavior, which of the following problems can be best approached using clustering?

Options :

6406532774829. ✖ Predicting the exact number of sales for each customer segment.

6406532774830. ✖ Classifying customers into segments based on labeled customer data.

6406532774831. ✔ Identifying inherent groupings in customer data without predefined labels.

6406532774832. ✖ Reducing the number of features in the customer dataset.

✔ **Official Answer Key: Option (C)**

MLF-Q245 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 7 July 2024*

Category: Week 1 - ML Foundations › ML Tasks

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 62 Question Id : 640653827519 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Consider a dataset from an e-commerce company such as Flipkart that has details of customer behavior, which of the following problems can be best approached using clustering?

Options :

6406532781353. ✖ Predicting the exact number of sales for each customer segment.

6406532781354. ✖ Classifying customers into segments based on labeled customer data.

6406532781355. ✔ Identifying inherent groupings in customer data without predefined labels.

6406532781356. ✖ Reducing the number of features in the customer dataset.

✔ **Official Answer Key: Option (C)**

MLF-Q258 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 7 July 2024(1) (+1 more papers)*

Category: Week 1 - ML Foundations › ML Tasks

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 62 Question Id : 640653827807 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

Consider a dataset from an e-commerce company such as Flipkart that has details of customer behavior, which of the following problems can be best approached using clustering?

Options :

6406532782149. ✖ Predicting the exact number of sales for each customer segment.

6406532782150. ✖ Classifying customers into segments based on labeled customer data.

6406532782151. ✔ Identifying inherent groupings in customer data without predefined labels.

6406532782152. ✖ Reducing the number of features in the customer dataset.

✔ **Official Answer Key: Option (C)**

MLF-Q260 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Tasks

Question Number : 111 Question Id : 6406531527840 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

A bank wants to decide whether a loan applicant should be given a loan or not, based on features like income, credit score and past records. Which of the following machine learning techniques is applicable for this scenario?

Options :

6406535115676. ✖ Regression

6406535115677. ✔ Classification

6406535115678. ✖ Clustering

6406535115679. ✖ Dimensionality Reduction

Sub-Section Number :	3
Sub-Section Id :	640653244923
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B)**

MLF-Q279 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Tasks

Question Number : 111 Question Id : 6406531528143 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

A bank wants to decide whether a loan applicant should be given a loan or not, based on features like income, credit score and past records. Which of the following machine learning techniques is applicable for this scenario?

Options :

6406535116519. ✖ Regression

6406535116520. ✔ Classification

6406535116521. ✖ Clustering

6406535116522. ✖ Dimensionality Reduction

Sub-Section Number :	3
Sub-Section Id :	640653245019
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B)**

MLF-Q298 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Tasks

Question Number : 111 Question Id : 6406531528805 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

A bank wants to decide whether a loan applicant should be given a loan or not, based on features like income, credit score and past records. Which of the following machine learning techniques is applicable for this scenario?

Options :

6406535118450. ✖ Regression

6406535118451. ✔ Classification

6406535118452. ✖ Clustering

6406535118453. ✖ Dimensionality Reduction

Sub-Section Number :	3
Sub-Section Id :	640653245204
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B)**

MLF-Q317 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Tasks

Question Number : 137 Question Id : 6406531525151 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

A bank wants to decide whether a loan applicant should be given a loan or not, based on features like income, credit score and past records. Which of the following machine learning techniques is applicable for this scenario?

Options :

6406535108151. ✖ Regression

6406535108152. ✔ Classification

6406535108153. ✖ Clustering

6406535108154. ✖ Dimensionality Reduction

Sub-Section Number :	3
Sub-Section Id :	640653244043
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B)**

MLF-Q336 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Tasks

Question Number : 137 Question Id : 6406531525778 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

A bank wants to decide whether a loan applicant should be given a loan or not, based on features like income, credit score and past records. Which of the following machine learning techniques is applicable for this scenario?

Options :

6406535109928. ✖ Regression

6406535109929. ✔ Classification

6406535109930. ✖ Clustering

6406535109931. ✖ Dimensionality Reduction

Sub-Section Number :	3
Sub-Section Id :	640653244236
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B)**

MLF-Q355 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Tasks

Question Number : 137 Question Id : 6406531526707 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

A bank wants to decide whether a loan applicant should be given a loan or not, based on features like income, credit score and past records. Which of the following machine learning techniques is applicable for this scenario?

Options :

6406535112573. ✖ Regression

6406535112574. ✔ Classification

6406535112575. ✖ Clustering

6406535112576. ✖ Dimensionality Reduction

Sub-Section Number :	3
Sub-Section Id :	640653244527
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B)

MLF-Q374 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA2 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Tasks

Question Number : 136 Question Id : 6406531534276 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

A bank wants to decide whether a loan applicant should be given a loan or not, based on features like income, credit score and past records. Which of the following machine learning techniques is applicable for this scenario?

Options :

6406535133761. ✖ Regression

6406535133762. ✔ Classification

6406535133763. ✖ Clustering

6406535133764. ✖ Dimensionality Reduction

Sub-Section Number :	3
Sub-Section Id :	640653246972
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B)

MLF-Q393 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA3 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Tasks

Question Number : 136 Question Id : 6406531534615 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

A bank wants to decide whether a loan applicant should be given a loan or not, based on features like income, credit score and past records. Which of the following machine learning techniques is applicable for this scenario?

Options :

6406535134731. ✖ Regression

6406535134732. ✔ Classification

6406535134733. ✖ Clustering

6406535134734. ✖ Dimensionality Reduction

Sub-Section Number :	3
Sub-Section Id :	640653247074
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B)

MLF-Q412 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA4 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Tasks

Question Number : 136 Question Id : 6406531535052 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

A bank wants to decide whether a loan applicant should be given a loan or not, based on features like income, credit score and past records. Which of the following machine learning techniques is applicable for this scenario?

Options :

6406535135877. ✖ Regression

6406535135878. ✔ Classification

6406535135879. ✖ Clustering

6406535135880. ✖ Dimensionality Reduction

Sub-Section Number :	3
Sub-Section Id :	640653247204
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B)

MLF-Q431 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Tasks

Question Number : 130 Question Id : 6406531535788 Question Type : MCQ
Correct Marks : 1
Question Label : Multiple Choice Question
A bank wants to decide whether a loan applicant should be given a loan or not, based on features like income, credit score and past records. Which of the following machine learning techniques is applicable for this scenario?
Options :
6406535138057. ✖ Regression
6406535138058. ✔ Classification
6406535138059. ✖ Clustering
6406535138060. ✖ Dimensionality Reduction

Sub-Section Number :	3
Sub-Section Id :	640653247430
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B)**

MLF-Q450 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Tasks

Question Number : 130 Question Id : 6406531536128 Question Type : MCQ
Correct Marks : 1
Question Label : Multiple Choice Question
A bank wants to decide whether a loan applicant should be given a loan or not, based on features like income, credit score and past records. Which of the following machine learning techniques is applicable for this scenario?
Options :
6406535139157. ✖ Regression
6406535139158. ✔ Classification
6406535139159. ✖ Clustering
6406535139160. ✖ Dimensionality Reduction

Sub-Section Number :	3
Sub-Section Id :	640653247532
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B)**

MLF-Q469 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Tasks

Question Number : 130 Question Id : 6406531536418 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

A bank wants to decide whether a loan applicant should be given a loan or not, based on features like income, credit score and past records. Which of the following machine learning techniques is applicable for this scenario?

Options :

6406535139988. ✖ Regression

6406535139989. ✔ Classification

6406535139990. ✖ Clustering

6406535139991. ✖ Dimensionality Reduction

Sub-Section Number :	3
Sub-Section Id :	640653247634
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B)

MLF-Q488 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*

Category: Week 1 - ML Foundations › ML Tasks

Question Number : 134 Question Id : 6406531527031 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

A bank wants to decide whether a loan applicant should be given a loan or not, based on features like income, credit score and past records. Which of the following machine learning techniques is applicable for this scenario?

Options :

6406535113450. ✖ Regression

6406535113451. ✔ Classification

6406535113452. ✖ Clustering

6406535113453. ✖ Dimensionality Reduction

Sub-Section Number :	3
Sub-Section Id :	640653244631
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (B)

MLF-Q507 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Tasks

Question Number : 134 Question Id : 6406531527310 Question Type : MCQ
Correct Marks : 1
Question Label : Multiple Choice Question
A bank wants to decide whether a loan applicant should be given a loan or not, based on features like income, credit score and past records. Which of the following machine learning techniques is applicable for this scenario?
Options :
6406535114208. ✖ Regression
6406535114209. ✔ Classification
6406535114210. ✖ Clustering
6406535114211. ✖ Dimensionality Reduction

Sub-Section Number :	3
Sub-Section Id :	640653244735
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B)**

MLF-Q526 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*
Category: Week 1 - ML Foundations › ML Tasks

Question Number : 134 Question Id : 6406531527589 Question Type : MCQ
Correct Marks : 1
Question Label : Multiple Choice Question
A bank wants to decide whether a loan applicant should be given a loan or not, based on features like income, credit score and past records. Which of the following machine learning techniques is applicable for this scenario?
Options :
6406535114966. ✖ Regression
6406535114967. ✔ Classification
6406535114968. ✖ Clustering
6406535114969. ✖ Dimensionality Reduction

Sub-Section Number :	3
Sub-Section Id :	640653244839
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (B)**

■ Category: Regression Basics (6 Questions)

MLF-Q068 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024*
Category: Week 1 - ML Foundations › Regression Basics

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 141 Question Id : 640653737421 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following is/are supervised learning task(s)?

Options :

6406532467997. ✔ Predicting future sales of a product based on the past sales data.

6406532467998. ✖ Reducing the dimensionality of images while retaining essential features, enabling efficient storage and transmission.

6406532467999. ✔ Recognizing and interpreting handwritten characters or words, often used in digit recognition or reading handwritten text.

6406532468000. ✔ Categorizing text documents into predefined classes, such as spam or non-spam emails.

Sub-Section Number :	6
Sub-Section Id :	640653107635
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A, C, D)

MLF-Q080 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024*

Category: Week 1 - ML Foundations › Regression Basics

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 141 Question Id : 640653737707 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following is/are supervised learning task(s)?

Options :

6406532468821. ✔ Predicting future sales of a product based on the past sales data.

6406532468822. ✖ Reducing the dimensionality of images while retaining essential features, enabling efficient storage and transmission.

6406532468823. ✔ Recognizing and interpreting handwritten characters or words, often used in digit recognition or reading handwritten text.

6406532468824. ✔ Categorizing text documents into predefined classes, such as spam or non-spam emails.

Sub-Section Number :	6
Sub-Section Id :	640653107748
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A, C, D)

MLF-Q092 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QDD4 25 Feb 2024*

Category: Week 1 - ML Foundations › Regression Basics

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 141 Question Id : 640653737993 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following is/are supervised learning task(s)?

Options :

6406532469645. ✔ Predicting future sales of a product based on the past sales data.

6406532469646. ✖ Reducing the dimensionality of images while retaining essential features, enabling efficient storage and transmission.

6406532469647. ✔ Recognizing and interpreting handwritten characters or words, often used in digit recognition or reading handwritten text.

6406532469648. ✔ Categorizing text documents into predefined classes, such as spam or non-spam emails.

Sub-Section Number :	6
Sub-Section Id :	640653107861
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A, C, D)

MLF-Q545 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct*

Category: Week 1 - ML Foundations › Regression Basics

Question Number : 119 Question Id : 640653668559 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following is/are supervised learning task(s)?

Options :

6406532239873. ✔ Predicting future weather conditions based on the past weather observations.

6406532239874. ✖ Reducing the dimensionality of images while retaining essential features, enabling efficient storage and transmission.

6406532239875. ✔ Identifying objects in images with multiple categories, such as cats, dogs, cars, and trees.

6406532239876. ✔ Categorizing news articles into topics like politics, sports, entertainment, and technology.

Sub-Section Number :	3
Sub-Section Id :	64065395172
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A, C, D)

MLF-Q561 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QPD3 29 Oct*
Category: Week 1 - ML Foundations › Regression Basics

Question Number : 119 Question Id : 640653668920 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following is/are supervised learning task(s)?

Options :

6406532240969. ✔ Predicting future weather conditions based on the past weather observations.

6406532240970. ✖ Reducing the dimensionality of images while retaining essential features, enabling efficient storage and transmission.

6406532240971. ✔ Identifying objects in images with multiple categories, such as cats, dogs, cars, and trees.

6406532240972. ✔ Categorizing news articles into topics like politics, sports, entertainment, and technology.

Sub-Section Number :	3
Sub-Section Id :	64065395277
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key: Option (A, C, D)**

MLF-Q577 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QPD4 29 Oct*

Category: Week 1 - ML Foundations › Regression Basics

Question Number : 119 Question Id : 640653669181 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following is/are supervised learning task(s)?

Options :

6406532241721. ✔ Predicting future weather conditions based on the past weather observations.

6406532241722. ✖ Reducing the dimensionality of images while retaining essential features, enabling efficient storage and transmission.

6406532241723. ✔ Identifying objects in images with multiple categories, such as cats, dogs, cars, and trees.

6406532241724. ✔ Categorizing news articles into topics like politics, sports, entertainment, and technology.

Sub-Section Number :	3
Sub-Section Id :	64065395376
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key: Option (A, C, D)**

■ WEEK 2 - MATHEMATICAL FOUNDATIONS

■ Category: Derivatives (9 Questions)

MLF-Q048 | [MSQ] | Marks: +2 | *Quiz 1 Jan23*

Category: Week 2 - Mathematical Foundations › Derivatives

Question Number : 123 Question Id : 640653470051 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction

Time : 0

Correct Marks : 3 Selectable Option : 0

Question Label : Multiple Select Question

Which of the following statements is/are true about function $f(x) = \begin{cases} \frac{|x|}{x^2}, x \neq 0 \\ 0, x = 0 \end{cases}$?

Options :

6406531562050. ✔ The function is differentiable for all the values of x except when $x = 0$.

6406531562051. ✖ The function is differentiable for all the values of x.

6406531562052. ✖ The function is continuous at $x = 0$.

6406531562053. ✔ The function is not continuous at $x = 0$.

Sub-Section Number :	3
Sub-Section Id :	64065367687
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A, D)

MLF-Q144 | [MCQ] | Marks: +2 | *MLF*

Category: Week 2 - Mathematical Foundations › Derivatives

Question Number : 3 Question Id : 6406531951385
Correct Marks : 2
Question Label : Multiple Choice Question

Consider the function $f : \mathbb{R} \to \mathbb{R}$:

$$f(x) = \begin{cases} x^2 + 2, & x \leqslant 1 \\ x + 2, & x > 1 \end{cases}$$

Which of the following is true?

Options :

6406536344513. ✔ f is continuous at $x = 1$, but not differentiable at $x = 1$.

6406536344514. ✖ f is both continuous and differentiable at $x = 1$.

6406536344515. ✖ f is neither continuous nor differentiable at $x = 1$.

6406536344516. ✖ f is differentiable at $x = 1$, but not continuous at $x = 1$.

Sub-Section Number :	3
Sub-Section Id :	640653360282
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A)**

MLF-Q154 | [MCQ] | Marks: +2 | *MLF*
Category: Week 2 - Mathematical Foundations › Derivatives

Question Number : 13 Question Id : 6406531951392

Correct Marks : 3

Question Label : Short Answer Question

Find the directional derivative of the function $f : \mathbb{R}^3 \to \mathbb{R}$ given by $f(x, y, z) = x^2y - yz^3 + z$ at the point

$(1, -2, 0)$ in the direction of the vector $\frac{1}{3}\begin{bmatrix} -2 \\ -1 \\ 2 \end{bmatrix}$. Your answer should be an integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

3

Sub-Section Number :	8
Sub-Section Id :	640653360287
Question Shuffling Allowed :	No

Question Id : 6406531951393 Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (14 to 15)

MLF-Q228 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024*

Category: Week 2 - Mathematical Foundations › Derivatives

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 58 Question Id : 640653825046 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following statements is true for a function f(x, y) that is differentiable at a point (a, b)?

Options :

6406532774819. ✔ The directional derivative exists in every direction.

6406532774820. ✖ The directional derivative is the same in every direction.

6406532774821. ✖ The directional derivative can only be calculated along the axes.

6406532774822. ✖ The directional derivative does not exist.

✔ **Official Answer Key: Option (A)**

MLF-Q241 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 7 July 2024*

Category: Week 2 - Mathematical Foundations › Derivatives

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 58 Question Id : 640653827515 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following statements is true for a function f(x, y) that is differentiable at a point (a, b)?

Options :

6406532781343. ✔ The directional derivative exists in every direction.

6406532781344. ✖ The directional derivative is the same in every direction.

6406532781345. ✖ The directional derivative can only be calculated along the axes.

6406532781346. ✖ The directional derivative does not exist.

✔ **Official Answer Key: Option (A)**

MLF-Q254 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 7 July 2024(1) (+1 more papers)*

Category: Week 2 - Mathematical Foundations › Derivatives

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 58 Question Id : 640653827803 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following statements is true for a function f(x, y) that is differentiable at a point (a, b)?

Options :

6406532782139. ✔ The directional derivative exists in every direction.

6406532782140. ✖ The directional derivative is the same in every direction.

6406532782141. ✖ The directional derivative can only be calculated along the axes.

6406532782142. ✖ The directional derivative does not exist.

✔ Official Answer Key: Option (A)

MLF-Q553 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct*

Category: Week 2 - Mathematical Foundations › Derivatives

Question Number : 127 Question Id : 640653668562 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Let $f(x) = \begin{cases} e^{-1/x^2}, & x > 0 \\ 0, & x \leq 0 \end{cases}$. Select which among the following are true?

Options :

6406532239882. ✖ f is discontinuous at $x = 0$.

6406532239883. ✔ f is differentiable.

6406532239884. ✖ f is not differentiable at $x = 0$.

6406532239885. ✔ $f'(0) = 0$

Sub-Section Number :	6
Sub-Section Id :	64065395175
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key: Option (B, D)**

MLF-Q569 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QPD3 29 Oct*

Category: Week 2 - Mathematical Foundations › Derivatives

Question Number : 127 Question Id : 640653668923 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Let $f(x) = \begin{cases} e^{-1/x^2}, & x > 0 \\ 0, & x \leq 0 \end{cases}$. Select which among the following are true?

Options :

6406532240978. ✖ f is discontinuous at $x = 0$.

6406532240979. ✔ f is differentiable.

6406532240980. ✖ f is not differentiable at $x = 0$.

6406532240981. ✔ $f'(0) = 0$

Sub-Section Number :	6
Sub-Section Id :	64065395280
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key: Option (B, D)**

MLF-Q585 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QPD4 29 Oct*

Category: Week 2 - Mathematical Foundations › Derivatives

Question Number : 127 Question Id : 640653669184 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Let $f(x) = \begin{cases} e^{-1/x^2}, & x > 0 \\ 0, & x \leq 0 \end{cases}$. Select which among the following are true?

Options :

6406532241730. ✖ f is discontinuous at $x = 0$.

6406532241731. ✔ f is differentiable.

6406532241732. ✖ f is not differentiable at $x = 0$.

6406532241733. ✔ $f'(0) = 0$

Sub-Section Number :	6
Sub-Section Id :	64065395379
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key: Option (B, D)**

■ Category: Functions (10 Questions)

MLF-Q006 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG1*

Category: Week 2 - Mathematical Foundations › Functions

Question Label : Comprehension

What is the linear approximation of $f(x_1, x_2) = 2x_1^2 + 2x_2^2$ around (1,1)?
If the approximation is $ax_1 + bx_2 + c$.

Based on the above data, answer the subquestions.

Sub questions

Question Number : 60 Question Type : SA

Correct Marks : 3

Question Label : Short Answer Question

Find the element $P_{1,2}$ of the projection matrix P that project on to the line through vector $v = \begin{bmatrix} 2 \\ 1 \\ -6 \end{bmatrix}$. (Assume matrix entries are indexed starting from (1,1)).

NOTE: Enter your answer in three decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

0.045 to 0.052

Question Type : COMPREHENSION

Question Numbers : (61 to 63)

Question Label : Comprehension

What is the linear approximation of $f(x_1, x_2) = 2x_1^2 + 2x_2^2$ around (1,1)?
If the approximation is $ax_1 + bx_2 + c$.

Based on the above data, answer the subquestions.

Sub questions

✔ Official Answer Key (NAT): -2

MLF-Q007 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ EXAM QPG1*

Category: Week 2 - Mathematical Foundations › Functions

Question Label : Comprehension

What is the linear approximation of $f(x_1, x_2) = 2x_1^2 + 2x_2^2$ around (1,1)?
If the approximation is $ax_1 + bx_2 + c$.

Based on the above data, answer the subquestions.

Sub questions

Question Number : 63 Question Type : SA

Correct Marks : 1

Question Label : Short Answer Question

Enter the value of c:

NOTE: Enter your answer to the nearest integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

-4

Question Type : COMPREHENSION

Question Numbers : (64 to 67)

✔ **Official Answer Key (NAT): 2.0 to 2.5**

MLF-Q046 | [MSQ] | Marks: +2 | *Quiz 1 Jan23*

Category: Week 2 - Mathematical Foundations › Functions

Question Number : 121 Question Id : 640653470047 Question Type : MSQ Is Question

Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3 Selectable Option : 0

Question Label : Multiple Select Question

Which of the following is/are true about loss after applying Encoder and Decoder functions to a given data?

Options :

6406531562040. ✖ The loss value can be equal to any real number.

6406531562041. ✔ The loss value cannot be negative.

6406531562042. ✖ A perfect encoder and decoder function will not exist for any data, that is loss value can never be equal to zero.

6406531562043. ✔ For some data, there can exist a perfect encoder and decoder functions, that is loss value can be equal to zero.

✔ **Official Answer Key: Option (B, D)**

MLF-Q109 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*
Category: Week 2 - Mathematical Foundations › Functions

Question Label : Comprehension

Consider a 3×3 matrix B given by

$$B = (-1)\begin{bmatrix} -4 & 3 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & -1 \end{bmatrix}.$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 95 Question Id : 6406531113789 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

If the threshold in the step function is changed to $z \geq 1.0$, what will happen to the misclassification rate?

Options :

6406533774244. ✔ Increase

6406533774245. ✖ Decrease

6406533774246. ✖ Remain same

6406533774247. ✖ Cannot be determined

Question Id : 6406531113790 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (96 to 97)

Question Label : Comprehension

Consider a 3×3 matrix B given by

$$B = (-1)\begin{bmatrix} -4 & 3 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & -1 \end{bmatrix}.$$

Based on the above data, answer the given subquestions.

Sub questions

✔ **Official Answer Key: Option (A)**

MLF-Q124 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*

Category: Week 2 - Mathematical Foundations › Functions

Question Label : Comprehension

Consider a 3×3 matrix B given by

$$B = (-1)\begin{bmatrix} -4 & 3 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & -1 \end{bmatrix}.$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 95 Question Id : 6406531114416 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

If the threshold in the step function is changed to $z \geq 1.0$, what will happen to the misclassification rate?

Options :

6406533776057. ✔ Increase

6406533776058. ✖ Decrease

6406533776059. ✖ Remain same

6406533776060. ✖ Cannot be determined

Question Id : 6406531114417 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (96 to 97)

Question Label : Comprehension

Consider a 3×3 matrix B given by

$$B = (-1)\begin{bmatrix} -4 & 3 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & -1 \end{bmatrix}.$$

Based on the above data, answer the given subquestions.

Sub questions

✔ **Official Answer Key: Option (A)**

MLF-Q139 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*

Category: Week 2 - Mathematical Foundations › Functions

Question Label : Comprehension

Consider a 3×3 matrix B given by

$$B = (-1)\begin{bmatrix} -4 & 3 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & -1 \end{bmatrix}.$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 95 Question Id : 6406531114129 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

If the threshold in the step function is changed to $z \geq 1.0$, what will happen to the misclassification rate?

Options :

6406533775228. ✔ Increase

6406533775229. ✖ Decrease

6406533775230. ✖ Remain same

6406533775231. ✖ Cannot be determined

Question Id : 6406531114130 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (96 to 97)

Question Label : Comprehension

Consider a 3×3 matrix B given by

$$B = (-1)\begin{bmatrix} -4 & 3 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & -1 \end{bmatrix}.$$

Based on the above data, answer the given subquestions.

Sub questions

✔ **Official Answer Key: Option (A)**

MLF-Q153 | [MCQ] | Marks: +2 | *MLF*

Category: Week 2 - Mathematical Foundations › Functions

Question Number : 12 Question Id : 6406531951391

Correct Marks : 2

Question Label : Short Answer Question

Consider the function $f : \mathbb{R} \backslash \{\frac{-1}{2}\} \to \mathbb{R}$ given by $f(x) = \frac{1}{(1+2x)^5}$. If the best linear approximation to f at $x = 0$ is given as $h(x) = a + bx$, find $a - b$. Your answer should be an integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

11

Sub-Section Number :	7
Sub-Section Id :	640653360286
Question Shuffling Allowed :	Yes

MLF-Q229 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024*

Category: Week 2 - Mathematical Foundations › Functions

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 59 Question Id : 640653825047 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Which of the following functions cannot be approximated linearly around $x = 0$?
Options :

6406532774823. ✖ $f(x) = \sin(x)$

6406532774824. ✖ $f(x) = \cos(x)$

6406532774825. ✔ $f(x) = \ln(x)$

6406532774826. ✖ $f(x) = x^2$

Sub-Section Number :	6
Sub-Section Id :	640653122730
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (C)**

MLF-Q242 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 7 July 2024*
Category: Week 2 - Mathematical Foundations › Functions

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 59 Question Id : 640653827516 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Which of the following functions cannot be approximated linearly around $x = 0$?
Options :

6406532781347. ✖ $f(x) = \sin(x)$

6406532781348. ✖ $f(x) = \cos(x)$

6406532781349. ✔ $f(x) = \ln(x)$

6406532781350. ✖ $f(x) = x^2$

Sub-Section Number :	6
Sub-Section Id :	640653123481
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (C)**

MLF-Q255 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 7 July 2024(1) (+1 more papers)*
Category: Week 2 - Mathematical Foundations › Functions

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 59 Question Id : 640653827804 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Which of the following functions cannot be approximated linearly around $x = 0$?
Options :

6406532782143. ✖ $f(x) = \sin(x)$

6406532782144. ✖ $f(x) = \cos(x)$

6406532782145. ✔ $f(x) = \ln(x)$

6406532782146. ✖ $f(x) = x^2$

Sub-Section Number :	6
Sub-Section Id :	640653123588
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (C)**

■ WEEK 3 - LINEAR ALGEBRA & SUBSPACES

■ Category: Dot Product & Norms (19 Questions)

MLF-Q143 | [MCQ] | Marks: +2 | *MLF*

Category: Week 3 - Linear Algebra & Subspaces › Dot Product & Norms

Question Number : 2 Question Id : 6406531951381

Correct Marks : 2

Question Label : Multiple Choice Question

Let the vectors $u = (1, a, 2), v = (2, -1, b)$ be orthogonal. Which of the following must be true?

Options :

6406536344498. ✔ $a - 2b = 2$

6406536344499. ✖ $a + 2 = 2b$

6406536344500. ✖ $2b - a = 2$

6406536344501. ✖ $a - 2b = 0$

✔ **Official Answer Key: Option (A)**

MLF-Q184 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 13 July 2025*

Category: Week 3 - Linear Algebra & Subspaces › Dot Product & Norms

Question Label : Comprehension

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

Sub questions

Question Number : 89 Question Id : 6406531297915 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following options is/are true?

Options :

6406534364706. ✖ $A^* = A$, where A^* is conjugate transpose of matrix A.

6406534364707. ✔ A is a symmetric matrix.

6406534364708. ✖ A is a unitary matrix.

6406534364709. ✔ Columns of A are not orthogonal.

✔ **Official Answer Key: Option (B, D)**

MLF-Q199 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 13 July 2025*

Category: Week 3 - Linear Algebra & Subspaces › Dot Product & Norms

Question Label : Comprehension

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

Sub questions

Question Number : 89 Question Id : 6406531298185 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following options is/are true?

Options :

6406534365477. ✖ $A^* = A$, where A^* is conjugate transpose of matrix A.

6406534365478. ✔ A is a symmetric matrix.

6406534365479. ✖ A is a unitary matrix.

6406534365480. ✔ Columns of A are not orthogonal.

✔ **Official Answer Key: Option (B, D)**

MLF-Q214 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025*

Category: Week 3 - Linear Algebra & Subspaces › Dot Product & Norms

Question Label : Comprehension

Consider a matrix $A = \begin{bmatrix} 1 & i & -i \\ i & 1 & 0 \\ -i & 0 & 1 \end{bmatrix}$. Based on this information, answer the given subquestions.

Sub questions

Question Number : 89 Question Id : 6406531298455 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following options is/are true?
Options :
6406534366248. ✖ $A^* = A$, where A^* is conjugate transpose of matrix A.
6406534366249. ✔ A is a symmetric matrix.
6406534366250. ✖ A is a unitary matrix.
6406534366251. ✔ Columns of A are not orthogonal.

✔ **Official Answer Key: Option (B, D)**

MLF-Q270 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*

Category: Week 3 - Linear Algebra & Subspaces › Dot Product & Norms

Question Id : 6406531527859 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Number : 121 Question Id : 6406531527857 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following is/are true?
Options :
6406535115713. ✔ A is an orthogonal matrix
6406535115714. ✔ A is invertible
6406535115715. ✖ A is symmetric

✔ **Official Answer Key: Option (A, B)**

MLF-Q289 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*

Category: Week 3 - Linear Algebra & Subspaces › Dot Product & Norms

Question Id : 6406531528162 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Number : 121 Question Id : 6406531528160 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following is/are true?
Options :
6406535116556. ✔ *A* is an orthogonal matrix
6406535116557. ✔ *A* is invertible
6406535116558. ✖ *A* is symmetric

✔ **Official Answer Key: Option (A, B)**

MLF-Q308 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*

Category: Week 3 - Linear Algebra & Subspaces › Dot Product & Norms

Question Id : 6406531528824 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Number : 121 Question Id : 6406531528822 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following is/are true?
Options :
6406535118487. ✔ *A* is an orthogonal matrix
6406535118488. ✔ *A* is invertible
6406535118489. ✖ *A* is symmetric

✔ **Official Answer Key: Option (A, B)**

MLF-Q327 | [MSQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*

Category: Week 3 - Linear Algebra & Subspaces › Dot Product & Norms

Question Label : Comprehension

There is no common data for this question. There are three independent sub-questions for which you have to answer True/False.

Question Number : 147 Question Id : 6406531525168 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following is/are true?

Options :

6406535108188. ✔ A is an orthogonal matrix

6406535108189. ✔ A is invertible

6406535108190. ✖ A is symmetric

✔ Official Answer Key: Option (A, B)

MLF-Q346 | [MSQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*

Category: Week 3 - Linear Algebra & Subspaces › Dot Product & Norms

Question Label : Comprehension

There is no common data for this question. There are three independent sub-questions for which you have to answer True/False.

Question Number : 147 Question Id : 6406531525795 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following is/are true?

Options :

6406535109965. ✔ A is an orthogonal matrix

6406535109966. ✔ A is invertible

6406535109967. ✖ A is symmetric

✔ Official Answer Key: Option (A, B)

MLF-Q365 | [MSQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*

Category: Week 3 - Linear Algebra & Subspaces › Dot Product & Norms

Question Label : Comprehension

There is no common data for this question. There are three independent sub-questions for which you have to answer True/False.

Question Number : 147 Question Id : 6406531526724 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following is/are true?

Options :

6406535112610. ✔ *A* is an orthogonal matrix

6406535112611. ✔ *A* is invertible

6406535112612. ✖ *A* is symmetric

✔ **Official Answer Key: Option (A, B)**

MLF-Q384 | [MSQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA2 26 Oct 2025*

Category: Week 3 - Linear Algebra & Subspaces › Dot Product & Norms

Question Label : Comprehension

Consider the matrix A given below:

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

where $0 < \theta < 180°$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 146 Question Id : 6406531534293 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following is/are true?

Options :

6406535133798. ✔ *A* is an orthogonal matrix

6406535133799. ✔ *A* is invertible

6406535133800. ✖ *A* is symmetric

✔ **Official Answer Key: Option (A, B)**

MLF-Q403 | [MSQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA3 26 Oct 2025*

Category: Week 3 - Linear Algebra & Subspaces › Dot Product & Norms

Question Label : Comprehension

Consider the matrix A given below:

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

where $0 < \theta < 180°$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 146 Question Id : 6406531534632 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following is/are true?

Options :

6406535134768. ✔ *A* is an orthogonal matrix

6406535134769. ✔ *A* is invertible

6406535134770. ✖ *A* is symmetric

✔ Official Answer Key: Option (A, B)

MLF-Q422 | [MSQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA4 26 Oct 2025*

Category: Week 3 - Linear Algebra & Subspaces › Dot Product & Norms

Question Label : Comprehension

Consider the matrix A given below:

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

where $0 < \theta < 180°$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 146 Question Id : 6406531535069 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following is/are true?
Options :
6406535135914. ✔ *A* is an orthogonal matrix
6406535135915. ✔ *A* is invertible
6406535135916. ✖ *A* is symmetric

✔ **Official Answer Key: Option (A, B)**

MLF-Q441 | [MSQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*
Category: Week 3 - Linear Algebra & Subspaces › Dot Product & Norms

Question Label : Comprehension
There is no common data for this question. There are three independent sub-questions for which you have to answer True/False.

Question Number : 140 Question Id : 6406531535805 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following is/are true?
Options :
6406535138094. ✔ *A* is an orthogonal matrix
6406535138095. ✔ *A* is invertible
6406535138096. ✖ *A* is symmetric

✔ **Official Answer Key: Option (A, B)**

MLF-Q460 | [MSQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*

Category: Week 3 - Linear Algebra & Subspaces › Dot Product & Norms

Question Label : Comprehension

There is no common data for this question. There are three independent sub-questions for which you have to answer True/False.

Question Number : 140 Question Id : 6406531536145 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following is/are true?

Options :

6406535139194. ✔ A is an orthogonal matrix

6406535139195. ✔ A is invertible

6406535139196. ✖ A is symmetric

✔ **Official Answer Key: Option (A, B)**

MLF-Q479 | [MSQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*

Category: Week 3 - Linear Algebra & Subspaces › Dot Product & Norms

Question Label : Comprehension

There is no common data for this question. There are three independent sub-questions for which you have to answer True/False.

Question Number : 140 Question Id : 6406531536435 Question Type : MSQ

Correct Marks : 2 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following is/are true?

Options :

6406535140025. ✔ A is an orthogonal matrix

6406535140026. ✔ A is invertible

6406535140027. ✖ A is symmetric

✔ **Official Answer Key: Option (A, B)**

MLF-Q498 | [MSQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*
Category: Week 3 - Linear Algebra & Subspaces › Dot Product & Norms

Question Id : 6406531527050 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Number : 144 Question Id : 6406531527048 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following is/are true?
Options :
6406535113487. ✔ A is an orthogonal matrix
6406535113488. ✔ A is invertible
6406535113489. ✖ A is symmetric

✔ **Official Answer Key: Option (A, B)**

MLF-Q517 | [MSQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*
Category: Week 3 - Linear Algebra & Subspaces › Dot Product & Norms

Question Id : 6406531527329 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Number : 144 Question Id : 6406531527327 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following is/are true?
Options :
6406535114245. ✔ A is an orthogonal matrix
6406535114246. ✔ A is invertible
6406535114247. ✖ A is symmetric

✔ **Official Answer Key: Option (A, B)**

MLF-Q536 | [MSQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*
Category: Week 3 - Linear Algebra & Subspaces › Dot Product & Norms

Question Id : 6406531527608 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Number : 144 Question Id : 6406531527606 Question Type : MSQ
Correct Marks : 2 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following is/are true?
Options :
6406535115003. ✔ *A* is an orthogonal matrix
6406535115004. ✔ *A* is invertible
6406535115005. ✖ *A* is symmetric

✔ Official Answer Key: Option (A, B)

■ Category: Matrices (1 Questions)

MLF-Q149 | [MCQ] | Marks: +2 | *MLF*
Category: Week 3 - Linear Algebra & Subspaces › Matrices

Question Number : 8 Question Id : 6406531951387
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following is/are true?
Options :
6406536344521. ✖ If A and B are matrices of the same size, then $\text{rank}(A+B) = \text{rank}(A) + \text{rank}(B)$.

6406536344522. ✖ The rank of a matrix is equal to the number of non-zero rows in it.

6406536344523. ✔ If A is a matrix which is obtained by permuting the rows of the 5×5 identity matrix, then the rank of A is 5.

6406536344524. ✔ If u is a non-zero column vector in $\mathbb{R}^{10}$, then the rank of the matrix uu^T is 1.

✔ Official Answer Key: Option (C, D)

■ Category: Span & Basis (1 Questions)

MLF-Q170 | [SA] | Marks: +2 | *Quiz1 May23*

Category: Week 3 - Linear Algebra & Subspaces › Span & Basis

Question Number : 130 Question Id : 640653577883 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

Let P_1 be the projection matrix onto the line spanned by the vector $a = [1, 2, 3]^T$. Let P_2 be the projection matrix that projects vectors onto the plane perpendicular to span of a. Find the determinant of $P_1 + P_2$.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

1

Sub-Section Number :	6
Sub-Section Id :	64065382599
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key (NAT): 1**

■ Category: Subspaces (70 Questions)

MLF-Q062 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 135 Question Id : 640653737413 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Find the row space of matrix *A*.

Options :

6406532467968. ✖ $\text{span}\left\{\begin{pmatrix}1\\-1\\2\end{pmatrix}, \begin{pmatrix}2\\-2\\5\end{pmatrix}\right\}$

6406532467969. ✔ $\text{span}\left\{\begin{pmatrix}1\\0\\3\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}\right\}$

6406532467970. ✖ $\text{span}\left\{\begin{pmatrix}1\\2\\3\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}\right\}$

6406532467971. ✖ $\text{span}\left\{\begin{pmatrix}1\\2\\3\end{pmatrix}, \begin{pmatrix}-1\\-2\\-3\end{pmatrix}, \begin{pmatrix}2\\5\\6\end{pmatrix}\right\}$

✔ **Official Answer Key: Option (B)**

MLF-Q063 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 136 Question Id : 640653737414 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Find the column space of matrix *A*.

Options :

6406532467972. ✔ $\text{span}\left\{\begin{pmatrix}1\\-1\\2\end{pmatrix}, \begin{pmatrix}2\\-2\\5\end{pmatrix}\right\}$

6406532467973. ✖ $\text{span}\left\{\begin{pmatrix}1\\0\\3\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}\right\}$

6406532467974. ✖ $\text{span}\left\{\begin{pmatrix}1\\-1\\2\end{pmatrix}, \begin{pmatrix}2\\-2\\5\end{pmatrix}, \begin{pmatrix}3\\-3\\6\end{pmatrix}\right\}$

6406532467975. ✖ $\text{span}\left\{\begin{pmatrix}1\\2\\3\end{pmatrix}, \begin{pmatrix}-1\\-2\\-3\end{pmatrix}, \begin{pmatrix}2\\5\\6\end{pmatrix}\right\}$

Sub-Section Number :	3
Sub-Section Id :	640653107632
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A)

MLF-Q074 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 135 Question Id : 640653737699 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Find the row space of matrix *A*.

Options :

6406532468792. ✖ $\text{span}\left\{\begin{pmatrix}1\\-1\\2\end{pmatrix}, \begin{pmatrix}2\\-2\\5\end{pmatrix}\right\}$

6406532468793. ✔ $\text{span}\left\{\begin{pmatrix}1\\0\\3\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}\right\}$

6406532468794. ✖ $\text{span}\left\{\begin{pmatrix}1\\2\\3\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}\right\}$

6406532468795. ✖ $\text{span}\left\{\begin{pmatrix}1\\2\\3\end{pmatrix}, \begin{pmatrix}-1\\-2\\-3\end{pmatrix}, \begin{pmatrix}2\\5\\6\end{pmatrix}\right\}$

✔ **Official Answer Key: Option (B)**

MLF-Q075 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 136 Question Id : 640653737700 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Find the column space of matrix *A*.

Options :

6406532468796. ✔ $\text{span}\left\{\begin{pmatrix}1\\-1\\2\end{pmatrix}, \begin{pmatrix}2\\-2\\5\end{pmatrix}\right\}$

6406532468797. ✖ $\text{span}\left\{\begin{pmatrix}1\\0\\3\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}\right\}$

6406532468798. ✖ $\text{span}\left\{\begin{pmatrix}1\\-1\\2\end{pmatrix}, \begin{pmatrix}2\\-2\\5\end{pmatrix}, \begin{pmatrix}3\\-3\\6\end{pmatrix}\right\}$

6406532468799. ✖ $\text{span}\left\{\begin{pmatrix}1\\2\\3\end{pmatrix}, \begin{pmatrix}-1\\-2\\-3\end{pmatrix}, \begin{pmatrix}2\\5\\6\end{pmatrix}\right\}$

Sub-Section Number :	3
Sub-Section Id :	640653107745
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A)

MLF-Q086 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QDD4 25 Feb 2024*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 135 Question Id : 640653737985 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Find the row space of matrix *A*.

Options :

6406532469616. ✖ $\text{span}\left\{\begin{pmatrix} 1 \\ -1 \\ 2 \end{pmatrix}, \begin{pmatrix} 2 \\ -2 \\ 5 \end{pmatrix}\right\}$

6406532469617. ✔ $\text{span}\left\{\begin{pmatrix} 1 \\ 0 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}\right\}$

6406532469618. ✖ $\text{span}\left\{\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} 0 \\ 1 \\ 0 \end{pmatrix}\right\}$

6406532469619. ✖ $\text{span}\left\{\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}, \begin{pmatrix} -1 \\ -2 \\ -3 \end{pmatrix}, \begin{pmatrix} 2 \\ 5 \\ 6 \end{pmatrix}\right\}$

✔ **Official Answer Key: Option (B)**

MLF-Q087 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QDD4 25 Feb 2024*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 136 Question Id : 640653737986 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Find the column space of matrix *A*.

Options :

6406532469620. ✔ $\text{span}\left\{\begin{pmatrix}1\\-1\\2\end{pmatrix}, \begin{pmatrix}2\\-2\\5\end{pmatrix}\right\}$

6406532469621. ✖ $\text{span}\left\{\begin{pmatrix}1\\0\\3\end{pmatrix}, \begin{pmatrix}0\\1\\0\end{pmatrix}\right\}$

6406532469622. ✖ $\text{span}\left\{\begin{pmatrix}1\\-1\\2\end{pmatrix}, \begin{pmatrix}2\\-2\\5\end{pmatrix}, \begin{pmatrix}3\\-3\\6\end{pmatrix}\right\}$

6406532469623. ✖ $\text{span}\left\{\begin{pmatrix}1\\2\\3\end{pmatrix}, \begin{pmatrix}-1\\-2\\-3\end{pmatrix}, \begin{pmatrix}2\\5\\6\end{pmatrix}\right\}$

Sub-Section Number :	3
Sub-Section Id :	640653107858
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ Official Answer Key: Option (A)

MLF-Q147 | [MCQ] | Marks: +2 | *MLF*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces

Question Number : 6 Question Id : 6406531951384
Correct Marks : 3
Question Label : Multiple Choice Question

Which of the following is a basis for the column space of the matrix A given below?

$$A = \begin{bmatrix} 1 & 1 & 2 \\ 2 & -1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$$

Options :

6406536344509. ✔ $\{(1,2,1),(1,-1,0)\}$

6406536344510. ✖ $\{(1,0,0),(0,1,0)\}$

6406536344511. ✖ $\{(1,2,1),(1,-1,0),(2,1,1)\}$

6406536344512. ✖ $\{(1,3,2),(1,-1,0)\}$

Sub-Section Number :	4
Sub-Section Id :	640653360283
Question Shuffling Allowed :	Yes

✔ **Official Answer Key: Option (A)**

MLF-Q161 | [MSQ] | Marks: +2 | *Quiz1 May23*
Category: Week 3 - Linear Algebra & Subspaces › Subspaces

Question Number : 121 Question Id : 640653577884 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following statements is/are true?

Options :

6406531929745. ✔ Row space of a matrix is orthogonal to the null space of the matrix.

6406531929746. ✖ Row space of a matrix is orthogonal to the null space of transpose of the matrix.

6406531929747. ✔ Dimension of row space of a matrix is always equal to the dimension of column space of the matrix.

6406531929748. ✖ Dimension of row space of a matrix need not be equal to the dimension of column space of the matrix.

✔ Official Answer Key: Option (A, C)

MLF-Q268 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*
Category: Week 3 - Linear Algebra & Subspaces › Subspaces

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 119 Question Id : 6406531527854 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following sets forms a basis for the row space of A?

Options :

6406535115705. ✖ {(1, 0, 2), (0, 1,−1)}

6406535115706. ✔ {(1, 0, 2), (0, 1,−1), (0, 0,−1)}

6406535115707. ✖ {(1, 0, 2), (0,−1, 1), (0,−1,−4)}

6406535115708. ✖ {(0, 1,−1), (0, 0,−4)}

✔ **Official Answer Key: Option (B)**

MLF-Q287 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 119 Question Id : 6406531528157 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following sets forms a basis for the row space of A?

Options :

6406535116548. ✖ {(1, 0, 2), (0, 1,−1)}

6406535116549. ✔ {(1, 0, 2), (0, 1,−1), (0, 0,−1)}

6406535116550. ✖ {(1, 0, 2), (0,−1, 1), (0,−1,−4)}

6406535116551. ✖ {(0, 1,−1), (0, 0,−4)}

✔ **Official Answer Key: Option (B)**

MLF-Q306 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 119 Question Id : 6406531528819 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following sets forms a basis for the row space of A?

Options :

6406535118479. ✖ {(1, 0, 2), (0, 1,−1)}

6406535118480. ✔ {(1, 0, 2), (0, 1,−1), (0, 0,−1)}

6406535118481. ✖ {(1, 0, 2), (0,−1, 1), (0,−1,−4)}

6406535118482. ✖ {(0, 1,−1), (0, 0,−4)}

✔ **Official Answer Key: Option (B)**

MLF-Q325 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 145 Question Id : 6406531525165 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following sets forms a basis for the row space of A?

Options :

6406535108180. ✖ {(1, 0, 2), (0, 1,−1)}

6406535108181. ✔ {(1, 0, 2), (0, 1,−1), (0, 0,−1)}

6406535108182. ✖ {(1, 0, 2), (0,−1, 1), (0,−1,−4)}

6406535108183. ✖ {(0, 1,−1), (0, 0,−4)}

✔ **Official Answer Key: Option (B)**

MLF-Q344 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 145 Question Id : 6406531525792 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following sets forms a basis for the row space of A?

Options :

6406535109957. ✖ {(1, 0, 2), (0, 1,−1)}

6406535109958. ✔ {(1, 0, 2), (0, 1,−1), (0, 0,−1)}

6406535109959. ✖ {(1, 0, 2), (0,−1, 1), (0,−1,−4)}

6406535109960. ✖ {(0, 1,−1), (0, 0,−4)}

✔ **Official Answer Key: Option (B)**

MLF-Q363 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 145 Question Id : 6406531526721 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following sets forms a basis for the row space of A?

Options :

6406535112602. ✖ {(1, 0, 2), (0, 1,−1)}

6406535112603. ✔ {(1, 0, 2), (0, 1,−1), (0, 0,−1)}

6406535112604. ✖ {(1, 0, 2), (0,−1, 1), (0,−1,−4)}

6406535112605. ✖ {(0, 1,−1), (0, 0,−4)}

✔ **Official Answer Key: Option (B)**

MLF-Q382 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA2 26 Oct 2025*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 144 Question Id : 6406531534290 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Which of the following sets forms a basis for the row space of A?
Options :
6406535133790. ✖ {(1, 0, 2), (0, 1,−1)}
6406535133791. ✔ {(1, 0, 2), (0, 1,−1), (0, 0,−1)}
6406535133792. ✖ {(1, 0, 2), (0,−1, 1), (0,−1,−4)}
6406535133793. ✖ {(0, 1,−1), (0, 0,−4)}

✔ **Official Answer Key: Option (B)**

MLF-Q401 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA3 26 Oct 2025*
Category: Week 3 - Linear Algebra & Subspaces › Subspaces

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 144 Question Id : 6406531534629 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Which of the following sets forms a basis for the row space of A?
Options :
6406535134760. ✖ {(1, 0, 2), (0, 1,−1)}
6406535134761. ✔ {(1, 0, 2), (0, 1,−1), (0, 0,−1)}
6406535134762. ✖ {(1, 0, 2), (0,−1, 1), (0,−1,−4)}
6406535134763. ✖ {(0, 1,−1), (0, 0,−4)}

✔ **Official Answer Key: Option (B)**

MLF-Q420 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA4 26 Oct 2025*
Category: Week 3 - Linear Algebra & Subspaces › Subspaces

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 144 Question Id : 6406531535066 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Which of the following sets forms a basis for the row space of A?
Options :
6406535135906. ✖ {(1, 0, 2), (0, 1,−1)}
6406535135907. ✔ {(1, 0, 2), (0, 1,−1), (0, 0,−1)}
6406535135908. ✖ {(1, 0, 2), (0,−1, 1), (0,−1,−4)}
6406535135909. ✖ {(0, 1,−1), (0, 0,−4)}

✔ **Official Answer Key: Option (B)**

MLF-Q439 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*
Category: Week 3 - Linear Algebra & Subspaces › Subspaces

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 138 Question Id : 6406531535802 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following sets forms a basis for the row space of A?

Options :

6406535138086. ✖ {(1, 0, 2), (0, 1,−1)}

6406535138087. ✔ {(1, 0, 2), (0, 1,−1), (0, 0,−1)}

6406535138088. ✖ {(1, 0, 2), (0,−1, 1), (0,−1,−4)}

6406535138089. ✖ {(0, 1,−1), (0, 0,−4)}

✔ **Official Answer Key: Option (B)**

MLF-Q458 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 138 Question Id : 6406531536142 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following sets forms a basis for the row space of A?

Options :

6406535139186. ✖ {(1, 0, 2), (0, 1,−1)}

6406535139187. ✔ {(1, 0, 2), (0, 1,−1), (0, 0,−1)}

6406535139188. ✖ {(1, 0, 2), (0,−1, 1), (0,−1,−4)}

6406535139189. ✖ {(0, 1,−1), (0, 0,−4)}

✔ **Official Answer Key: Option (B)**

MLF-Q477 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 138 Question Id : 6406531536432 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following sets forms a basis for the row space of A?

Options :

6406535140017. ✖ {(1, 0, 2), (0, 1,−1)}

6406535140018. ✔ {(1, 0, 2), (0, 1,−1), (0, 0,−1)}

6406535140019. ✖ {(1, 0, 2), (0,−1, 1), (0,−1,−4)}

6406535140020. ✖ {(0, 1,−1), (0, 0,−4)}

✔ **Official Answer Key: Option (B)**

MLF-Q496 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 142 Question Id : 6406531527045 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Which of the following sets forms a basis for the row space of A?
Options :
6406535113479. ✖ {(1, 0, 2), (0, 1,−1)}
6406535113480. ✔ {(1, 0, 2), (0, 1,−1), (0, 0,−1)}
6406535113481. ✖ {(1, 0, 2), (0,−1, 1), (0,−1,−4)}
6406535113482. ✖ {(0, 1,−1), (0, 0,−4)}

✔ **Official Answer Key: Option (B)**

MLF-Q515 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*
Category: Week 3 - Linear Algebra & Subspaces › Subspaces

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 142 Question Id : 6406531527324 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following sets forms a basis for the row space of A?

Options :

6406535114237. ✖ {(1, 0, 2), (0, 1,−1)}

6406535114238. ✔ {(1, 0, 2), (0, 1,−1), (0, 0,−1)}

6406535114239. ✖ {(1, 0, 2), (0,−1, 1), (0,−1,−4)}

6406535114240. ✖ {(0, 1,−1), (0, 0,−4)}

✔ **Official Answer Key: Option (B)**

MLF-Q534 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces

Question Label : Comprehension

Consider the following matrix

$$A = \begin{bmatrix} 1 & 0 & 2 \\ 3 & 1 & 5 \\ 2 & -1 & 1 \end{bmatrix}$$

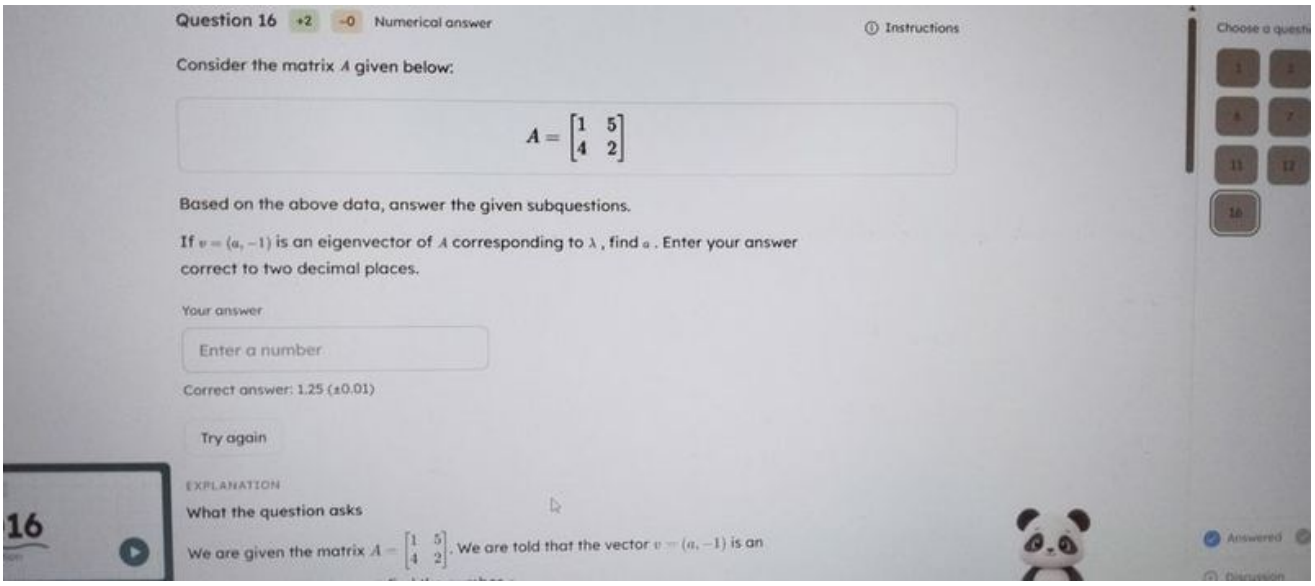
Based on the above data, answer the given subquestions.

Sub questions

Question Number : 142 Question Id : 6406531527603 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Which of the following sets forms a basis for the row space of A?
Options :
6406535114995. ✖ {(1, 0, 2), (0, 1,−1)}
6406535114996. ✔ {(1, 0, 2), (0, 1,−1), (0, 0,−1)}
6406535114997. ✖ {(1, 0, 2), (0,−1, 1), (0,−1,−4)}
6406535114998. ✖ {(0, 1,−1), (0, 0,−4)}

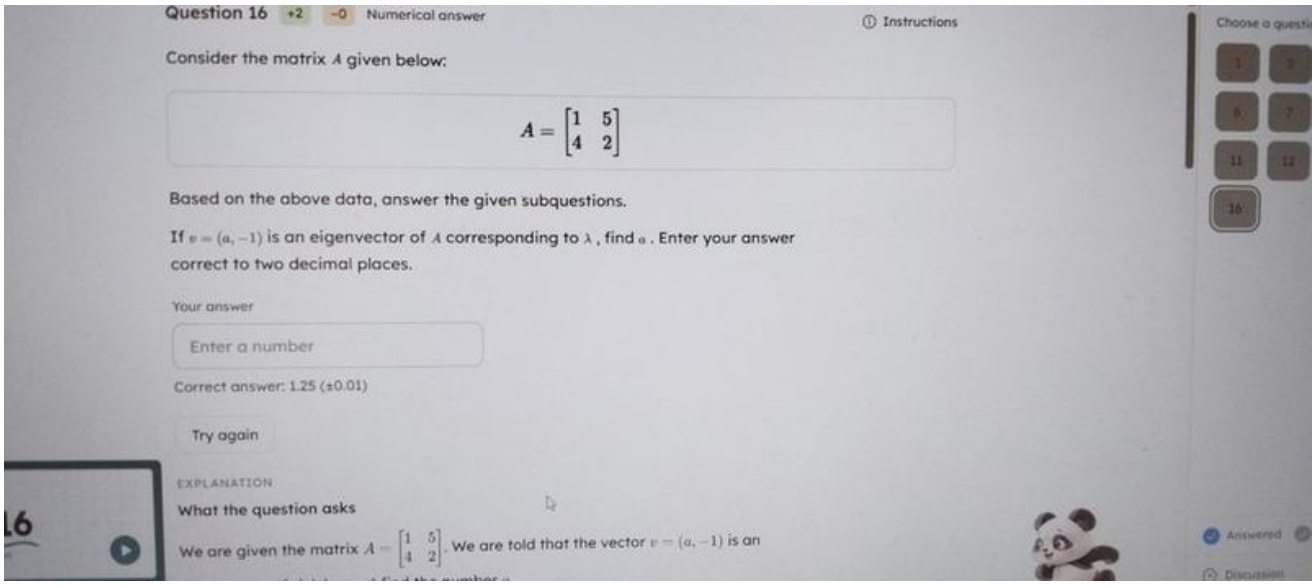
✔ **Official Answer Key: Option (B)**

MLF-Q631 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*
Category: Week 3 - Linear Algebra & Subspaces › Subspaces



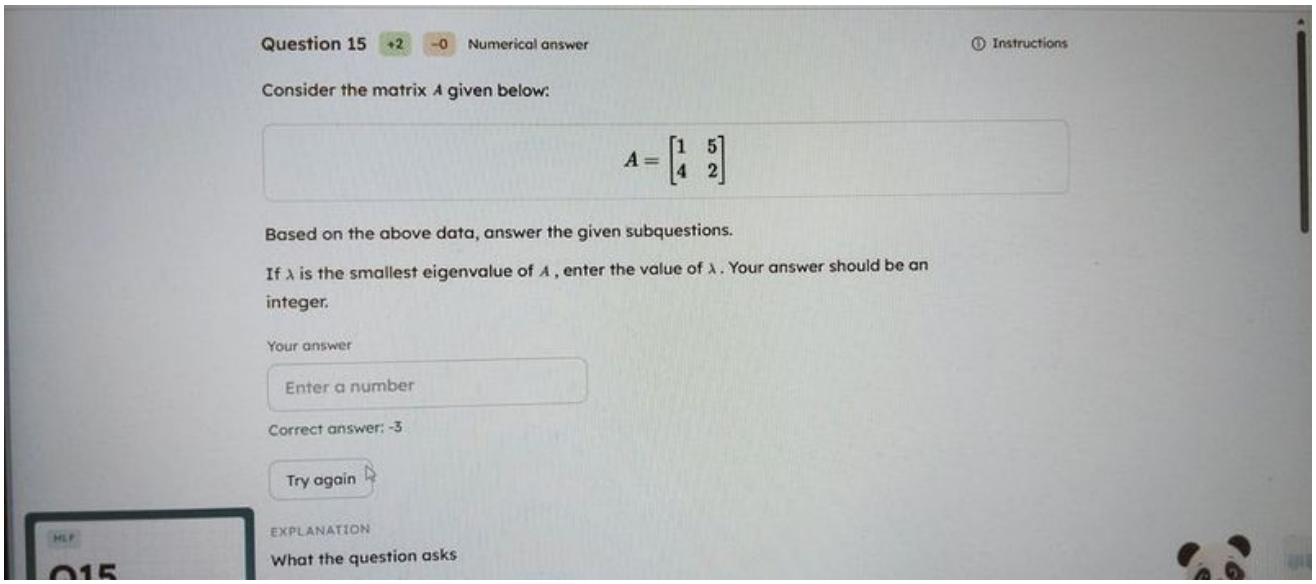
MLF-Q632 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



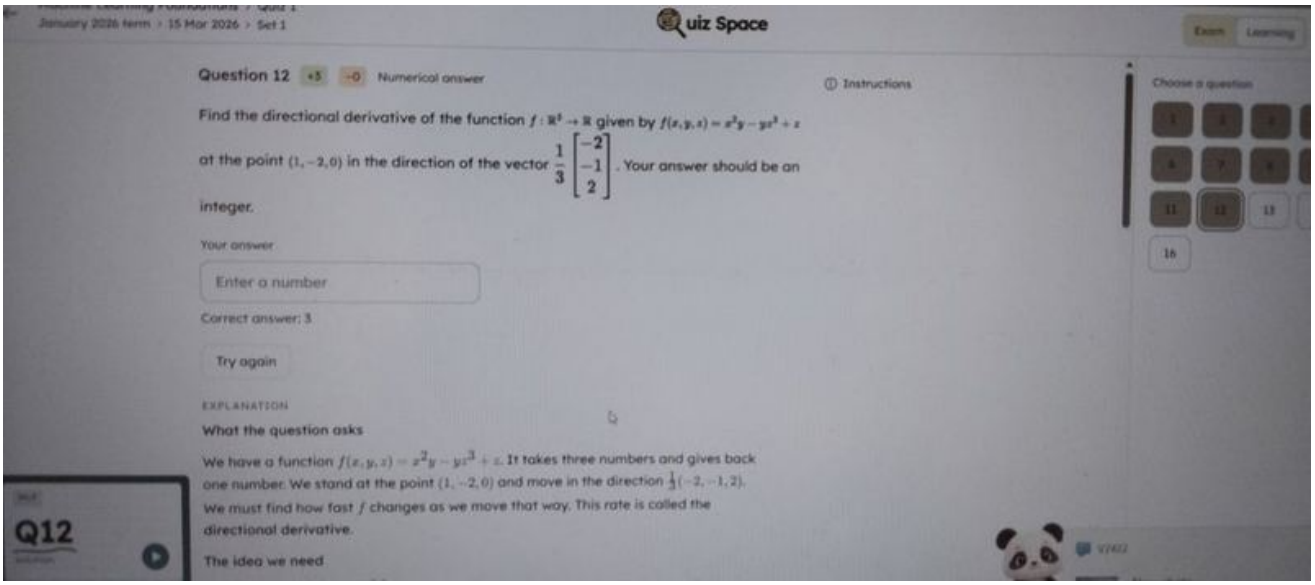
MLF-Q633 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



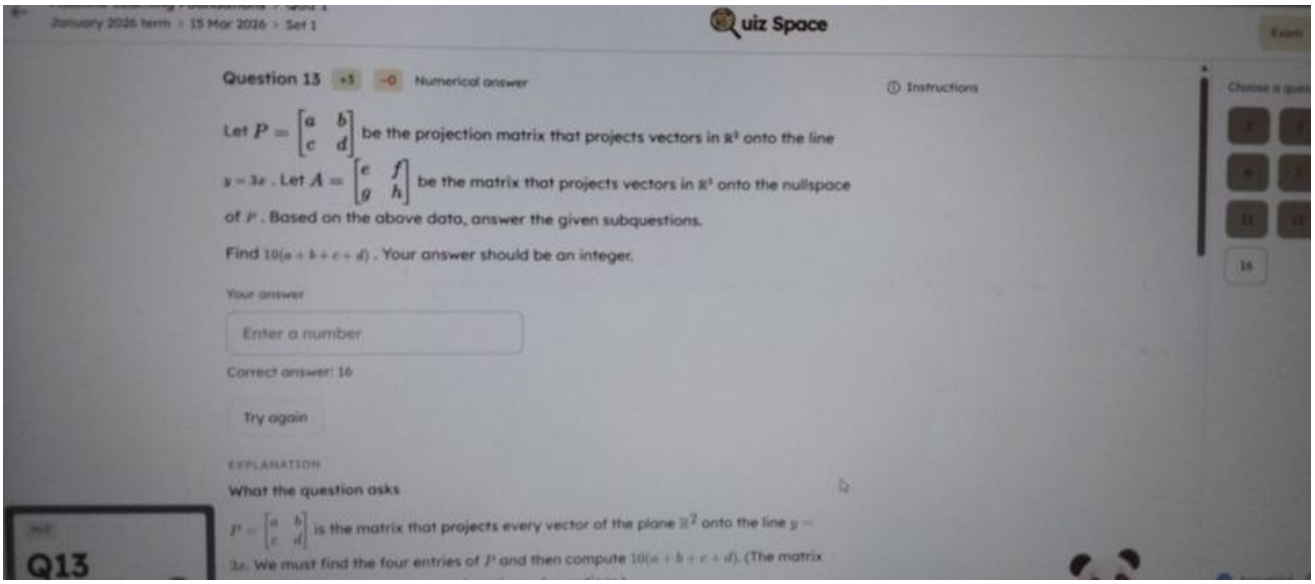
MLF-Q634 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



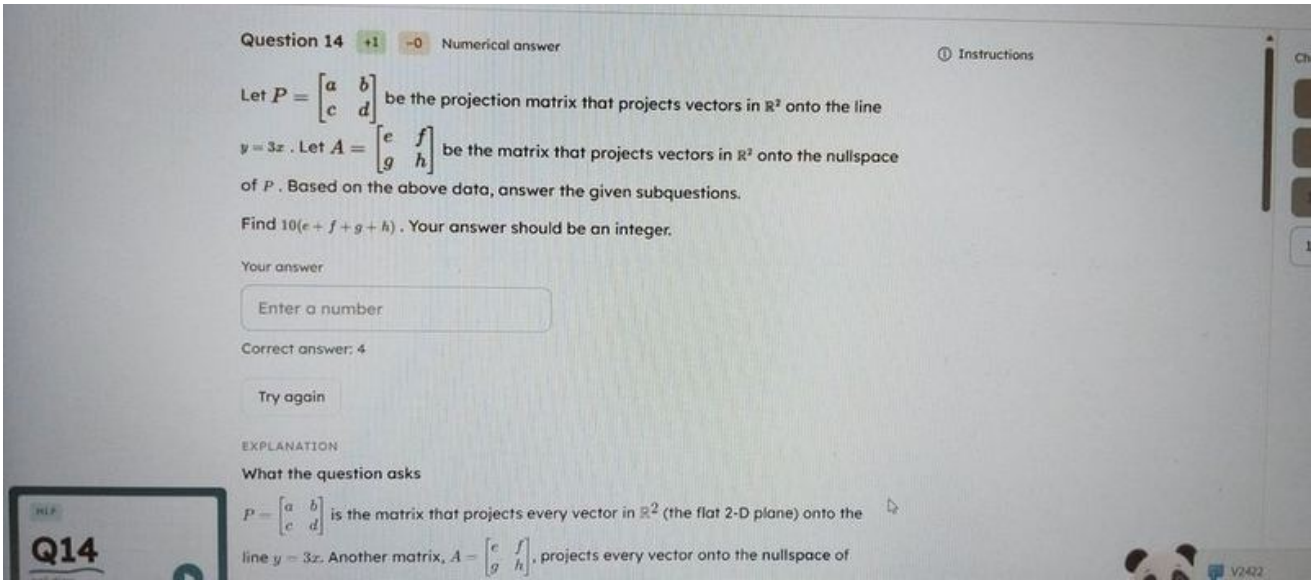
MLF-Q635 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



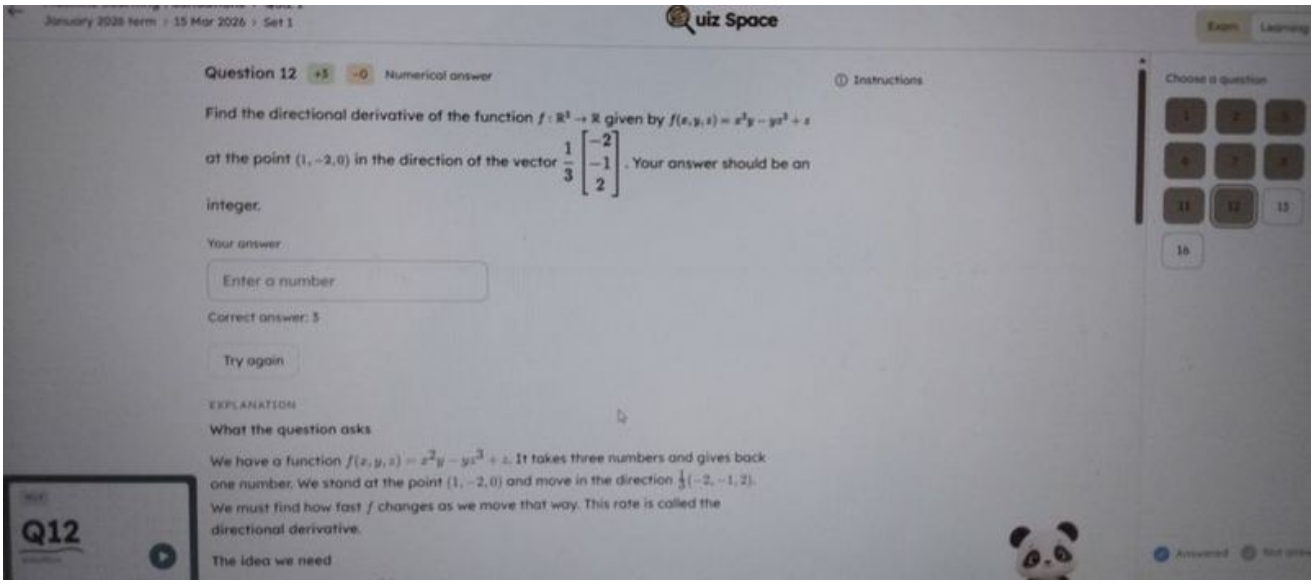
MLF-Q636 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



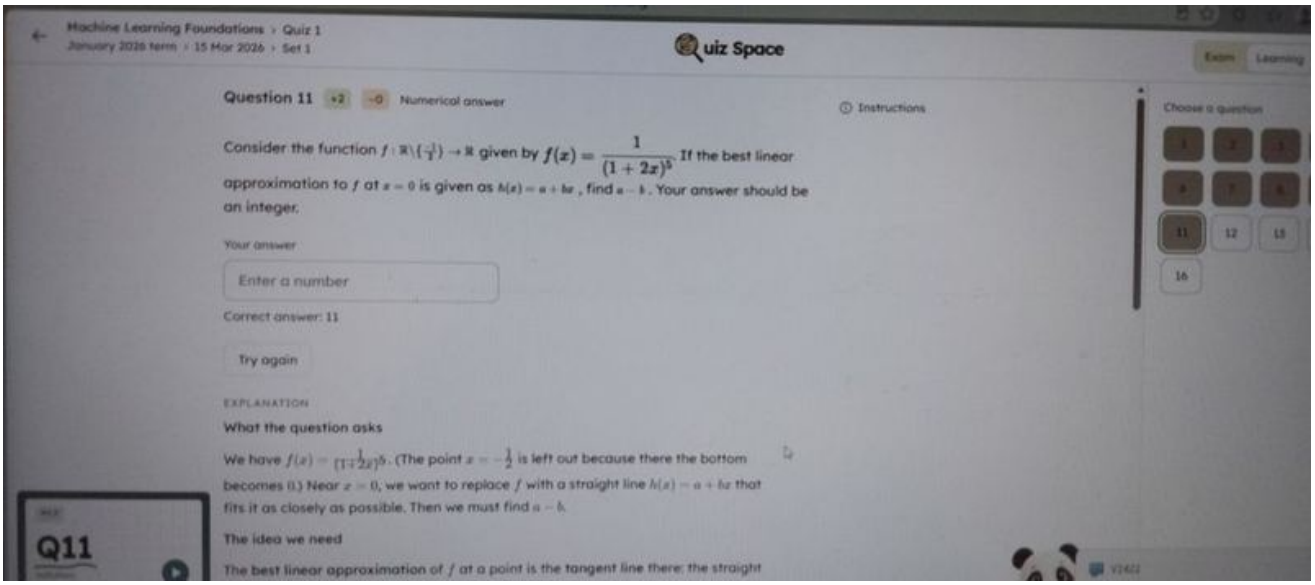
MLF-Q637 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



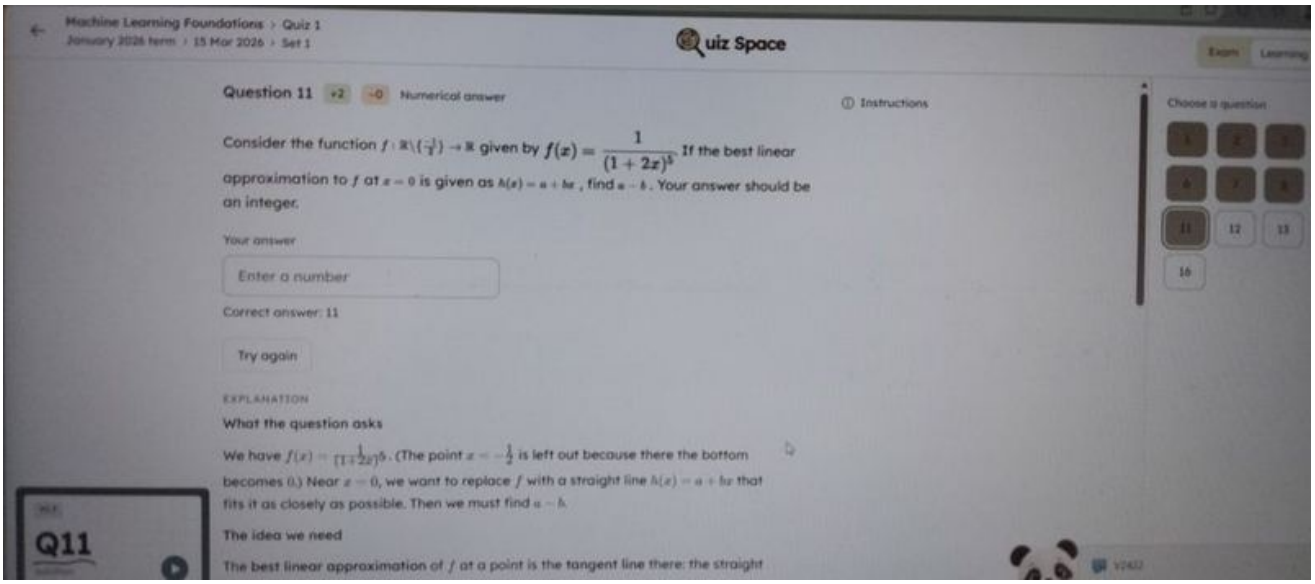
MLF-Q638 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



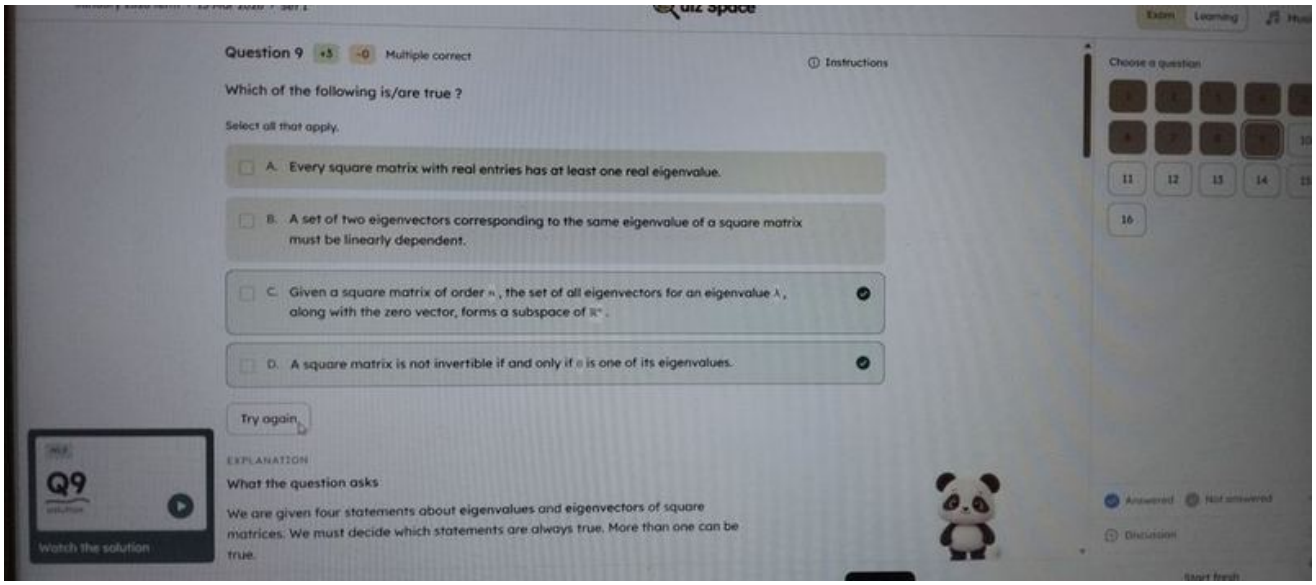
MLF-Q639 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



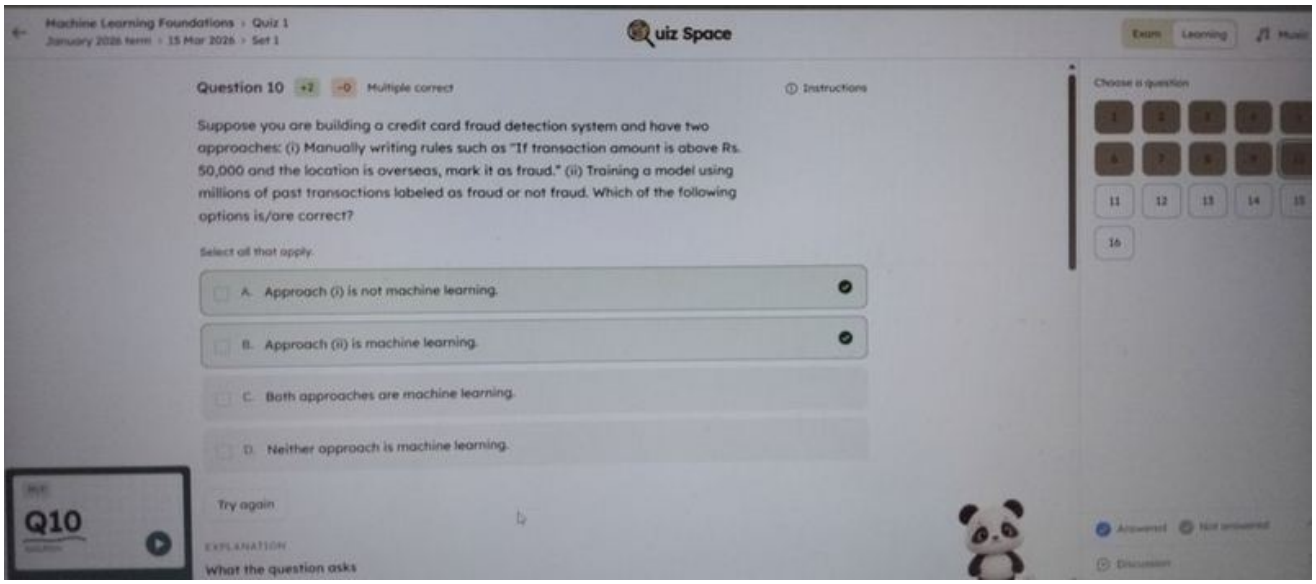
MLF-Q640 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



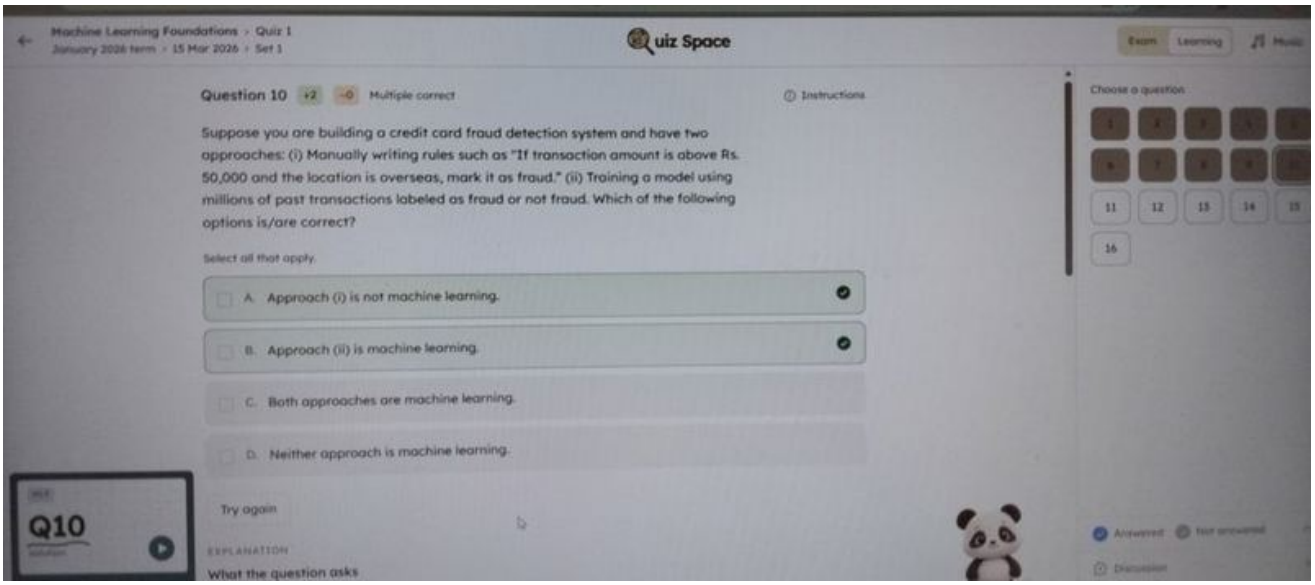
MLF-Q641 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



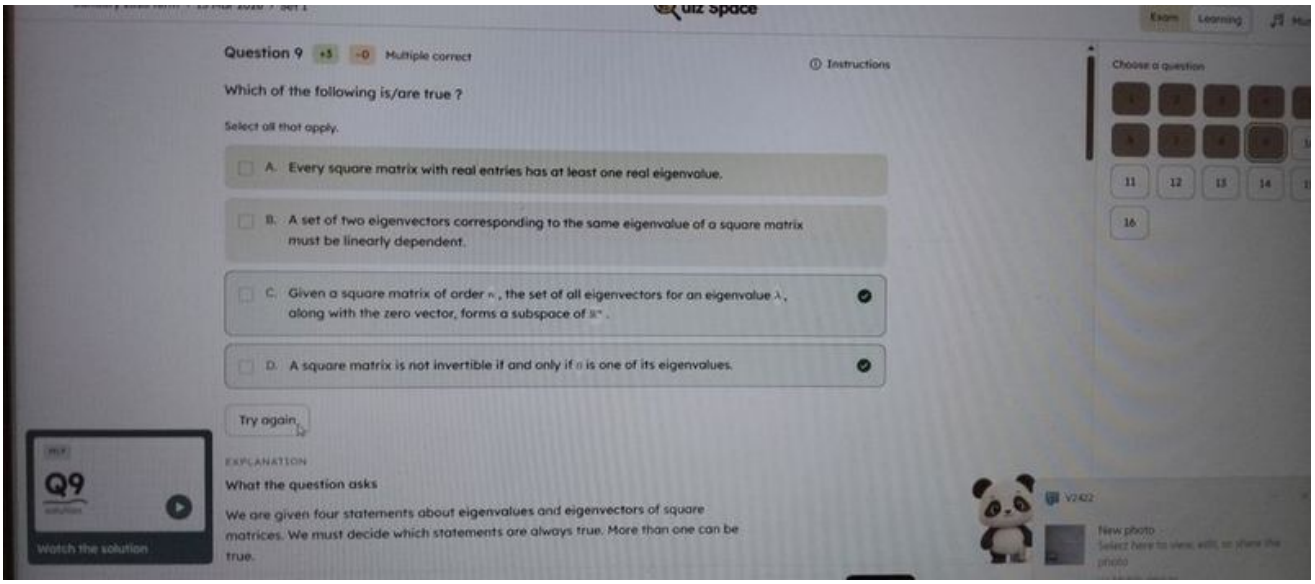
MLF-Q642 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



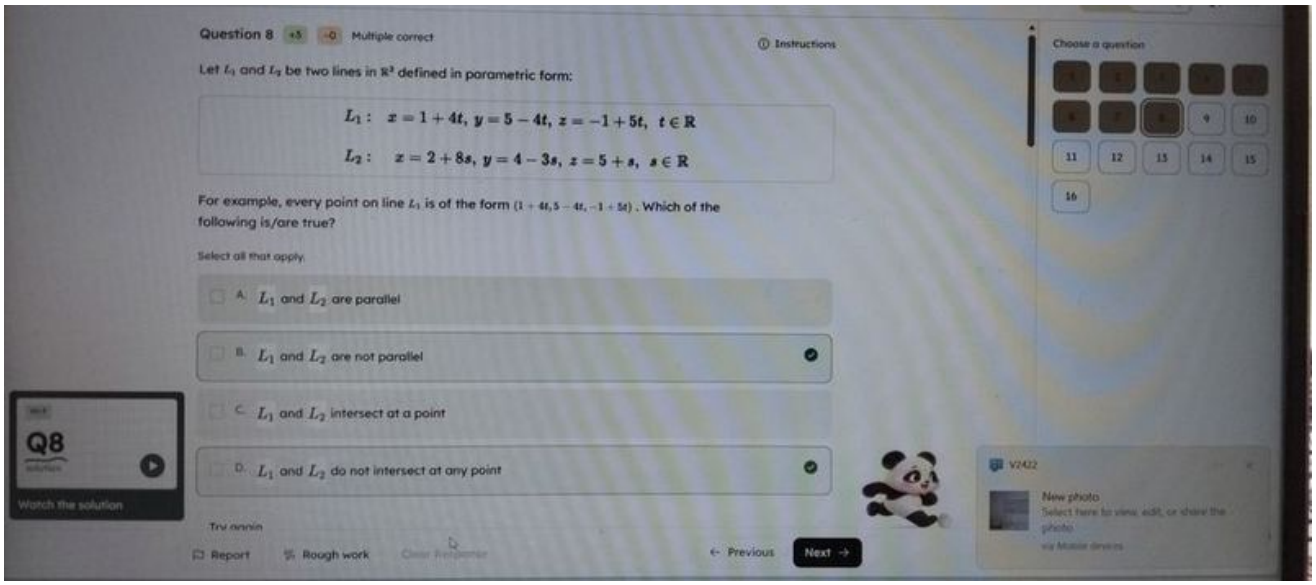
MLF-Q643 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



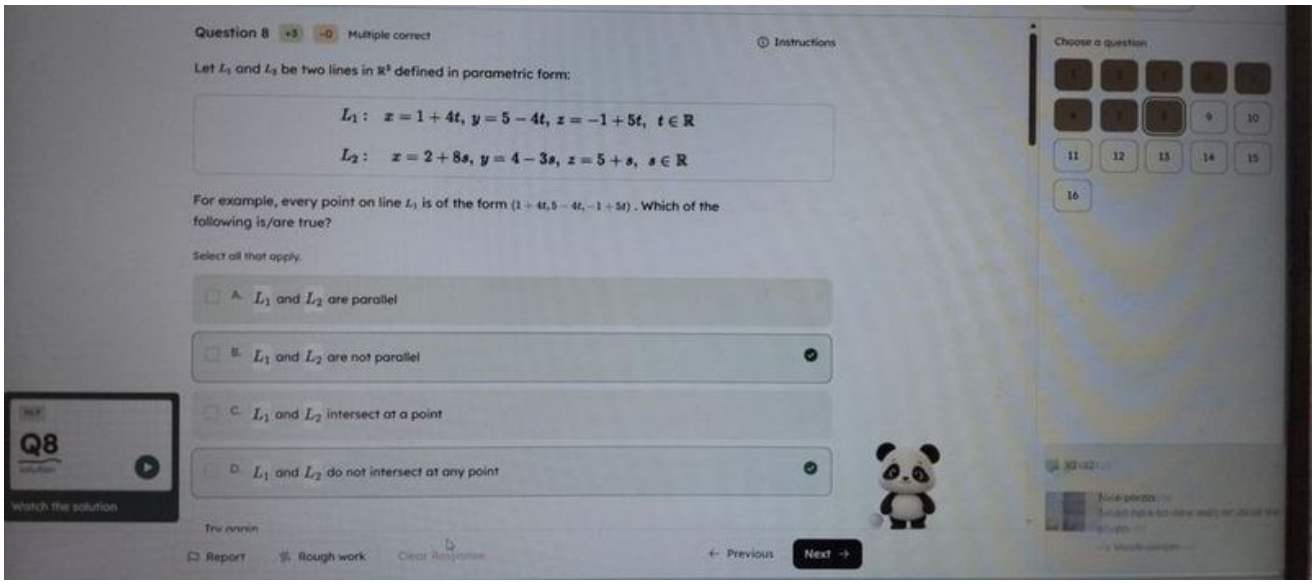
MLF-Q644 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



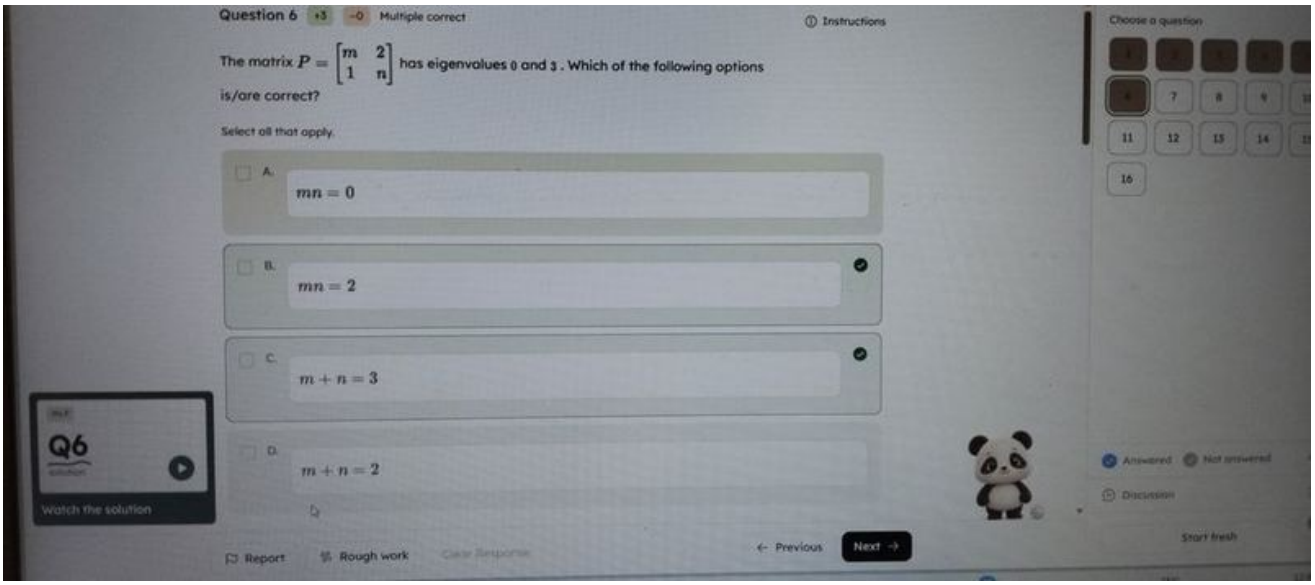
MLF-Q645 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



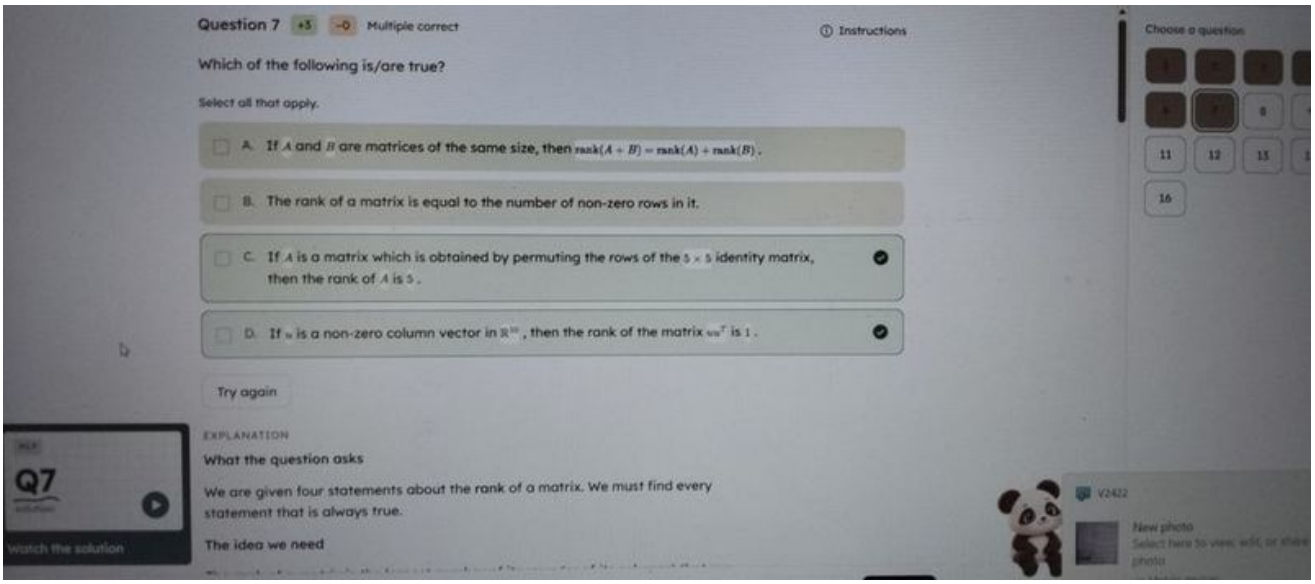
MLF-Q646 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



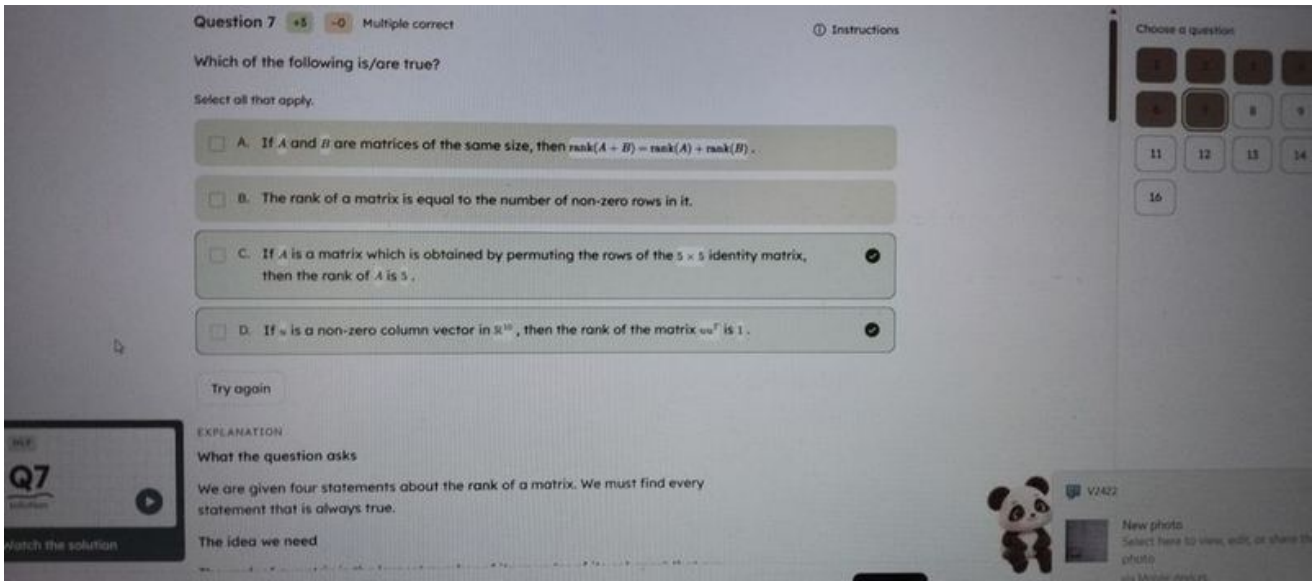
MLF-Q647 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



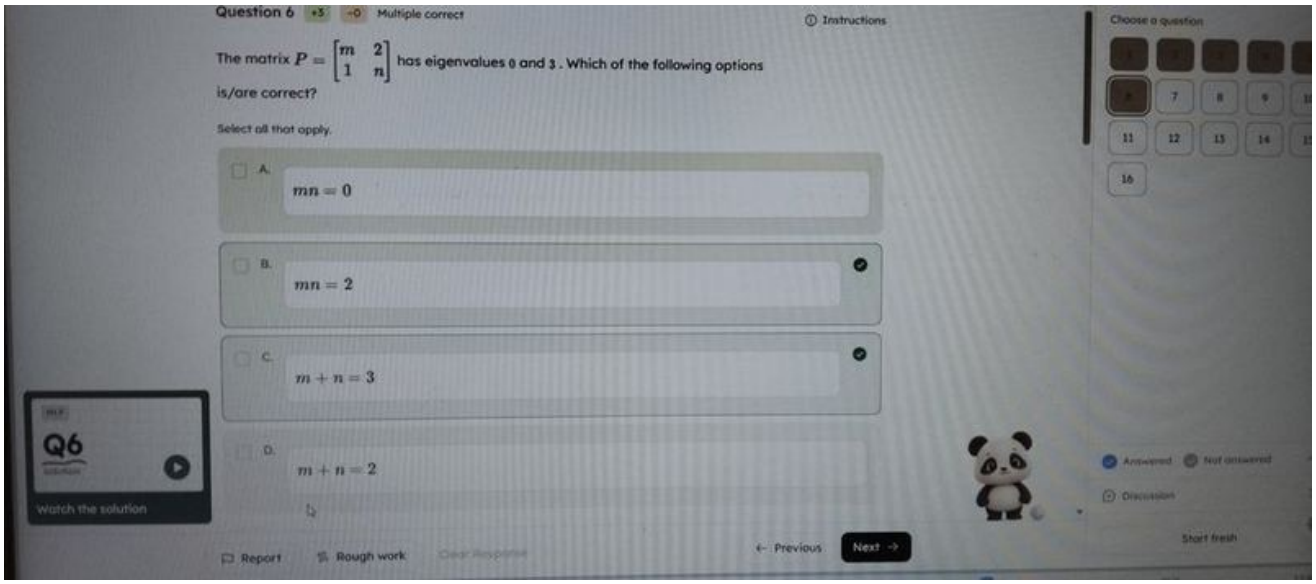
MLF-Q648 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



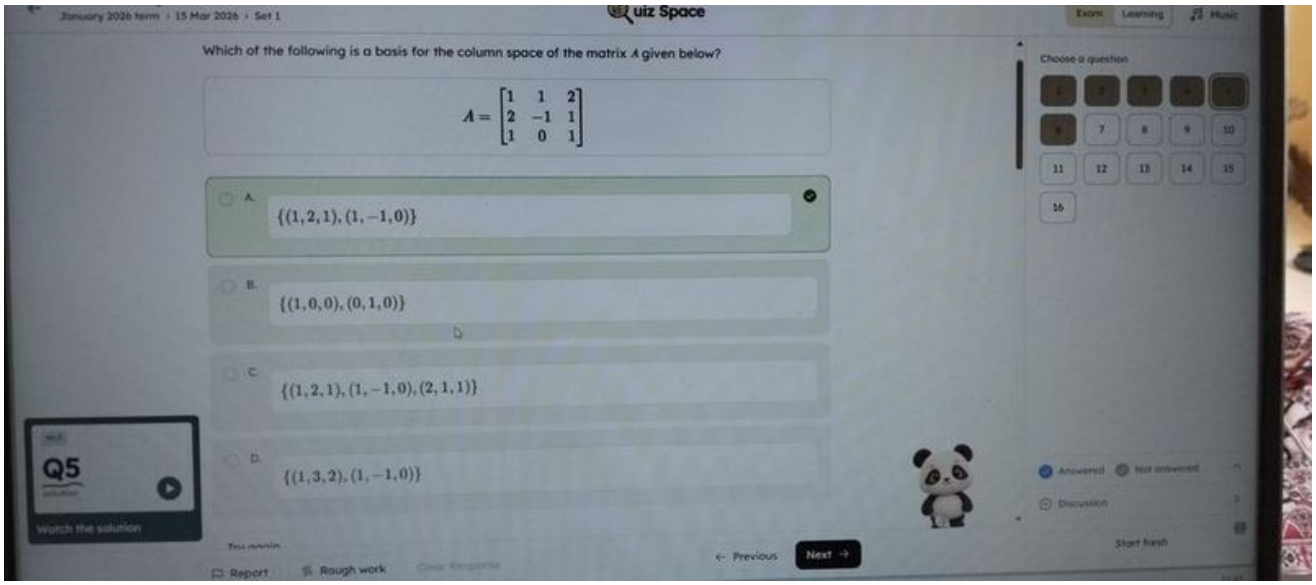
MLF-Q649 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



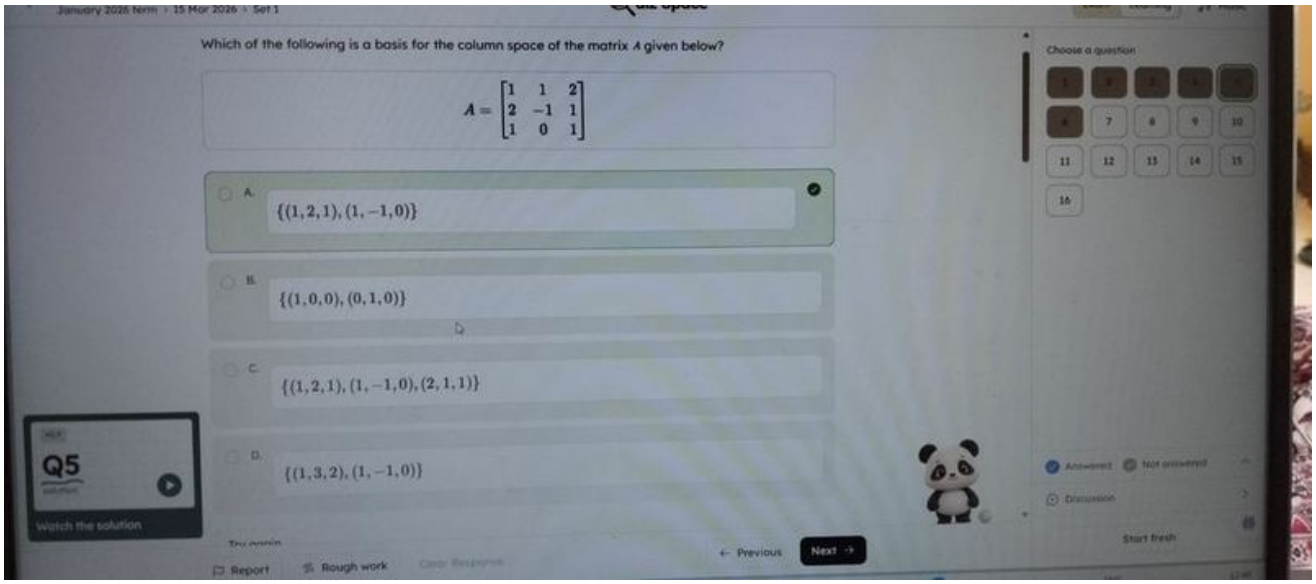
MLF-Q650 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



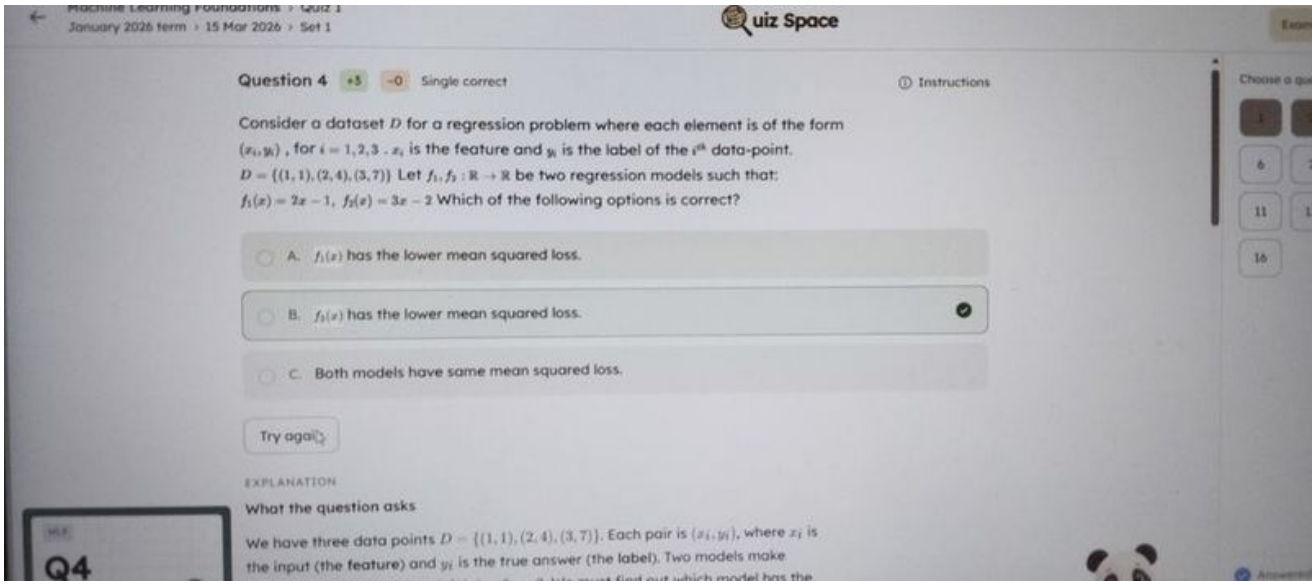
MLF-Q651 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



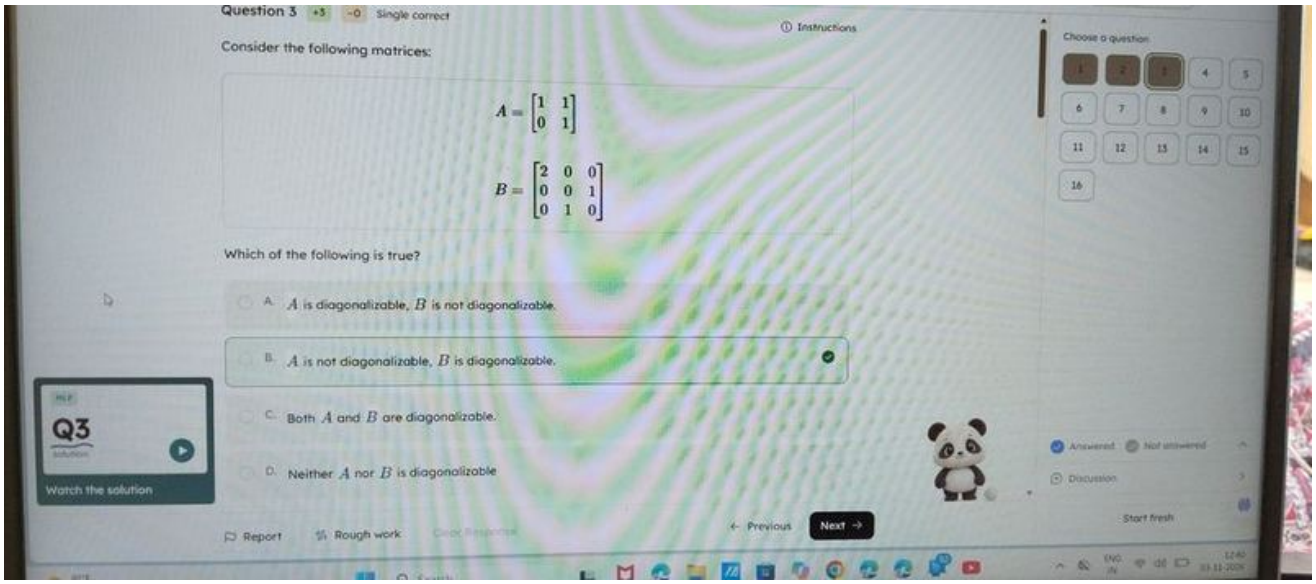
MLF-Q652 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



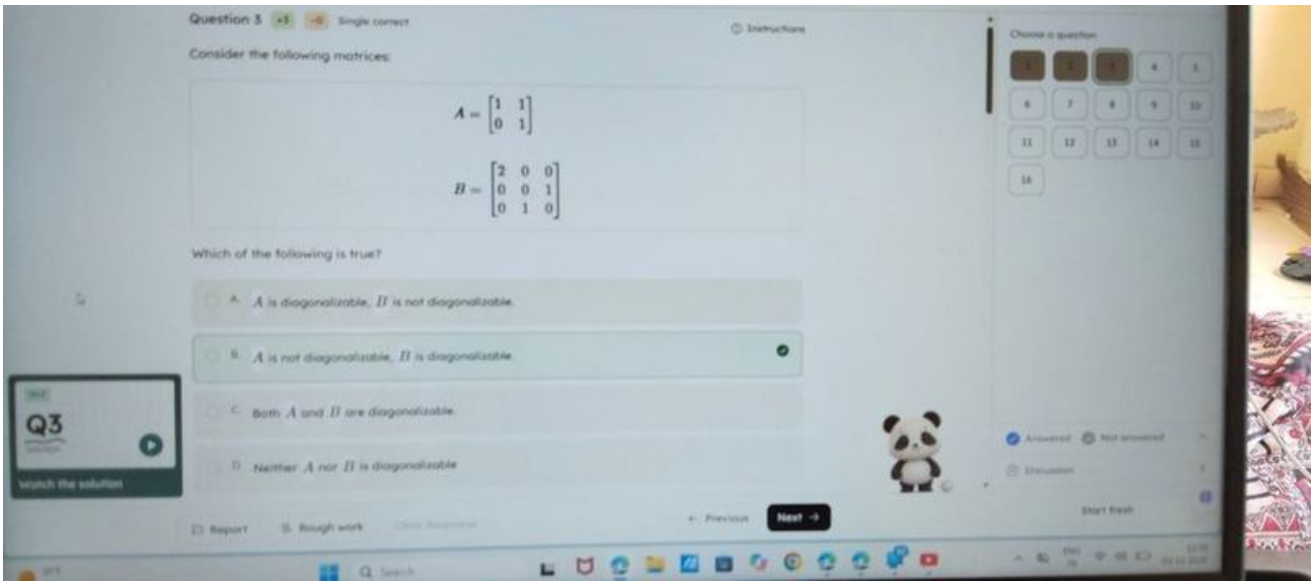
MLF-Q653 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



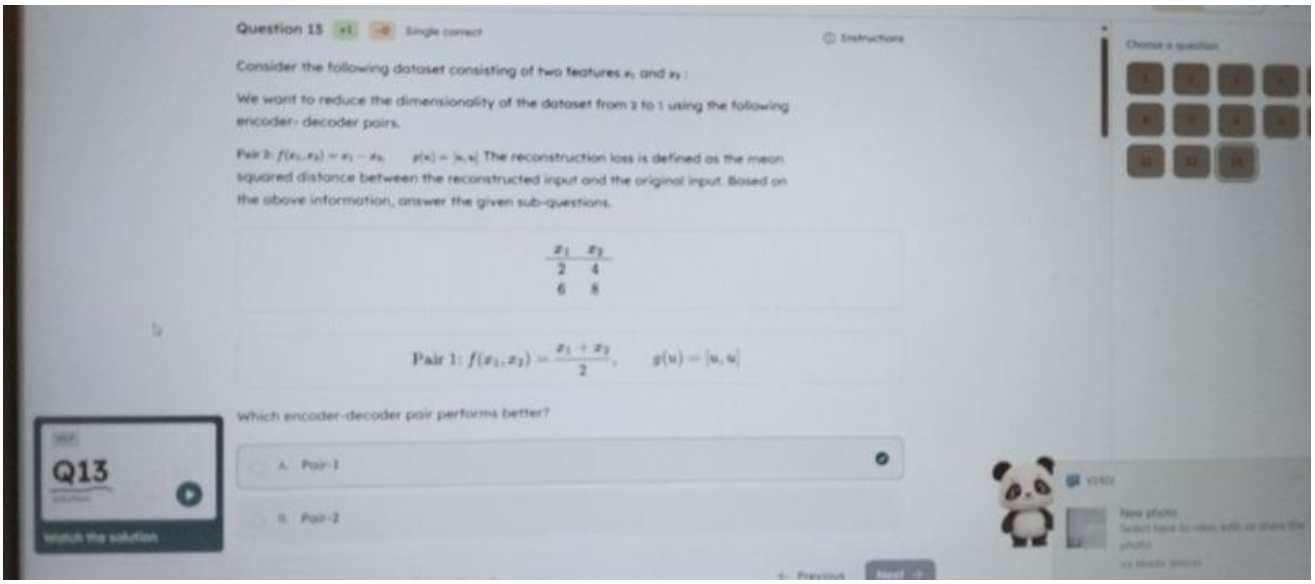
MLF-Q654 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



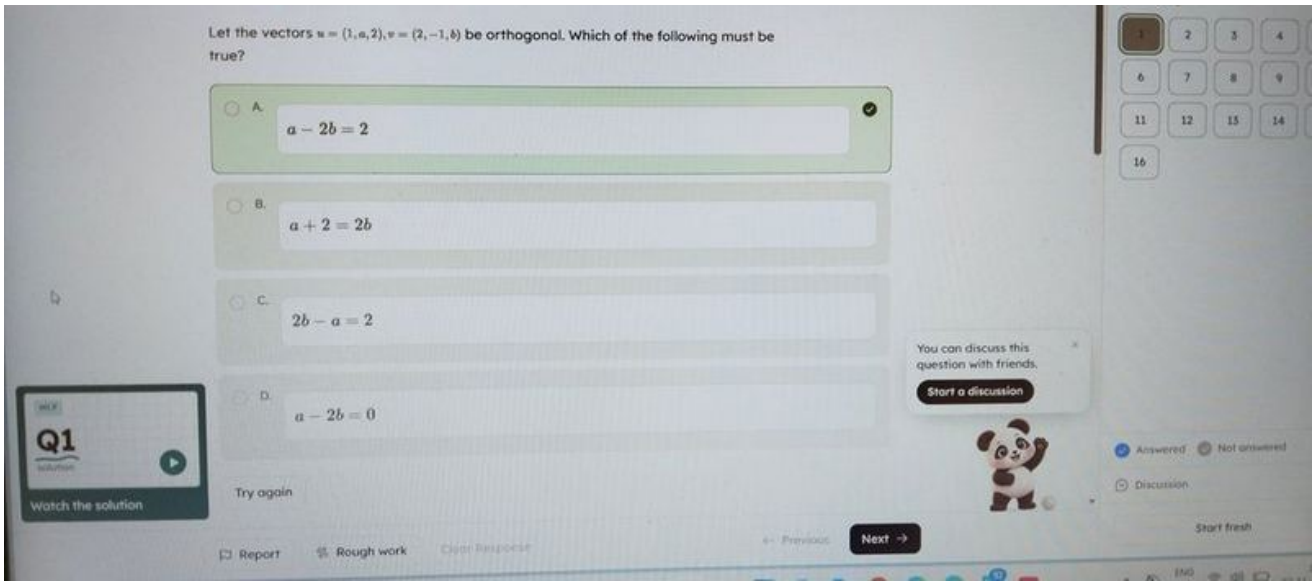
MLF-Q655 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



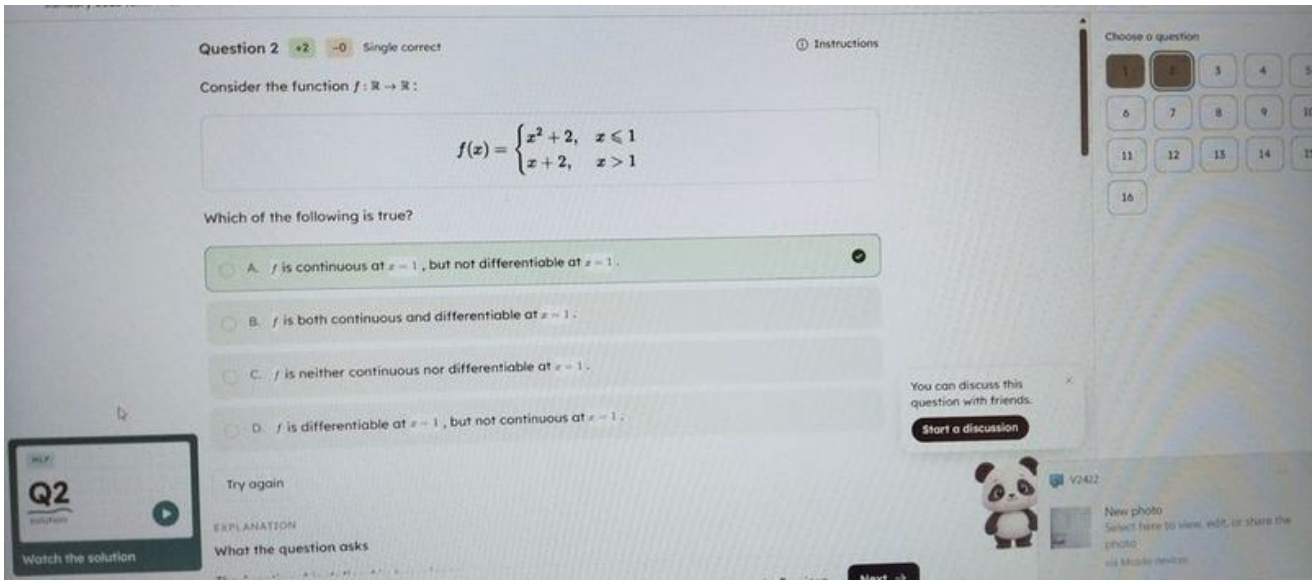
MLF-Q656 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



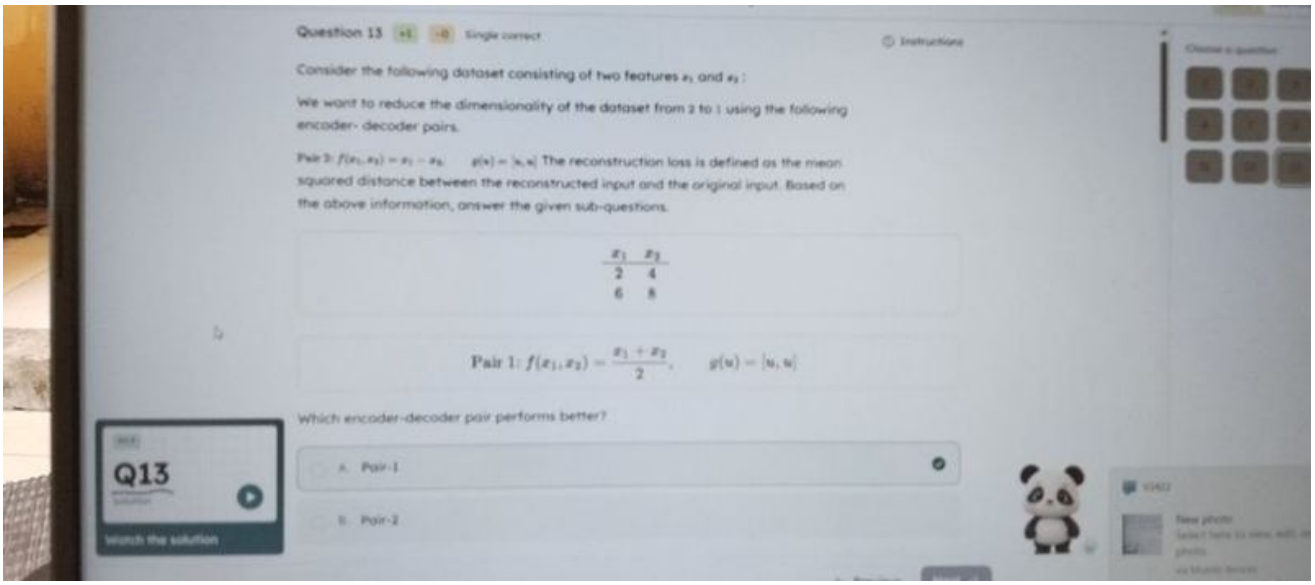
MLF-Q657 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



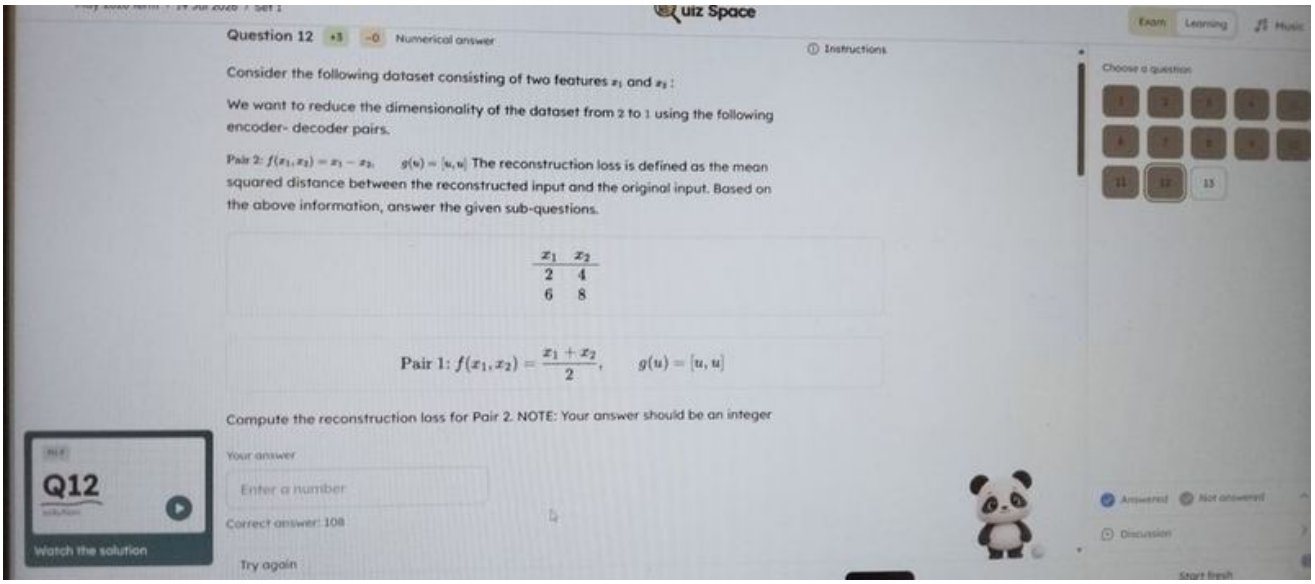
MLF-Q658 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



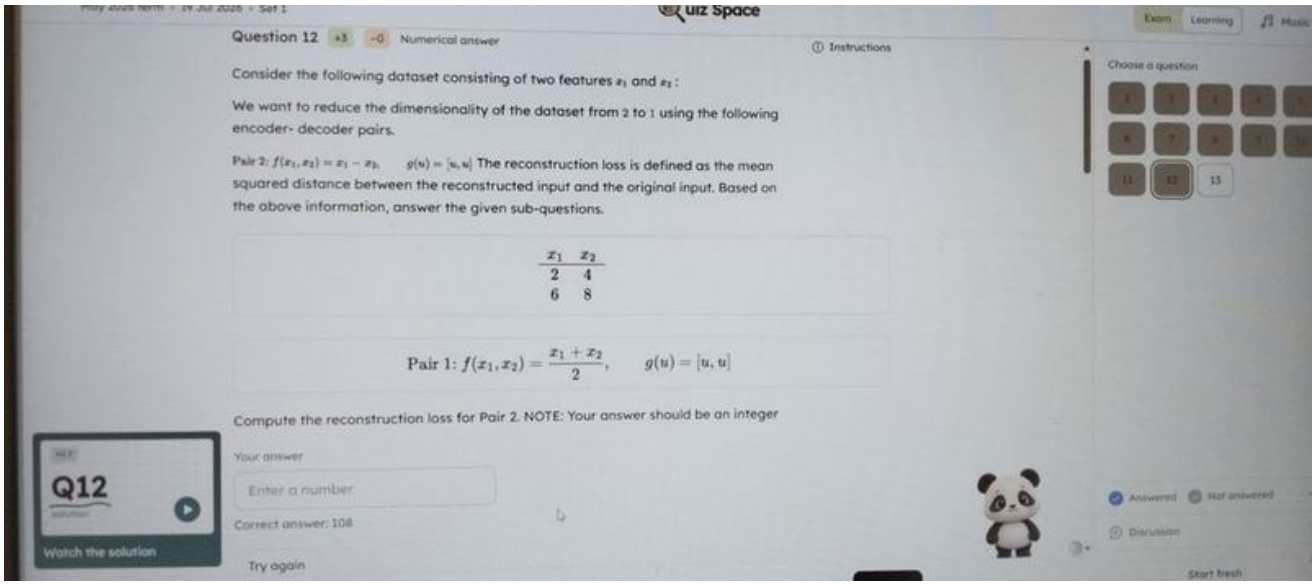
MLF-Q659 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



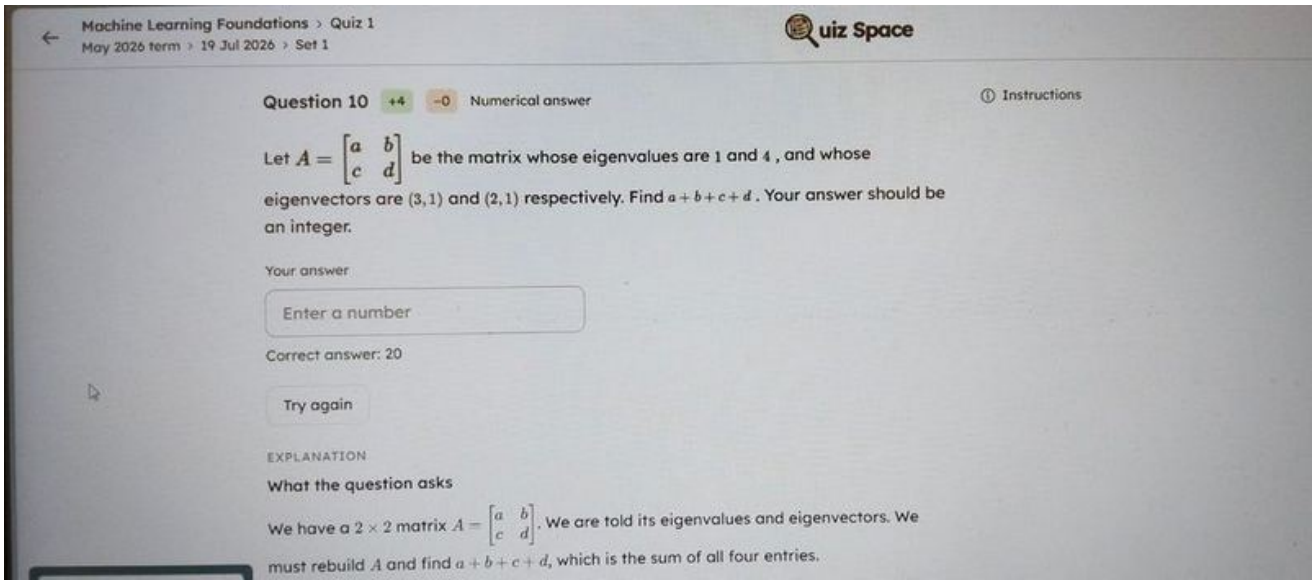
MLF-Q660 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



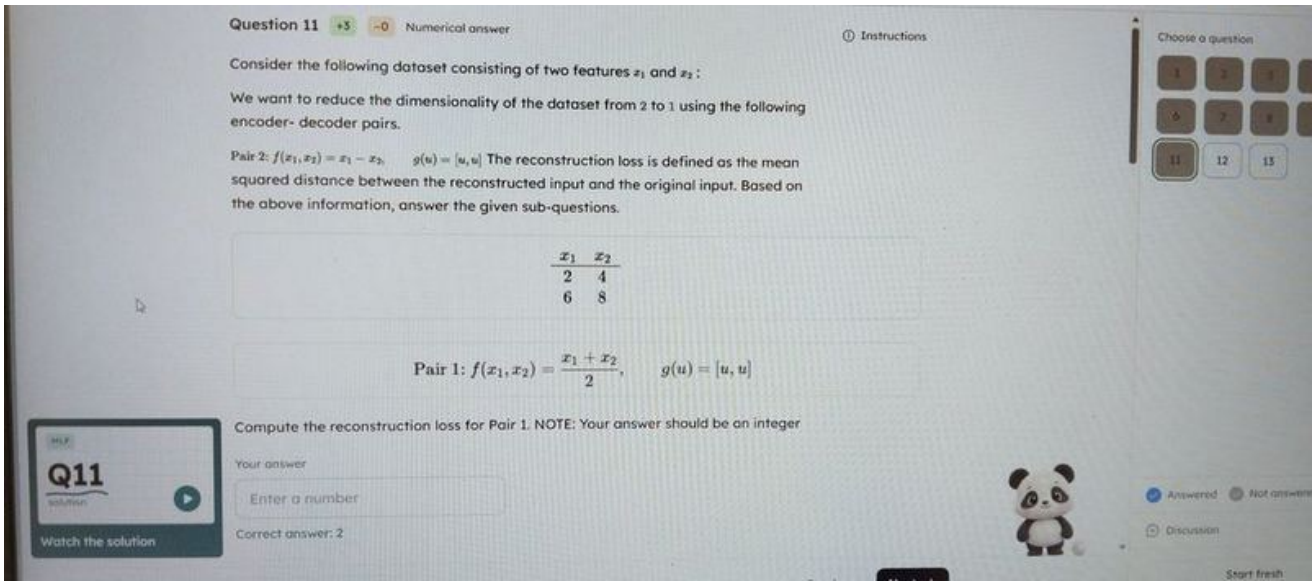
MLF-Q661 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



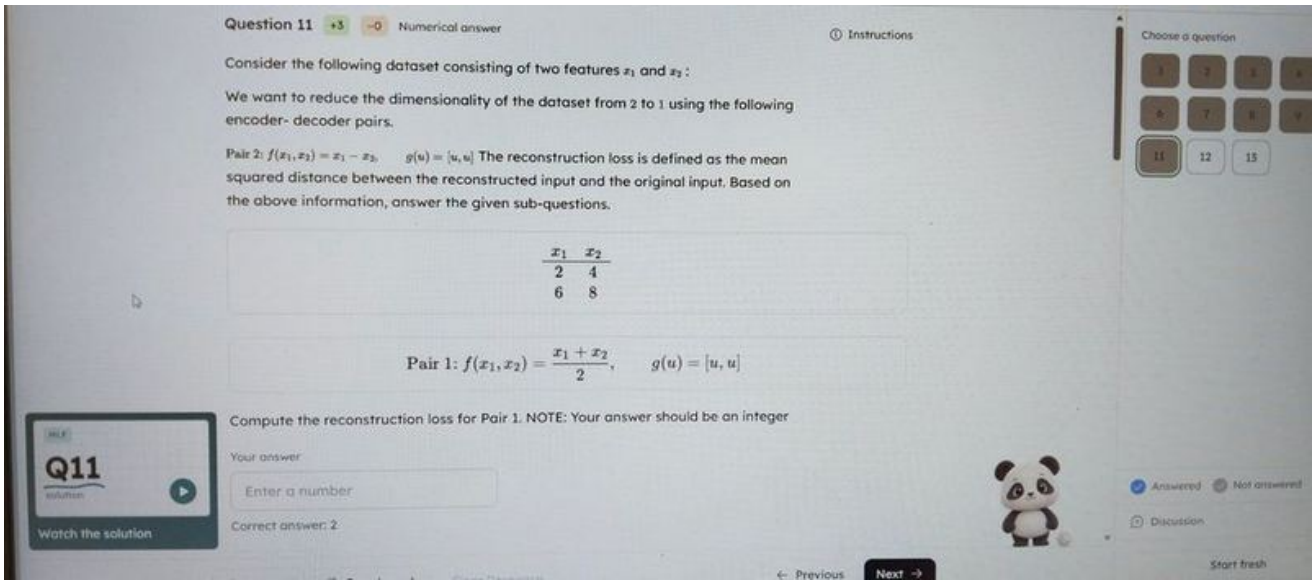
MLF-Q662 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



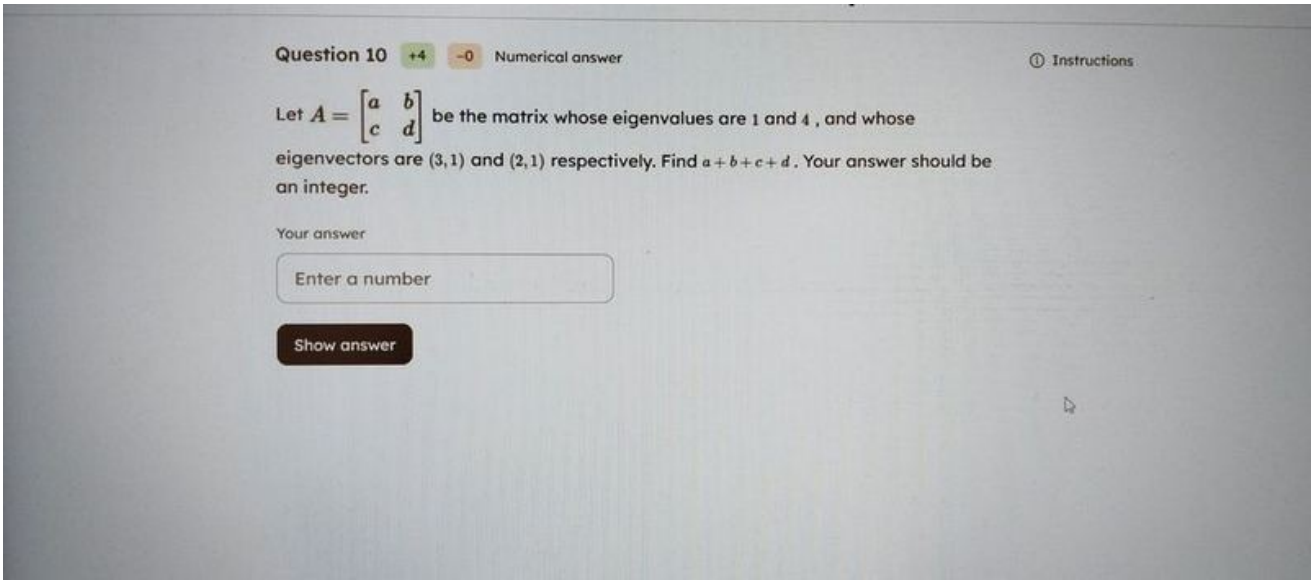
MLF-Q663 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



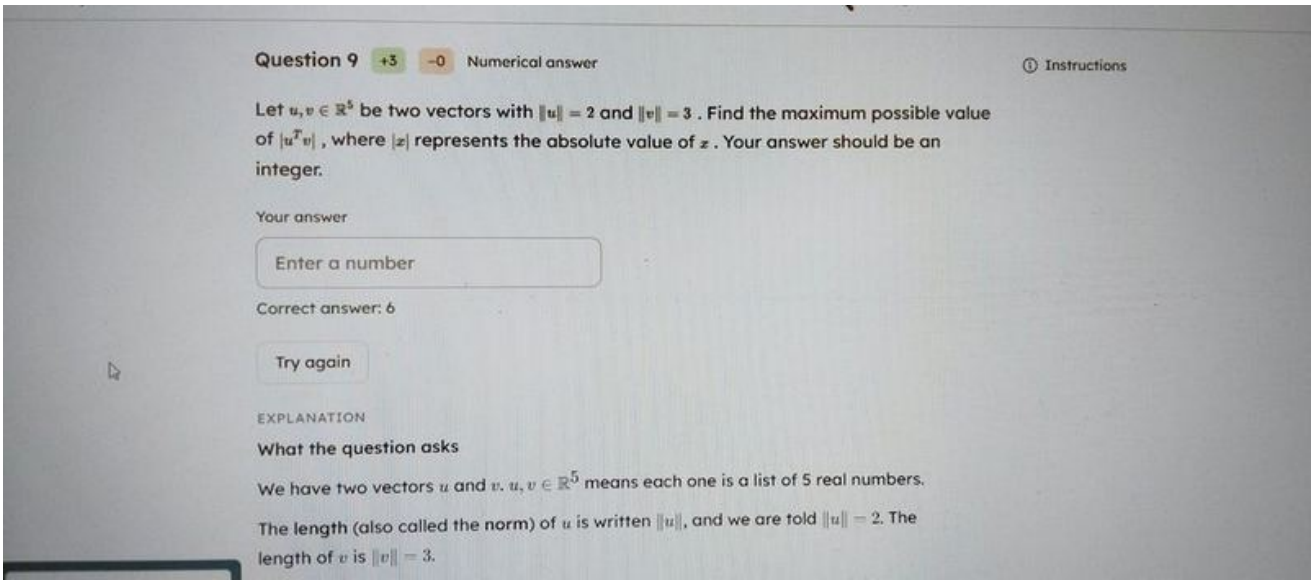
MLF-Q664 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



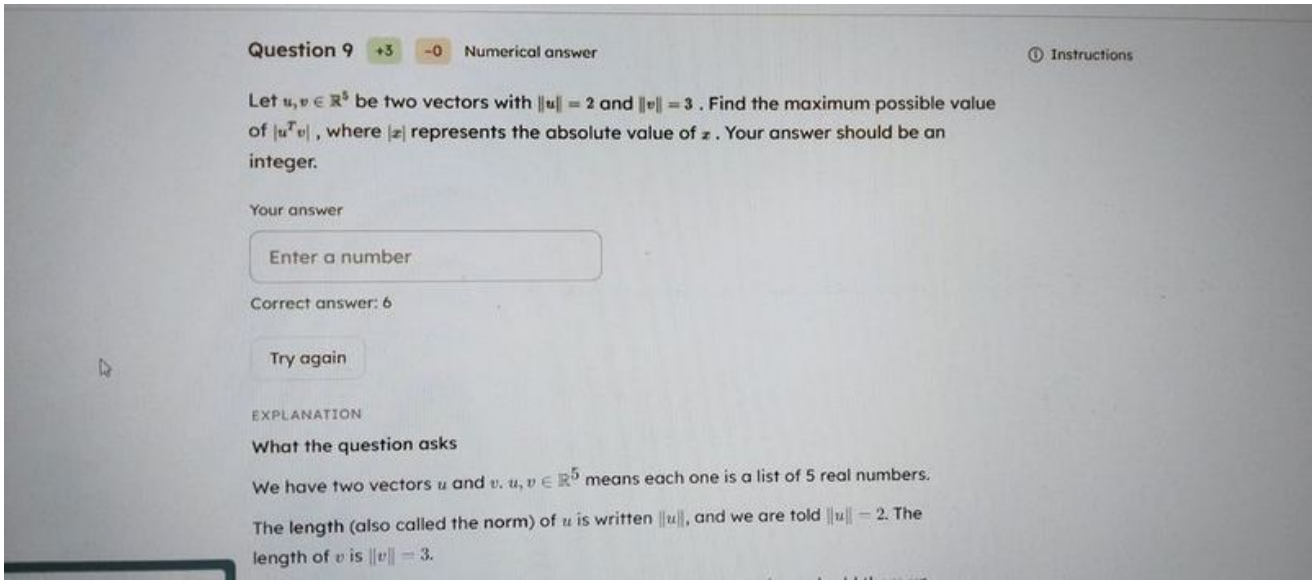
MLF-Q665 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



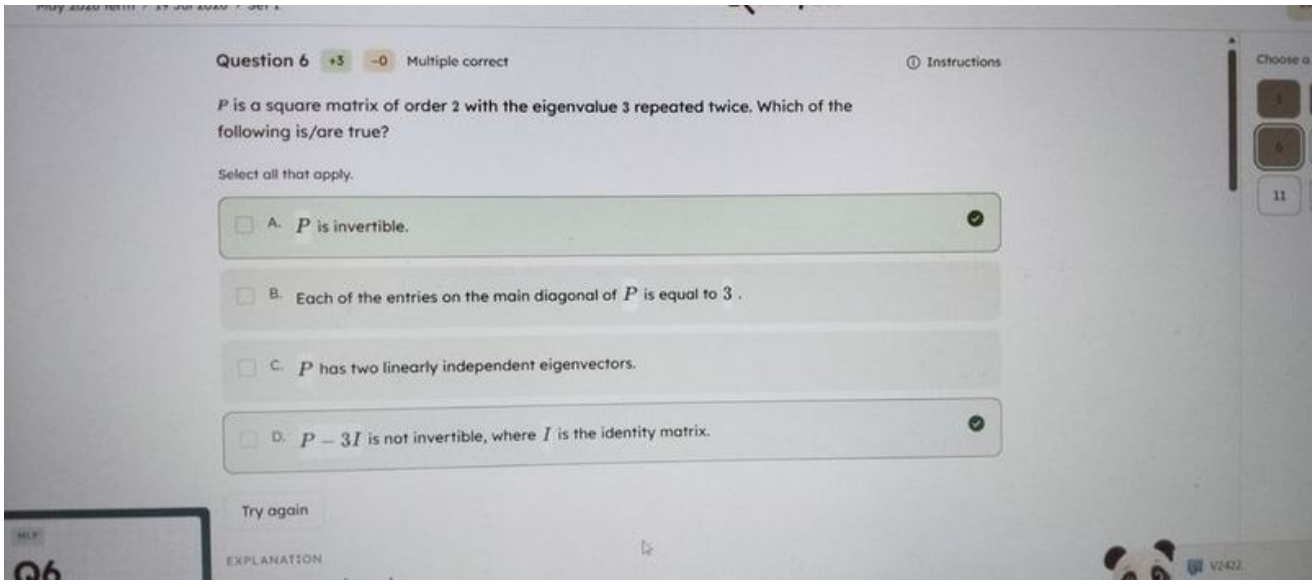
MLF-Q666 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



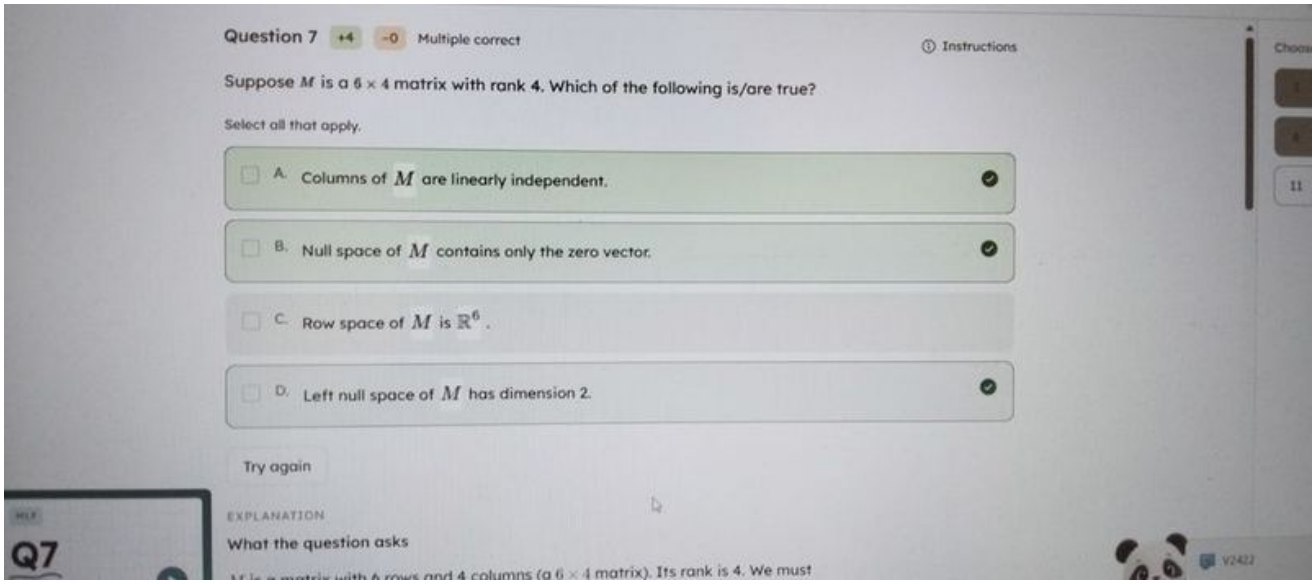
MLF-Q667 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



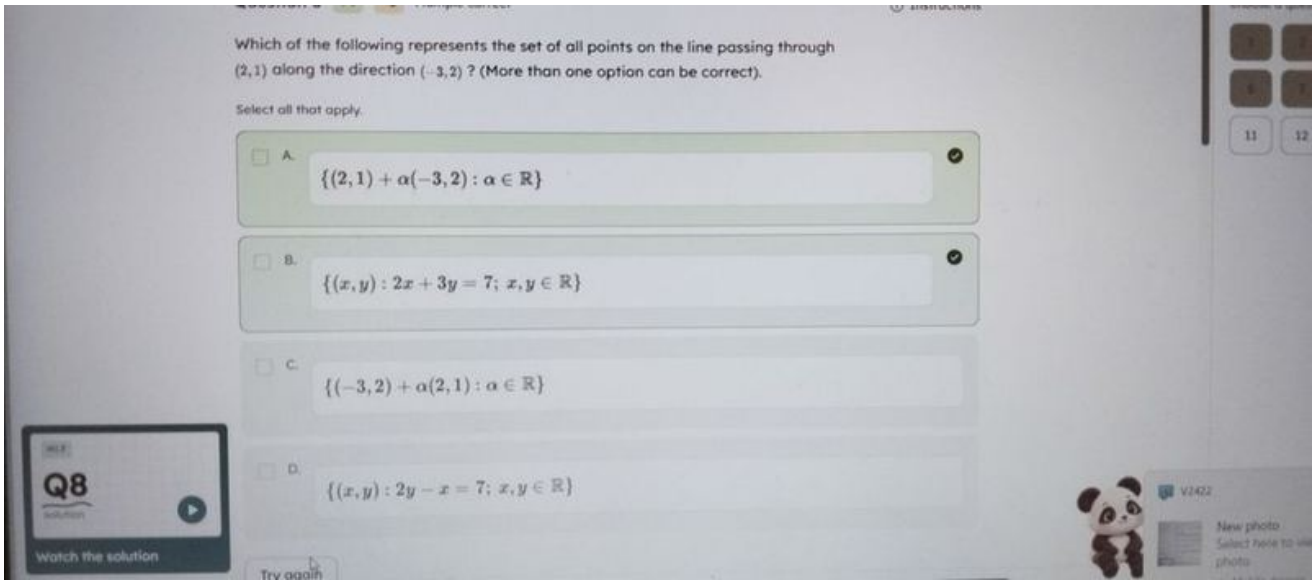
MLF-Q668 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



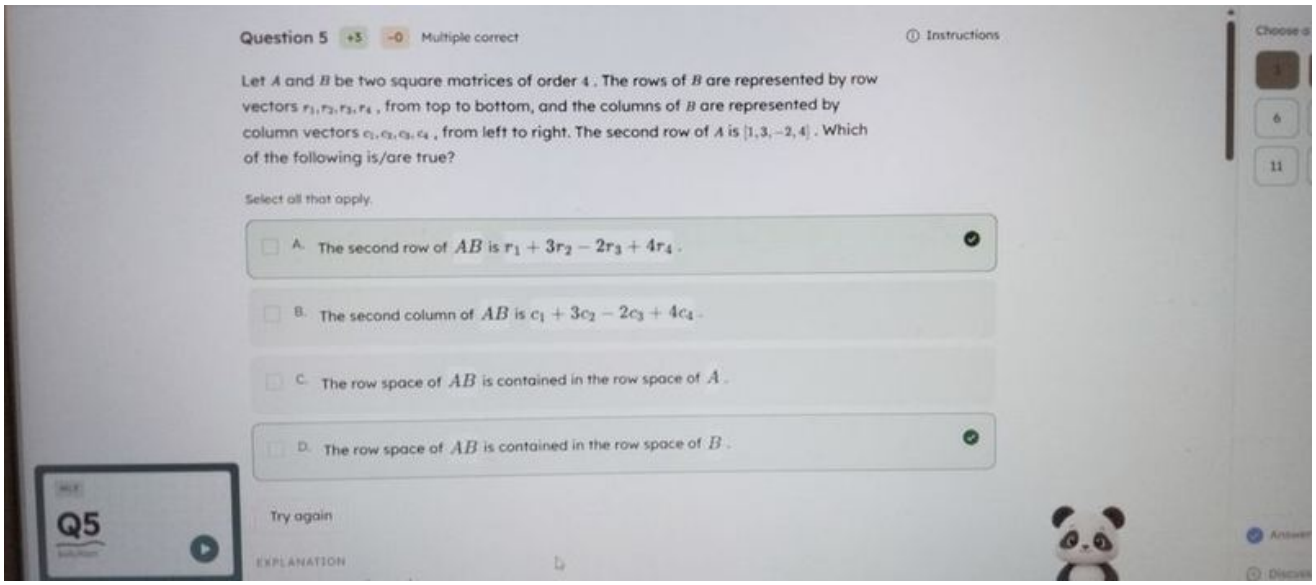
MLF-Q669 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



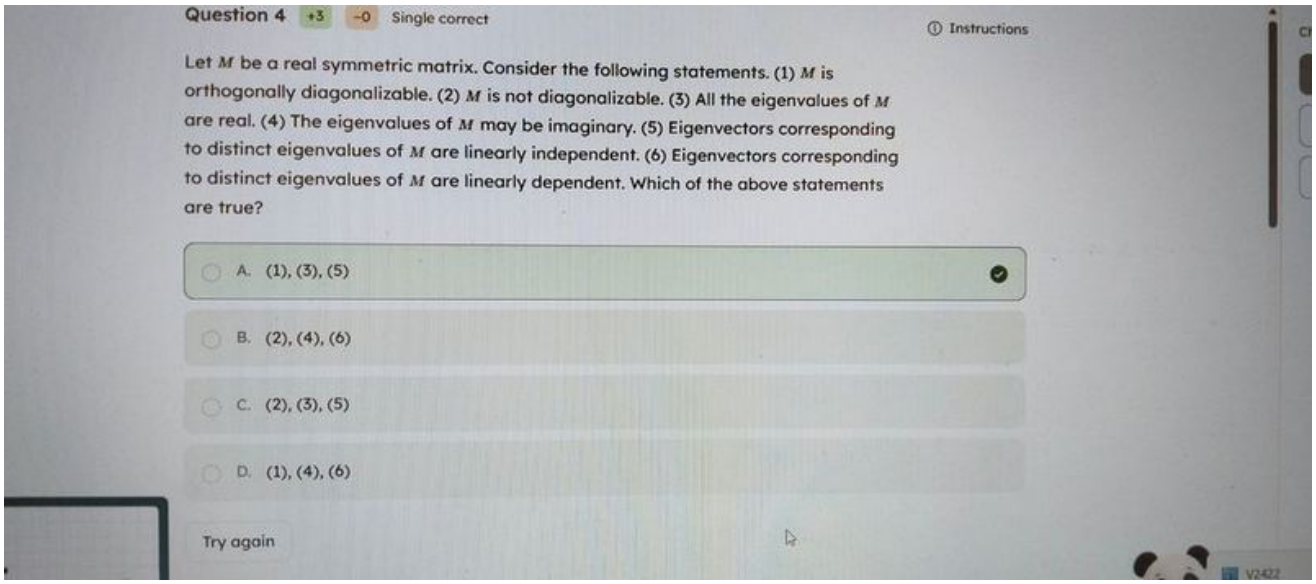
MLF-Q670 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



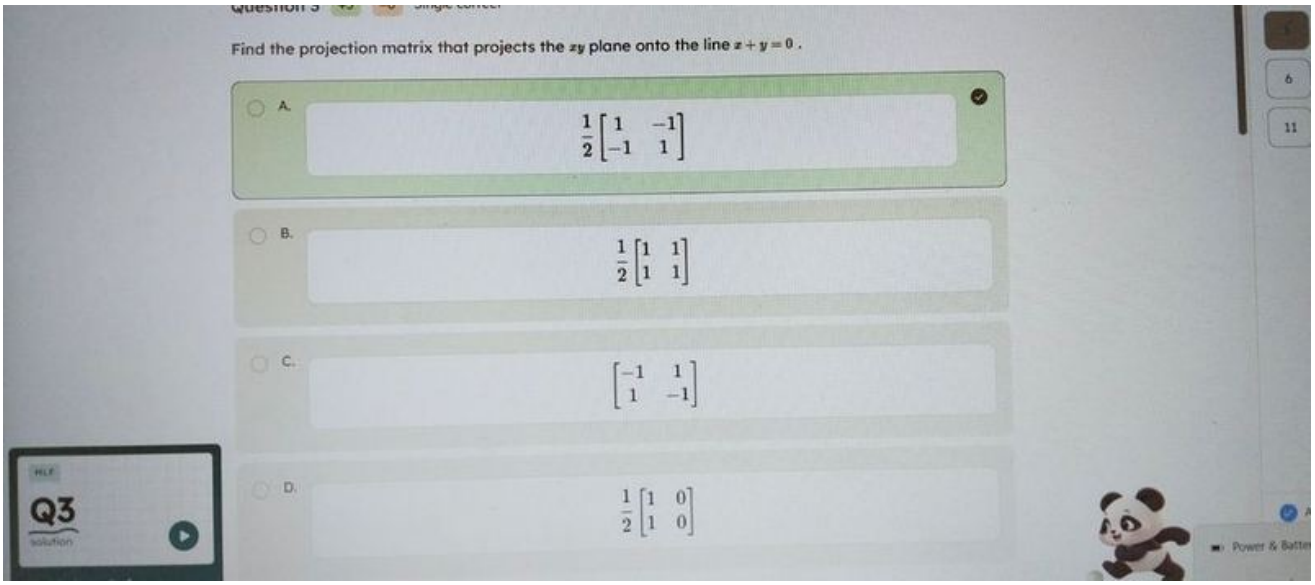
MLF-Q671 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



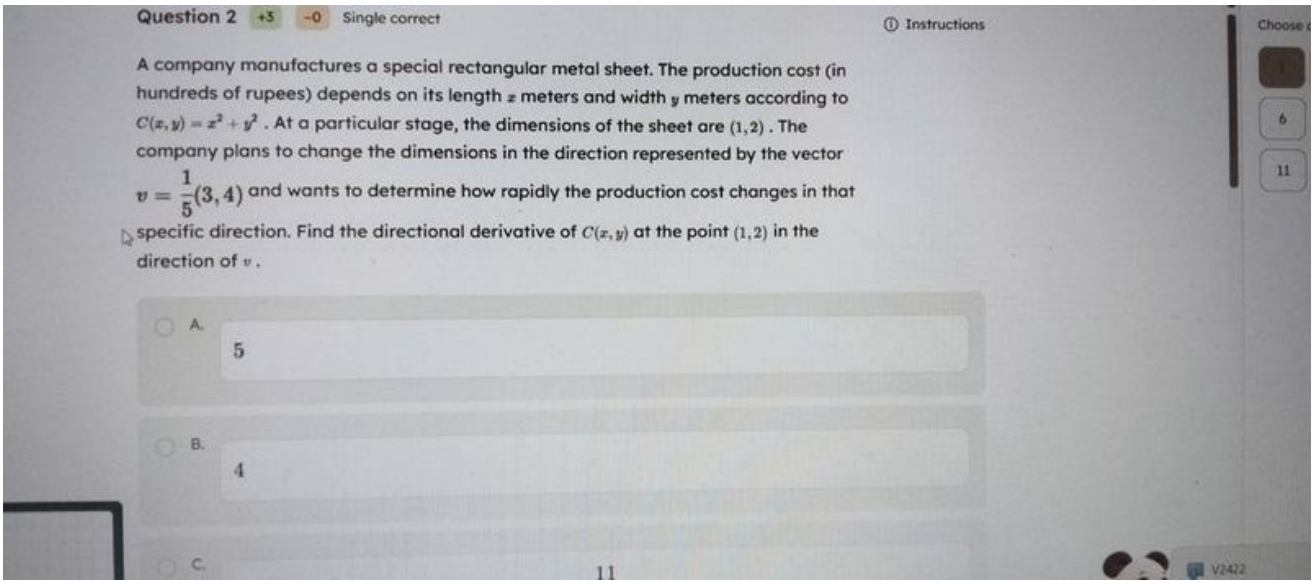
MLF-Q672 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



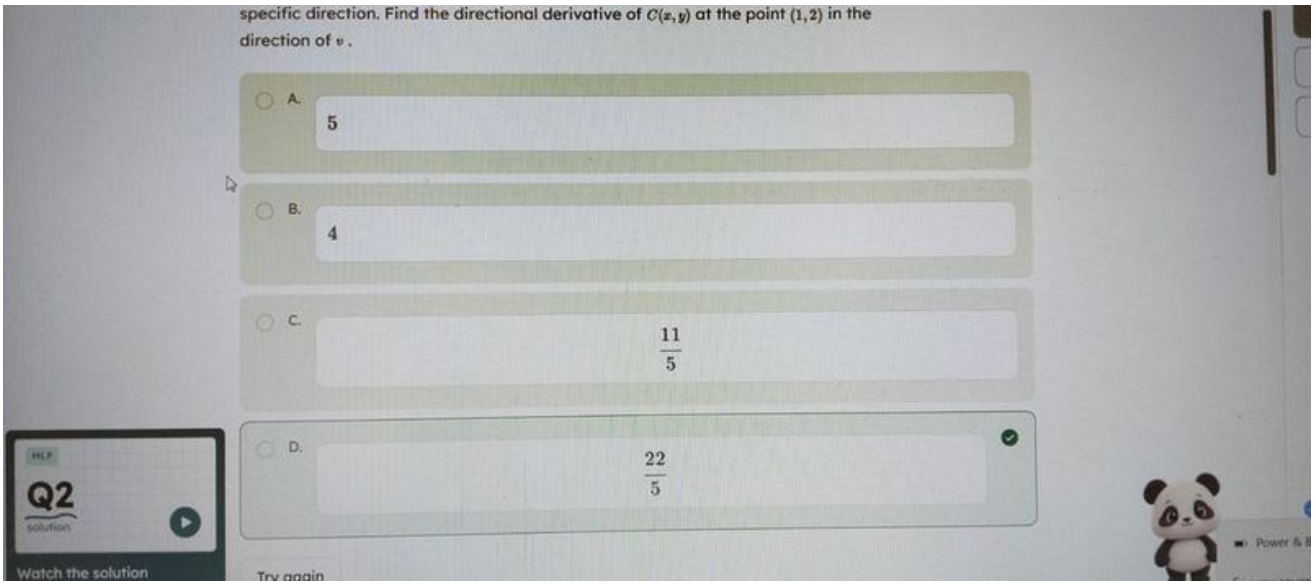
MLF-Q673 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



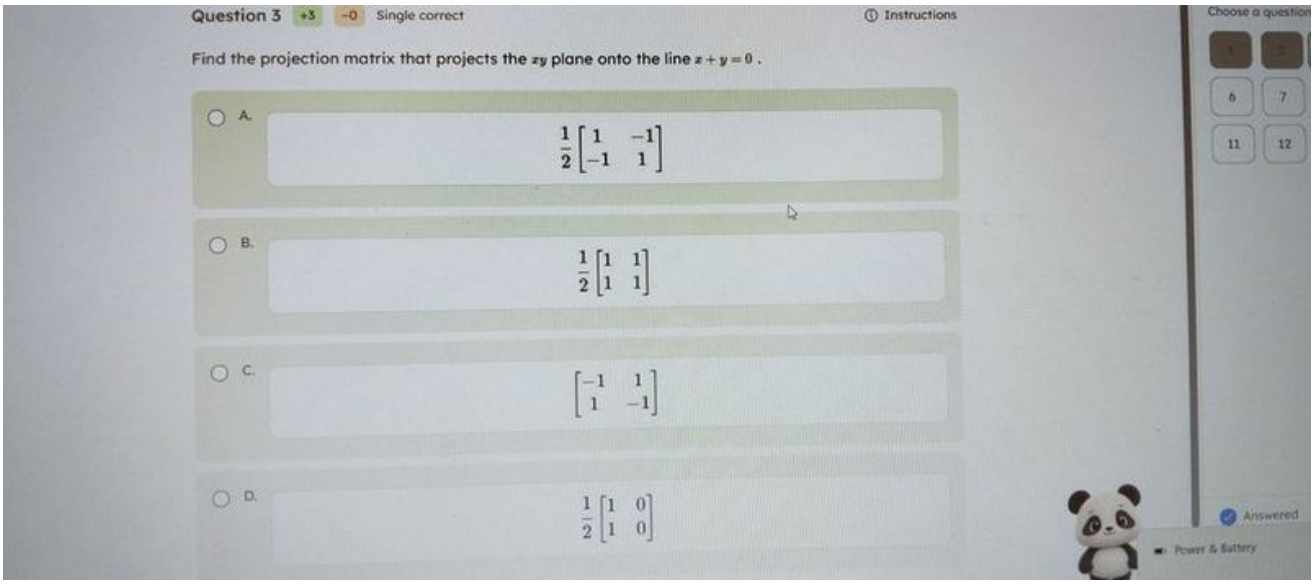
MLF-Q674 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



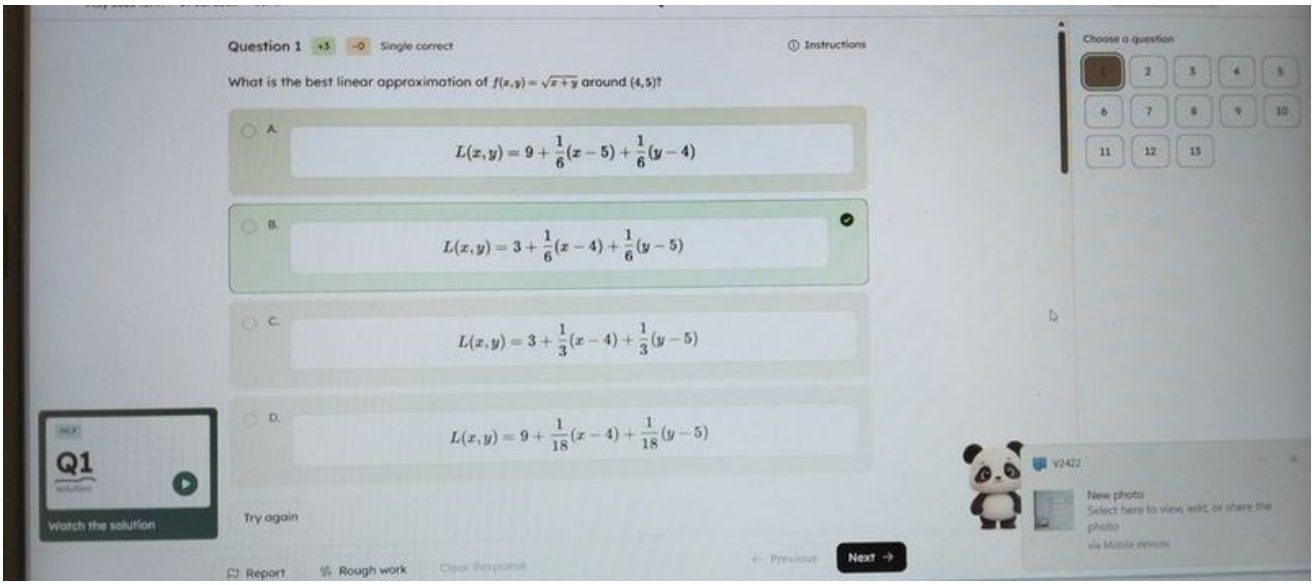
MLF-Q675 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



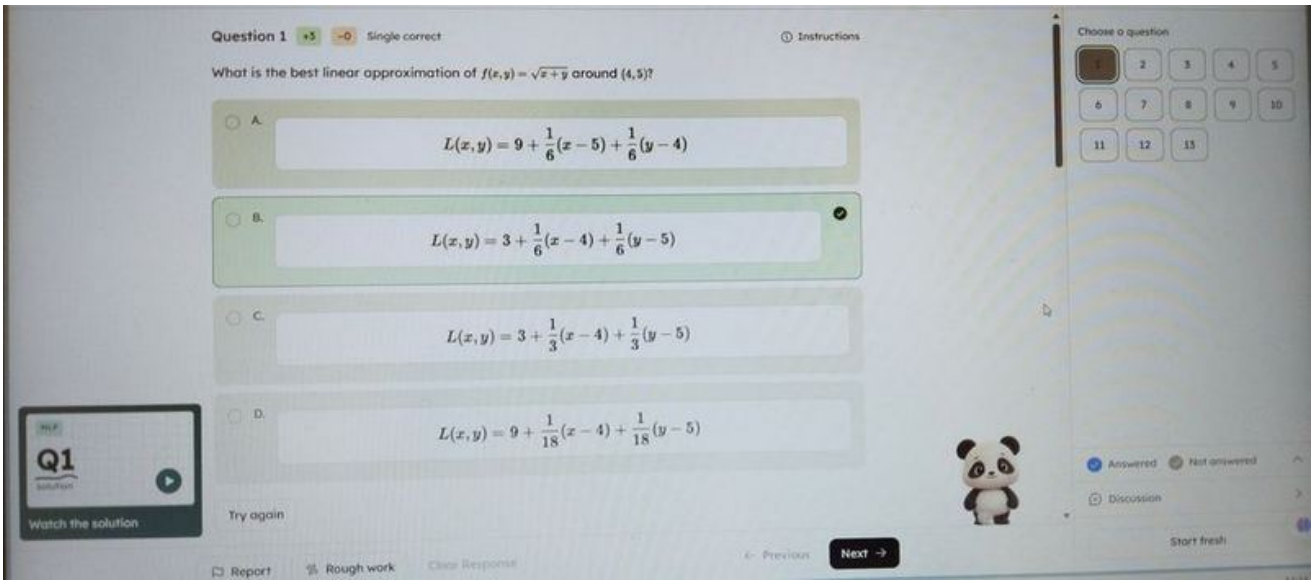
MLF-Q676 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



MLF-Q677 | [MCQ] | Marks: +2 | *PDF_COM_202611031249451070*

Category: Week 3 - Linear Algebra & Subspaces › Subspaces



■ WEEK 4 - REGRESSION & EIGENVALUES

■ Category: Eigenvalues (88 Questions)

MLF-Q054 | [MCQ] | Marks: +2 | *Quiz 1 Jan23*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Number : 129 Question Id : 640653470060 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

The characteristic polynomial of a matrix with eigenvalues 1, 2, 3 and 4 is

Options :

6406531562081. ✔ $\lambda^4 - 10\lambda^3 + 35\lambda^2 - 50\lambda + 24$

6406531562082. ✖ $\lambda^4 + 10\lambda^3 + 35\lambda^2 + 50\lambda + 24$

6406531562083. ✖ $\lambda^4 + 10\lambda^3 - 35\lambda^2 + 50\lambda - 24$

6406531562084. ✖ $2\lambda^4 - 20\lambda^3 + 35\lambda^2 - 100\lambda + 48$

Sub-Section Number :	5
Sub-Section Id :	64065367689
Question Shuffling Allowed :	Yes
Is Section Default? :	null

✔ **Official Answer Key: Option (A)**

MLF-Q069 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 142 Question Id : 640653737415 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a matrix $B = A^4 - 3A - 2I$, where $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}$. Which of the following options is/are true?

Options :

6406532467976. ✔ Eigenvalues of *A* are 1, 2, and 3.

6406532467977. ✖ Eigenvalues of *A* are 1, 2, and 2.

6406532467978. ✖ Eigenvalues of *B* are 2, 20, and 78.

6406532467979. ✔ Matrix *B* is diagonalizable.

✔ **Official Answer Key: Option (A, D)**

MLF-Q070 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 143 Question Id : 640653737416 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Let A and B be symmetric matrices such that $AB = BA$. Then which of the following options is/are true?

Options :

6406532467980. ✔ AB is a symmetric matrix.

6406532467981. ✔ AB is orthogonally diagonalizable.

6406532467982. ✔ BA has real eigenvalues.

6406532467983. ✖ AB is not a symmetric matrix.

✔ **Official Answer Key: Option (A, B, C)**

MLF-Q081 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 142 Question Id : 640653737701 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a matrix $B = A^4 - 3A - 2I$, where $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}$. Which of the following options is/are true?

Options :

6406532468800. ✔ Eigenvalues of *A* are 1, 2, and 3.

6406532468801. ✖ Eigenvalues of *A* are 1, 2, and 2.

6406532468802. ✖ Eigenvalues of *B* are 2, 20, and 78.

6406532468803. ✔ Matrix *B* is diagonalizable.

✔ **Official Answer Key: Option (A, D)**

MLF-Q082 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 143 Question Id : 640653737702 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Let A and B be symmetric matrices such that $AB = BA$. Then which of the following options is/are true?

Options :

6406532468804. ✔ AB is a symmetric matrix.

6406532468805. ✔ AB is orthogonally diagonalizable.

6406532468806. ✔ BA has real eigenvalues.

6406532468807. ✖ AB is not a symmetric matrix.

✔ **Official Answer Key: Option (A, B, C)**

MLF-Q093 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QDD4 25 Feb 2024*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 142 Question Id : 640653737987 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a matrix $B = A^4 - 3A - 2I$, where $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}$. Which of the following options is/are true?

Options :

6406532469624. ✔ Eigenvalues of *A* are 1, 2, and 3.

6406532469625. ✖ Eigenvalues of *A* are 1, 2, and 2.

6406532469626. ✖ Eigenvalues of *B* are 2, 20, and 78.

6406532469627. ✔ Matrix *B* is diagonalizable.

✔ **Official Answer Key: Option (A, D)**

MLF-Q094 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QDD4 25 Feb 2024*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 143 Question Id : 640653737988 Question Type : MSQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Let A and B be symmetric matrices such that $AB = BA$. Then which of the following options is/are true?

Options :

6406532469628. ✔ AB is a symmetric matrix.

6406532469629. ✔ AB is orthogonally diagonalizable.

6406532469630. ✔ BA has real eigenvalues.

6406532469631. ✖ AB is not a symmetric matrix.

✔ **Official Answer Key: Option (A, B, C)**

MLF-Q099 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*
Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Number : 85 Question Id : 6406531113778 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the matrix M given by:

$$M = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$$

Which of the following options is true?

Options :

6406533774210. ✖ Eigenvectors corresponding to the two different eigenvalues of *M* are orthogonal.

6406533774211. ✖ The matrix *M* is not diagonalizable.

6406533774212. ✖ The eigenvalues of *M* are 4 and 1

6406533774213. ✔ The rank of *M* is 1.

✔ Official Answer Key: Option (D)

MLF-Q110 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*
Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider a 3×3 matrix B given by

$$B = (-1)\begin{bmatrix} -4 & 3 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & -1 \end{bmatrix}.$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 96 Question Id : 6406531113791 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

If B is diagonalizable, what is the invertible matrix P such that $B = PDP^{-1}$?

Options :

6406533774248. ✖ $\begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

6406533774249. ✖ $\begin{bmatrix} 3 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$

6406533774250. ✖ $\begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

6406533774251. ✔ $\begin{bmatrix} 3 & 1 & 0 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

6406533774252. ✖ B is not diagonalizable.

✔ Official Answer Key: Option (D)

MLF-Q111 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 23 Feb 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider a 3×3 matrix B given by

$$B = (-1)\begin{bmatrix} -4 & 3 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & -1 \end{bmatrix}.$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 97 Question Id : 6406531113792 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

If B is diagonalizable, which of the following is a diagonal matrix D such that $B = PDP^{-1}$? Use the matrix P obtained in the previous question.

Options :

6406533774253. ✔ $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

6406533774254. ✖ $\begin{bmatrix} 4 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

6406533774255. ✖ $\begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

6406533774256. ✖ $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

6406533774257. ✖ B is not diagonalizable.

✔ **Official Answer Key: Option (A)**

MLF-Q114 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Number : 85 Question Id : 6406531114405 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Consider the matrix M given by:

$$M = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$$

Which of the following options is true?

Options :

6406533776023. ✖ Eigenvectors corresponding to the two different eigenvalues of *M* are orthogonal.

6406533776024. ✖ The matrix *M* is not diagonalizable.

6406533776025. ✖ The eigenvalues of *M* are 4 and 1

6406533776026. ✔ The rank of *M* is 1.

✔ Official Answer Key: Option (D)

MLF-Q125 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider a 3×3 matrix B given by

$$B = (-1)\begin{bmatrix} -4 & 3 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & -1 \end{bmatrix}.$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 96 Question Id : 6406531114418 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

If B is diagonalizable, what is the invertible matrix P such that $B = PDP^{-1}$?

Options :

6406533776061. ✖ $\begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

6406533776062. ✖ $\begin{bmatrix} 3 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$

6406533776063. ✖ $\begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

6406533776064. ✔ $\begin{bmatrix} 3 & 1 & 0 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

6406533776065. ✖ B is not diagonalizable.

✔ **Official Answer Key: Option (D)**

MLF-Q126 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 23 Feb 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider a 3×3 matrix B given by

$$B = (-1)\begin{bmatrix} -4 & 3 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & -1 \end{bmatrix}.$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 97 Question Id : 6406531114419 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

If B is diagonalizable, which of the following is a diagonal matrix D such that $B = PDP^{-1}$? Use the matrix P obtained in the previous question.

Options :

6406533776066. ✔ $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

6406533776067. ✖ $\begin{bmatrix} 4 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

6406533776068. ✖ $\begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

6406533776069. ✖ $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

6406533776070. ✖ B is not diagonalizable.

✔ **Official Answer Key: Option (A)**

MLF-Q129 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*
Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Number : 85 Question Id : 6406531114118 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question

Consider the matrix M given by:

$$M = \begin{bmatrix} 1 & 2 \\ 3 & 6 \end{bmatrix}$$

Which of the following options is true?

Options :

6406533775194. ✖ Eigenvectors corresponding to the two different eigenvalues of *M* are orthogonal.

6406533775195. ✖ The matrix *M* is not diagonalizable.

6406533775196. ✖ The eigenvalues of *M* are 4 and 1

6406533775197. ✔ The rank of *M* is 1.

✔ Official Answer Key: Option (D)

MLF-Q140 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*
Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider a 3×3 matrix B given by

$$B = (-1)\begin{bmatrix} -4 & 3 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & -1 \end{bmatrix}.$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 96 Question Id : 6406531114131 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

If B is diagonalizable, what is the invertible matrix P such that $B = PDP^{-1}$?

Options :

6406533775232. ✖ $\begin{bmatrix} 4 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

6406533775233. ✖ $\begin{bmatrix} 3 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 0 & 1 \end{bmatrix}$

6406533775234. ✖ $\begin{bmatrix} 1 & 1 & 1 \\ 2 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

6406533775235. ✔ $\begin{bmatrix} 3 & 1 & 0 \\ 2 & 1 & 0 \\ 1 & 0 & 1 \end{bmatrix}$

6406533775236. ✖ B is not diagonalizable.

✔ **Official Answer Key: Option (D)**

MLF-Q141 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 23 Feb 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider a 3×3 matrix B given by

$$B = (-1)\begin{bmatrix} -4 & 3 & 0 \\ -2 & 1 & 0 \\ -1 & 1 & -1 \end{bmatrix}.$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 97 Question Id : 6406531114132 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

If B is diagonalizable, which of the following is a diagonal matrix D such that $B = PDP^{-1}$? Use the matrix P obtained in the previous question.

Options :

6406533775237. ✔ $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

6406533775238. ✖ $\begin{bmatrix} 4 & 0 & 0 \\ 0 & -1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

6406533775239. ✖ $\begin{bmatrix} 3 & 0 & 0 \\ 0 & 2 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

6406533775240. ✖ $\begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix}$

6406533775241. ✖ B is not diagonalizable.

✔ Official Answer Key: Option (A)

MLF-Q145 | [MCQ] | Marks: +2 | *MLF*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Number : 4 Question Id : 6406531951382
Correct Marks : 3
Question Label : Multiple Choice Question
Consider the following matrices:

$$A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}$$

$$B = \begin{bmatrix} 2 & 0 & 0 \\ 0 & 0 & 1 \\ 0 & 1 & 0 \end{bmatrix}$$

Which of the following is true?
Options :

6406536344502. ✖ A is diagonalizable, B is not diagonalizable.

6406536344503. ✔ A is not diagonalizable, B is diagonalizable.

6406536344504. ✖ Both A and B are diagonalizable.

6406536344505. ✖ Neither A nor B is diagonalizable

✔ **Official Answer Key: Option (B)**

MLF-Q151 | [MCQ] | Marks: +2 | *MLF*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Number : 10 Question Id : 6406531951390

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following is/are true ?

Options :

6406536344533. ✖ Every square matrix with real entries has at least one real eigenvalue.

6406536344534. ✖ A set of two eigenvectors corresponding to the same eigenvalue of a square matrix must be linearly dependent.

6406536344535. ✔ Given a square matrix of order n, the set of all eigenvectors for an eigenvalue λ, along with the zero vector, forms a subspace of $\mathbb{R}^n$.

6406536344536. ✔ A square matrix is not invertible if and only if 0 is one of its eigenvalues.

Sub-Section Number :	5
Sub-Section Id :	640653360284
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (C, D)

MLF-Q157 | [MCQ] | Marks: +2 | *MLF*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider the matrix A given below:

$$A = \begin{bmatrix} 1 & 5 \\ 4 & 2 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 16 Question Id : 6406531951397

Correct Marks : 2

Question Label : Short Answer Question

If λ is the smallest eigenvalue of A, enter the value of λ. Your answer should be an integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Equal

Text Areas : PlainText

Possible Answers :

-3

MLF-Q177 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 13 July 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Number : 82 Question Id : 6406531297918 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following options is/are true regarding the projection matrix?
Options :
6406534364718. ✔ It is a symmetric matrix.
6406534364719. ✖ It is an orthogonal matrix.
6406534364720. ✔ It is orthogonally diagonalizable.
6406534364721. ✖ It is always a rank 1 matrix.

✔ Official Answer Key: Option (A, C)

MLF-Q192 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 13 July 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Number : 82 Question Id : 6406531298188 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question
Which of the following options is/are true regarding the projection matrix?
Options :
6406534365489. ✔ It is a symmetric matrix.
6406534365490. ✖ It is an orthogonal matrix.
6406534365491. ✔ It is orthogonally diagonalizable.
6406534365492. ✖ It is always a rank 1 matrix.

✔ Official Answer Key: Option (A, C)

MLF-Q207 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Number : 82 Question Id : 6406531298458 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following options is/are true regarding the projection matrix?

Options :

6406534366260. ✔ It is a symmetric matrix.

6406534366261. ✖ It is an orthogonal matrix.

6406534366262. ✔ It is orthogonally diagonalizable.

6406534366263. ✖ It is always a rank 1 matrix.

✔ Official Answer Key: Option (A, C)

MLF-Q223 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 7 July 2024*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 53 Question Id : 640653825041 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following options is/are true?

Options :

6406532774799. ✔ *A* is a symmetric matrix.

6406532774800. ✖ *A* is a diagonal matrix.

6406532774801. ✔ *A* is orthogonally diagonalizable.

6406532774802. ✔ Eigenvalues of *A* are -1, -1 and 2.

✔ **Official Answer Key: Option (A, C, D)**

MLF-Q236 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 7 July 2024*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 53 Question Id : 640653827510 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following options is/are true?

Options :

6406532781323. ✔ *A* is a symmetric matrix.

6406532781324. ✖ *A* is a diagonal matrix.

6406532781325. ✔ *A* is orthogonally diagonalizable.

6406532781326. ✔ Eigenvalues of *A* are -1, -1 and 2.

✔ **Official Answer Key: Option (A, C, D)**

MLF-Q249 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 7 July 2024(1) (+1 more papers)*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Let $A = \begin{pmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 1 & 1 & 0 \end{pmatrix}$ be a 3×3 matrix. Use this information to answer the given subquestions

Sub questions

Question Number : 53 Question Id : 640653827798 Question Type : MSQ

Correct Marks : 4 Max. Selectable Options : 0

Question Label : Multiple Select Question

Which of the following options is/are true?

Options :

6406532782119. ✔ *A* is a symmetric matrix.

6406532782120. ✖ *A* is a diagonal matrix.

6406532782121. ✔ *A* is orthogonally diagonalizable.

6406532782122. ✔ Eigenvalues of *A* are -1, -1 and 2.

✔ **Official Answer Key: Option (A, C, D)**

MLF-Q271 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Id : 6406531527859 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Number : 122 Question Id : 6406531527858 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Which of the following is true?
Options :
6406535115716. ✔ *A* does not have real eigenvalues
6406535115717. ✖ *A* has two distinct real eigenvalues
6406535115718. ✖ *A* has two real eigenvalues, not necessarily distinct

Question Id : 6406531527859 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (123 to 125)

✔ Official Answer Key: Option (A)

MLF-Q272 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 123 Question Id : 6406531527860 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
If a square matrix A is diagonalizable, then A is invertible.
Options :
6406535115719. ✖ True
6406535115720. ✔ False

✔ Official Answer Key: Option (B)

MLF-Q273 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 124 Question Id : 6406531527861 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The eigenvalues of a real symmetric matrix are real.

Options :

6406535115721. ✔ True

6406535115722. ✖ False

✔ Official Answer Key: Option (A)

MLF-Q274 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 125 Question Id : 6406531527862 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
A square matrix *A* of shape *n* x *n* is diagonalizable if and only if it has *n* distinct eigenvalues.
Options :
6406535115723. ✖ True
6406535115724. ✔ False

Sub-Section Number :	8
Sub-Section Id :	640653244928
Question Shuffling Allowed :	No

Question Id : 6406531527849 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (126 to 128)
Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.
Sub questions

✔ **Official Answer Key: Option (B)**

MLF-Q290 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Id : 6406531528162 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Number : 122 Question Id : 6406531528161 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Which of the following is true?
Options :
6406535116559. ✔ *A* does not have real eigenvalues
6406535116560. ✖ *A* has two distinct real eigenvalues
6406535116561. ✖ *A* has two real eigenvalues, not necessarily distinct

Question Id : 6406531528162 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (123 to 125)

✔ Official Answer Key: Option (A)

MLF-Q291 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 123 Question Id : 6406531528163 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
If a square matrix *A* is diagonalizable, then *A* is invertible.
Options :
6406535116562. ✖ True
6406535116563. ✔ False

✔ Official Answer Key: Option (B)

MLF-Q292 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 124 Question Id : 6406531528164 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The eigenvalues of a real symmetric matrix are real.

Options :

6406535116564. ✔ True

6406535116565. ✖ False

✔ Official Answer Key: Option (A)

MLF-Q293 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 125 Question Id : 6406531528165 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
A square matrix *A* of shape *n* x *n* is diagonalizable if and only if it has *n* distinct eigenvalues.
Options :
6406535116566. ✖ True
6406535116567. ✔ False

Sub-Section Number :	8
Sub-Section Id :	640653245024
Question Shuffling Allowed :	No

Question Id : 6406531528152 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (126 to 128)
Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.
Sub questions

✔ **Official Answer Key: Option (B)**

MLF-Q309 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Id : 6406531528824 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Number : 122 Question Id : 6406531528823 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Which of the following is true?
Options :
6406535118490. ✔ *A* does not have real eigenvalues
6406535118491. ✖ *A* has two distinct real eigenvalues
6406535118492. ✖ *A* has two real eigenvalues, not necessarily distinct

Question Id : 6406531528824 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (123 to 125)

✔ **Official Answer Key: Option (A)**

MLF-Q310 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 123 Question Id : 6406531528825 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
If a square matrix *A* is diagonalizable, then *A* is invertible.
Options :
6406535118493. ✖ True
6406535118494. ✔ False

✔ **Official Answer Key: Option (B)**

MLF-Q311 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 124 Question Id : 6406531528826 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The eigenvalues of a real symmetric matrix are real.

Options :

6406535118495. ✔ True

6406535118496. ✖ False

✔ Official Answer Key: Option (A)

MLF-Q312 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 125 Question Id : 6406531528827 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
A square matrix *A* of shape *n* x *n* is diagonalizable if and only if it has *n* distinct eigenvalues.
Options :
6406535118497. ✖ True
6406535118498. ✔ False

Sub-Section Number :	8
Sub-Section Id :	640653245209
Question Shuffling Allowed :	No

Question Id : 6406531528814 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (126 to 128)
Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.
Sub questions

✔ **Official Answer Key: Option (B)**

MLF-Q328 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*
Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

There is no common data for this question. There are three independent sub-questions for which you have to answer True/False.

Question Number : 148 Question Id : 6406531525169 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following is true?

Options :

6406535108191. ✔ *A* does not have real eigenvalues

6406535108192. ✖ *A* has two distinct real eigenvalues

6406535108193. ✖ *A* has two real eigenvalues, not necessarily distinct

Question Id : 6406531525170 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (149 to 151)

Question Label : Comprehension

There is no common data for this question. There are three independent sub-questions for which you have to answer True/False.

Sub questions

✔ **Official Answer Key: Option (A)**

MLF-Q329 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 149 Question Id : 6406531525171 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

If a square matrix *A* is diagonalizable, then *A* is invertible.

Options :

6406535108194. ✖ True

6406535108195. ✔ False

✔ **Official Answer Key: Option (B)**

MLF-Q330 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 150 Question Id : 6406531525172 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The eigenvalues of a real symmetric matrix are real.

Options :

6406535108196. ✔ True

6406535108197. ✖ False

✔ Official Answer Key: Option (A)

MLF-Q331 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 151 Question Id : 6406531525173 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

A square matrix *A* of shape *n* x *n* is diagonalizable if and only if it has *n* distinct eigenvalues.

Options :

6406535108198. ✖ True

6406535108199. ✔ False

Sub-Section Number :	8
Sub-Section Id :	640653244048
Question Shuffling Allowed :	No

Question Id : 6406531525160 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (152 to 154)

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

✔ Official Answer Key: Option (B)

MLF-Q347 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*
Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

There is no common data for this question. There are three independent sub-questions for which you have to answer True/False.

Question Number : 148 Question Id : 6406531525796 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following is true?

Options :

6406535109968. ✔ *A* does not have real eigenvalues

6406535109969. ✖ *A* has two distinct real eigenvalues

6406535109970. ✖ *A* has two real eigenvalues, not necessarily distinct

Question Id : 6406531525797 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (149 to 151)

Question Label : Comprehension

There is no common data for this question. There are three independent sub-questions for which you have to answer True/False.

Sub questions

✔ **Official Answer Key: Option (A)**

MLF-Q348 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 149 Question Id : 6406531525798 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

If a square matrix *A* is diagonalizable, then *A* is invertible.

Options :

6406535109971. ✖ True

6406535109972. ✔ False

✔ **Official Answer Key: Option (B)**

MLF-Q349 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 150 Question Id : 6406531525799 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The eigenvalues of a real symmetric matrix are real.

Options :

6406535109973. ✔ True

6406535109974. ✖ False

✔ Official Answer Key: Option (A)

MLF-Q350 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 151 Question Id : 6406531525800 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

A square matrix *A* of shape *n* x *n* is diagonalizable if and only if it has *n* distinct eigenvalues.

Options :

6406535109975. ✖ True

6406535109976. ✔ False

Sub-Section Number :	8
Sub-Section Id :	640653244241
Question Shuffling Allowed :	No

Question Id : 6406531525787 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (152 to 154)

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

✔ Official Answer Key: Option (B)

MLF-Q366 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*
Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

There is no common data for this question. There are three independent sub-questions for which you have to answer True/False.

Question Number : 148 Question Id : 6406531526725 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following is true?

Options :

6406535112613. ✔ *A* does not have real eigenvalues

6406535112614. ✖ *A* has two distinct real eigenvalues

6406535112615. ✖ *A* has two real eigenvalues, not necessarily distinct

Question Id : 6406531526726 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (149 to 151)

Question Label : Comprehension

There is no common data for this question. There are three independent sub-questions for which you have to answer True/False.

Sub questions

✔ **Official Answer Key: Option (A)**

MLF-Q367 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 149 Question Id : 6406531526727 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

If a square matrix *A* is diagonalizable, then *A* is invertible.

Options :

6406535112616. ✖ True

6406535112617. ✔ False

✔ **Official Answer Key: Option (B)**

MLF-Q368 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 150 Question Id : 6406531526728 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The eigenvalues of a real symmetric matrix are real.

Options :

6406535112618. ✔ True

6406535112619. ✖ False

✔ Official Answer Key: Option (A)

MLF-Q369 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 151 Question Id : 6406531526729 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

A square matrix *A* of shape *n* x *n* is diagonalizable if and only if it has *n* distinct eigenvalues.

Options :

6406535112620. ✖ True

6406535112621. ✔ False

Sub-Section Number :	8
Sub-Section Id :	640653244532
Question Shuffling Allowed :	No

Question Id : 6406531526716 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (152 to 154)

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

✔ **Official Answer Key: Option (B)**

MLF-Q385 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA2 26 Oct 2025*
Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider the matrix A given below:

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

where $0 < \theta < 180°$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 147 Question Id : 6406531534294 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following is true?

Options :

6406535133801. ✔ *A does not have real eigenvalues*

6406535133802. ✖ *A has two distinct real eigenvalues*

6406535133803. ✖ *A has two real eigenvalues, not necessarily distinct*

Question Id : 6406531534295 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (148 to 150)

✔ **Official Answer Key: Option (A)**

MLF-Q386 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA2 26 Oct 2025*
Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 148 Question Id : 6406531534296 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
If a square matrix A is diagonalizable, then A is invertible.
Options :
6406535133804. ✖ True
6406535133805. ✔ False

✔ Official Answer Key: Option (B)

MLF-Q387 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA2 26 Oct 2025*
Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 149 Question Id : 6406531534297 Question Type : MCQ
Correct Marks : 1
Question Label : Multiple Choice Question
The eigenvalues of a real symmetric matrix are real.
Options :
6406535133806. ✔ True
6406535133807. ✖ False

✔ Official Answer Key: Option (A)

MLF-Q388 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA2 26 Oct 2025*
Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 150 Question Id : 6406531534298 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
A square matrix *A* of shape *n* x *n* is diagonalizable if and only if it has *n* distinct eigenvalues.
Options :
6406535133808. ✖ True
6406535133809. ✔ False

Sub-Section Number :	8
Sub-Section Id :	640653246977
Question Shuffling Allowed :	No

Question Id : 6406531534285 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (151 to 153)
Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.
Sub questions

✔ **Official Answer Key: Option (B)**

MLF-Q404 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA3 26 Oct 2025*
Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider the matrix A given below:

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

where $0 < \theta < 180°$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 147 Question Id : 6406531534633 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following is true?

Options :

6406535134771. ✔ *A* does not have real eigenvalues

6406535134772. ✖ *A* has two distinct real eigenvalues

6406535134773. ✖ *A* has two real eigenvalues, not necessarily distinct

Question Id : 6406531534634 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix Question Numbers : (148 to 150)

✔ **Official Answer Key: Option (A)**

MLF-Q405 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA3 26 Oct 2025*
Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 148 Question Id : 6406531534635 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
If a square matrix A is diagonalizable, then A is invertible.
Options :
6406535134774. ✖ True
6406535134775. ✔ False

✔ **Official Answer Key: Option (B)**

MLF-Q406 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA3 26 Oct 2025*
Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 149 Question Id : 6406531534636 Question Type : MCQ
Correct Marks : 1
Question Label : Multiple Choice Question
The eigenvalues of a real symmetric matrix are real.
Options :
6406535134776. ✔ True
6406535134777. ✖ False

✔ **Official Answer Key: Option (A)**

MLF-Q407 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA3 26 Oct 2025*
Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 150 Question Id : 6406531534637 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
A square matrix *A* of shape *n* x *n* is diagonalizable if and only if it has *n* distinct eigenvalues.
Options :
6406535134778. ✘ True
6406535134779. ✔ False

Sub-Section Number :	8
Sub-Section Id :	640653247079
Question Shuffling Allowed :	No

Question Id : 6406531534624 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (151 to 153)
Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.
Sub questions

✔ **Official Answer Key: Option (B)**

MLF-Q423 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA4 26 Oct 2025*
Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider the matrix A given below:

$$A = \begin{bmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{bmatrix}$$

where $0 < \theta < 180°$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 147 Question Id : 6406531535070 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following is true?

Options :

6406535135917. ✔ *A does not have real eigenvalues*

6406535135918. ✖ *A has two distinct real eigenvalues*

6406535135919. ✖ *A has two real eigenvalues, not necessarily distinct*

Question Id : 6406531535071 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (148 to 150)

✔ **Official Answer Key: Option (A)**

MLF-Q424 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA4 26 Oct 2025*
Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 148 Question Id : 6406531535072 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
If a square matrix A is diagonalizable, then A is invertible.
Options :
6406535135920. ✖ True
6406535135921. ✔ False

✔ **Official Answer Key: Option (B)**

MLF-Q425 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA4 26 Oct 2025*
Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 149 Question Id : 6406531535073 Question Type : MCQ
Correct Marks : 1
Question Label : Multiple Choice Question
The eigenvalues of a real symmetric matrix are real.
Options :
6406535135922. ✔ True
6406535135923. ✖ False

✔ **Official Answer Key: Option (A)**

MLF-Q426 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA4 26 Oct 2025*
Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 150 Question Id : 6406531535074 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
A square matrix *A* of shape *n* x *n* is diagonalizable if and only if it has *n* distinct eigenvalues.
Options :
6406535135924. ✖ True
6406535135925. ✔ False

Sub-Section Number :	8
Sub-Section Id :	640653247209
Question Shuffling Allowed :	No

Question Id : 6406531535061 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (151 to 153)
Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.
Sub questions

✔ **Official Answer Key: Option (B)**

MLF-Q442 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*
Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

There is no common data for this question. There are three independent sub-questions for which you have to answer True/False.

Question Number : 141 Question Id : 6406531535806 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following is true?

Options :

6406535138097. ✔ *A* does not have real eigenvalues

6406535138098. ✖ *A* has two distinct real eigenvalues

6406535138099. ✖ *A* has two real eigenvalues, not necessarily distinct

Question Id : 6406531535807 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (142 to 144)

Question Label : Comprehension

There is no common data for this question. There are three independent sub-questions for which you have to answer True/False.

Sub questions

✔ **Official Answer Key: Option (A)**

MLF-Q443 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 142 Question Id : 6406531535808 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

If a square matrix *A* is diagonalizable, then *A* is invertible.

Options :

6406535138100. ✖ True

6406535138101. ✔ False

✔ **Official Answer Key: Option (B)**

MLF-Q444 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 143 Question Id : 6406531535809 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The eigenvalues of a real symmetric matrix are real.

Options :

6406535138102. ✔ True

6406535138103. ✖ False

✔ Official Answer Key: Option (A)

MLF-Q445 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 144 Question Id : 6406531535810 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

A square matrix *A* of shape *n* x *n* is diagonalizable if and only if it has *n* distinct eigenvalues.

Options :

6406535138104. ✖ True

6406535138105. ✔ False

Sub-Section Number :	8
Sub-Section Id :	640653247435
Question Shuffling Allowed :	No

Question Id : 6406531535797 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (145 to 147)

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

✔ **Official Answer Key: Option (B)**

MLF-Q461 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*
Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

There is no common data for this question. There are three independent sub-questions for which you have to answer True/False.

Question Number : 141 Question Id : 6406531536146 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following is true?

Options :

6406535139197. ✔ *A* does not have real eigenvalues

6406535139198. ✖ *A* has two distinct real eigenvalues

6406535139199. ✖ *A* has two real eigenvalues, not necessarily distinct

Question Id : 6406531536147 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (142 to 144)

Question Label : Comprehension

There is no common data for this question. There are three independent sub-questions for which you have to answer True/False.

Sub questions

✔ **Official Answer Key: Option (A)**

MLF-Q462 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 142 Question Id : 6406531536148 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

If a square matrix *A* is diagonalizable, then *A* is invertible.

Options :

6406535139200. ✖ True

6406535139201. ✔ False

✔ **Official Answer Key: Option (B)**

MLF-Q463 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 143 Question Id : 6406531536149 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The eigenvalues of a real symmetric matrix are real.

Options :

6406535139202. ✔ True

6406535139203. ✖ False

✔ Official Answer Key: Option (A)

MLF-Q464 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 144 Question Id : 6406531536150 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

A square matrix *A* of shape *n* x *n* is diagonalizable if and only if it has *n* distinct eigenvalues.

Options :

6406535139204. ✖ True

6406535139205. ✔ False

Sub-Section Number :	8
Sub-Section Id :	640653247537
Question Shuffling Allowed :	No

Question Id : 6406531536137 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (145 to 147)

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

✔ **Official Answer Key: Option (B)**

MLF-Q480 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

There is no common data for this question. There are three independent sub-questions for which you have to answer True/False.

Question Number : 141 Question Id : 6406531536436 Question Type : MCQ

Correct Marks : 3

Question Label : Multiple Choice Question

Which of the following is true?

Options :

6406535140028. ✔ *A* does not have real eigenvalues

6406535140029. ✖ *A* has two distinct real eigenvalues

6406535140030. ✖ *A* has two real eigenvalues, not necessarily distinct

Question Id : 6406531536437 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (142 to 144)

Question Label : Comprehension

There is no common data for this question. There are three independent sub-questions for which you have to answer True/False.

Sub questions

✔ **Official Answer Key: Option (A)**

MLF-Q481 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 142 Question Id : 6406531536438 Question Type : MCQ

Correct Marks : 2

Question Label : Multiple Choice Question

If a square matrix *A* is diagonalizable, then *A* is invertible.

Options :

6406535140031. ✖ True

6406535140032. ✔ False

✔ **Official Answer Key: Option (B)**

MLF-Q482 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 143 Question Id : 6406531536439 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The eigenvalues of a real symmetric matrix are real.

Options :

6406535140033. ✔ True

6406535140034. ✖ False

✔ Official Answer Key: Option (A)

MLF-Q483 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 144 Question Id : 6406531536440 Question Type : MCQ
Correct Marks : 2

Question Label : Multiple Choice Question

A square matrix *A* of shape *n* x *n* is diagonalizable if and only if it has *n* distinct eigenvalues.

Options :

6406535140035. ✖ True

6406535140036. ✔ False

Sub-Section Number :	8
Sub-Section Id :	640653247639
Question Shuffling Allowed :	No

Question Id : 6406531536427 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (145 to 147)

Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.

Sub questions

✔ **Official Answer Key: Option (B)**

MLF-Q499 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Id : 6406531527050 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Number : 145 Question Id : 6406531527049 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Which of the following is true?
Options :
6406535113490. ✔ *A* does not have real eigenvalues
6406535113491. ✖ *A* has two distinct real eigenvalues
6406535113492. ✖ *A* has two real eigenvalues, not necessarily distinct

Question Id : 6406531527050 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (146 to 148)

✔ **Official Answer Key: Option (A)**

MLF-Q500 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 146 Question Id : 6406531527051 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
If a square matrix *A* is diagonalizable, then *A* is invertible.
Options :
6406535113493. ✖ True
6406535113494. ✔ False

✔ **Official Answer Key: Option (B)**

MLF-Q501 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 147 Question Id : 6406531527052 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The eigenvalues of a real symmetric matrix are real.

Options :

6406535113495. ✔ True

6406535113496. ✖ False

✔ Official Answer Key: Option (A)

MLF-Q502 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 148 Question Id : 6406531527053 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
A square matrix *A* of shape *n* x *n* is diagonalizable if and only if it has *n* distinct eigenvalues.
Options :
6406535113497. ✖ True
6406535113498. ✔ False

Sub-Section Number :	8
Sub-Section Id :	640653244636
Question Shuffling Allowed :	No

Question Id : 6406531527040 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (149 to 151)
Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.
Sub questions

✔ **Official Answer Key: Option (B)**

MLF-Q518 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Id : 6406531527329 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Number : 145 Question Id : 6406531527328 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Which of the following is true?
Options :
6406535114248. ✔ *A* does not have real eigenvalues
6406535114249. ✖ *A* has two distinct real eigenvalues
6406535114250. ✖ *A* has two real eigenvalues, not necessarily distinct

Question Id : 6406531527329 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (146 to 148)

✔ Official Answer Key: Option (A)

MLF-Q519 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 146 Question Id : 6406531527330 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
If a square matrix *A* is diagonalizable, then *A* is invertible.
Options :
6406535114251. ✖ True
6406535114252. ✔ False

✔ Official Answer Key: Option (B)

MLF-Q520 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 147 Question Id : 6406531527331 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The eigenvalues of a real symmetric matrix are real.

Options :

6406535114253. ✔ True

6406535114254. ✖ False

✔ Official Answer Key: Option (A)

MLF-Q521 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 148 Question Id : 6406531527332 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
A square matrix *A* of shape *n* x *n* is diagonalizable if and only if it has *n* distinct eigenvalues.
Options :
6406535114255. ✖ True
6406535114256. ✔ False

Sub-Section Number :	8
Sub-Section Id :	640653244740
Question Shuffling Allowed :	No

Question Id : 6406531527319 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (149 to 151)
Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.
Sub questions

✔ **Official Answer Key: Option (B)**

MLF-Q537 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Id : 6406531527608 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Number : 145 Question Id : 6406531527607 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Which of the following is true?
Options :
6406535115006. ✔ *A* does not have real eigenvalues
6406535115007. ✖ *A* has two distinct real eigenvalues
6406535115008. ✖ *A* has two real eigenvalues, not necessarily distinct

Question Id : 6406531527608 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix

Question Numbers : (146 to 148)

✔ Official Answer Key: Option (A)

MLF-Q538 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 146 Question Id : 6406531527609 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
If a square matrix *A* is diagonalizable, then *A* is invertible.
Options :
6406535115009. ✖ True
6406535115010. ✔ False

✔ Official Answer Key: Option (B)

MLF-Q539 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 147 Question Id : 6406531527610 Question Type : MCQ

Correct Marks : 1

Question Label : Multiple Choice Question

The eigenvalues of a real symmetric matrix are real.

Options :

6406535115011. ✔ True

6406535115012. ✖ False

✔ Official Answer Key: Option (A)

MLF-Q540 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Label : Comprehension

Question Number : 148 Question Id : 6406531527611 Question Type : MCQ
Correct Marks : 2
Question Label : Multiple Choice Question
A square matrix *A* of shape *n* x *n* is diagonalizable if and only if it has *n* distinct eigenvalues.
Options :
6406535115013. ✖ True
6406535115014. ✔ False

Sub-Section Number :	8
Sub-Section Id :	640653244844
Question Shuffling Allowed :	No

Question Id : 6406531527598 Question Type : COMPREHENSION Sub Question Shuffling Allowed : No Group Comprehension Questions : No Question Pattern Type : NonMatrix
Question Numbers : (149 to 151)
Question Label : Comprehension

Consider $f : \mathbb{R}^2 \to \mathbb{R}$ and $x = (x_1, x_2)$:

$$f(x) = e^{x_1} + x_2$$

Let $L_{(0,0)}[f](x)$ be the linear approximation of f around the point $(0, 0)$.

Based on the above data, answer the given subquestions.
Sub questions

✔ **Official Answer Key: Option (B)**

MLF-Q594 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Number : 3 Question Id : 640653995428 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a matrix $B = A^4 - 3A - 2I$, where $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}$. Which of the following options is/are true?

Options :

6406533360589. ✔ *B* has eigenvalues -4, 8, 70.

6406533360590. ✖ *A* has eigenvalues 1, 16, 81.

6406533360591. ✔ *B* is an upper triangular matrix.

6406533360592. ✖ *B* is a lower triangular matrix.

Sub-Section Number :	3
Sub-Section Id :	640653146771
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A, C)

MLF-Q607 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Number : 3 Question Id : 640653995710 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a matrix $B = A^4 - 3A - 2I$, where $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}$. Which of the following options is/are true?

Options :

6406533361373. ✔ *B* has eigenvalues -4, 8, 70.

6406533361374. ✖ *A* has eigenvalues 1, 16, 81.

6406533361375. ✔ *B* is an upper triangular matrix.

6406533361376. ✖ *B* is a lower triangular matrix.

Sub-Section Number :	3
Sub-Section Id :	640653146860
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A, C)

MLF-Q620 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4*

Category: Week 4 - Regression & Eigenvalues › Eigenvalues

Question Number : 3 Question Id : 640653995992 Question Type : MSQ

Correct Marks : 3 Max. Selectable Options : 0

Question Label : Multiple Select Question

Consider a matrix $B = A^4 - 3A - 2I$, where $A = \begin{bmatrix} 1 & 1 & 1 \\ 0 & 2 & 1 \\ 0 & 0 & 3 \end{bmatrix}$. Which of the following options is/are true?

Options :

6406533362157. ✔ *B* has eigenvalues -4, 8, 70.

6406533362158. ✖ *A* has eigenvalues 1, 16, 81.

6406533362159. ✔ *B* is an upper triangular matrix.

6406533362160. ✖ *B* is a lower triangular matrix.

Sub-Section Number :	3
Sub-Section Id :	640653146949
Question Shuffling Allowed :	Yes

✔ Official Answer Key: Option (A, C)

■ Category: Eigenvectors (1 Questions)

MLF-Q158 | [MCQ] | Marks: +2 | *MLF*

Category: Week 4 - Regression & Eigenvalues › Eigenvectors

Question Label : Comprehension

Consider the matrix A given below:

$$A = \begin{bmatrix} 1 & 5 \\ 4 & 2 \end{bmatrix}$$

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 17 Question Id : 6406531951398
Correct Marks : 2
Question Label : Short Answer Question
If $v = (a, -1)$ is an eigenvector of A corresponding to λ, find a. Enter your answer correct to two decimal places.
Response Type : Numeric
Evaluation Required For SA : Yes
Show Word Count : Yes
Answers Type : Range
Text Areas : PlainText
Possible Answers :
1.24 to 1.26

■ Category: Error Calculations (23 Questions)

MLF-Q032 | [SA] | Marks: +2 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*
Category: Week 4 - Regression & Eigenvalues › Error Calculations

Question Number : 124 Question Id : 640653345259 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Short Answer Question

$f(x_1, x_2, x_3) = \frac{x_1+x_2+x_3}{3}$ is used as encoder function and $g(u) = [u, 2u, 3u]$ is used as decoder function for dimensionality reduction of following data set.

X
[1,2,3]
[2,3,4]

Give the reconstruction error for this encoder decoder pair. The reconstruction error is the mean of the squared distance between the reconstructed input and input.

NOTE: Enter your answer to the nearest integer.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

24 to 25

✔ **Official Answer Key (NAT): 24 to 25**

MLF-Q146 | [MCQ] | Marks: +2 | *MLF*

Category: Week 4 - Regression & Eigenvalues › Error Calculations

Question Number : 5 Question Id : 6406531951383
Correct Marks : 3
Question Label : Multiple Choice Question

Consider a dataset D for a regression problem where each element is of the form (x_i, y_i), for $i = 1, 2, 3$. x_i is the feature and y_i is the label of the i^{th} data-point.

$$D = \{(1, 1), (2, 4), (3, 7)\}$$

Let $f_1, f_2 : \mathbb{R} \to \mathbb{R}$ be two regression models such that:

$$f_1(x) = 2x - 1, \; f_2(x) = 3x - 2$$

Which of the following options is correct?
Options :

6406536344506. ✖ $f_1(x)$ has the lower mean squared loss.

6406536344507. ✔ $f_2(x)$ has the lower mean squared loss.

6406536344508. ✖ Both models have same mean squared loss.

✔ **Official Answer Key: Option (B)**

MLF-Q181 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 13 July 2025*
Category: Week 4 - Regression & Eigenvalues › Error Calculations

Question Number : 86 Question Id : 6406531297925 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

The table below shows the original values and predicted values obtained by some machine learning algorithm for multiclass classification problems. Compute the squared error loss and 0-1 loss. Which of the following options regarding the loss Function is/are correct?

Original values	−1	0	1	2	3	1	0	3
Predicted values	1	1	1	2	1	0	0	3

Options :

6406534364743. ✔ Square error loss ≤ 0-1 loss

6406534364744. ✖ Square error loss ≥ 0-1 loss

6406534364745. ✖ The 0-1 loss is 0.25

6406534364746. ✔ The square error loss is 1.25

✔ Official Answer Key: Option (A, D)

MLF-Q196 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 13 July 2025*
Category: Week 4 - Regression & Eigenvalues › Error Calculations

Question Number : 86 Question Id : 6406531298195 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

The table below shows the original values and predicted values obtained by some machine learning algorithm for multiclass classification problems. Compute the squared error loss and 0-1 loss. Which of the following options regarding the loss Function is/are correct?

Original values	−1	0	1	2	3	1	0	3
Predicted values	1	1	1	2	1	0	0	3

Options :

6406534365514. ✔ Square error loss ≤ 0-1 loss

6406534365515. ✖ Square error loss ≥ 0-1 loss

6406534365516. ✖ The 0-1 loss is 0.25

6406534365517. ✔ The square error loss is 1.25

✔ Official Answer Key: Option (A, D)

MLF-Q211 | [MSQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 13 July 2025*
Category: Week 4 - Regression & Eigenvalues › Error Calculations

Question Number : 86 Question Id : 6406531298465 Question Type : MSQ
Correct Marks : 3 Max. Selectable Options : 0
Question Label : Multiple Select Question

The table below shows the original values and predicted values obtained by some machine learning algorithm for multiclass classification problems. Compute the squared error loss and 0-1 loss. Which of the following options regarding the loss Function is/are correct?

Original values	−1	0	1	2	3	1	0	3
Predicted values	1	1	1	2	1	0	0	3

Options :

6406534366285. ✔ Square error loss ≤ 0-1 loss

6406534366286. ✖ Square error loss ≥ 0-1 loss

6406534366287. ✖ The 0-1 loss is 0.25

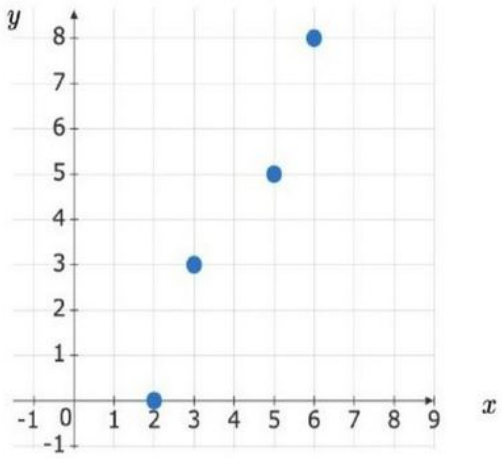
6406534366288. ✔ The square error loss is 1.25

✔ Official Answer Key: Option (A, D)

MLF-Q266 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD2 26 Oct 2025*
Category: Week 4 - Regression & Eigenvalues › Error Calculations

Question Label : Comprehension

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 117 Question Id : 6406531527847 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Which of these models has the smallest squared loss on this training dataset?

Options :

6406535115691. ✔ $f(x) = 2x - 4$

6406535115692. ✖ $f(x) = 4$

6406535115693. ✖ $f(x) = 8$

6406535115694. ✖ $f(x) = 4 - 2x$

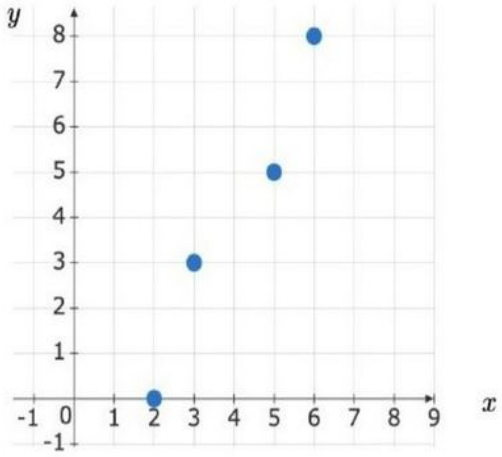
✔ Official Answer Key: Option (A)

MLF-Q285 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD3 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Error Calculations

Question Label : Comprehension

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 117 Question Id : 6406531528150 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Which of these models has the smallest squared loss on this training dataset?

Options :

6406535116534. ✔ $f(x) = 2x - 4$

6406535116535. ✖ $f(x) = 4$

6406535116536. ✖ $f(x) = 8$

6406535116537. ✖ $f(x) = 4 - 2x$

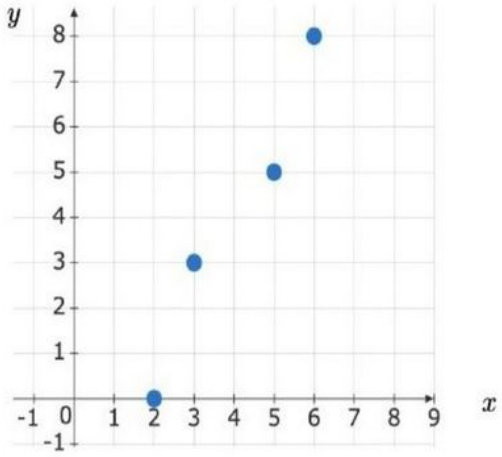
✔ Official Answer Key: Option (A)

MLF-Q304 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN EXAM QDD4 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Error Calculations

Question Label : Comprehension

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 117 Question Id : 6406531528812 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Which of these models has the smallest squared loss on this training dataset?

Options :

6406535118465. ✔ $f(x) = 2x - 4$

6406535118466. ✖ $f(x) = 4$

6406535118467. ✖ $f(x) = 8$

6406535118468. ✖ $f(x) = 4 - 2x$

✔ Official Answer Key: Option (A)

MLF-Q323 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF2 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Error Calculations

Question Label : Comprehension

Question Number : 143 Question Id : 6406531525158 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Which of these models has the smallest squared loss on this training dataset?
Options :

6406535108166. ✔ $f(x) = 2x - 4$

6406535108167. ✖ $f(x) = 4$

6406535108168. ✖ $f(x) = 8$

6406535108169. ✖ $f(x) = 4 - 2x$

✔ **Official Answer Key: Option (A)**

MLF-Q342 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF3 26 Oct 2025*
Category: Week 4 - Regression & Eigenvalues › Error Calculations

Question Label : Comprehension

Question Number : 143 Question Id : 6406531525785 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Which of these models has the smallest squared loss on this training dataset?
Options :

6406535109943. ✔ $f(x) = 2x - 4$

6406535109944. ✖ $f(x) = 4$

6406535109945. ✖ $f(x) = 8$

6406535109946. ✖ $f(x) = 4 - 2x$

✔ **Official Answer Key: Option (A)**

MLF-Q361 | [MCQ] | Marks: +2 | *IIT M FOUNDATION AN EXAM QDF4 26 Oct 2025*
Category: Week 4 - Regression & Eigenvalues › Error Calculations

Question Label : Comprehension

Question Number : 143 Question Id : 6406531526714 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Which of these models has the smallest squared loss on this training dataset?
Options :

6406535112588. ✔ $f(x) = 2x - 4$

6406535112589. ✖ $f(x) = 4$

6406535112590. ✖ $f(x) = 8$

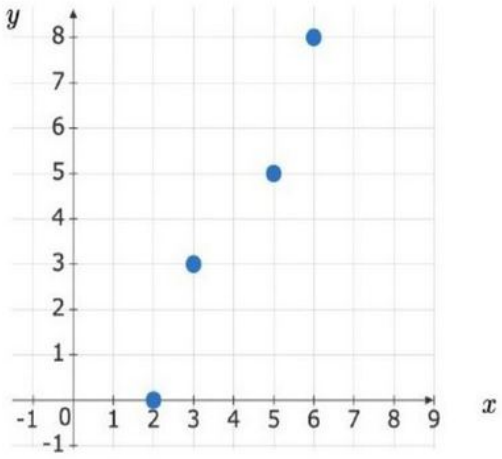
6406535112591. ✖ $f(x) = 4 - 2x$

✔ **Official Answer Key: Option (A)**

MLF-Q380 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA2 26 Oct 2025*
Category: Week 4 - Regression & Eigenvalues › Error Calculations

Question Label : Comprehension

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 142 Question Id : 6406531534283 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Which of these models has the smallest squared loss on this training dataset?

Options :

6406535133776. ✔ $f(x) = 2x - 4$

6406535133777. ✖ $f(x) = 4$

6406535133778. ✖ $f(x) = 8$

6406535133779. ✖ $f(x) = 4 - 2x$

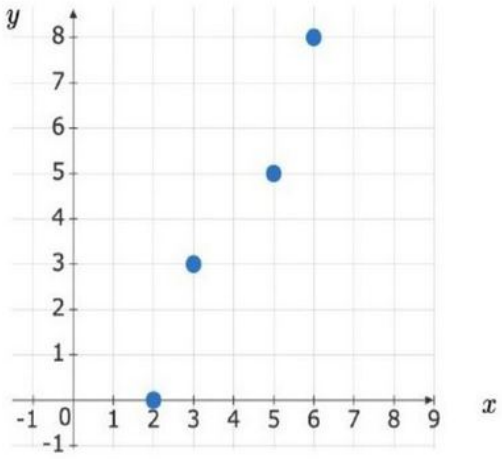
✔ Official Answer Key: Option (A)

MLF-Q399 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA3 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Error Calculations

Question Label : Comprehension

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 142 Question Id : 6406531534622 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Which of these models has the smallest squared loss on this training dataset?

Options :

6406535134746. ✔ $f(x) = 2x - 4$

6406535134747. ✖ $f(x) = 4$

6406535134748. ✖ $f(x) = 8$

6406535134749. ✖ $f(x) = 4 - 2x$

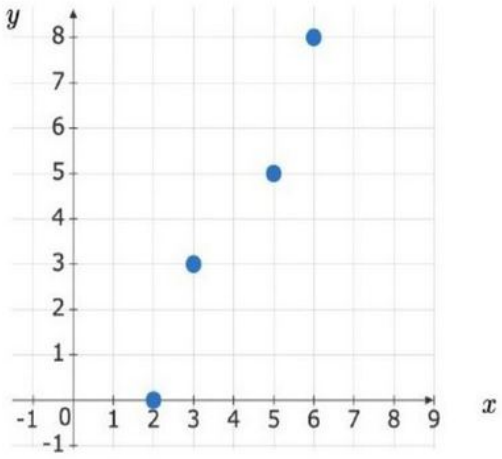
✔ Official Answer Key: Option (A)

MLF-Q418 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIA4 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Error Calculations

Question Label : Comprehension

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 142 Question Id : 6406531535059 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Which of these models has the smallest squared loss on this training dataset?

Options :

6406535135892. ✔ $f(x) = 2x - 4$

6406535135893. ✖ $f(x) = 4$

6406535135894. ✖ $f(x) = 8$

6406535135895. ✖ $f(x) = 4 - 2x$

✔ Official Answer Key: Option (A)

MLF-Q437 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB2 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Error Calculations

Question Label : Comprehension

Question Number : 136 Question Id : 6406531535795 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Which of these models has the smallest squared loss on this training dataset?
Options :

6406535138072. ✔ $f(x) = 2x - 4$

6406535138073. ✖ $f(x) = 4$

6406535138074. ✖ $f(x) = 8$

6406535138075. ✖ $f(x) = 4 - 2x$

✔ **Official Answer Key: Option (A)**

MLF-Q456 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB3 26 Oct 2025*
Category: Week 4 - Regression & Eigenvalues › Error Calculations

Question Label : Comprehension

Question Number : 136 Question Id : 6406531536135 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Which of these models has the smallest squared loss on this training dataset?
Options :

6406535139172. ✔ $f(x) = 2x - 4$

6406535139173. ✖ $f(x) = 4$

6406535139174. ✖ $f(x) = 8$

6406535139175. ✖ $f(x) = 4 - 2x$

✔ **Official Answer Key: Option (A)**

MLF-Q475 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIB4 26 Oct 2025*
Category: Week 4 - Regression & Eigenvalues › Error Calculations

Question Label : Comprehension

Question Number : 136 Question Id : 6406531536425 Question Type : MCQ
Correct Marks : 3
Question Label : Multiple Choice Question
Which of these models has the smallest squared loss on this training dataset?
Options :

6406535140003. ✔ $f(x) = 2x - 4$

6406535140004. ✖ $f(x) = 4$

6406535140005. ✖ $f(x) = 8$

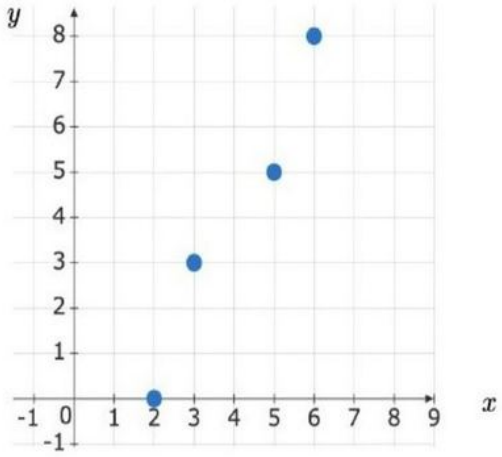
6406535140006. ✖ $f(x) = 4 - 2x$

✔ **Official Answer Key: Option (A)**

MLF-Q494 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM2 26 Oct 2025*
Category: Week 4 - Regression & Eigenvalues › Error Calculations

Question Label : Comprehension

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 140 Question Id : 6406531527038 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Which of these models has the smallest squared loss on this training dataset?

Options :

6406535113465. ✔ $f(x) = 2x - 4$

6406535113466. ✖ $f(x) = 4$

6406535113467. ✖ $f(x) = 8$

6406535113468. ✖ $f(x) = 4 - 2x$

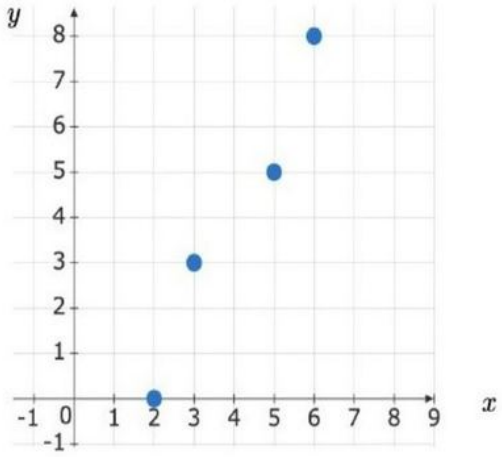
✔ Official Answer Key: Option (A)

MLF-Q513 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM3 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Error Calculations

Question Label : Comprehension

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 140 Question Id : 6406531527317 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Which of these models has the smallest squared loss on this training dataset?

Options :

6406535114223. ✔ $f(x) = 2x - 4$

6406535114224. ✖ $f(x) = 4$

6406535114225. ✖ $f(x) = 8$

6406535114226. ✖ $f(x) = 4 - 2x$

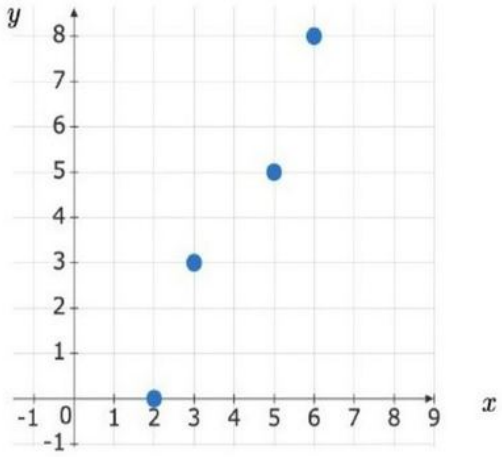
✔ Official Answer Key: Option (A)

MLF-Q532 | [MCQ] | Marks: +2 | *IIT M IMPROVEMENT AN EXAM QIM4 26 Oct 2025*

Category: Week 4 - Regression & Eigenvalues › Error Calculations

Question Label : Comprehension

Consider a regression problem that has 4 points in its training dataset as displayed below. The graph is drawn to scale.



Based on the above data, answer the given subquestions.

Sub questions

Question Number : 140 Question Id : 6406531527596 Question Type : MCQ
Correct Marks : 3

Question Label : Multiple Choice Question

Which of these models has the smallest squared loss on this training dataset?

Options :

6406535114981. ✔ $f(x) = 2x - 4$

6406535114982. ✖ $f(x) = 4$

6406535114983. ✖ $f(x) = 8$

6406535114984. ✖ $f(x) = 4 - 2x$

✔ Official Answer Key: Option (A)

MLF-Q546 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct*

Category: Week 4 - Regression & Eigenvalues › Error Calculations

Question Number : 120 Question Id : 640653668560 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Consider the following input data points:

X	Y
$[0, 1]$	$+1$
$[0, 2]$	$+1$
$[1, 3]$	$+1$
$[3, 4]$	-1
$[4, 5]$	-1
$[7, 6]$	-1

Which among the following models has the least misclassification error?

Options :

6406532239877. ✖ $Sign(x_1 - 2)$

6406532239878. ✖ $Sign(x_2 + 2)$

6406532239879. ✖ $Sign(x_1 + x_2)$

6406532239880. ✔ $Sign(x_2 - x_1)$

✔ **Official Answer Key: Option (D)**

MLF-Q562 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QPD3 29 Oct*

Category: Week 4 - Regression & Eigenvalues › Error Calculations

Question Number : 120 Question Id : 640653668921 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Consider the following input data points:

X	Y
[0, 1]	+1
[0, 2]	+1
[1, 3]	+1
[3, 4]	-1
[4, 5]	-1
[7, 6]	-1

Which among the following models has the least misclassification error?

Options :

6406532240973. ✖ $Sign(x_1 - 2)$

6406532240974. ✖ $Sign(x_2 + 2)$

6406532240975. ✖ $Sign(x_1 + x_2)$

6406532240976. ✔ $Sign(x_2 - x_1)$

✔ Official Answer Key: Option (D)

MLF-Q578 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QPD4 29 Oct*

Category: Week 4 - Regression & Eigenvalues › Error Calculations

Question Number : 120 Question Id : 640653669182 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 2

Question Label : Multiple Choice Question

Consider the following input data points:

X	Y
[0, 1]	+1
[0, 2]	+1
[1, 3]	+1
[3, 4]	-1
[4, 5]	-1
[7, 6]	-1

Which among the following models has the least misclassification error?

Options :

6406532241725. ✖ $Sign(x_1 - 2)$

6406532241726. ✖ $Sign(x_2 + 2)$

6406532241727. ✖ $Sign(x_1 + x_2)$

6406532241728. ✔ $Sign(x_2 - x_1)$

✔ Official Answer Key: Option (D)

■ Category: Least Squares (11 Questions)

MLF-Q044 | [MCQ] | Marks: +2 | *IIT M DIPLOMA QUIZ1 EXAM QPF1*

Category: Week 4 - Regression & Eigenvalues › Least Squares

Question Number : 136 Question Id : 640653345268 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Consider the data set with input x and output y as follows.

x	y
1	2
2	1
1	3
3	4

For the above given data set Raj wants to compute a least square approximation that best fits the dataset with minimum error. Which of the following is the correct equation for the best fit line computed by Raj?

Options :

6406531148416. ✖ $\hat{y} = 0.412x + 0.215$

6406531148417. ✖ $\hat{y} = 1.335x + 2.245$

6406531148418. ✔ $\hat{y} = 0.545x + 1.545$

6406531148419. ✖ $\hat{y} = 1.713x + 1.175$

6406531148420. ✖ $\hat{y} = 2.225x + 3.665$

✔ Official Answer Key: Option (C)

MLF-Q066 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QDD2 25 Feb 2024*

Category: Week 4 - Regression & Eigenvalues › Least Squares

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 139 Question Id : 640653737418 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Find the best fit line for the dataset given below using least squares method.

x	0	2	3	6	8
y	-2	5	3	1	-3

Options :

6406532467988. ✖ $\hat{y} = 0.6x - 3.2$

6406532467989. ✔ $\hat{y} = -0.6x + 3.2$

6406532467990. ✖ $\hat{y} = -x + 2.8$

6406532467991. ✖ $\hat{y} = -0.4x + 2.8$

✔ **Official Answer Key: Option (B)**

MLF-Q078 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QDD3 25 Feb 2024*

Category: Week 4 - Regression & Eigenvalues › Least Squares

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 139 Question Id : 640653737704 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Find the best fit line for the dataset given below using least squares method.

x	0	2	3	6	8
y	-2	5	3	1	-3

Options :

6406532468812. ✖ $\hat{y} = 0.6x - 3.2$

6406532468813. ✔ $\hat{y} = -0.6x + 3.2$

6406532468814. ✖ $\hat{y} = -x + 2.8$

6406532468815. ✖ $\hat{y} = -0.4x + 2.8$

✔ **Official Answer Key: Option (B)**

MLF-Q090 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QDD4 25 Feb 2024*

Category: Week 4 - Regression & Eigenvalues › Least Squares

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & 2 & 3 \\ -1 & -2 & -3 \\ 2 & 5 & 6 \end{pmatrix}$ be an 3 x 3 matrix. Based on this information, answer the given subquestions:

Sub questions

Question Number : 139 Question Id : 640653737990 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 4

Question Label : Multiple Choice Question

Find the best fit line for the dataset given below using least squares method.

x	0	2	3	6	8
y	-2	5	3	1	-3

Options :

6406532469636. ✖ $\hat{y} = 0.6x - 3.2$

6406532469637. ✔ $\hat{y} = -0.6x + 3.2$

6406532469638. ✖ $\hat{y} = -x + 2.8$

6406532469639. ✖ $\hat{y} = -0.4x + 2.8$

✔ **Official Answer Key: Option (B)**

MLF-Q174 | [SA] | Marks: +2 | *Quiz1 May23*

Category: Week 4 - Regression & Eigenvalues › Least Squares

Question Label : Comprehension

Let $A = \begin{pmatrix} 1 & -1 \\ 1 & 1 \\ 1 & 2 \end{pmatrix}$ and $b = \begin{pmatrix} 1 \\ 1 \\ 3 \end{pmatrix}$.

Based on the above data, answer the given subquestions.

Sub questions

Question Number : 134 Question Id : 640653577882 Question Type : SA Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Short Answer Question

Using the least square approximation, find the length of the projection vector p of b onto the column space of A. Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

3.12 to 3.16

✔ **Official Answer Key (NAT): 3.12 to 3.16**

MLF-Q558 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct*

Category: Week 4 - Regression & Eigenvalues › Least Squares

Question Label : Comprehension

Consider a subspace $U \subseteq \mathbb{R}^3$ and a vector $b \subseteq \mathbb{R}^3$, where

$$U = \text{span}\left\{\begin{bmatrix}1\\1\\2\end{bmatrix}, \begin{bmatrix}1\\-1\\1\end{bmatrix}\right\}, \quad b = \begin{bmatrix}2\\0\\-4\end{bmatrix}$$

Based on the above data, answer the given subquestions.

Question Number : 132 Question Id : 640653668569 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Find the projection of *b* onto U.

Options :

6406532239898. ✖ $\begin{bmatrix}-1\\-1\\2\end{bmatrix}$

6406532239899. ✔ $\begin{bmatrix}-1\\-1\\-2\end{bmatrix}$

6406532239900. ✖ $\begin{bmatrix}1\\1\\-2\end{bmatrix}$

6406532239901. ✖ $\begin{bmatrix}1\\-1\\-2\end{bmatrix}$

✔ Official Answer Key: Option (B)

MLF-Q559 | [SA] | Marks: +2 | *IIT M DIPLOMA AN2 EXAM QPD2 29 Oct*

Category: Week 4 - Regression & Eigenvalues › Least Squares

Question Label : Comprehension

Consider a subspace $U \subseteq \mathbb{R}^3$ and a vector $b \subseteq \mathbb{R}^3$, where

$$U = \text{span}\left\{\begin{bmatrix}1\\1\\2\end{bmatrix}, \begin{bmatrix}1\\-1\\1\end{bmatrix}\right\}, \quad b = \begin{bmatrix}2\\0\\-4\end{bmatrix}$$

Based on the above data, answer the given subquestions.

Question Number : 133 Question Id : 640653668570 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

Find the length of the error vector $e = b - p$, where p is the projection of b onto U. Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

3.70 to 3.78

✔ **Official Answer Key (NAT): 3.70 to 3.78**

MLF-Q574 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QPD3 29 Oct*

Category: Week 4 - Regression & Eigenvalues › Least Squares

Question Label : Comprehension

Consider a subspace $U \subseteq \mathbb{R}^3$ and a vector $b \subseteq \mathbb{R}^3$, where

$$U = \text{span}\left\{\begin{bmatrix}1\\1\\2\end{bmatrix}, \begin{bmatrix}1\\-1\\1\end{bmatrix}\right\}, \quad b = \begin{bmatrix}2\\0\\-4\end{bmatrix}$$

Based on the above data, answer the given subquestions.

Question Number : 132 Question Id : 640653668930 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Find the projection of *b* onto U.

Options :

6406532240994. ✖ $\begin{bmatrix}-1\\-1\\2\end{bmatrix}$

6406532240995. ✔ $\begin{bmatrix}-1\\-1\\-2\end{bmatrix}$

6406532240996. ✖ $\begin{bmatrix}1\\1\\-2\end{bmatrix}$

6406532240997. ✖ $\begin{bmatrix}1\\-1\\-2\end{bmatrix}$

✔ Official Answer Key: Option (B)

MLF-Q575 | [SA] | Marks: +2 | *IIT M DIPLOMA AN3 EXAM QPD3 29 Oct*

Category: Week 4 - Regression & Eigenvalues › Least Squares

Question Label : Comprehension

Consider a subspace $U \subseteq \mathbb{R}^3$ and a vector $b \subseteq \mathbb{R}^3$, where

$$U = \text{span}\left\{\begin{bmatrix}1\\1\\2\end{bmatrix}, \begin{bmatrix}1\\-1\\1\end{bmatrix}\right\}, \quad b = \begin{bmatrix}2\\0\\-4\end{bmatrix}$$

Based on the above data, answer the given subquestions.

Question Number : 133 Question Id : 640653668931 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

Find the length of the error vector $e = b - p$, where p is the projection of b onto U. Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

3.70 to 3.78

✔ **Official Answer Key (NAT): 3.70 to 3.78**

MLF-Q590 | [MCQ] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QPD4 29 Oct*

Category: Week 4 - Regression & Eigenvalues › Least Squares

Question Label : Comprehension

Consider a subspace $U \subseteq \mathbb{R}^3$ and a vector $b \subseteq \mathbb{R}^3$, where

$$U = \text{span}\left\{\begin{bmatrix}1\\1\\2\end{bmatrix}, \begin{bmatrix}1\\-1\\1\end{bmatrix}\right\}, \quad b = \begin{bmatrix}2\\0\\-4\end{bmatrix}$$

Based on the above data, answer the given subquestions.

Question Number : 132 Question Id : 640653669191 Question Type : MCQ Is Question Mandatory : No Calculator : None Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 3

Question Label : Multiple Choice Question

Find the projection of *b* onto U.

Options :

6406532241746. ✖ $\begin{bmatrix}-1\\-1\\2\end{bmatrix}$

6406532241747. ✔ $\begin{bmatrix}-1\\-1\\-2\end{bmatrix}$

6406532241748. ✖ $\begin{bmatrix}1\\1\\-2\end{bmatrix}$

6406532241749. ✖ $\begin{bmatrix}1\\-1\\-2\end{bmatrix}$

✔ Official Answer Key: Option (B)

MLF-Q591 | [SA] | Marks: +2 | *IIT M DIPLOMA AN4 EXAM QPD4 29 Oct*

Category: Week 4 - Regression & Eigenvalues › Least Squares

Question Label : Comprehension

Consider a subspace $U \subseteq \mathbb{R}^3$ and a vector $b \subseteq \mathbb{R}^3$, where

$$U = \text{span}\left\{\begin{bmatrix}1\\1\\2\end{bmatrix}, \begin{bmatrix}1\\-1\\1\end{bmatrix}\right\}, \quad b = \begin{bmatrix}2\\0\\-4\end{bmatrix}$$

Based on the above data, answer the given subquestions.

Question Number : 133 Question Id : 640653669192 Question Type : SA Calculator : None

Response Time : N.A Think Time : N.A Minimum Instruction Time : 0

Correct Marks : 1

Question Label : Short Answer Question

Find the length of the error vector $e = b - p$, where p is the projection of b onto U. Enter the answer correct to two decimal places.

Response Type : Numeric

Evaluation Required For SA : Yes

Show Word Count : Yes

Answers Type : Range

Text Areas : PlainText

Possible Answers :

3.70 to 3.78

✔ **Official Answer Key (NAT): 3.70 to 3.78**